Global Innovation Institute

GII Research | White Paper | World Models, Robotics AI & Restaurant Commercialization

The Restaurant as a Physical AI Platform

Location-Based Economics of Physical AI | World Models | Robotics Hardware & Software | Intelligent Kitchens | Commercialization

The Site Is the Unit of Economics. The Task Is the Unit of Learning. The Network Is the Unit of Scale.

GII Independent Research | Physical AI Commercialization

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For World Model researchers, robotics hardware/software companies, restaurant operating leaders, corporate development teams and Physical AI investors. Independent investment research; hypothesis, issuer claim, vendor demonstration and measured return are identified separately.

Executive Summary

Restaurants combine food production, dining, pickup and local delivery at a single operating site. That makes them a useful test for World Model developers and robotics companies. This report examines where Physical AI can improve the work and how restaurant networks could extend that value beyond the four walls. Buildings, franchise contracts and operating routines determine which sites qualify.

The commercial case starts with a useful task that the site can support. It grows when the next location needs less adaptation and has dependable service. Drone pilots establish bounded feasibility; customer preference, employee workload and store-manager acceptance still need to be tested. Reliable food and service, workable jobs and positive site economics must precede a broad rollout.



Picture 1 | The restaurant as an integrated Physical AI platform

AI-generated concept illustration for GII; equipment and settings are illustrative.

Reader Navigation — A Commercialization Map for World Model and Robotics AI Teams

Technical readers can begin with World Models and learned manipulation in Chapter 4, robot hardware in Chapters 6 and 9, and production coordination in Chapter 7. Chapters 1–3 establish the operating and property context; Chapters 10–15 assess deployment evidence, suppliers and commercial decisions. The three investment modules show the financial calculations. The linked contents and exhibit lists provide access to individual topics.

Table 1 | Restaurant economy and commercialization opportunity

Measure	Reported value or model	Interpretation and boundary
Restaurant & foodservice outlets	More than 1 million	Industry footprint, not immediately eligible robot locations
2026 foodservice sales	$1.36 trillion	NRA benchmark-revised U.S. forecast [S01]
Employment base	15.8 million projected	NRA 2026 workforce forecast; distinct from payroll establishment counts
Top 20 restaurant brand units	124,185	2026 QSR 50 (2025 data) [S04]
Main commercialization constraint	Task economics × adoption	Physical task feasibility is not signed customer demand
Investor underwriting benchmark	NPV versus best alternative	Five-year GII illustrative case, Module II

Source: NRA benchmark revision [S01], outlet footprint (NRA background statement) [S93] and 2026 employment forecast [S03], and QSR 50 brand data [S04]. Data periods and economic definitions are not interchangeable; industry sales do not constitute Physical AI TAM.

Table 2 | Research Chapters and Recommended Reading Path

Research section	Reader purpose and location
Executive	Executive summary, author perspective, investment thesis
Part I — The Opportunity	1 Scale of restaurant physical economy · 2 Task economics · 3 Real estate and supply-chain constraints
Part II — World To Action	4 World models · 5 Robot-learning data · 6 Intelligent kitchens/AIoT · 7 Omnichannel order orchestration
Part III — Implementation	8 Cloud and edge · 9 Robotics and employee adoption
Part IV — Market Evidence	10 Major-chain deployments · 11 Technology supplier industry structure
Part V — Value Creation	12 TAM/SAM/SOM & DCF · 13 Stakeholders and corporate development · 14 Commercialization outlook
Part VI — Builders Playbook	15 Technical-to-commercial market-entry framework
Investment Decision Tools	Module I Feasibility · II Restaurant DCF · III Supplier value capture
Supporting Evidence	Appendices A–N: original sources, model checks, rights review and 2026 technical sources

Financial-model figures and 2030 scenarios are illustrative unless explicitly described as company-reported. A source does not automatically establish causality or technology-specific revenue. All figures retain their stated periods and geographic denominators.

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Executive Summary — Commercializing Physical AI in the Restaurant Economy

Restaurants give World Model developers and robotics companies a demanding commercial test. Food must be prepared correctly, staff must manage a busy shift and customers must receive what they ordered. A useful system has to improve that work at a cost the site can support. This report examines which tasks meet that test and what changes when a supplier moves from one installation to a restaurant network.

Physical scale is a task-screening input. The National Restaurant Association reduced its 2026 U.S. restaurant and foodservice sales forecast from $1.55 trillion to $1.36 trillion after the October 2026 government benchmark revision; its October 19 release retained projected growth of 4.3% nominal and 0.8% real [S01]. The broad industry comprises over a million outlets [S93], while the Top 20 sample in the 2026 QSR 50 (2025 data) accounts for 124,185 U.S. locations [S04]. These populations have different scopes. Developers must narrow from locations to repeatable tasks, eligible formats, data rights, customer budgets and achievable adoption.

POS, ordering and loyalty systems describe demand. They rarely record the physical detail needed to teach a robot how to grasp an ingredient, recover from a spill or verify a completed task. Robot learning needs observations aligned with actions and outcomes, including interruptions and failures. Collecting those records requires operating permission and appropriate protection for employees and customers.

The robot arrives in a restaurant that already has equipment, software and established ways of working. Installation therefore includes kitchen integration, staff training, recovery procedures and continuing service. McDonald's restaurant Edge program, Chick-fil-A's Kubernetes deployment, Chipotle's automation pilots and Starbucks' order orchestration illustrate different parts of that work [S09, S10, S98, S99]. Their disclosures establish specific capabilities, rather than a general-purpose restaurant world model operating at scale.

Restaurant technology serves several businesses. Ordering platforms, kitchen scheduling, connected equipment, manipulation robots and autonomous delivery have different purchasers and cost structures. Their revenues cannot simply be added to restaurant sales, equipment investment or venture funding to produce a Physical AI market estimate. Supplier receipts and pass-through payments need particular care to avoid double counting.

Unit economics decide. GII's illustrative five-year restaurant AI model requires approximately $44,325 in mature annual gross benefits to reach zero NPV on an $80,000 initial investment under stated ramp, operating-cost, tax, refresh and discount assumptions. The estimated hurdle is neither a verified customer result nor a universal robot price. Lower-cost process redesign or sensing-first alternatives sometimes offer greater value and therefore belong in every investment comparison.

The U.S. drone cases document bounded pilots and limited early services. They establish feasibility on particular routes, with particular handling arrangements. Broad restaurant adoption remains a separate question. Customers may decline the service, and employees or store managers may find that dispatch and recovery add work. Repeat use, failed orders and willingness to continue matter as much as a successful demonstration.

The benefits are divided among several parties. A franchisee may gain productive capacity while the franchisor collects a technology fee. Employees may spend less time on a repetitive task yet inherit more exception handling. Customers benefit only if food and service improve. For the supplier, the decisive result is a retained paying account with sufficient margin after installation and support. Contracts determine how much of the operating gain each party keeps.

The opportunity is a network of useful operating locations. An eligible restaurant can prepare food, serve diners and support pickup or delivery, earning sales inside and beyond the four walls. Repeated tasks and varied sites could also improve robot learning. That network becomes commercially valuable when the work is accepted locally and the next installation earns its cost. An expansion announcement does not settle either question.

Table 3 | Commercialization questions by reader

Decision-maker	Core question	Decision-grade metric
Restaurant owner	What measurable loss is reduced?	Incremental unit NPV versus non-AI alternative
Technology provider	Does deployment become active recurring use?	Gross profit/site, integration cost, retention
Franchise system	Who pays and who gets the benefits?	Franchisor/franchisee incidence, site readiness
Investor / Corp Dev	What is defensible, scalable cash flow?	Customer concentration, contract control, ROIC
Customer / workforce	Is service and work measurably better?	Food quality, wait time, worker burden and privacy

The starting point is the restaurant's work. Identify an operating problem, the person who can authorize a purchase and a physical task that the site permits. A world model earns a role if it improves the system's reliability or adaptation. A robot

earns a purchase if the completed work fits the shift and improves the economics. This gives builders a practical research agenda without requiring a prediction about which model architecture will dominate.

Evidence discipline: Developer-published model capabilities are not restaurant-scale deployments. Sources [S01–S130] retain the October 26, 2026 core cutoff; the location-network, ODSC and drone supplement [S131–S138] and the Appendix N technical supplement [S139–S154] were checked October 28, 2026. The principal five-year economics case remains GII Module II.

A Three-Unit Theory for Restaurant Physical AI

The site is the unit of economics. Every restaurant has a particular rent burden, kitchen geometry, electrical and network capacity, employee mix, menu, average ticket, local demand curve, and owner who authorizes capital spending. Those conditions determine whether a seemingly impressive technical capability deserves an equipment purchase, SaaS contract, or operational redesign. The meaningful denominator is therefore not the number of robots a supplier could imagine selling; it is the number of locations at which the complete system delivers positive incremental cash flow and acceptable service quality.

Deployment judgment. Bounded production tasks provide the clearest near-term route; learned manipulation needs validation under restaurant service conditions. The site is the binding commercial test, while network transfer remains a conditional advantage. Sections 4.7–4.8, 6.10–6.12 and 7.11–7.13 connect current robot evidence to embodiment, peak capacity, staffing and order execution; Appendix N supplies the technical exhibits.

The task is the unit of learning. Robot hardware and software teams need to identify the precise physical state to observe, the permitted actions, the data required to evaluate a policy, and the intervention and safety thresholds. An order-ahead application supplies a demand signal, but generally does not supply labeled gripper trajectories, food-contact safety states, or verified physical task completions. Conversely, a sophisticated manipulation demonstration has limited operator value unless it completes a commercially meaningful task under restaurant shift conditions.

The network is the unit of scale. Multi-unit restaurants connect kitchens, customer demand, pickup points, delivery catchments, equipment procurement and regional service. They can provide repeated tasks and varied sites for permitted data and transfer tests. Franchise ownership, layouts, local demand and human preferences create limits: useful scale requires the next site to remain accepted and economically viable. A restaurant location can amplify both embodied learning and commercial reach; a count of stores alone proves neither.

Begin by observing the local work and identifying the buyer. Then specify the learning data and test throughput, recovery and food quality during service. Only after that result is credible should the analysis move to fleet economics: hardware, installation, maintenance, training and support. This is GII's proposed route to commercialization; general-purpose restaurant autonomy remains unproven.

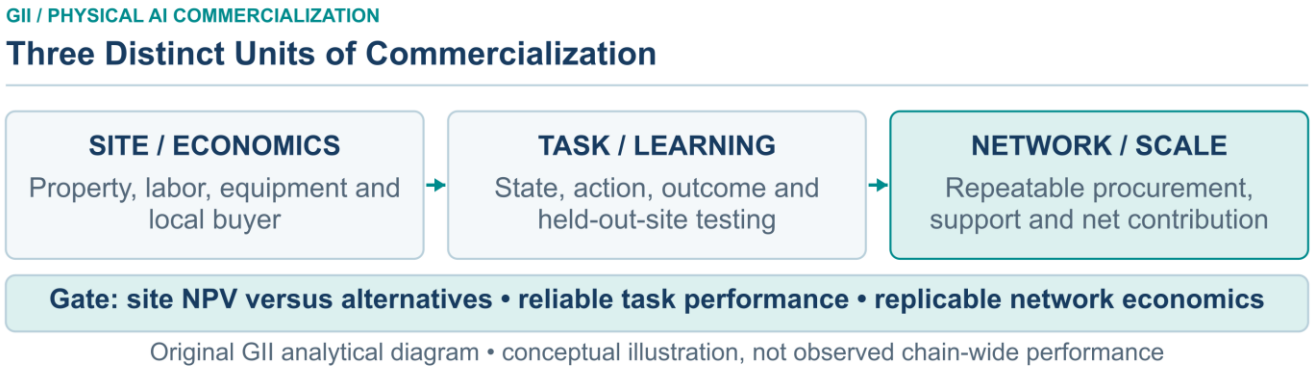


Figure 1 | The site, task and network are separate, necessary units of commercialization
GII original industry-economics synthesis. Conceptual framework; no vendor claims or market sizing are implied.

Author's Note — Why a Restaurant Business Educator Is Writing About Physical AI

AUTHOR PERSPECTIVE | INTERDISCIPLINARY COMMERCIALIZATION RESEARCH

Robot learning, simulation and embodied reasoning have produced increasingly capable demonstrations. For a restaurant operator, the next question is practical: can that capability complete useful work during service, and will the people running the restaurant choose to keep it? Those decisions connect the technical research to a physical market.

I approach this question through restaurant operations, franchising, commercial real estate and technology adoption. One location may combine food production, dining, QSR pickup, courier dispatch and a potential drone handoff. My interest is in how those functions can earn their place in the business, create sales inside and beyond the four walls, and make a larger location network useful to Physical AI developers.

Readers seeking the engineering detail can proceed to Sections 4.7–4.8, 6.10–6.12 and 7.11–7.13, then Appendix N. These sections connect learned manipulation and embodiment to production capacity, skilled labor and order execution.

World models and robot-learning systems may eventually learn from the coordinated activity of restaurant staff, machines, ingredients and consumers. But a video of a kitchen is not necessarily action-ready training data, and a robot that completes a manipulation benchmark has not thereby demonstrated a profitable restaurant deployment. The evidence must connect physical capability to safety, employee adoption, system integration and returns on invested capital.

Buildings and contracts shape adoption as surely as model performance does. The brand may select the technology while a franchisee pays for installation and employees absorb the change in routine. I therefore ask who can authorize the work, who bears the costs and whether customers and staff will keep using the result. A second site that is easier to install and support provides more convincing evidence of scale than a second announcement.

My contribution is to examine those deployment decisions through hospitality operations, real estate and corporate finance. The report asks where embodied intelligence can become a durable part of food production and service, and what evidence would justify the capital committed to it.

Executive Investment Conclusions — From Embodied Capability to Restaurant Cash Flow

GII ORIGINAL INVESTMENT SYNTHESIS | RESEARCH CUTOFF: OCTOBER 26, 2026

Restaurant Physical AI becomes a business when a useful physical task leads to reliable service and measurable cash returns. That path runs through the restaurant's property constraints, staffing and customer relationships. Shipment counts and model benchmarks describe parts of the opportunity; the investment decision belongs at the operating site.

GII brings operating economics, property underwriting and capital allocation to this technical discussion. A large restaurant footprint creates places to investigate. It does not make every task viable. Digital ordering and equipment dashboards can improve the business, while World Models require a more specific account of how learned predictions support physical action.

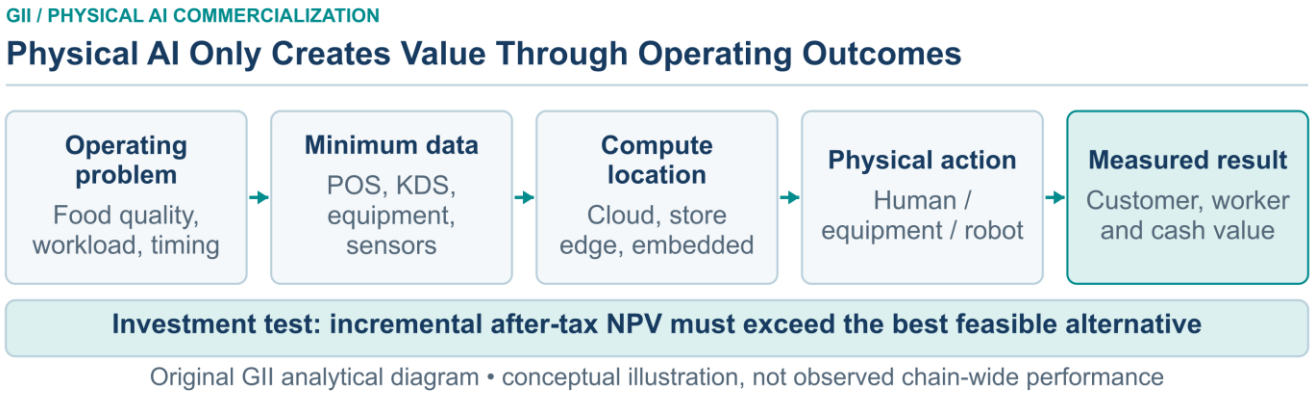


Figure 2 | Restaurant-first physical intelligence value chain

Original GII framework; conceptual, not measured company performance.

Seven Investment Conclusions to Test — Not Seven Product Predictions

Table 4 | Investment conclusions and falsification tests

ID / Thesis	Primary public evidence	What would change the view	Investor consequence
IC-1 | The market is the set of economically actionable restaurant tasks, not a macro headline	NRA $1.36T revised 2026 sales; >1M broad outlets; QSR 124,185 Top-20 sites [S01,S04,S93]	Format/task-level data show far fewer economically eligible sites or weak customer budgets	Segment task frequency, physical constraints, purchaser rights and fee capture before TAM
IC-2 | Capital allocation must clear the best alternative, not merely clear zero NPV	Module II analyst scenario: Edge/AI +$1.5K, redesign +$14.8K five-year NPV [M1]	Verified materially larger incremental cash flows or longer durable life from AI	Benchmark against non-AI and sensing-first alternatives
IC-3 | Computing location is part of restaurant operating design	Chick-fil-A chain-wide k3s edge clusters [S88]; no unit ROI disclosed	Actual cloud resilience/cost outperforms edge for required workloads	Underwrite latency, uptime and TCO by application
IC-4 | The restaurant operating platform can hold more bargaining power than the robot	Toast FY25 81.9% fintech revenue; PAR 36% top-3 customer exposure [S90-S91]	Vendor proves stand-alone AI economics and pricing power	Normalize software, payments, hardware and buyer concentration
IC-5 | Deployment, operating improvement and consumer demand must be measured independently	Wingstop Q2 2026 domestic comps -7.5% [S84]; tech rollout evidence is a different scope	Controlled evidence that deployment raises transactions or contribution	Measure throughput and guest demand separately
IC-6 | The most promising Physical AI may be invisible to diners	Chipotle 2024 named Autocado / Hyphen pilots [S87]; Sweetgreen Spyce transaction [S110]	Durable and superior economics demonstrated for general robots	Compare specialized equipment, licensing, and ownership
IC-7 | Human outcomes are financial inputs, not decorative ESG language	NRA 2026 operator survey: 94% no permanent job eliminations from recent tech [S93]	Representative controlled workforce and customer studies show a different balance	Track employee burden and consumer experience, not just labor substitution

Evidence convention: [S01–S154] identifies the report's source register; individual appendix headings retain their own source ranges. Illustrative results derive from stated GII assumptions, with the principal five-year case in Module II (Tables 117–121). None of these statements is a security buy/sell recommendation.

IC-1 — The market is the set of economically actionable restaurant tasks, not a macro headline

The United States has a large physical restaurant network with substantial sales, labor, property and equipment investment. Nevertheless, neither more than one million broad foodservice outlets nor $1.36 trillion of forecast industry spending is a technology revenue pool. A robotics developer must segment by repeatable task, format, installed equipment, site constraints, willingness to pay, workforce acceptance and purchaser authority. Economic losses such as order defects or food waste are possible use cases, not TAM denominators.



Figure 3 | U.S. restaurant physical-economy scale

Original GII synthesis of NRA and QSR sources [S01], [S03], [S04], [S93]. The four measures cover different populations and are not additive.

For market sizing, follow a task from the restaurant footprint to feasible sites, accessible purchasers and active deployments. Then deduct the cost of serving those accounts. Operator savings, supplier revenue and enterprise value measure different results and require separate calculations.

IC-2 — Capital allocation must clear the best alternative, not merely clear zero NPV

An operator may address the same restaurant problem through staff training, station redesign or conventional software. In the Module II hypothetical comparison, the $80,000 Edge/AI case produces about $1,507 NPV at a 12% discount rate, while a separately assumed $15,000 workflow redesign produces about $14,848. The ranking follows the assigned benefits and costs; it does not establish that redesign generally outperforms AI. Table 121 is a decision procedure using uncalibrated analyst assumptions. A product decision requires comparable task objectives and defensible benefit estimates.

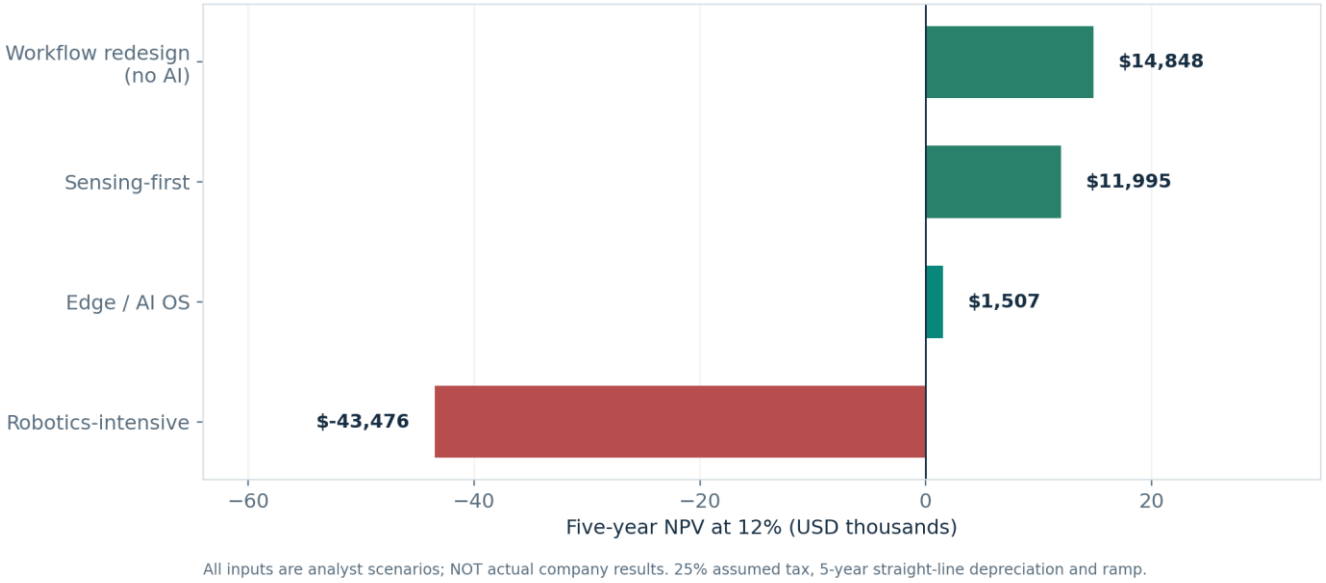


Figure 4 | Illustrative investment alternatives and after-tax five-year NPV

GII Module II illustrative comparison. Returns depend on Table 121's stated benefits and reconstructed, unconfirmed refresh inputs; they are not observed product returns.

The benefit hurdle is useful before selecting a product. Under the stated five-year life, ramp, tax policy, operating cost and Year-4 replacement, an $80K investment needs roughly $44.3K in mature annual gross benefit to reach zero NPV. Higher investment raises that requirement. Transaction volume and peak demand determine whether the restaurant can turn released capacity into cash.

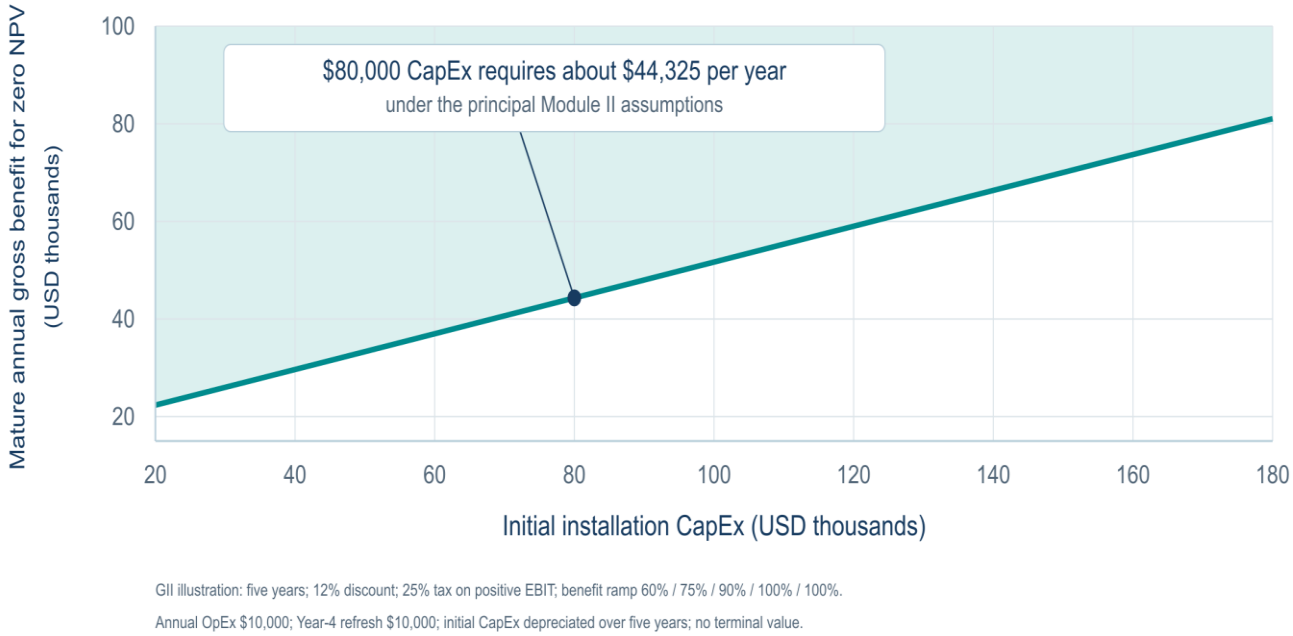


Figure 5 | Project-level break-even benefit frontier

GII Module II model: five years, 12% hurdle, 25% assumed cash tax, $10K annual OpEx, $10K Year-4 refresh; no terminal value.

Budget approval should identify how the benefit reaches cash: changed paid hours, avoidable waste, retained customers or additional orders served at a contribution margin. A positive NPV still needs comparison with simpler uses of the same capital.

IC-3 — Computing location is part of restaurant operating design

A restaurant can function as a distributed node of intelligent production without becoming a miniature hyperscale data center. Chick-fil-A publicly describes a chain-wide edge platform with at least three Kubernetes nodes per store and support for additional GPU-oriented use cases. This validates the physical feasibility of local compute and the importance of resilience; it does not disclose incremental customer economics or imply that every restaurant format needs the same system. McDonald's multi-year NEXT strategy likewise combines operational redesign, accuracy, predictive maintenance and AI, rather than an AI-only deployment thesis. Physical AI architecture therefore belongs in site-development and multi-unit operations analysis, not just in a corporate IT budget.

Compute placement belongs in operating design. Compare the latency and outage consequences of each workload, then include networking, power, installation, refresh and site disruption in total cost. Franchise contracts determine who can approve and pay for the chosen arrangement.

IC-4 — The restaurant operating platform can hold more bargaining power than the robot

Control of ordering, kitchen workflows and franchise procurement can be as commercially important as the robot itself. Toast's FY2025 GAAP revenue was $6.153 billion: $5.037 billion from financial-technology solutions (81.9%), $936 million from subscription services (15.2%), and $180 million from hardware and professional services (2.9%). These rounded shares describe reported revenue categories, not Physical AI revenue [S90]. GPV measures the value of payments processed through the platform; it is not Toast's revenue.

PAR's FY2025 Form 10-K attributes 21% of revenue to McDonald's, 8% to Yum! and 7% to Dairy Queen. Those three customers account for approximately 36%; the remaining customers contribute approximately 64%. The percentages are rounded and describe customer concentration, not customer AI spending [S91]. For an acquirer, the practical questions concern renewals, switching costs, uptime obligations and who controls the data. Restaurant acquirers should examine platform entitlements and franchisee economics before assigning value to proprietary AI.

IC-5 — Deployment, operating improvement and consumer demand must be measured independently

A restaurant can gain capacity while demand weakens. Wingstop's supplier-reported Smart Kitchen rollout and its issuer-reported 7.5% decline in Q2 2026 U.S. comparable sales measure different things. Their coexistence does not establish a causal relationship. The useful sequence of evidence is installation, active use, operating improvement, additional transactions and cash returns. Each step requires its own support.

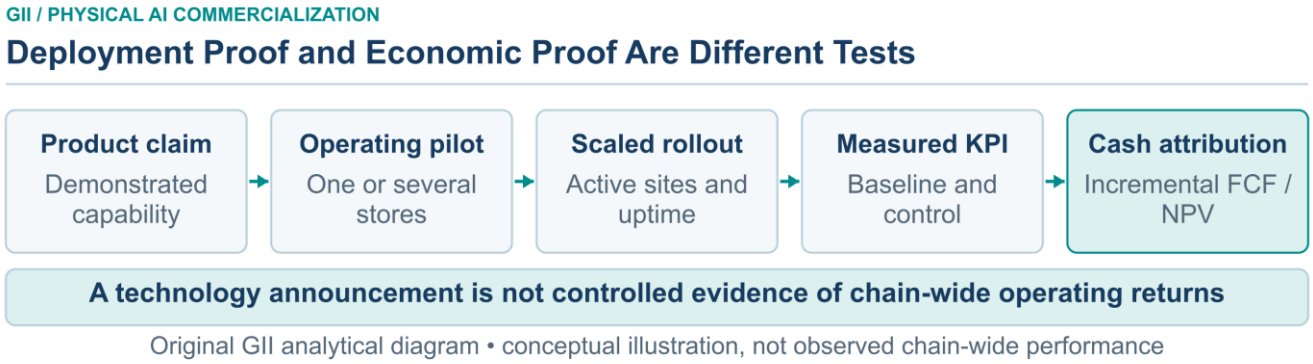


Figure 6 | Evidence ladder from demonstration to attributed cash flow

Synthesis based on named primary disclosures [S84], [S87], [S88], [S86]. Illustrative stage descriptions.

Revenue attribution needs a baseline that accounts for price, menu mix, new stores, promotions, seasonality and consumer demand. Where comparable controls are unavailable, the evidence supports an association rather than a causal return.

IC-6 — The most promising Physical AI may be invisible to diners

Food quality, order sequencing, equipment diagnostics and peak kitchen throughput can be improved by sensors, local prediction and well-designed staff interfaces without a humanoid robot. Chipotle's Autocado and Hyphen augmented makeline tests provide examples of specialized task engineering; one machine processes ingredients, and people continue other preparation steps. The 2025 Sweetgreen sale of Spyce technology to Wonder, with continuing supply and licensing

arrangements, illustrates an additional economic question: a restaurant chain may benefit from deployment without owning a robotics development company. Ownership is a capital-allocation decision, not a badge of innovation.

An acquisition case may rest on access to workflows, distribution or data rights. It also needs an account of hardware investment, installation, maintenance and equipment replacement. Continuing supply and licensing contracts can be as consequential as ownership of the underlying technology.

IC-7 — Human outcomes are financial inputs, not decorative ESG language

A restaurant combines production and hospitality; removing one physical task may lower walking distance while increasing error-handling or surveillance concern. The National Restaurant Association's 2026 operator research reports **that 94% of operators saw no permanent job elimination from recent technology investments, underscoring why headcount reduction is a poor default benchmark**. Technology can create value through lower injuries, faster learning, reduced turnover, higher food quality, less guest friction and more dependable shifts—even where payroll does not contract. The employee perspective should be gathered through consent-based feedback, not unvalidated emotion inference from continuous camera monitoring.



Figure 7 | The four-stakeholder investment test

Original GII framework; conceptual, outcomes require separate measurement.

Measure a guest outcome and an employee or operating outcome alongside EBITDA and free cash flow. A project that damages food quality or shifts unpriced work onto staff may struggle to sustain its accounting gains.

Cross-Chapter Synthesis — One Economic Theory, Fifteen Research Chapters

The report follows the commercial decision from the restaurant site to the supplier business. Chapters 1–3 establish market scale, tasks and property constraints. Chapters 4–7 examine learning, equipment and order execution; Chapters 8–11 assess compute, deployments and suppliers. Chapters 12–15 return to market sizing, human outcomes and capital allocation. The investment modules provide the calculations behind those judgments.

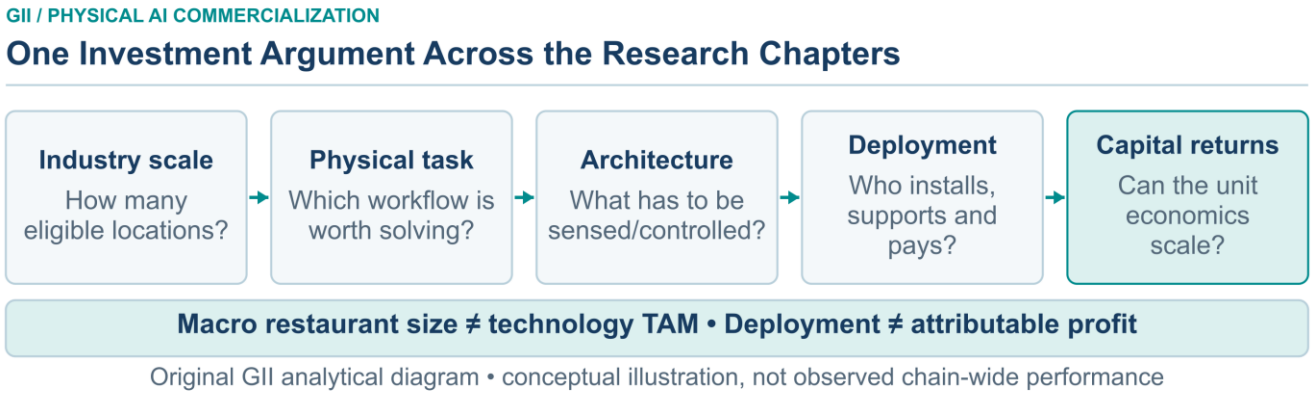


Figure 8 | Industrial-economics and investment logic across chapters

Original GII chapter synthesis; concepts and implications, not external market data.

Table 5 | Chapters and investment decisions

Evidence tier	Relevant chapters	Economics question	Deliverable to investors
Industry / task opportunity	1–3	What repeatable restaurant jobs and property footprints are commercially addressable?	Task–site–buyer funnel with format boundaries
Model and data feasibility	4–5	Can permitted data support action prediction and generalization across sites?	State/action/outcome protocol, benchmark and rights file
Physical execution	6–9	Can intelligent equipment and robots operate safely with normal kitchen workflows?	Equipment cost, exception handling and TCO
Deployment evidence	10–11	Are pilots repeated at paying stores, and who holds the software and customer relationship?	Chain evidence ladder and supplier industry structure
Investable economics	12–14	Do customer and supplier cash flows beat no-AI alternatives?	TAM/SAM/SOM, unit DCF and bargaining analysis
Commercial market entry	15 and Modules I–III	What buyer, contract, capital structure and scale conditions make a durable business?	Five-gate market-entry plan and downside tests

Source: GII original cross-chapter synthesis; no corporate data implied by this mapping.

Investment Committee Checklist — What Would Reverse the Thesis?

Table 6 | Diligence, disproving evidence and action

Decision area	Disproving or weakening evidence	Investor / operator action
Market size	Technology spend is already bundled in existing POS contract; eligible sites far below reported unit universe	Reduce TAM; use active paid locations and attach rates
Operating effectiveness	No repeatable difference in food quality, peak throughput, labor burden or waste	Pause rollouts; compare staffing and low-tech process alternatives
Deployment quality	Pilots work but real-store uptime and support costs deteriorate after rollout	Stage-gate rollout; require service-level and field-maintenance economics
Financial returns	Project NPV negative at reasonable adoption, refresh, taxes and cash conversion	Do not approve on gross savings or demo metrics
Supplier durability	AI feature is bundled for free, margin declines or key account churn rises	Stress revenue retention, customer concentration and pricing power
Consumer / employee acceptance	Measured guest trust, food quality or worker autonomy deteriorates	Modify use case, privacy practices, and human-workflow design
Real estate feasibility	Retrofit cost, lease approval or power/network constraint blocks site economics	Segment sites; pursue new builds or embedded low-power solutions

Investment committee actions are analyst recommendations, not representations of actions taken by named issuers.

I would judge restaurant Physical AI by what changes during a shift and what remains valuable after the pilot. Better food, more dependable service and a workable job for employees are operating results that can support a return on capital. The preferred robot or model should follow from that evidence.

All market estimates, management targets, supplier claims and analyst scenarios remain separately labeled in the underlying source ledger. The graphical exhibits in this executive section are original GII syntheses or transparent computations of cited figures; they are not unlicensed restaurant installation photographs.

1. The Restaurant Economy as a Physical AI Deployment Network

1.1 Revised market-size baseline

A credible industry economics report begins by separating the overall restaurant-and-foodservice revenue pool from narrower NAICS or limited-service chain measures. The October 2026 benchmark revision reduces absolute market levels across the 2021–2026 time series.

- The revised NRA chart supplies a six-year nominal sales series; the stated 2026 growth rates use the association's underlying, unrounded estimates.
- Restaurant and drinking-place activity has a narrower scope than foodservice in schools, lodging, recreation, retail and other host channels. Revised category data are needed to reconcile those populations.
- Nominal sales combine price and volume effects. Traffic and real consumer demand therefore require separate measures.

Research limitation: revised 2021–2026 limited-service and full-service category series under a common benchmark methodology remain outstanding.

Sources: [S01]–[S03], [S27], [S28]

Table 7 | Revised restaurant and foodservice sales baseline

Fiscal year	Earlier view ($B)	Revised ($B)	Nominal revised YoY	Data status
2021	1,060	940	—	Actual/revised
2022	1,220	1,060	+12.8%	Actual/revised
2023	1,340	1,190	+12.3%	Actual/revised
2024	1,410	1,250	+5.0%	Actual/revised
2025	1,480	1,310	+4.8%	Actual/revised
2026	1,550	1,360	+4.3% official*	Forecast; rounded chart implies +3.8%*

Chart-rounded amounts, not internal NRA unrounded series. Revised annual growth rates based on these rounded levels are approximate. Sources: [S01],[S36]

1.2 The U.S. Top 20 limited-service brand sample

The QSR 50 provides a valuable common-scope company dataset. It is a brand sales ranking across U.S. limited-service operations rather than a list of the 20 largest consolidated restaurant companies.

- The brand profile retains 2025 sales, reported AUV, franchise/license units, corporate units and changes in units.
- Estimated brand statistics and parent-company portfolios require separate labels because a brand ranking is not a consolidated-company financial sample.
- Historical comparisons use exact prior-year figures where available; missing 2024 observations are not interpolated.

Sources: [S04], [S05]

Table 8 | Top 20 U.S. limited-service brands, 2025

Rank	Brand	U.S. sales $M	AUV $k	Units	Franchise %	YoY units
01	McDonald's	55,061	4,088	13,706	95.3%	+147
02	Starbucks*	30,250	1,800	16,860	40.4%	-75
03	Chick-fil-A*	23,918	7,700	3,287	97.9%	+178
04	Taco Bell	17,247	2,216	7,784	91.5%	+180
05	Dunkin'	13,106	1,400	9,999	99.6%	+231
06	Wendy's	11,898	2,001	5,969	92.9%	+36
07	Chipotle	11,679	3,104	3,938	0.0%	+294
08	Burger King	11,074	1,660	6,649	83.0%	-52
09	Domino's	9,953	1,385	7,186	96.4%	+172
10	Subway*	9,200	490	18,773	100.0%	-729
11	Panda Express	6,408	2,646	2,607	7.1%	+102
12	Panera*	5,894	2,611	2,214	50.0%	+29
13	Popeyes	5,698	1,800	3,196	97.0%	+48
14	Raising Cane's	5,487	6,580	913	2.4%	+92
15	Wingstop	5,343	2,000	2,586	97.8%	+382
16	Sonic Drive-In	5,242	1,500	3,412	91.4%	-49
17	Pizza Hut	5,094	808	6,307	98.8%	-250
18	Dairy Queen	5,013	1,218	4,115	100.0%	-97
19	KFC	4,682	1,329	3,523	97.4%	-146
20	Whataburger	4,311	3,914	1,161	21.8%	+76

Starred brand names indicate an estimate flag in the original QSR ranking. Franchise % calculated from licensed/franchised units ÷ total units; QSR ownership definitions may vary and require brand-specific validation. Sources: [S04]; GII arithmetic

The sample throughout this report is the 2026 QSR 50 (2025 data). Year labels in the tables identify the data period; the ranking is a brand-level limited-service sample, not a consolidated-company or whole-industry census.

Table 9 | Top 20 concentration and operating footprint — derived

Metric	Result	Computation / interpretation
2025 Top 20 U.S. system sales	$246.558B	Sum 20 QSR brand sales; not GAAP parent revenue
2025 Top 20 U.S. stores	124,185	Sum of brand-level year-end units
Top 5 system sales	$139.582B	56.6% of Top 20 sales; not total market share
Top 10 system sales	$193.386B	78.4% of Top 20 sales
Simple sales / year-end store	$1,985K	Derived ratio; NOT QSR AUV and not weighted annual comp-store AUV
Universe definition	U.S. limited-service brands	Excludes many full-service operators and independent restaurants

Sources: [S04]; GII calculations

1.3 Industry formats and operating economics

Restaurant formats have materially different service requirements, kitchen standardization, capital intensity, and opportunities for automation. A single adoption rate applied to all locations would be analytically weak.

- QSR burger, pizza, chicken, beverage, fast-casual and full-service formats differ in transaction frequency, average check, peak queues and property footprint.
- Normalize revenue per restaurant, contribution margin, labor hours per transaction, food costs, occupancy and asset age where disclosures permit.
- Franchisor financial statements describe a different economic entity from the unit-level cash flows that determine a franchisee's technology budget.

Research limitation: comparable public metrics by format remain incomplete, and restaurant-level margin data often remain private.

Table 10 | Restaurant-format economics and readiness template

Format	Demand pattern	Operational repeatability	Key physical constraint	Best early AI question
Burger QSR	Drive-through and peak meals	High to medium	Drive-through queue / fry station	Order accuracy and timed cooking
Pizza	High off-premise share	High	Oven/dispatch / production sequencing	Production and dispatch synchronization
Chicken QSR	Peak-driven product flow	High to medium	Holding quality and fry capacity	Quality and station orchestration
Coffee/beverage	Short, variable personalized jobs	Medium	Barista workload and machine uptime	Equipment reliability and order staging
Fast casual	Visible assembly and customization	Medium	Makeline bottleneck and portions	Portion consistency and digital assembly
Full service	Longer visits / more service steps	Lower	Human service and space circulation	Staff augmentation, not blanket automation

Qualitative GII framework, not measured industry-wide scoring.

1.4 Market concentration and scale economies

Large chains may spread AI software and data-integration costs across multiple units, but a franchisor does not automatically internalize the store-level operating benefits. Contractual governance affects investment incentives.

- Measure sales concentration within the ranked limited-service sample and selected category groupings.
- Examine data network effects: does broader installed scale produce distinctive predictions, faster experimentation, and supply-chain coordination?
- Large chains and independent operators differ in bargaining power, integration budgets and upgrade cycles; national location counts alone do not capture those differences.

Table 11 | Top 20 category revenue concentration (within sample)

QSR category	Brands	2025 U.S. sales $B	Share of Top 20
Burger	6	92.599	37.6%
Snack	2	43.356	17.6%
Chicken	5	45.128	18.3%
Global	3	35.334	14.3%
Pizza	2	15.047	6.1%
Sandwich	2	15.094	6.1%

Sources: [S04] categories; GII calculations

1.5 Open research tasks

Market sizing must remain vintage-controlled. The $1.36T revised national projection and the QSR Top 20 sales are useful but do not form a clean industry-wide denominator for technology penetration until differences in industry scope are reconciled.

- The remaining establishment-count work requires NAICS-aligned Census and industry measures with explicit outlet definitions.
- The revised NRA series also needs a restaurant-versus-other-foodservice breakout before it can support segment-level technology penetration estimates.
- BLS wages, occupational mix, establishment counts and wage inflation provide distinct labor-market denominators; their integration by restaurant segment remains a research task.

Open research agenda: reconcile Census/BLS series and price deflators, document the 2026 AIES benchmark, and add a comparable full-service chain sample. These gaps limit segmentation rather than invalidate the broad industry baseline.

1.6 Market-baseline discipline: exact growth versus rounded revenue

The October 19, 2026 NRA revision exposes a measurement risk that directly affects investment conclusions. The revised 2025 and 2026 chart points ($1.31 trillion and $1.36 trillion) imply approximately +3.8% when divided as printed. However, the association states nominal 2026 growth of +4.3% and inflation-adjusted growth of +0.8%; this difference arises because the graphical dollar series is rounded. The report should use official published growth rates for growth claims and chart values only for rounded industry-scale comparisons. [S01]

The 2025 Top 20 limited-service brands collectively reported $246.558 billion of U.S. systemwide sales across 124,185 stores [S04]. These sums are useful as a technology deployment sample, not as a share of the broader NRA restaurant-and-foodservice universe. The latter includes different establishments and foodservice spending categories. Outlet counts also require reconciliation because a location count, Census establishment count, and restaurant operator unit count need not describe the same denominator.

Table 12 | National restaurant measures and their limits

Metric	Observed baseline	Period	Use	Do not infer
Broad U.S. foodservice sales	$1.36T forecast	2026 NRA revised	Macro market scale	Annual AI vendor market
Nominal U.S. foodservice growth	+4.3% official	2026 forecast	Demand context	Exact rate from rounded chart
Real U.S. foodservice growth	+0.8% official	2026 forecast	Volume/real-spend context	Store-level real sales growth
Top 20 limited-service system sales	$246.558B	2025 QSR brand ranking	Chain sample purchasing power	GAAP parent corporate revenue
Top 20 limited-service U.S. units	124,185	2025 year-end QSR	Bottom-up eligible store screen	All U.S. restaurant outlets

Sources: [S01],[S04]. All figures have differing definitions. Any market share must be calculated using a consistent denominator and time period.

1.7 The Top 20 operating universe: a bounded deployment opportunity

The 2026 QSR 50 (2025 data) is a useful census of leading limited-service brands, but not an exhaustive count of U.S. restaurant establishments. The 20 brands in the existing table total 124,185 U.S. restaurants and $246.558 billion of 2025 U.S. systemwide sales. These figures support a market-sensitivity exercise without assuming that all stores have equivalent technology needs, ownership rights or purchasing budgets.

Financial significance depends on adoption and payment—not physical site count alone. To prevent inflated estimates, the analysis below treats incremental software receipts as a hypothetical annual third-party revenue pool, holding implementation fees, hardware, payments processing and existing bundled software outside the estimate. It must not be called forecast ARR, current spend or total Physical AI TAM.

Table 13 | Illustrative Top 20 software receipts, USD millions

Share of 124,185 stores	At $500/year ($M)	At $1,500/year ($M)	At $3,000/year ($M)
10%	$6.2	$18.6	$37.3
30%	$18.6	$55.9	$111.8
50%	$31.0	$93.1	$186.3

Pure sensitivity using the bounded Top 20 sample; no observed market price or industry TAM is asserted.

Calculated by GII from 2025 top-20 restaurant unit counts in Table 8: 124,185 x adoption x incremental annual supplier fee. Entirely hypothetical. No inference about actual AI adoption, prices, vendor margins or payment processing revenue.

1.8 Commercial opportunity starts with location heterogeneity

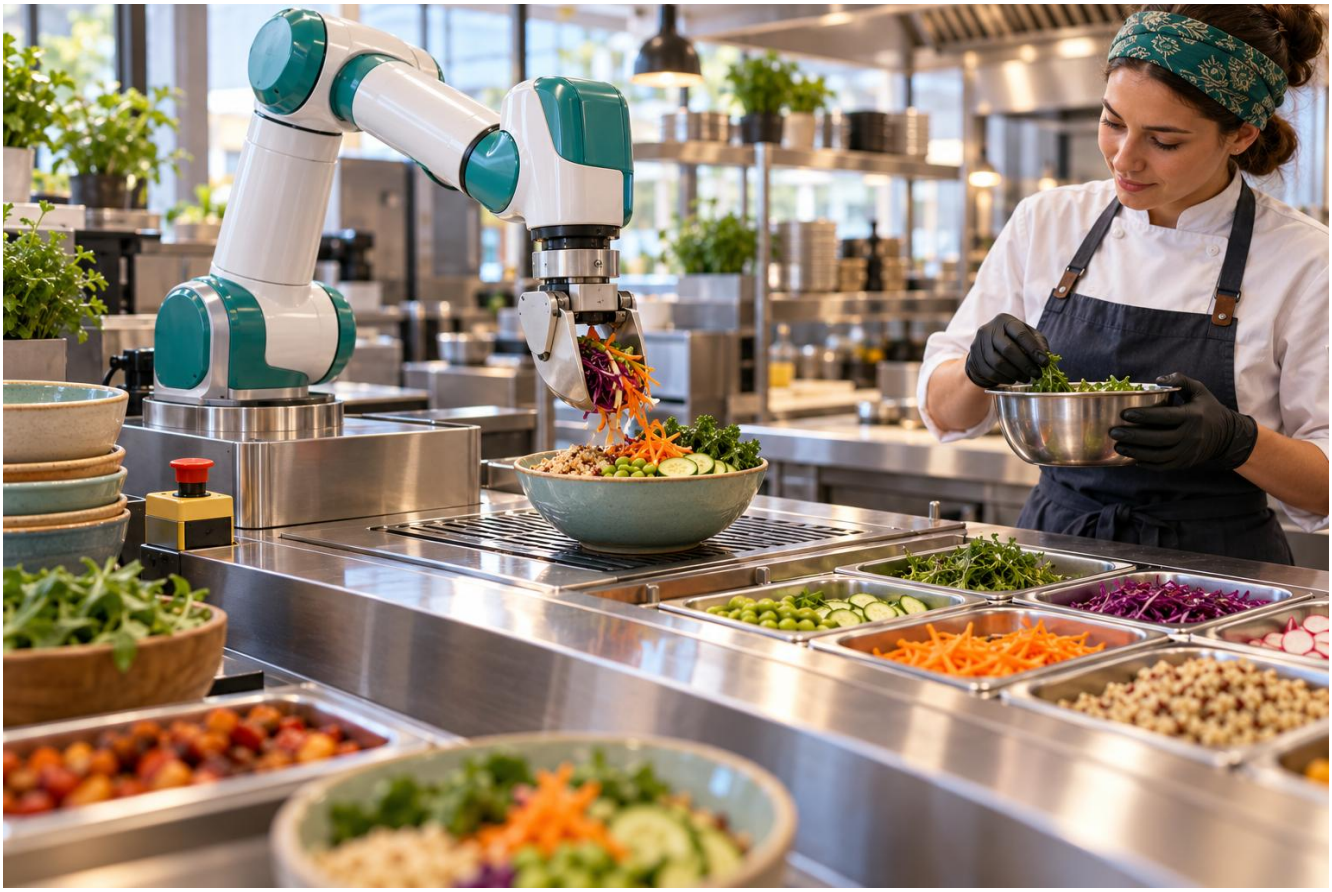
The industry should be counted by the relevant decision unit. A broad foodservice-outlet statistic includes formats with very different menus, staffing, operating hours and ownership. An issuer's systemwide unit count may include franchised restaurants for which the parent does not buy or finance a new robot. A delivery-only kitchen may share production space with other brands, while a coffee pickup unit may have little dining area. The correct commercialization denominator must identify country, format, channel mix, production equipment and purchasing authority—not just a logo on a map.

Restaurant economics are local even when software is centralized. A store with high utilization can have a valuable bottleneck intervention that matters little in a low-volume sister location. Restaurant sales describe the importance of the physical economy, but the potential value of an AI intervention derives from the marginal effect of a specific task and its surrounding operation. In one location improved batching may reduce queue delay; in another the same equipment may be idle and fail to produce even its subscription fee. Every pilot needs a site-level baseline before extrapolating to a chain.

For robotics developers, this creates a disciplined partnership rationale. Restaurant companies can provide variation across store layouts, traffic periods and menu mixes, and can potentially support standardized testing and maintenance at scale. In exchange, the technology provider must demonstrate task-level reliability, predictable installation, safe exception handling and a traceable economic bridge. The relevant question is not how many existing stores could technically fit a machine, but how many possess the necessary task volume, site infrastructure, operator budget and route to repeated deployment.

2. Task Economics: Restaurant Problems Worth Automating

Physical AI applications should compete against real operating alternatives rather than against a technology hype benchmark. Each category below is a measurable operating problem with an identifiable counterfactual, data requirement, and stakeholder incidence. The aim is to estimate a practical economic opportunity, not to add theoretical efficiencies into an inflated technology TAM.



Picture 2 | Food preparation as a complete human and robot workflow
AI-generated concept illustration for GII; equipment and settings are illustrative.

2.1 Food quality, safety and product consistency

Consumer repeat behavior depends first on whether restaurant output meets product expectations. AI can support consistency through sensors, predictive failure alerts and standard-process feedback, but data completeness has no value unless it changes operational decisions.

- Quantify remake cost, portion variance, safety incidents, equipment uptime, and product-loss rates.
- Determine which quality problems are observable through existing KDS, temperature, weight or camera systems.
- Investigate whether staff training and process simplification outperform automation.

2.2 Throughput and queueing

Peak-period constraints, not total annual capacity, may determine whether intelligent scheduling increases revenue. Distinguish customers who abandon purchases from customers who willingly wait.

- A site-level queueing study needs station and task maps, peak arrival rates, service times and bottleneck throughput.
- Erlang-type approximations are useful only when their arrival, service and queue assumptions fit the operating process.
- Additional orders create financial value only to the extent that attributable contribution profit exceeds the incremental cost of fulfillment.

2.3 Workforce and human outcomes

Reduced movement, fewer interruptions, better station information and predictable equipment condition may improve employee experiences even without immediate payroll savings. Conversely, new monitoring and exception-handling duties may undermine adoption.

- Workforce outcomes require paid hours, overtime, turnover, injuries, employee-reported frustration and task-interruption measures.
- Separately analyze perceived workload, realized compensation cost, employee turnover and customer-facing service quality.
- Consent-based interviews can document worker experience without inferring employee emotions from camera surveillance.

2.4 Consumer welfare and demand

A restaurant is a food-and-service business. Marginal utility of three minutes saved in a drive-through differs from the utility of better taste or greater perceived fairness. Consumer surplus should be assessed directly where possible.

- Consumer research can test willingness to pay for consistent food, reliable service, shorter waits and privacy using explicit survey questions.
- Measure actual return visits if access is available; avoid treating self-reported technology interest as realized demand.
- Consider whether the benefits vary by age, dining format, party size and transaction channel.

Table 14 | Restaurant-first pain-point economic map

Pain point	Base KPI	Available physical data	Action / alternative	Economic test
Inconsistent food	Remake %; quality rating	Weight, temperature, cook time	Alerts / better SOP / automation	Waste + retention contribution
Peak station bottleneck	P95 ticket time; orders/hour	KDS events; station queue	Flow balancing / schedule change	Lost orders × contribution
Labor overload	Paid hours; overtime; turnover	Task timestamps; opt-in survey	Staffing forecast / better tools	Realized paid hours + turnover
Machine failure	Hours down; repair bill	Equipment sensors / event logs	Predictive maintenance	Downtime + lost margin avoided
Waiting / incorrect orders	P95 wait; accuracy	Arrival times; order/weight signals	Order verify / pickup sequencing	Repeat purchases + fewer refunds
Inefficient capital	ROIC; comp-store FCF	Store P&L; asset registry	Defer, redesign, outsource, AI	Incremental NPV vs best substitute

Framework prompts; no aggregate U.S. value imputed without representative data.

Table 15 | Operating-data evidence checklist

Outcome	Recommended measurement	Minimum data	Identification challenge
Taste / food consistency	Blind panel scores; repeat-complaint rates	Recipe/temp/portion; customer feedback	Menu changes and staffing confounds
Peak throughput	Orders/hour; realized queue abandonment	KDS POS timestamps; arrivals	Shift composition and promotions
Food waste	Waste $ / food sales; SKU-level loss	Waste logs; ingredient cost	Seasonality, inflation
Worker impact	Turnover; overtime; fatigue survey	HR records; consented survey	Worker self-selection and churn
Revenue / retention	Comp transactions; visit interval	Loyalty and transaction cohorts	Price increases vs volume
Customer privacy	Data collected, retention, access	Data dictionary and contract	Visibility into third-party sharing

Measures are planned; restaurant-level primary datasets have not been obtained.

Table 16 | Hypothetical restaurant operating opportunity

Operating lever	Illustrative per-unit assumption	Formula for annual benefit	Required validation
Remake reduction	8 incidents/day × $6 variable cost × 20% reduction	8×365×6×0.20 = $3,504	Observed incident, cost and reduction rates
Realized labor overtime	3 paid overtime hours/week × $30/hr all-in	3×52×30 = $4,680	Actual payroll reduction vs redirected time
Peak incremental orders	4 extra paid orders/day × $7 contribution	4×365×7 = $10,220	Demand constrained by prior capacity
Machine downtime	8 avoided hours/year × $350 contribution/hr	8×350 = $2,800	Real outage frequency and revenue lost

Hypothetical operating opportunity; this is not a measured market estimate.

Illustrative, arithmetic-only figures. Do not add benefits without testing overlap, causality and fixed vs variable cost classification.

2.5 Economic prioritization

The appropriate ranking criterion is incremental expected value relative to the cost of intervention and uncertainty. A camera system that measures employee movement may be expensive yet fail to reveal the actual quality bottleneck; a fryer-temperature feed may be far more actionable.

- Create a task-by-task Value-at-Stake estimate.
- Economic prioritization also depends on whether operating data exist and whether the proposed intervention controls the relevant failure mode.
- Each AI investment needs a counterfactual based on the cheapest feasible alternative that addresses the same operating objective.

2.6 Resource efficiency as one task-level measurement example

Food quality and waste are a useful test of a restaurant-first Physical AI approach precisely because the largest visible problem is not necessarily a problem that a robot can solve. ReFED estimates that U.S. restaurants and broader foodservice operations generated 12.5 million tons of surplus food during 2024, with an estimated economic value of $157 billion. The same report says roughly 12.4 million tons ultimately became food waste and 9.73 million tons went to landfill. This is a modeled national value of food and food-related inputs, not the annual income statement loss of an investable universe of restaurant chains. [S51],[S52]

The composition matters. ReFED attributes approximately 8.72 million tons (69.6%) to plate waste, 1.49 million tons (11.9%) to overproduction, and additional surplus to catering and other sources. Sensors, stock forecasting, and accurate portioning might be more relevant to preventable production losses; portion choice, product quality, price, and guest preferences are often more relevant to plate waste. A robot provider cannot reasonably claim that the entire $157 billion is its addressable TAM, and a kitchen-only intervention should not claim savings on restaurant leftovers outside its causal reach. [S52]

Table 17 | ReFED foodservice surplus-food economic baseline, 2024

Measure	2024 reported figure	Scope / denominator	Investment interpretation	Source
Surplus food generated	12.5 million tons	U.S. foodservice including broader restaurants, catering and institutional activity	Physical system scale, not sales of technology	[S51],[S52]
Estimated economic value	$157 billion	Modeled value of surplus food, not guaranteed recoverable cash	Large friction pool with heterogeneous causes	[S51]
Food ultimately wasted	12.4 million tons	Of sector 12.5M-ton surplus estimate	Preservation/disposition is part of the solution	[S51]
Landfilled	9.73 million tons	77.6% of total surplus; 78.4% of waste in ReFED source framing	Environmental and operational exposure	[S51],[S52]
Plate waste	8.72 million tons; 69.6%	Share of 12.5M-ton surplus, ReFED driver classification	Consumer and product-quality issue—not simply kitchen robotic task	[S52]
Overproduction	1.49 million tons; 11.9%	Share of total foodservice surplus	Potential sensing, demand prediction and batch control target	[S52]
U.S. foodservice employment	15.7M for 2025; 15.8M forecast 2026	NRA foodservice category; not BLS NAICS 722 only	Workforce scale, not direct robot-replaceable roles	[S55]

ReFED model years are 2024; do not compare to 2026 NRA nominal sales as if numerator and denominator shared the same year or sector definition.

Percentage shares and tonnages are reproduced as published by ReFED; ratios calculated from the displayed tonnages differ. The source does not explain that difference here.

Table 18 | Food waste decision matrix: minimum necessary intelligence

Loss driver	Minimum data	Low-tech alternative	AI / Physical intervention	Attribution test
Plate waste	Portion returned, menu item, guest feedback	Smaller sides, portion choice, quality fixes	Opt-in waste measurement and product analysis	Plate grams/customer, repeat purchase, privacy acceptance
Overproduction	Batch timing, sales by slot, hold time, expiration	Smaller batches, production boards	Edge demand forecasts, smart scales, temperature alerts	Prepared/unsold units, labor, gross margin
Wrong order / remake	KDS events, correction reason, order error	Expeditor training and menu simplification	Vision check at handoff, order-to-item matching	Remakes per 1k orders, service delay, false alerts
Spoilage	Temperature, storage time, deliveries	FIFO labels and audits	Sensor alarms and conditional cooler optimization	Inventory shrinkage and food safety incidents
Equipment failure	Fault codes, ambient temperature, maintenance records	Preventive checklists	Predictive alert plus technician workflow	Downtime, repair cost, avoided discard

Solution presence does not prove causal returns; pilot sites must be compared against matched operations or an explicit counterfactual.

2.7 Labor-market conditions and the technology substitution fallacy

The labor-shortage story is time-dependent. In the NRA April 2026 staffing report, **only 22% of operators said they lacked enough staff to meet existing demand in 2025, compared with 32% in 2024 and 78% in 2021**. That decline does not mean hiring and retention ceased to matter; it means a sales pitch premised on an unchanging universal labor shortage is economically incomplete. Restaurant technologies should be underwritten against current local wage, staffing, turnover, service, and throughput data. [S55]

The NRA also reports that new hourly employees took an average of 31.8 days to become net positive in terms of value contribution, versus 72.2 days for managers and salaried employees. These figures motivate broader Physical AI propositions—better onboarding, safer stations, simpler kitchen execution, reduced peak-hour cognitive load—without pretending that each improvement produces immediately observable payroll reductions. [S55]

Table 19 | Restaurant workforce and AI return considerations

NRA observation	Period / statistic	What it may signal	Not justified without more evidence	Source
Insufficient staff to meet demand	2021: 78%; 2024: 32%; 2025: 22%	Labor scarcity differs across economic cycles	Permanent national labor shortage assumption	[S55]
Hourly staff net-positive time	31.8 days on average	Onboarding and quality of training matter	Direct incremental AI ROI	[S55]
Manager/salaried net-positive time	72.2 days on average	Leadership is a scarce operating capability	Assumption robotics substitutes managers	[S55]
Technology did not permanently eliminate jobs	94% of operators over last 2–3 years	Augmentation remains common in observed self-reports	No labor redistribution, task displacement, or future change	[S55]
Use of tools incorporating AI	26% all; FSR 28%; LSR 24%	Self-recognized use remains a minority in broad operator group	Rejects narrower vs broader survey estimates as false	[S55]

Source: NRA Research Insight: Hiring & Staffing, April 2026. Staffing percentages reflect survey responses, not labor-market census totals.

2.8 Translate the kitchen into a robot-ready task portfolio

Physical AI teams should begin with an **operating-task census** rather than a humanoid concept image. Break each workflow into observable actions, required physical dexterity, elapsed and active human time, quality-control points, exceptional states, equipment interfaces, and who owns the work. A ten-second task repeated hundreds of times may be commercially attractive, but only when the automation reduces a real bottleneck or avoidable cash expense rather than simply moves an employee elsewhere for the same paid shift.

A practical screening equation is **expected incremental annual benefit = eligible task volume × actually monetizable improvement per task − new operational burdens.** The monetizable improvement may be higher throughput during sold-out peaks, fewer remakes, fewer paid hours at unchanged service level, reduced equipment downtime, or better retention. Seconds saved should not be converted mechanically into payroll savings. If a crew cannot be rescheduled or sales do not increase, the value may remain operational flexibility rather than immediate cash.

The technical frontier and economic frontier are different. Delicate food handling may be difficult to automate but important to a high-volume kitchen. A task like passive temperature alerting may be easier and already addressable with connected equipment. A broad manipulator may eventually complete more tasks, yet current deployment economics can favor a purpose-built station with fewer degrees of freedom, simpler cleaning, bounded movements and certified interfaces. The correct first product may be perception, orchestration, or an equipment retrofit rather than a new robot chassis.

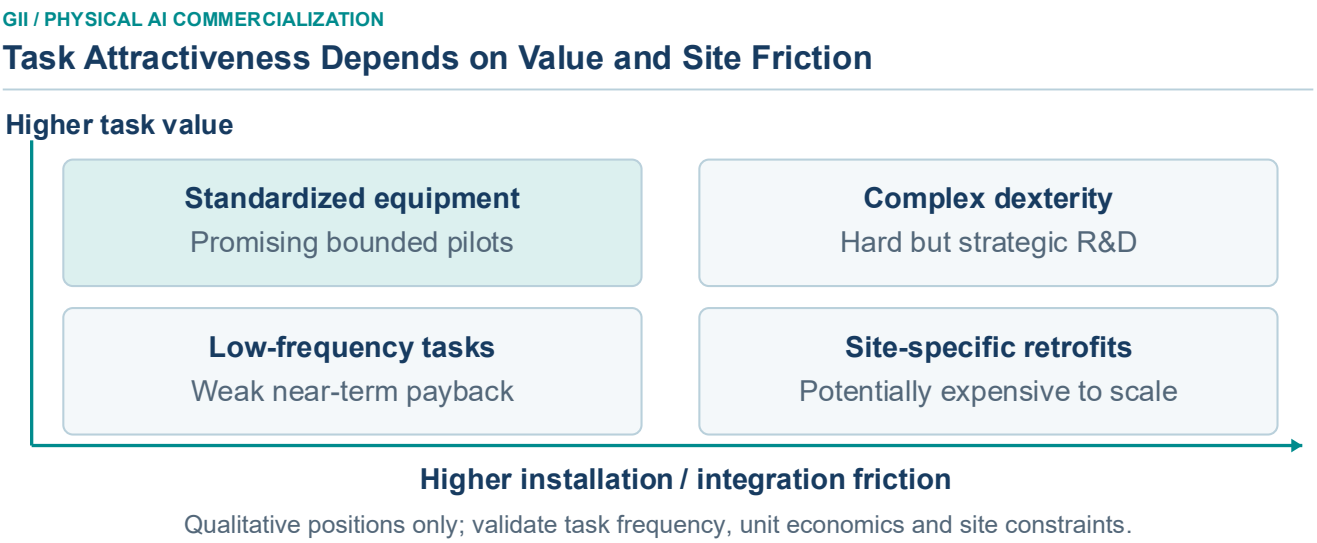


Figure 9 | A conceptual map of task importance versus restaurant site friction

Positions are GII qualitative hypotheses, not benchmarked technical performance or financial rankings.

2.9 A task scorecard for pilots and procurement

The same technical performance can have different commercial results across a station, shift and restaurant format. The following scorecard turns robot task selection into an accountable engineering and purchasing discussion. The proposed tests need observed denominators and should record failed attempts rather than averaging them out.

Table 20 | Task engineering and operating screen

Task family	Learning/physical complexity	Site economics to verify
Kitchen transport	Navigation, corridor conflicts and human handoffs	Loaded/unloaded time; worker travel displaced
Batch fry/cook	Heat, timing, sanitation and equipment interfaces	Conforming items per hour; downtime; intervention
Ingredient portioning	Deformability, variable weights, cleanability	Yield consistency; throughput at constrained station
Order orchestration	Event time, partial observability, capacity constraints	Late orders, remake rate, courier dwell
Equipment condition	Telemetry, fault detection, validated control	Avoided interruptions and maintenance cost
General manipulation	Cross-task dexterity and sim-to-real transfer	Use across tasks net of reset, training and service

Source: GII original task taxonomy; illustrative questions, not observed task outcomes.

3. Restaurants as Real Estate and Supply-Chain Operating Nodes

Physical AI exists within a leased or owned restaurant with electrical capacity, grease exhaust systems, safety constraints, fire codes, circulation paths, retrofit limitations and available floor area. An investment case that omits these property-level conditions will systematically underestimate the real cost of deployment.



Picture 3 | The building and utilities shape a restaurant robot installation
AI-generated concept illustration for GII; equipment and settings are illustrative.

3.1 Site-level physical feasibility

A real estate-informed review begins before the robot purchase: layout, store age, site access, network segmentation, power, HVAC, kitchen equipment and landlord rights.

- The proposed site walk-through covers QSR drive-through, mall-inline, freestanding fast-casual and full-service formats.
- The installation budget should include labor, downtime, IT commissioning, device footprint and utility modifications.
- Lease consents and site-specific safety constraints require review by the relevant property and technical professionals.

Table 21 | Physical AI site-feasibility checklist

Site element	Operator questions	Likely cost/risk	Evidence source
Circulation and aisle width	Can customers/staff and moving equipment coexist?	Workflow and safety constraints	Site measurements; photos with permission
Electrical / panels	Can added devices and equipment be served?	Circuit work, peak demand, backup	Engineer / utility records
Network and cybersecurity	Are secure local and cloud links viable?	Switches, SD-WAN, IoT segmentation	IT survey
Back-of-house space	Where will hardware, charging and service sit?	Lost productive floor area	Lease drawings / site measurements
Kitchen equipment	Can devices expose trustworthy machine telemetry?	OEM integration and retrofit	Equipment list / OEM interfaces
Landlord and code	Who authorizes modifications?	Permit and consent delays	Lease / jurisdiction professional review
Operational downtime	Can installation occur outside peaks?	Contribution lost during fit-out	Schedule and P&L

3.2 Retrofit versus purpose-built kitchen

Legacy restaurants often carry hidden constraints that reduce the addressable market for equipment-intensive technology. New builds can integrate sensors and infrastructure during construction at lower marginal installation cost, though site economics vary.

- The older-store and new-build budgets below are hypothetical; contractor and equipment-vendor evidence is needed before using them for a project decision.
- Downtime cost depends on lost site contribution after catch-up sales, rather than the full value of daily sales.
- Alternative kitchen layouts, including modular workstations, may change both installation cost and task feasibility.

Table 22 | Illustrative site retrofit costs — scope to validate

Cost element	Legacy retrofit	New-build marginal cost	Status
Site survey/design	$2,500	$1,000	Illustrative
Wiring/network	$8,000	$3,000	Illustrative
Computer/sensor install	$18,000	$18,000	Illustrative
Mechanical/space alteration	$12,000	$4,000	Illustrative
Commissioning/training	$5,000	$3,500	Illustrative
Downtime contribution loss	$6,000	$0	Illustrative
Total	$51,500	$29,500	Illustrative — NOT actual supplier estimate

Scenario placeholders; validate local labor costs, safety constraints and scope before using in underwriting.

3.3 Franchise and property cost allocation

Even a positive systemwide business case may fail to deploy if the franchisor requires local operators to bear hardware costs while headquarters captures most of the digital data and strategic benefits.

- Franchise fees and mandated technology upgrades can change who funds a project and who receives its benefits.
- Landlord, franchisor, franchisee, OEM and managed-provider responsibilities need explicit contractual allocation.
- The remaining lease term constrains the period over which an operator can recover investment in AI hardware and site improvements.

Table 23 | Technology costs and benefits by stakeholder

Item	Likely payer	Benefit recipient	Contractual diligence
Restaurant electrical retrofit	Tenant / franchisee, case dependent	Restaurant / sometimes landlord	Lease and franchise agreement
Edge processing hardware	Franchisee or chain IT budget	Store and chain	Hardware ownership and refresh
Cloud platform subscription	Brand, franchisee or vendor bundle	Chain/store/vendor	Price escalation, data access
Kitchen automation machine	Store / equipment lessor	Store labor + product quality	Lease, maintenance and warranty
Training / integration	Store + chain + vendor	Employees + service quality	Training obligation, acceptance
Cross-chain data products	Parent platform investment	Parent / participating stores	Data rights and revenue sharing

3.4 Real estate value implications

A technology investment may increase a tenant's cash flow without increasing landlord NOI or property value. Restaurant enterprise value and property capitalization therefore require separate cash-flow measures.

- Equipment ownership and leasehold-improvement rights determine which party retains the asset and any residual value.
- Upgrades may support rent-paying capacity and tenant credit only if the operating benefit is sustainable.
- Building capitalization rates cannot be applied mechanically to incremental restaurant operating profit.

3.5 The physical site is the deployment filter

An attractive national market is meaningless for a given equipment product if the target restaurants lack the power, circulation width, ventilation clearances, landlord permissions or technical support needed to use the technology. Restaurant real estate imposes location-specific costs that do not disappear when a software vendor standardizes its offering. For franchise systems, the party paying for installation may differ from the owner of the customer or operating data.

An investment-ready deployment funnel should move from total units to technically eligible units, commercially feasible units, authorized buyers and funded projects. The resulting funnel is a better basis for SAM/SOM than applying a generic robot adoption percentage to the entire U.S. foodservice outlet count.

Table 24 | Physical site underwriting funnel

Step	Eligible-site requirement	Validation evidence	Economic implication
1. Format	Task repeats sufficiently within menu/service format	Menu mix and throughput distribution	Exclude low-repeatability formats
2. Physical space	Clearances, travel paths, food safety and egress	Survey, CAD/as-built, site walk	Retrofit and downtime
3. Electrical and network	Available circuits, Wi-Fi/fiber, uptime	MEP and IT survey	Infra capital and maintenance
4. Data and integration	POS/KDS/OEM permission and compatible APIs	Vendor contracts and cybersecurity review	Technical and legal switching costs
5. Economic viability	Net benefit > fully loaded deployment costs	Matched restaurant cash-flow model	Funded SOM rather than theoretical TAM
6. Commercial authority	Buyer, franchisor, landlord and privacy approvals	Executed agreements and consent	Sales cycle and bargaining power

GII underwriting checklist; no store is assumed eligible absent measured proof.

3.6 Vertical integration versus outsourced platforms: supplier customer concentration

A rarely discussed public-company investment risk is now quantitatively observable. PAR Technology disclosed 2025 customer revenue concentrations of approximately 21% from McDonald's, 8% from Yum! Brands and 7% from Dairy Queen, or roughly 36% together (based on individually rounded percentages). The McDonald's share increased from approximately 15% in 2024 to 21% in 2025. This illustrates the counterparty power that can accompany large-chain restaurant digital infrastructure agreements. It does not prove that a chain will terminate vendor relationships or internalize PAR's products. [S83]

Table 25 | PAR Technology named-customer concentration (FY2025 10-K)

Customer	2025 revenue %	2024 revenue %	Approx. FY2025 revenue	Underwriting relevance
McDonald's	21%	15%	~$95.7M	Major chain concentration
Yum! Brands	8%	9%	~$36.4M	Internal and vendor platforms can coexist
Dairy Queen	7%	8%	~$31.9M	Brand counterpart exposure
Named customers combined	36%	32%	~$164.0M	Revenue concentration is not operating profit at risk

SOURCE / LIMITATION Customer percentages are rounded and not contract-level disclosures; estimated dollar exposure = reported FY2025 revenue $455.547M × rounded share. The sum is approximate, and concentration figures do not disclose margins, churn probabilities or forward commitments. [S83]

Table 26 | Illustrative PAR revenue-loss sensitivity

Scenario (not prediction)	Assumed lost % of MCD-linked 2025 revenue	Est. lost annual revenue	PAR consolidated FY25 revenue effect	Critical missing information
No loss	0%	$0.0M	0.0%	Contract term, renewal, vendor service scope
10% attrition	10%	$9.6M	2.1%	Substitution versus cross-sell revenue; stream margin
20% attrition	20%	$19.1M	4.2%	Product-level replacement costs, contribution margin
50% attrition	50%	$47.8M	10.5%	Timing, alternatives, potential offsetting contracts

Hypothetical partial revenue loss; no customer termination or forecast is asserted.

SOURCE / LIMITATION All sensitivities are mechanical static illustrations using rounded 21% revenue exposure and ignoring offsets, volume growth, timing, gross margin, or price effects. They are not forecasts or a probability of vendor displacement.

For buyers and investors, concentration is dual-edged. Winning a national franchise group can reduce customer-acquisition costs and demonstrate product-market fit; it can also confer bargaining power on the customer and expose a supplier to a single contract renewal. Diligence should examine product modules, multi-year renewal schedules, integrations, customer data portability, gross profit by account, and potential platform overlap between large-chain internal software and vendor capabilities. [S83]

3.7 A restaurant location is a physical contract, not a robotics blank slate

A retrofit begins with possession and site rights. The operator may lease the building, lease some equipment, have a franchisor-imposed specification, and depend on a landlord, utility or inspector for alterations. Electrical loads, grease extraction, drainage, washdown procedures, accessible circulation, fire and life-safety rules, network reliability and overnight installation windows can determine whether the robot is feasible before any model inference cost is considered. The facility owner and the robotics purchasing entity may not be the same party.

For a multi-unit chain, rollout strategy must consider the distribution of physical configurations. A technology company that designs only for the newest flagship store risks overlooking the older installed base that creates volume. Conversely, designing for the worst legacy site can impose excessive hardware cost everywhere. A site-classification protocol—format, footprint, utilities, workflow layout, equipment vintage, order-channel mix and local demand—permits a staged addressability estimate and reduces installation surprises.

Supply-chain information also has physical consequences at the restaurant boundary. An order promise is not executable if a required ingredient has not arrived, a cold item is out of temperature, an equipment component is down, or a batch has not completed its cooking cycle. Restaurant intelligence must reconcile planned demand with inventory availability, vendor deliveries, station readiness and physical handoff. An integrated system should recognize when human intervention is safer or more economical than attempting autonomous recovery.

GII / PHYSICAL AI COMMERCIALIZATION

Every Restaurant Is a Site-Specific Multi-Asset System

PROPERTY	Lease, power, floor plan, physical routing	Landlord / franchisee
EQUIPMENT	Fryers, refrigeration, kitchen and maintenance	Vendor / owner
PEOPLE	Task skills, staffing, safety and hospitality	Employees / operator
DEMAND	Order clocks, customer preferences and pickup	Brand / marketplace

Incremental project FCF(t) = gross cash benefit(t) − cash OpEx(t) − cash tax(t) − project CapEx(t)

Module II: installation at t=0; refresh in Year 4; financing and working capital excluded.

Figure 10 | Restaurants are multi-asset, site-specific operating businesses

GII original representation of the economic and contracting layers of a restaurant location.

3.8 The geography of scale and service

Location matters again after the robot is installed. Service engineers, spare-parts depots, sanitation suppliers and installers travel across a geographic network. If the installed fleet is thinly dispersed, the service cost per active machine may consume recurring software margin; a dense deployment cluster can support shorter service routes and consolidated spare-parts inventory. This creates a second market filter beyond restaurant count: the geographic density at which the vendor can support its promised uptime.

The analyst should model per-site fully installed capex and fleet-level support separately. A vendor may report thousands of contracted locations while only a subset is live, productive, renewed, and economically serviceable. Chain procurement advantages can be offset by concentrated buyer bargaining power, customization requests, and difficult service-level penalties. Neither the restaurant group nor the supplier should call a location live until installation, acceptance, training and recurring operations are documented.

The airport configuration in Section 6.7 makes the site contract concrete: production equipment, customer access and service responsibilities have to fit the same location. The supplemental cases are used to examine those interfaces; they do not add units to the U.S. market denominator.

3.9 One location can support several production and fulfillment functions

A restaurant location can operate as a small food-production facility, a dining venue, a QSR pickup point and a dispatch interface for couriers, ground robots or drones. These are potential operating functions of an eligible site, not an assertion that every restaurant can or should support all of them. Kitchen capacity, frontage, parking, utilities, food-quality windows and the owner's rights determine which combination is feasible.

Inside the four walls, connected equipment and bounded kitchen automation may improve production consistency or release a bottleneck. Beyond the four walls, first-party ordering, third-party marketplaces, catering and delivery can connect that production to nearby demand. The same kitchen serves both. Additional channels consume packing space, staff attention and production capacity; an order shifted from dining to delivery is not automatically incremental revenue.

This location-based perspective connects restaurant and franchise management with commercial real estate and technology adoption. A landlord controls parts of the physical envelope; a franchisor may set operating standards; a franchisee often bears local capital and labor costs; a platform may control the customer interface. A viable deployment must align these parties rather than assume that a large brand's store count is a centrally controlled robotics installation base.

Network scale becomes useful when standardized recipes, equipment, event interfaces, installation methods and regional field service reduce the cost of the next accepted location. Permissioned task data and tests at unfamiliar sites can then help developers investigate generalization. The advantage is conditional: layout variation, local demand, unwilling users and expensive support can offset the benefits of a large network. Appendix M maps the functions, evidence and acceptance questions.

4. World Models for Restaurant Environments: A Research-to-Deployment Bridge

PART II | WORLD MODELS, ACTION DATA AND RESTAURANT INTELLIGENCE

Technical question. World models are relevant because restaurant work is dynamic and contingent: orders arrive unexpectedly, ingredients vary in geometry and texture, equipment heats or cools, and humans share tight spaces with moving objects. The useful technical problem is not recognizing a “kitchen” in video. It is representing sufficient world state to predict what an action will change, detect failure, and adapt safely to variation.

A world foundation model, a **vision-language-action (VLA) policy**, a process simulator, an industrial digital twin and a rule-based kitchen dispatcher are not synonyms. World models aim to represent or predict evolving environment states under actions; VLA models convert multimodal observations and instructions into actions; digital twins replicate a bounded facility or process; scheduling systems allocate tasks under constraints. A particular commercial system may combine these components, but their evidence requirements differ.

NVIDIA’s May 2026 Cosmos 3 technical blog describes an architecture spanning physical reasoning, world and action generation [S96]. Google DeepMind’s July 2026 Gemini Robotics On-Device 2 model card describes on-device general-purpose manipulation and limited tester access [S97]. These are useful technology reference points, not confirmation of a scaled restaurant-specific World Model deployment.



Picture 4 | Predicting physical outcomes and testing transfer between kitchens
AI-generated concept illustration for GII; equipment and settings are illustrative.

4.1 State, action, outcome and feedback requirements

The modeled restaurant environment should represent local geometry, staff and robot proximity, ingredient state, food-service order state, equipment telemetry, action permissions, physical safety interlocks and the consequences of timing. Action-conditioned prediction becomes commercially valuable only when it helps select safe and productive actions that outperform deterministic control or human execution.

Table 27 | World Model research capabilities

World-model input / task	Restaurant representation	Evaluation metric
Scene and geometry	Work-cell bounds, station locations, aisle width, reachability	Collision-free completion; generalization across layouts
Materials and objects	Variable food state, liquid levels, packaging, utensils	Manipulation success; product quality; rework
Dynamic equipment	Cook time, temperatures, motor state, system faults	State prediction error; safe actuation latency
Human collaboration	Worker location, task queues, handoffs, supervision	Interruptions; intervention rate; safe handoff
Order and demand	Station backlog, due time, courier arrival	On-time fulfillment; weighted tardiness
Outcome and failure	Quality result, exception, human correction	Recovery time; repeated-error reduction

GII conceptual synthesis. Categories are proposed measurements, not observed industry benchmarks. Technical references: [S96–S97].

4.2 Sim-to-real transfer and realistic commercial benchmarks

Proof of concept should not be graded only by task completion in a curated demonstration. A credible restaurant benchmark needs held-out stores, unseen equipment variations, difficult ingredient conditions, human interruptions and recovery protocols. Demonstrating 95% success on one prepared task may still be unacceptable in food preparation if the remaining failures create safety, sanitation or service disruptions.

Research gate. Specify the physical task boundary, environmental variations, a safety stop protocol, an intervention taxonomy, latency tolerance and a measured baseline against trained human or conventional automation. Report performance and cost per successful, acceptable restaurant task rather than raw actions or training-video hours.

Primary evidence: NVIDIA Cosmos 3 [S96]; Google DeepMind model card [S97]

4.3 A precise technical map: world model, planner, VLA and controller

A world model is best treated as a **representation or predictor of relevant environmental state and transitions under actions or external events**. A robot policy, including a vision-language-action model, maps observations and goals toward actions. An embodied reasoner may plan or decompose tasks, while a deterministic controller enforces critical equipment constraints. These components may be combined in an architecture but should not be conflated; impressive video generation is not evidence of food-safe closed-loop manipulation in an operating kitchen.

NVIDIA describes Cosmos 3 as integrating physical reasoning, world generation and action prediction within a foundation-model family [S103]. Google DeepMind describes Gemini Robotics ER 2 as an embodied reasoning model for spatial understanding, multistep planning and multi-robot collaboration [S104]. Those announcements establish developer capabilities and technical direction. They do not establish a restaurant-specific benchmark, validated general kitchen autonomy, or business returns. The white paper deliberately bridges published platform descriptions to a proposed restaurant test program rather than making deployment claims.

Restaurant environments expose difficult state-estimation problems: reflective stainless steel, transparent containers, steam, hands occluding ingredients, deformable foods, temperature-dependent texture, noisy equipment, and multiple people entering and leaving a station. A useful model may need to represent both continuous phenomena, such as heating and fluid movement, and discrete events, such as whether an order has been accepted or a fryer is unavailable. A general visual model may need equipment-specific telemetry and calibrated sensing to avoid confusing plausible predictions with operational truth.

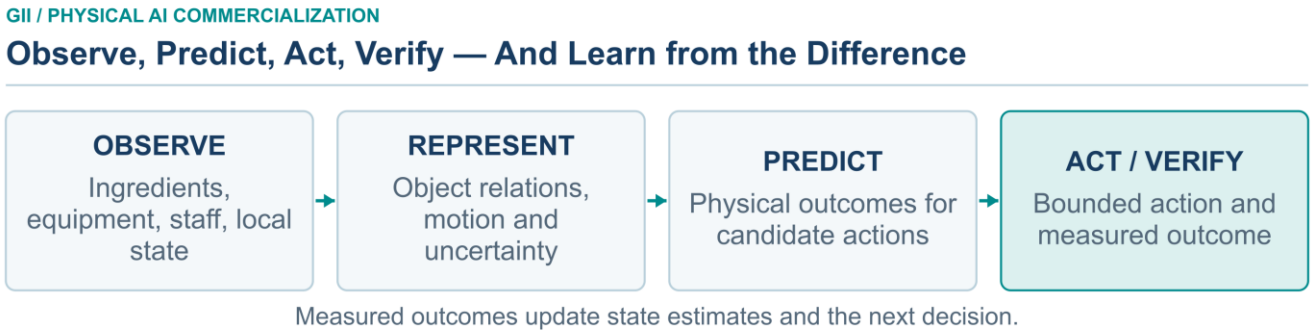


Figure 11 | State, prediction, action and measured feedback form a proposed restaurant world-model loop
Adapted conceptually from physical-AI model research [S103–S104]; no chainwide restaurant world-model deployment is implied.

4.4 From sim-to-real to site-to-site transfer

Restaurant service tests a capability repeatedly, under changing conditions. Menus change, equipment wears and staff intervene during a rush. Synthetic trajectories can expand training coverage, but their value still needs to be tested against measured physical outcomes. A useful benchmark states which conditions remain fixed and which vary across restaurants.

A falsifiable research program could compare three approaches on a bounded preparation task: a fixed-rule machine, a perception-enhanced specialized robot, and a model-based adaptable policy. Each would receive identical task objectives and operational constraints. Report first-pass success, time per successful unit, time to recover from failures, cleaning/reset burden, human interventions, food-quality conformity, service uptime and cost per acceptable output. The study should specify which failures trigger a human handoff and which abort a task immediately.

The most interesting economic prospect is reusable intelligence across a network of locations. If a policy transfers from one kitchen to another with less retraining, commissioning and exception engineering, a robotics supplier may reduce fleet acquisition and support costs. That benefit is not guaranteed: every new robot body, end-effector, product class or safety certification may limit transfer. A commercial research paper should show the transfer boundary explicitly rather than extrapolating one laboratory success to an entire restaurant system.

Table 28 | Restaurant robotics commercialization benchmark

Dimension	Bench test	Operational acceptance evidence
Task success	Conforming result across object variation	Success per shift at an actual operating site
Speed	Cycle time excluding reset	End-to-end capacity including load, clean, handoff
Generalization	Unseen recipe/equipment/layout scenarios	Performance across held-out kitchens and shifts
Safety	Defined stop and human-override behavior	Logged near misses, violations and interventions
Recovery	Failure detection and retry	Minutes of interruption and safe restoration
Economics	Compute and bill of materials	Installed and supported cost per accepted output

Source: GII proposed benchmark; no claim of an existing harmonized industry-standard score.

4.5 The technical interface: sensing, reasoning, policy and safety authority

The strongest technical claim a restaurant vendor can make is not that its system has a large world model. It is that a complete chain from observation to safe action is specified, measurable and recoverable at a real site. Camera observations alone may miss temperature, equipment mode, allergen-control state, employee intent or tool wear. A predictive model should report uncertainty when its view is incomplete; the planner should be able to defer or request human review rather than assume the kitchen matches its training scenes.

A useful engineering separation has five interfaces: (1) sensors and store-event systems establish time-stamped physical and operational state; (2) a representation or world model predicts relevant consequences and uncertainties; (3) embodied reasoning or task planning selects permitted goals and substeps; (4) a robot policy converts the chosen task into motion or machine commands; and (5) local deterministic safety controls enforce physical constraints, isolation, emergency stops and safe recovery. Actual products may combine layers, but the safety and measurement responsibilities do not disappear when the software stack is unified.

A current illustration of the distinction is Google DeepMind’s Gemini Robotics ER 2. The official July 30, 2026 model card identifies it as a vision-language model that produces text, supporting spatial reasoning, orchestration and success detection. Google’s product description explains that it hands motor execution to a lower-level vision-language-action model. That is different from claiming ER 2 directly controls gripper actuators. NVIDIA’s Cosmos 3 spans world simulation, reasoning and action-model development, but its public benchmarks should not be presented as restaurant-specific safety or operating validation. [S103–S104, S112]

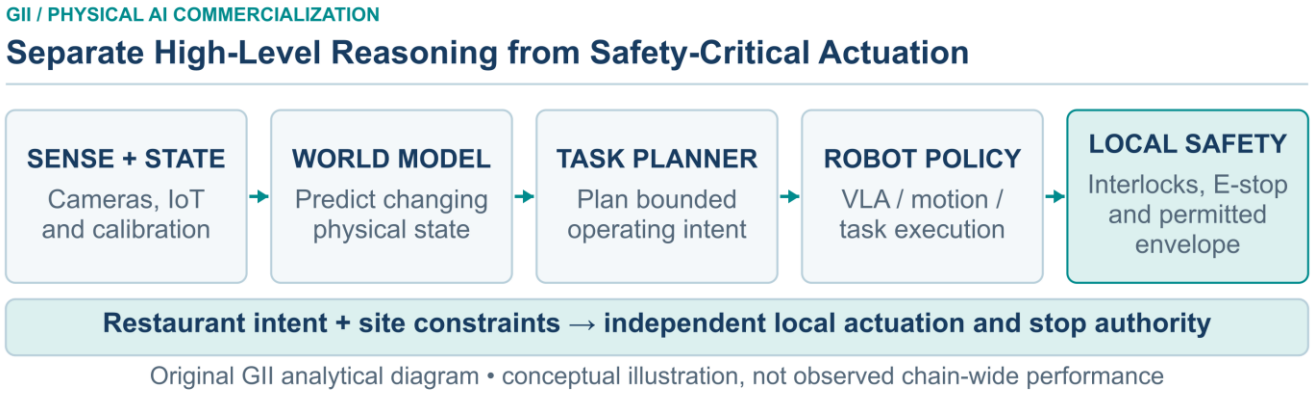


Figure 12 | A proposed sensing-to-action architecture with an explicit local safety boundary

GII original; technical classification supported by developer descriptions [S103–S104, S112]. This is not an approved machinery safety architecture, certification, or product-specific implementation.

The interface contract should name its owner, update rate, sensor calibration procedure, stale-data behavior and failure handling. In food contact, a model hallucination is not merely a wrong answer: it can become a contaminated product, damaged equipment or interruption of service. For that reason, the critical result of model uncertainty should often be a bounded no-op, machine stop or operator handoff. A cloud outage, site Wi-Fi loss or missing customer-event message must not bypass a local machinery safety function.

The measurement protocol must also separate predictive quality from operating quality. Model predictions can be scored against future sensed states; action policies need held-out task episodes; the integrated restaurant system additionally needs shift-level accepted output, latency distributions, intervention minutes, sanitation cycle time, uptime and incident logs. No single composite “robot success rate” can replace these denominators.

Table 29 | Pilot interfaces and acceptance evidence

Layer / owner	Primary output	Required evidence	No-go boundary
Sensor + site systems	Synchronized physical and KDS state	Timestamp coverage, calibration, missingness	Unknown food/equipment state
World model / predictor	Expected future state and uncertainty	Held-out prediction and uncertainty tests	Confident but unsafe predictions
Embodied planner	Permitted task and sequence	Plan feasibility, tool and state constraints	Unapproved task or kitchen sequence
VLA / robot policy	Bodied motor actions	Action success and interventions by condition	Unrecoverable manipulation errors
Local controller / safety	Physical authority and interlocks	Stop, isolation and recovery testing	Unsafe operation on model/cloud failure
Operating site / owner	Acceptable service output	Shift-level quality, uptime, labor and cash	Negative site economics or unsafe service

GII proposal. Vendor/model capability statements should be verified independently before use as restaurant deployment or safety claims.

4.6 Skild S1 and the commercial kitchen pilot boundary

Skild AI’s September 10, 2026 founder announcement identifies a pilot with Mitsui using S1-powered general-purpose robots in commercial kitchens. The statement does not disclose a named retail restaurant fleet, task-level production results or restaurant unit returns. Mitsui’s meal-supply scale describes the partner’s supply chain rather than robotic output. The appropriate evidence stage is a named commercial-kitchen pilot [S130].

The developer question is how a generalist policy earns an advantage over a specialized cell under restaurant operating conditions. A meaningful comparison would hold the task and quality requirement constant, then measure setup effort, successful completions, cycle time and interventions as tools, ingredient conditions or equipment change. A model that adapts quickly could reduce engineering effort, but that value needs to survive sanitation, safety authority and recovery requirements at the site. This is a proposed comparison, not a measured S1 result.

The Mitsui pilot can be interpreted using the architecture in Table 29. A foundation robot policy, a predictive world model and a local safety controller perform different functions; a product announcement does not establish all three. Commercial evidence should identify which component is being evaluated and who remains responsible for stopping or recovering the

machine. The production and service cases in Sections 6.6–6.9 and 9.10–9.12 provide other deployment settings, rather than a common leaderboard for unlike robot categories.

4.7 Restaurant manipulation provides the relevant model evidence

Restaurant robotics offers direct evidence about learned physical execution. The useful comparison is between a task-specific system and a learned policy performing the same accepted task under the same operating conditions. A robot should be evaluated on the complete drink, bowl or service episode, including recovery, rather than on the most successful movement in a demonstration. Table 147 separates the technical approach from its current implementation stage.

A vision-language-action model uses visual observations, an instruction and robot state to produce actions. Its action layer may generate joint commands or end-effector targets, sometimes as action chunks. A task planner can call bounded skills above that motor-action layer. A learned policy need not contain an explicit predictive world model. Its role also differs from the production scheduler that assigns work and the local controller that enforces physical limits. Google's On-Device 2 card explicitly identifies numerical robot actions as outputs; ER reasoning has a different role [S97,S148].

Artly describes learning beverage movements from human demonstrations, composing learned skills and adapting with VLA models. Its Robot School architecture includes robot-specific teleoperation data, embodiment adapters and force/torque constraints [S140,S141]. A useful acceptance test therefore changes a cup, pitcher or ingredient position while holding drink quality constant, records the installed hardware and policy version, and measures both completion and assistance. The vendor's general architecture should not be assumed present in identical form at every cafe.

Skild's S1 study uses video demonstrations as task context and shows food-related manipulation. At the largest tested pretraining scale (100,000 hours), its internal study reports approximately 96% for seen tasks and 66% for unseen tasks. These are cumulative per-step measures on 4–8-minute tasks. The protocol allows human recovery, used mainly for the language-conditioned VLA baseline; the results are not autonomous restaurant order-completion rates [S142]. The separate Mitsui disclosure establishes commercial-kitchen pilots, while the partner's wider meal volume remains outside the robot-production denominator [S130].

Physical Intelligence's April 2026 pi0.7 work explores richer prompts and skill composition, including kitchen appliances. Its October 13 API beta provides managed fine-tuning and hosted inference and acknowledges connectivity challenges [S150,S149]. Together with the local-execution route in On-Device 2, these illustrate deployment choices: model access, adaptation effort and inference location must be tested against the task's response-time and outage requirements.

Training a new policy, fine-tuning a pretrained policy and prompting a frozen policy with a demonstration are different processes. A claim of learning from one example can depend on extensive prior training. The commercial test is the total effort to reach accepted performance at the next site: demonstrations, tuning, validation, local configuration and subsequent maintenance. Versioned rollout and rollback should follow evaluation, rather than allowing an unvalidated policy update during service.

4.8 The generalist case still has to pass a restaurant service test

The generalist case deserves serious consideration. A reusable policy that transfers across recipes, tools and robot bodies could reduce repeated programming and make smaller task volumes economical. This would weaken the advantage of a narrowly scripted cell and increase the value of broad, permissioned task data. The relevant counter-test is whether the next installation needs less engineering while preserving quality, speed and safe recovery.

The unresolved work is physical and operational. Steam and glare can obscure state; food can deform, stick or spill; contact forces can vary with containers; a long task can lose track of a completed step. A recovery action may require discarding food, changing tools or asking a worker to intervene. These conditions should be represented in test episodes, including novel combinations and held-out restaurants, rather than addressed only through an average benchmark score.

Task success also compounds over a complete job. In a deliberately simplified example, twenty independent steps each succeeding 98% of the time yield only about 66.8% uninterrupted completion. Actual failures are often correlated, so this is a teaching calculation rather than a deployment prediction. Measure whole-task completion under a time limit, intervention minutes, rework and accepted output, with explicit denominators. Table 149 provides the release scorecard.

My current assessment is that bounded food-production tasks offer the most credible near-term implementation path. Site fit, human recovery and sustained peak performance remain the binding commercial tests. Transfer across a network is a valuable hypothesis that must be demonstrated with the same frozen evaluation policy before local retuning. General capability can lower adaptation cost without removing those site obligations.

Methodological context: Pavone's ODSC session and the Alpamayo driving paper remain background on reasoning and action research [S138,S131]. They provide no restaurant manipulation or service validation; the restaurant evidence above carries that argument.

5. Real-World Restaurant Data: From Human Demonstrations to Robot Learning

PART II | DATA RIGHTS, ROBOT LEARNING AND DEPLOYMENT EVIDENCE

Restaurants offer rich sensory and process data, yet much of it is unsuitable for learning physical actions without further instrumentation. POS events describe orders and payments; KDS logs describe production milestones; AIoT telemetry captures equipment status; camera streams reveal some activity. Robot learning requires synchronized observations, action representations, resulting states, task success and failure labels, and a valid right to collect and use the records.



Picture 5 | Capturing a permitted demonstration of a skilled beverage task

AI-generated concept illustration for GII; equipment and settings are illustrative.

5.1 A minimum viable action-data specification

Table 30 | Restaurant action-data diligence

Required field	Example restaurant observation	Commercialization and governance test
Task definition / boundary	Prepare, transfer or inspect one SKU or bounded food process	Can task value and failure cost be measured?
Timestamped state	Camera, tool pose, equipment temperature, order status	Are signals synchronized and quality assured?
Action and agent	Worker demonstration, robot control, equipment command	Was the action authorized and safely performed?
Outcome / intervention	Accepted plate, spill, timeout, manual correction	Can acceptable output be measured independently?
Variation metadata	Store format, item type, equipment model, shift conditions	Will learned behavior transfer across sites?
Consent and license	Worker notice, business permissions, customer exclusions	Who may commercialize data and for how long?

GII proposed collection protocol; not an assertion about any named firm's internal practices.

5.2 Robot-training data versus operating intelligence

A robotics laboratory may value diverse recordings of how humans handle food, equipment or containers. A restaurant operator may instead value a small set of timely diagnostics, lower order errors or fewer staff interruptions. The same data collection project may therefore generate unequal benefits. Contracts should specify permitted uses, worker and guest protections, derived model rights, deletion or retention, and whether participating restaurants gain access to useful operating outputs.

5.3 A practical consent-based pilot protocol

Begin with a non-customer-facing work cell and a limited approved task. Seek written site and employee consent, prevent incidental customer recording, minimize identifiable data, document sanitation and safety requirements, and pre-register metrics. Compare with the existing standard operating procedure. Include complete human intervention time, supervision, machine cleaning, exception recovery and output-quality testing in costs. A large quantity of footage is not success if the trained behavior cannot operate safely and economically.

Commercialization insight. The most strategically valuable dataset may be small, carefully labeled and strongly linked to an economic task. More video alone does not create model generalization, restaurant acceptance or recurring supplier revenue.

5.4 Design action-ready observations instead of collecting video indiscriminately

Robot learning needs data aligned to a task objective and an action vocabulary. In addition to images, useful records may require synchronized depth, object pose, contact or force measurements, equipment state, tool trajectory, order or recipe state, operator corrections and time-stamped success labels. A camera-only recording might help discover activity patterns but cannot necessarily reconstruct grip force, motor commands, food temperature or the conditions that made an action safe. Data volume is not a substitute for a usable schema.

A restaurant chain should negotiate whether recordings are limited to internal quality improvement, permitted for vendor model training, licensed for cross-customer learning, or retained after contract termination. Employees and customers may be captured incidentally, so consent, notices, access controls, retention limits, revocation mechanisms and security governance are critical. Worker performance monitoring should be governed separately from permissible robot-training use. A vendor has no automatic proprietary right to every sensor stream produced inside a restaurant.

From the operator's perspective, the exchange should be explicit: the robotics company may obtain permitted demonstrations, failure examples and operating constraints, while the restaurant should receive a defined service, data protections, operational reporting and potentially commercial terms recognizing its contribution. A supplier that monetizes customer-derived training data while leaving all sanitation, labor and installation costs with the restaurant may face adoption barriers even if the underlying model improves.

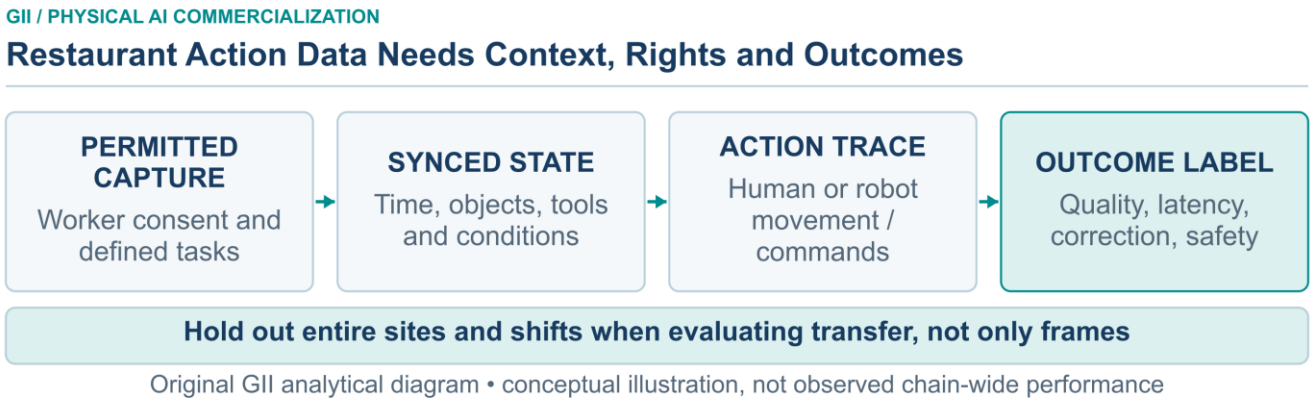


Figure 13 | A synchronized action-data schema and the governance boundary around it

GII research design. Fields are proposed; no identifiable restaurant employees or collected records are supplied.

5.5 A practical data-collection and benchmark agreement

Begin with a narrowly scoped task for which the restaurant can define acceptable output. Obtain a written site protocol covering permitted cameras and sensors, staff consent, operating hours, food-contact boundaries, exception handling, work interruption, data storage locations, third-party processors, deletion and model-training restrictions. Collect baseline human task observations without disrupting operations, then run supervised prototypes that record every intervention and rejected output. Independent quality checks should be built into the measurement, not added only after favorable results emerge.

Dataset documentation should identify the units of generalization: number of locations, kitchen layouts, menu items, workers, robot hardware variants, shifts and environmental conditions. A thousand similar trajectories at one carefully staged bench do not establish robustness across hundreds of operating restaurants. Sample construction must include rush periods, ingredient variability, device failures, network disruptions, cleaning and real handoffs. Data splits must avoid leakage from nearly identical successive frames into both training and evaluation subsets.

Before declaring commercial readiness, connect model metrics to restaurant KPIs. A manipulation success rate alone is insufficient if the station still requires an employee to load ingredients, clear faults and inspect each finished order. Report successful output per paid employee-hour, quality-conforming items per operating hour, intervention count per shift, adverse events, and total service cost per accepted item. The same log supports model improvement and unit-level financial underwriting.

Table 31 | Action dataset governance register

Dataset element	Required record	Commercial or safety reason
Context	Store, station, date/time, menu, equipment version	Define transfer and scenario coverage
Sensor observation	Permissioned video/depth/telemetry, sync	Reliable state estimation; access control
Actions	Human demonstration or robot commands	Supervised task policy training
Outcome label	Quality and task completion/failure	Evaluate results rather than visual plausibility
Intervention	Human correction and root cause	Field reliability and human-time economics
Rights and deletion	Usage purpose, training grant, retention, audit	Employee privacy and vendor IP boundary

Source: GII proposed action-data schema; collection requires legally appropriate permissions and governance.

5.6 Data quality, licensing and transferable task intelligence

The value of restaurant action data is not proportional to its file size. A developer may have thousands of hours of kitchen video yet lack robot joint states, object-contact changes, tool identity, food condition, permitted action classes or an independently observable completion signal. Each recorded task episode should therefore have a versioned task description, equipment and site configuration, synchronized clocks, input conditions, actual actions, outcome labels, exception causes and a record of any human intervention. The restaurant should be able to audit whether the resulting dataset reflects a meaningful work problem rather than casual surveillance.

For site-to-site transfer, predefine three splits: within-store held-out shifts, held-out equipment or ingredient conditions, and completely held-out restaurants with differing layouts and staff. The last split is especially commercially important. A performance gain demonstrated only after each location receives custom calibration, engineer intervention or new hand labeling may create a consulting project rather than a scalable Physical AI product.

Table 32 | Action-data package for permitted research

Data field	Why robotics needs it	Restaurant / worker safeguard	Validation test
State timestamps + sensor quality	Reconstruct action-conditioned transitions	Minimize incidental customer capture	Clock offset and missing-data audit
Task/action vocabulary	Match demonstrations to permitted behaviors	No inferred employee-performance scoring	Reviewer agreement on labels
Equipment / geometry version	Transfer across layouts and machines	Vendor-independent asset naming	Held-out site and hardware split
Failure + rescue trace	Train safe recovery and handoff	No punitive use of worker errors	Intervention minutes / severity log
Usage rights + retention	Reproducible training and model lineage	Site, worker and customer authorization	Deletion, export and derivative-rights audit

Research access requires permitted use; this package is not a blanket camera-collection license.

GII research protocol. Scope, labor requirements and privacy obligations are jurisdiction- and contract-specific; qualified counsel and site operators must review collection.

The contracting structure matters as much as the labeling schema. Data may belong to the restaurant, technology provider, employees or third parties under different legal frameworks; the report does not assume universal ownership. A commercial collaboration should specify field-of-use restrictions, model derivative rights, training and evaluation access, auditability, retention, revocation procedures and whether the participating operator receives dashboard insight, reduced deployment fees or another measurable benefit. The preferred pilot is narrow, documented and consent-based—not a blanket entitlement to record a workplace.

5.7 Using a commercial beverage task to test action data

Artly's beverage workflow provides a concrete setting for the action-data specification. Its public materials describe learning from barista demonstrations; Section 6.8 separates those supplier descriptions from the company's cafe operating disclosures [S119–S121]. A proposed GII study would select one permitted drink-preparation task and record the initial physical state, action trace, completion criteria and every intervention. Customer orders and cups served would remain operational measures, separate from training episodes.

The evaluation should distinguish a new recipe from a changed physical condition. A held-out test could vary cup geometry, pitcher identity or ingredient placement while keeping the required beverage quality fixed. The record would identify the hardware and policy version, permitted sensor fields, failed attempts and any local retuning. Transfer should be assessed before site-specific adjustments are mixed into the result. These are research-design proposals, not claims about Artly's internal tests.

Demonstrator permissions and cafe telemetry need separate agreements. Access to a worker's recorded motions does not automatically permit capture of guests or unrestricted model reuse. A useful commercial dataset would document permitted training uses and retain enough context to distinguish policy error from equipment, ingredient or service failures. Table 142 extends the measurement plan to the operating site and supplier.

6. Intelligent Kitchens and AIoT: Modern Equipment Before General Humanoids

PART II | AIOT, KITCHEN ASSETS AND PHYSICAL EXECUTION

Restaurant equipment is a long-lived physical capital base. Ovens, fryers, refrigeration, beverage machines, scales, food-handling stations and kitchen control systems can become richer sources of state information and more responsive actuators. But not every networked device needs a world model; reliable rule-based controls or conventional maintenance analytics may be the least-cost solution.

6.1 Four implementation levels

Table 33 | Restaurant physical-intelligence maturity

Level	What it does	Restaurant example	Evidence needed
Monitor	Records status or alerts	Refrigeration condition, fryer cycle	Data reliability and useful operator response
Recommend	Models suggest a step	Equipment fault prediction	Avoided failures, accepted recommendations
Closed-loop control	System safely changes setpoints/actions	Bounded cooking control	Food quality, safety, manual overrides
Robotic manipulation	Machine handles physical objects	Autocado, automated makeline pilot	Full workflow economics, generalization, maintenance

Digital monitoring does not establish closed-loop autonomy. GII interpretation supported by company use-case evidence [S99].

6.2 Deployment costs that laboratory teams routinely miss

A successful kitchen robot must fit existing equipment lines, fire and food-safety constraints, utility capacity, cleaning schedules, staff circulation, noise, maintenance access, work-permit conditions and franchisee capital plans. Hardware bill of materials is only one line item; other costs include installation, integration, training, lost operating hours, inspection, network support, remote monitoring, spare parts, upgrades and incident liability.

Chipotle's September 2024 release identified specific restaurant pilots for Autocado and an Augmented Makeline, and described how crews continue to handle related tasks [S99]. This is useful site-level evidence of targeted automation, not scaled adoption proof. The commercialization question is whether each complete restaurant workflow—including loading, cleaning, exception recovery and quality control—produces positive incremental value.

6.3 Asset ownership, replacement and supplier strategy

A restaurant chain may acquire equipment, license a service, co-develop with a startup, make a minority investment or divest technology development while retaining service access. The optimal structure depends on proprietary data, installed capital, maintenance responsibility, replacement cycles, franchise terms and potential supplier concentration. The restaurant's willingness to buy arises from unit-level return, not the vendor's fundraising valuation.

Primary evidence: Chipotle named restaurant pilots [S99]; Sweetgreen retained technology access [S110]; Miso dated fleet and service disclosures [S113].

6.4 Why intelligent equipment is a credible first path

The first economically feasible Physical AI application may be an intelligent controller on an existing appliance, not a mobile humanoid. Kitchen assets already constrain temperature, motion, sanitation cycles and production geometry; additional sensors, diagnostics and higher-level decision systems may improve reliability without introducing unbounded manipulation. A connected fryer can report oil condition and cooking cycles; a refrigerator can report fault states; a dispatch controller can sequence machines against incoming orders. Whether any of these becomes Physical AI depends on its degree of closed-loop decision and physical control.

The engineering stack should maintain a strict authority boundary. A prediction service can recommend a cook schedule or identify abnormal temperature trajectories, but safety-critical shutoffs and food-contact control interlocks should remain in validated equipment controllers. Cloud inference latency, wireless failure or model hallucination must not defeat those constraints. Physical execution requires a defined state machine for safe stop, restart, operator override, audit and maintenance. A dependable integration is often more valuable to a chain than a broadly capable policy without a certified appliance interface.

Modernization costs extend beyond hardware. Appliance interface access, cleaning clearances, power and network installation, foodservice code requirements, field technician training, software version compatibility, cybersecurity, vendor warranties and franchise approval can dominate a project's practical timeline. To estimate total cost of ownership, add

hardware or retrofit cost, installation, required downtime, recurring licensing, equipment service, consumables, insurance and periodic replacement. Compare with a conventional smart-appliance program before attributing the entire improvement to robotics AI.

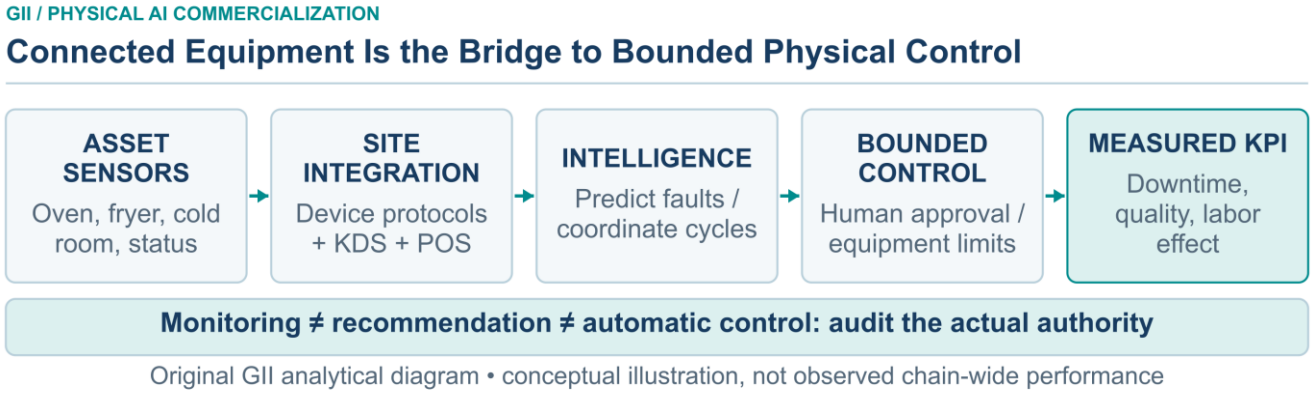


Figure 14 | From connected kitchen assets to bounded, measured physical control

GII architecture; monitoring, decision support and autonomous execution are intentionally separated.

6.5 Procurement and field service should shape robot architecture

A restaurant buyer will favor a system that an ordinary kitchen manager can use, clean and recover after an incident. Hardware engineers should test sanitation compatibility, tool-change time, maintenance access, consumable supply and available floor area as early as dexterity. Software engineers should test remote support, offline recovery, version rollback and auditability. World-model teams should establish whether sensors provide enough physical state for safe operation and whether inference belongs on a local controller, an edge cluster or a cloud service.

The winning architecture may be modular: intelligence layered above existing equipment where practical, specialized robotic stations for high-volume repetitive tasks, and humans retained for judgment, exceptions and hospitality. This layered path should not be interpreted as evidence against humanoids. It is a staged economic strategy: deploy what currently meets quality and cash hurdles while collecting consented evidence that may later justify more capable general-purpose systems.

Table 34 | Restaurant equipment integration and lifecycle audit

Layer	Engineering verification	Owner/accountability
Facility	Electric capacity, plumbing, ventilation, footprint	Landlord/restaurant/franchisee allocation
Machine control	Safety-rated states, food-contact limits	OEM and integrator responsibility
AI intelligence	Detection, recommendations, action bounds	Software vendor and equipment owner
Sanitation	Cleanability, washdown, tool replacement	Shift staff and maintenance partner
Support	Spare parts, technician routes, remote diagnostics	Contracted service-level obligations
Economic measure	Whole-station outputs and paid labor hours	Operator acceptance and reporting

Source: GII original commercial engineering diligence matrix; requirements depend on site and jurisdiction.

6.6 Miso and the deployment economics of a fry station

Miso offers a documented example of a specialized manipulation product in operating foodservice environments. Its FY2025 Form 1-K reports ten third-generation Flippy units across White Castle, Insert Coin and Levy Restaurants as of March 31, 2026, following withdrawal of earlier generations. This combined count is a dated deployment disclosure. It is not the number installed at White Castle alone or a current fleet census [S113].

The filing describes hardware sales and leases alongside subscriptions, installation and support. Miso retains ownership of leased units and describes covered repair and maintenance responsibilities. These contractual obligations make service execution relevant to the supplier's economics as well as the restaurant's. Management's prospective operator-benefit statements remain claims rather than measured site cash returns [S113].

The physical task gives developers a bounded engineering problem. GII's proposed evaluation would follow one fry-basket cycle from the accepted production instruction through handling, completion and release to the next worker. The measurement boundary should specify which food staging, replenishment, finishing and cleaning tasks remain outside automated execution. Those residual tasks are proposed diligence fields, not assertions that every installation uses the same staffing arrangement.

A task-specific product can gain value from stable equipment interfaces. The research question is whether added model flexibility reduces setup or intervention cost enough to justify its own compute, validation and support burden. A useful test would introduce a defined equipment or product change, compare the existing configuration with the proposed policy, and account for engineering hours as well as task success. The comparison should retain the same food-quality requirement and the same local stop authority.

For the restaurant, investment recovery depends on utilization and the route by which operating improvements become cash. Faster task execution may protect peak capacity, reduce overtime or improve quality; each channel has a different measurement requirement. If the next station remains the bottleneck, additional basket capacity may have little sales value. A purchase-versus-lease comparison should include installation downtime, recurring fees, removal obligations and the remaining period of authorized site use.

For the supplier, deployment count should be accompanied by support demand and retained contribution. GII proposes tracking installations completed, active paid units, service visits, parts consumption and technician hours by equipment generation. These observations would reveal whether a more capable product also becomes easier to support. They should be kept separate from restaurant sales and from the value of demonstration or training data.

Miso is consequently useful as an integrated product and service case. It does not calibrate the illustrative Module II inputs. The principal model and the alternative DCF cases retain their stated assumptions; a named restaurant return would require its own documented costs and benefit bridge. Table 139 records the evidence stage, while Table 142 identifies the measurements needed for a stronger underwriting case.



Picture 6 | Bounded fry automation and the surrounding human work
AI-generated concept illustration for GII; equipment and settings are illustrative.

Miso also reports using NVIDIA cuRobo and Isaac Sim in product development, connecting motion development and simulation to a commercial kitchen task [S113]. This is issuer-reported tooling use; it does not establish Alpamayo deployment, independent kitchen-policy generalization or measured restaurant returns.

6.7 Donatos and the airport production and pickup system

Donatos' May 2025 announcement names Appetronix, Agápe Automation and HMSHost in an automated pizza operation planned for John Glenn Columbus International Airport. The brand's FAQ identifies the airport Pepp-Tron machine and describes centrally prepared dough in its wider operating system. Airport planning material describes digital ordering and locker pickup. Together these sources identify a bounded production and retail configuration; they do not establish profitable fleet expansion [S116–S118].

The case connects the restaurant's demand interface to physical execution. For a developer, the relevant system boundary begins with an accepted order and ends when the correct customer can retrieve the completed product. A successful manipulation step is only part of that service. GII proposes testing order identity, recipe execution, readiness notification and pickup access as one transaction, including what happens when payment succeeds but production or retrieval fails.

An airport location also makes property and channel constraints visible. The eligible task depends on available utilities, food storage, customer access and the right to occupy the space. A compact automated station can change where food is offered, but a smaller production footprint does not by itself establish a lower total occupancy or service cost. The concessionaire, brand, equipment provider and airport may each control different parts of the operating agreement.

Prepared inputs are an important part of the technical boundary. Standardizing a food input can simplify local manipulation while shifting work to a supplying facility. The economic comparison should therefore follow preparation, transport, cold storage and replenishment, rather than count only the motions visible to the customer. This is a proposed full-workflow accounting boundary, not a finding about the allocation of costs in the Donatos contract.

Performance should be measured as completed orders over a stated demand window, together with unsuccessful transactions, interruptions and the human effort required to restore service. First-order preparation time and steady-state throughput are different measures. Published promotional timings are not used here as an observed operating benchmark. The system may also need to be available when demand is low, making utilization and scheduled service coverage material to a site comparison.

GII's proposed underwriting would compare the installed station with a feasible staffed or conventional automated alternative at the same location. It would identify gross contribution per attributable order, occupancy charges, replenishment labor, sanitation, consumables, software and field service. Undisclosed equipment and concession economics remain open inputs. Conventional Donatos franchise investment ranges should not be substituted for the cost of this robotic configuration.

The commercial lesson is that a production robot can become part of a retail format. Its opportunity rests on the complete order-to-pickup service, including the distribution of responsibilities among partners. The case informs Chapters 3 and 7 without changing the U.S. Top-20 sample or the principal financial model. Tables 139 and 141 distinguish the documented configuration from measurements still needed.



Picture 7 | Connecting automated production to airport order collection
AI-generated concept illustration for GII; equipment and settings are illustrative.

6.8 Artly and learning a skilled beverage task

Artly's May 17, 2025 company release reported fifteen cafe locations and nearly one million cumulative cups since its first model. It also described a Barista Bot using a vision-language-action model trained with human motion-capture demonstrations. These are company-reported operating and technical statements. Cumulative cups do not establish current throughput, per-site profitability or performance on unfamiliar tasks [S119].

A separate March 2025 announcement described an upcoming hand upgrade. MANUS, a commercial technology supplier, describes the demonstration-capture workflow. These sources help identify a proposed learning pathway, but do not establish that every operating cafe uses the newly announced hand or the same policy version. Supplier terminology about general intelligence is not treated here as an independently validated technical classification [S120–S121].

For World Model and robotics developers, the value of the case is its connection between a commercially recognizable task and action data. A beverage has an observable service objective, while manipulation quality depends on the initial state, tools and material behavior. GII proposes separating the recipe specification from the motor execution required on a particular embodiment. That separation would allow a study to identify whether adaptation occurs in the planner, learned policy, equipment settings or human preparation.

The first research comparison should hold product quality constant. A policy that completes more motions but changes drink quality or requires additional preparation has not demonstrated an equivalent service. Testing should record failures, discarded ingredients, cleaning effort and supervision alongside successful completions. For transfer, a new cafe should be evaluated before local tuning is included in the result. Section 5.7 provides the corresponding data and consent protocol.

Hardware versioning is particularly important when a company develops new hands and uses an installed commercial base. A field record should identify the arm, end effector, sensors and policy actually used for each observation. Otherwise a product demonstration can be inadvertently combined with operating results from another configuration. The same discipline applies to a menu expansion: adding a named beverage does not reveal how much engineering, teaching or fixture preparation was required.

A cafe may also earn value from the experience of watching preparation. That commercial hypothesis should be tested separately from labor productivity. Repeat purchases, quality consistency and contribution after service costs provide different information from first-time interest. A hardware/software vendor and an owner-operated cafe may recognize revenue from different activities; beverage sales should not be read as external robot subscription demand.

Artly links a described demonstration-learning approach with reported cafe activity, while leaving two distinct tests open: quality-equivalent transfer to a new physical condition and retained site contribution after the full operating workload. Tables 139, 141 and 142 distinguish those measures. The public operating counts and supplier descriptions do not calibrate the illustrative GII financial model.



Picture 8 | Beverage manipulation and human quality control
AI-generated concept illustration for GII; equipment and settings are illustrative.

6.9 Other preparation systems and their operating boundaries

Richtech Robotics reported more than 16,000 drinks served at Clouffee & Tea by June 17, 2025. The company described the Las Vegas cafe, opened February 9, as its own flagship concept using ADAM. This is evidence of company-reported activity in a customer-facing setting. It does not establish external chain procurement, paid drink volume or site profitability [S124].

The ADAM comparison is useful for examining the combination of manipulation and customer interaction. A proposed study would measure successful order fulfillment and operator effort while separately testing whether the visible robot changes demand or repeat visits. Any data used for a learned policy would need an explicit task and permission boundary. A customer-facing performance should not be used to infer a predictive world model or unrestricted autonomy.

January 2026 firsthand reporting on iWok describes an automated cooking setup and a separate boba arm ahead of its Houston opening. The operator provides named locations, but its live pages do not provide a reliable dated active-fleet series. The equipment supplier and control stack remain unconfirmed in the sources used here [S126–S127]. The sources identify a cooking configuration without establishing its installed control architecture or current fleet.

For research, a wok system makes the upstream preparation boundary important. A proposed comparison would account for portioned inputs, storage, local handling and customer service. It would identify which variations the equipment actually accepts and which are removed through menu or process design. Standardized preparation can be a sensible commercial choice; the amount of local autonomy should be stated in relation to that prepared environment.

Cafe X provides a further retail comparison through an airport-host listing of a robotic coffee bar at San Francisco International Airport [S128]. The listing identifies a named setting and service configuration. It supplies neither a historical fleet census nor a measured return. Its main use here is to focus attention on site access, customer pickup and the responsibilities surrounding a consumer-facing station.

These beverage and cooking configurations illustrate how prepared inputs, customer interfaces and the allocation of human work shape a deployable task. Table 141 compares public system boundaries, including the existing Chipotle examples. Changes to a menu, food-contact surface or customer handoff may create more deployment work than a successful arm motion suggests. GII's measurement plan therefore treats preparation and exceptions as part of the product's operating cost.

6.10 Choose the robot body for the permitted restaurant task

A tray-delivery robot, a wheeled mobile manipulator and a bipedal humanoid are different equipment categories. A tray carrier transports prepared items but may leave loading and unloading to people. A wheeled manipulator adds arms and tool interaction. A humanoid upper body can also use wheels, so humanoid should not be treated as a synonym for biped. Table 148 compares the engineering consequences of the base choice.

A wheeled base needs sufficient route width, suitable floors, docking accuracy and stable positioning at the workstation. A biped may reach across steps or changes in level, but must coordinate foot placement, balance and arm motion under load. Both need collision avoidance, reachability and payload validation. Carrying hot liquid adds acceleration and spill constraints; a biped also needs a demonstrated response to loss of balance that is acceptable around guests and kitchen staff.

Mobile ALOHA provides original research on wheeled bimanual cooking and kitchen manipulation [S154]. Google's whole-body work demonstrates another route while identifying continuing dexterity challenges [S148]. These support comparing embodiments by task, rather than assuming that walking is necessary for skilled food work. For a flat-floor location, a fixed cell or wheeled manipulator is a sensible first comparator. A biped should justify its additional system complexity through a measured access or workflow advantage.

The body choice does not settle sanitation or maintainability. Grippers, covers, joints and tools must tolerate the specified cleaning process; food-contact parts and operating configuration need the appropriate review. Battery changes, charging, emergency recovery and service access must fit the shift. Evaluate the complete installation with people and equipment in place, including the manual route when the robot is unavailable.

6.11 Recent kitchen systems reveal different implementation paths

Miso's October 20, 2026 announcement describes a laboratory demonstration of a new 3D vision system, with high-resolution color cameras, learned recognition and GPU inference at the fryer. The demonstration included distinguishing similar foods and rejecting a foreign utensil. First shipments were expected in December [S139]. This separates a new perception upgrade from the existing Flippy fleet. The disclosure supports a vision-and-control development; it does not establish VLA control, fleetwide installation of the upgrade or an independent safety result.

Ricoh and Hyphen's October 15, 2026 partnership addresses installation and lifecycle support. The described makeline accepts digital orders and assembles bowls below the counter while employees serve from above [S144]. This is a concrete division of work between software, equipment and people. The support agreement is evidence of a scaling mechanism, while the number and performance of resulting active installations require later observation.

Chef Robotics supplies a food-manufacturing comparator: its public material describes learned manipulation across variable ingredients, portions and containers [S145]. Its service offering includes hardware, software, maintenance, parts and monitoring, and it reports NSF/ANSI 169 certification for robotic module C-001748 [S146]. That statement requires configuration-specific confirmation in procurement. A prepared-meal production line has different demand, sanitation and labor conditions from a restaurant; it should not be folded into a restaurant deployment count.

These paths can coexist in one location. A constrained appliance may use conventional controls; a vision-guided arm may handle bounded manipulation; a learned policy may adapt a skill; and an orchestration service may schedule the work. Procurement should identify the exact task and installed generation before assigning a technology label or estimating savings.

6.12 Measure peak service and recovery before expanding autonomy

The release decision should use accepted output during the intended service window. Record first-pass task success, complete-job success without assistance, intervention minutes, cycle-time distribution, downtime, cleaning, changeover and food-quality failures. Denominators must include unsuccessful and interrupted attempts. A corrected trial and an uninterrupted autonomous trial belong in separate fields. Table 149 connects these measures to an operating decision.

An annual economic calculation cannot set the safety limit or the lunch-rush service promise. A system can generate enough average value while failing during the only interval in which capacity is scarce. Tests should therefore include peak load, replenishment, staff breaks, network loss and recovery after a stop. Critical hazards require their own acceptance criteria; a favorable average success rate cannot offset an unacceptable hazardous failure.

The evidence currently favors defining a narrow execution boundary, retaining a workable human fallback and testing additional autonomy against that baseline. The restaurant buys reliable finished work. The supplier must also deliver an installation that can be cleaned, restored and supported with ordinary site resources.

7. Omnichannel Demand-to-Physical Execution: The Invisible Physical AI Opportunity

PART II | ORDERING, SCHEDULING AND PHYSICAL FULFILLMENT

Order-ahead applications, kiosks, third-party marketplaces and catering platforms have already digitized demand. The next operating challenge is physical coordination: when should an order begin, what kitchen station is capacity-constrained, how long can prepared food wait, and when will a guest, courier, or drone actually arrive? The restaurant cannot realize digital demand without a reliable physical production and handoff process.

7.1 One kitchen, heterogeneous demand channels

The same employees and equipment may serve walk-ins, drive-through, scheduled first-party pickup, Uber Eats/DoorDash delivery and large corporate catering. Each channel has a distinct readiness expectation, promised time, packaging requirement and customer-data boundary. “More digital orders” may increase queue interference rather than incremental contribution if sequencing is not coordinated.



Figure 15 | Original GII omnichannel order-to-execution architecture

Conceptual; individual brands may have different systems and ownership boundaries.

7.2 What an intelligent sequencing system actually optimizes

A practical objective is to minimize the weighted costs of late promises, poor handoff timing, temperature or freshness degradation, station congestion and additional employee workload, subject to equipment capacity and service/safety constraints. A decision engine may be rule-based, operations-research-based or AI-assisted; the business value arises from accuracy and robust execution rather than the branding of the algorithm.

Table 35 | Multichannel orders and physical execution

Order channel	Critical state inputs	Primary physical decision	Performance measurement
In-store / drive-thru	Arrivals, station loads, customer queue	Start order; station assignment	Guest wait, abandonments, order accuracy
First-party pickup	Chosen pickup window, forecast service time	Timed make-ready; pickup shelf allocation	On-time ready, dwell, customer return
Third-party courier	Platform ETA, courier arrival uncertainty	Hold/release cooking, courier handoff	Cold-food dwell, courier waiting, wrong pickup
Catering / group order	Batch size, advance notice, packaging	Batch plan, preproduction and staffing	On-time large-order completion, contribution
Autonomous delivery	Payload, dispatch ETA, safe transfer	Packaging, launch staging, fallback	Doorstep service time and exception costs

Illustrative use-case taxonomy; not a claim of deployed capabilities for all named brands.

7.3 Leading real-world examples and their limits

Starbucks describes Smart Queue as an order-sequencing capability across café, drive-through, mobile and delivery [S98]. DoorDash introduced DashOS on September 30, 2026, combining customer relationship and restaurant-commerce tools [S101]. Uber announced a $2.3 billion cash acquisition agreement for ezCater on October 6, 2026, subject to closing conditions; the acquired business had reported more than 140,000 restaurant relationships and average orders above $400 [S100]. These cases show how customer channels shape physical task demand and channel bargaining power; none independently demonstrates a generalized robot world model.

Autonomous fulfillment makes the last physical meters unusually visible. DoorDash Air's September 30 announcement emphasized packaging, merchant ground equipment and kitchen-to-aircraft handoffs, with Northern California pilots involving restaurant partners [S102]. A robotics developer must treat the restaurant loading interface and operational fallback as part of the product, not just the aircraft or mobile robot.

Primary evidence: Starbucks Smart Queue [S98]; DoorDash DashOS [S101]; Uber/ezCater [S100]; DoorDash Air [S102]

7.4 A digital order becomes a real manufacturing deadline

A restaurant can receive a walk-in order, drive-through request, first-party mobile pre-order, third-party delivery ticket and scheduled catering order within minutes. These are different service promises competing for a shared preparation line, equipment capacity and employee time. Orders have different promised handoff times, batch sizes, product quality windows, courier arrival uncertainty and customer notification rules. The optimization problem is not simply first-in, first-out; a naïve rule can cause prepared food to age while another order misses its deadline.

An orchestration system must translate order events into production states: order accepted; item availability checked; equipment and station workload inferred; prep release time assigned; job moved to a human or automated station; quality check completed; pickup notified; handoff confirmed; exception reconciled. In addition to minimizing late orders, the objective may include reducing abandoned pickup time, courier waiting, station switching, food remakes, overtime and customer dissatisfaction. We should avoid claiming that any one algorithm optimizes all of those objectives without measured trade-offs.

Starbucks says its Smart Queue balances café, drive-through and mobile-order flows and supports scheduled five-minute pickup windows [S106]. Chipotle's digital pickup formats connect first-party ordering with physical site design and handoff; Luckin's 2025 SEC filing describes a cashier-less app-supported model and reports that 99.1% of company-operated stores were pickup-format stores [S107]. These are powerful examples of digital-to-physical integration, but not claims that autonomous general-purpose robots perform the beverage or meal preparation.

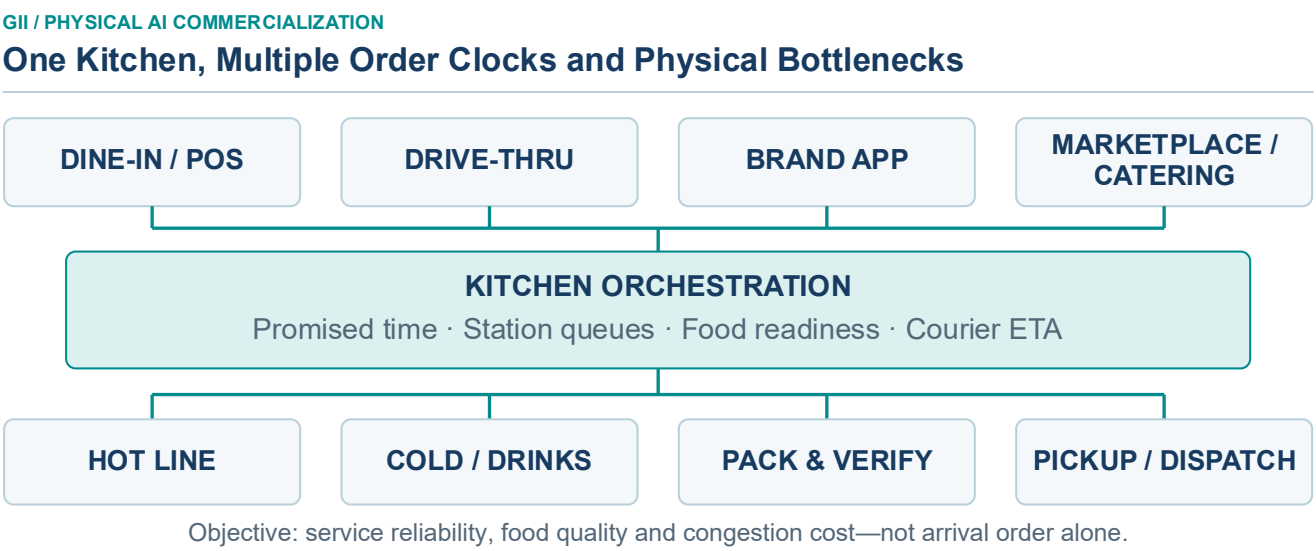


Figure 16 | One shared production system must serve heterogeneous order clocks and physical handoffs
GII original sequencing framework; Starbucks product description [S106] and Luckin SEC disclosure [S107] support selected examples, not the hypothetical full architecture.

7.5 The physical pickup interface is an economic system

Physical fulfillment determines whether a digital promise becomes customer value. A pickup shelf, self-service cubby, delivery window, drive-through lane or catering loading area consumes store real estate and affects employee walking, security, product temperature and queue behavior. If the customer arrives late, the system must balance holding time against product quality; if a courier arrives early, a kitchen may be pressured to interrupt more valuable production. These costs should be measured by handoff type rather than absorbed into generic delivery-platform fees.

Uber announced an agreement on October 6, 2026 to acquire ezCater for $2.3 billion; the transaction was not yet completed as of this report's cutoff [S100]. The commercial implication is not an immediate boost in restaurant revenue. Large scheduled group orders have different batch-prep, fulfillment and customer-acquisition needs, creating potential new orchestration tasks and technology demand. The research question is whether restaurants can profitably integrate that demand without reducing service for existing customers.

The most promising near-term restaurant Physical AI may be mostly invisible. Adaptive kitchen release schedules, connected equipment and better preparation-state estimates can improve operations without a conspicuous robot. The report treats these as software and systems opportunities; a physical world model becomes relevant when predicting the consequences of physical actions, not merely when an app forecasts demand.

Table 36 | One order, many clocks: fulfillment event schema

Event	Time/physical state recorded	Why it matters
Customer order	Placement, channel, promised pickup	External demand and service obligation
Production release	Station and actual start time	Queueing and readiness optimization
Food completion	Quality pass and ready timestamp	Dwell, rework and service quality
Courier/customer arrival	Physical ETA and actual arrival	Cross-channel synchronization
Handoff	Proof, pickup point, exceptions	Lost-item and customer experience
Feedback	Delay cause, refund, repeat behavior	Learning and economic attribution

Source: GII proposed event-to-physical-service data structure; not a disclosure by any one chain.

7.6 Why demand sequencing becomes nonlinear at the kitchen bottleneck

A kitchen is not one processor: parallel preparation stations, finite hot holding, human handoffs and different customer promises create an interdependent queueing network. Nonetheless, a one-station teaching illustration explains why order timing becomes disproportionately costly at peak utilization. Let orders arrive at rate λ per hour and one bottleneck complete μ acceptable units per hour. Under an intentionally simplified stable M/M/1 queue with Poisson arrivals and exponential service times, expected queue waiting time $W_q = \lambda / [\mu(\mu - \lambda)]$. This is a mathematical demonstration, not a calibration of any restaurant or a proof of AI improvement.

Table 37 | Peak workload and waiting-time sensitivity

Illustrative arrival λ	Bottleneck utilization	Expected queue wait Wq	Model boundary
60/hour	60%	0.9 minutes	Station capacity fixed at 100/hour
80/hour	80%	2.4 minutes	Station capacity fixed at 100/hour
90/hour	90%	5.4 minutes	Station capacity fixed at 100/hour
95/hour	95%	11.4 minutes	Station capacity fixed at 100/hour

GII calculation under M/M/1 assumptions only: μ = 100 units/hour; no abandonment, multiple classes, setup, batching, service failures or parallel station effects. Not a measured restaurant outcome.

As arrivals increase from 90 to 95 orders per hour in this simple model, expected queue delay approximately doubles—from 5.4 to 11.4 minutes—while the capacity assumption is unchanged. Actual restaurants have multiple queues and priority rules, so a deployment model must measure each constrained station. The commercial opportunity for intelligent sequencing is not simply predicting demand; it is using reliable order visibility, machine and labor state, pickup promises and courier ETA to prevent avoidable congestion without compromising food quality.

A useful order-orchestration pilot should measure ready-at-promise, actual customer collection, courier dwell, preparation rework, hot-hold duration, late orders by channel and employee interruption burden. A scheduling improvement may shift inconvenience from one channel to another, so channel-level and total-system performance are both needed. Starbucks states that Smart Queue balances café, drive-through and mobile orders and supports scheduled pickup windows [S106]; this is a company-reported architecture, not GII evidence of causal operating gains.

Section 6.7 applies the order-to-pickup boundary to the Donatos airport system. An evaluation should follow the customer transaction through physical completion and retrieval, including the recovery path when a production or pickup step fails.

7.7 U.S. drone delivery cases show bounded feasibility

The following cases are useful proofs of technical and workflow feasibility, comprising a single-store pilot, an early-access launch and limited-area services. They are not evidence of large restaurant-chain rollouts. The cited announcements retain their original dates and scope; they do not establish current October 2026 store counts, broad customer preference, employee acceptance, manager approval or positive restaurant unit returns. Appendix M compares these evidence boundaries.

Wendy’s, Wing and DoorDash — Christiansburg, Virginia. DoorDash’s March 21, 2024 announcement identifies a pilot at the Wendy’s at 2355 N. Franklin Street. Eligible customers could choose drone fulfillment through DoorDash; the restaurant prepared and packaged the order [S133]. This establishes a named preparation-to-air handoff. It does not establish a national Wendy’s program or independently measured acceptance and returns at the pilot store.

Chipotle and Zipline — Rowlett, Texas. Chipotle's August 21, 2025 announcement starts early access at 3109 Lakeview Parkway for a limited group of Zipline users. Employees load prepared orders into a Zipping Point pickup interface [S134]. The case connects digital orders, human packing and an aerial collection point. Early access and an announced future expansion do not establish broad store deployment or satisfactory everyday work for the store team.

Papa Johns and The Brass Tap, Flytrex and DoorDash — Little Elm and Frisco, Texas. The June 26, 2025 launch announcement names these restaurants within a limited-area delivery service and reports a preceding pilot [S135]. Marketplace integration can connect several kitchens with one delivery network. The announcement's reachable population and participating restaurants are not a dated count of installed restaurant dispatch sites or evidence of chainwide adoption.

Brookfield Properties, Wing and DoorDash — Stonebriar Centre and Hulen Mall, Texas. Wing's December 18, 2024 announcement names two shopping centers and discusses trial use of property space, including rooftops or excess parking [S136]. This makes the landlord's role visible. Its statement that more than 50 merchants would be available describes merchant eligibility; it does not establish 50 equipped restaurant launch sites or a broad property-network rollout.

These examples support examining the first and last physical handoff around a delivery system. They do not show that customers prefer aerial delivery to a courier, that employees welcome packing and loading changes, or that store managers consider the service worth its interruptions. A technology can work and still fail an adoption test. Those outcomes require separate evidence, with unsuccessful orders, refusals and discontinuance included rather than only successful flights.

7.8 Sales beyond the four walls must survive fulfillment costs

A restaurant's address can connect dining demand, pickup traffic and a wider delivery catchment to one production system. The appropriate comparison is the incremental contribution from each channel after food, packaging, platform and delivery fees, additional paid work, quality failures and equipment or property costs. A larger delivery radius has little value if orders arrive cold, displace more profitable demand or require unplanned staff intervention.

Track orders that are genuinely additional separately from orders shifted between first-party, marketplace, pickup and dining channels. Record production congestion and the effect on existing guests. Third-party platforms may supply demand while retaining customer relationships and charging fees; the restaurant, franchisee, landlord and delivery provider need distinct value-and-cost accounts. The existing GII financial scenarios remain unchanged; this is a measurement framework for testing a proposed channel.

7.9 The property and the handoff are part of the delivery product

The shopping-center example illustrates why a drone is only one element of a local operating system [S136]. The analyst must identify who controls the dispatch space, access route, loading equipment, electrical service and maintenance. Roof access or parking may have an opportunity cost. Restaurant employees may have to move packaged orders to a shared dispatch point, creating a new task rather than eliminating the entire handoff.

Compare direct restaurant loading with a shared property hub and conventional courier pickup. Evaluate actual walking and waiting, food holding, failed collections, weather downtime, recovery responsibility and effects on adjacent tenants. Proposed layouts require site-specific permissions and operating review. Concept renderings illustrate possible spatial arrangements; they are not approved flight operations, safety drawings or depictions of the named companies' installations.

7.10 Customer, employee and store-manager acceptance are separate gates

Technical feasibility is only the first gate. Customers may dislike delivery restrictions, noise, packaging, collection instructions or the loss of a familiar service interaction. Employees may face more lifting, walking, monitoring, exception handling or concern about future roles. Store managers may reject disruption during peak production, unclear accountability, extra training or costs they cannot recover. These are testable risks, not findings that all users of the four cases were dissatisfied.

Measure voluntary selection and repeat use, refusal and complaint reasons, food condition and recovery after failed delivery. For employees, measure workload across the complete shift, confidence, training and assistance required, alongside confidential feedback. For store managers, record peak-hour impact, extra paid work, service escalation, margin and willingness to continue after the pilot. A favorable result for one group does not settle the others. Table 146 provides a proposed acceptance register.

Hospitality research reinforces the need to assess continued use. Tuomi and colleagues interviewed eight suppliers and ten strategic decision-makers about planned continuation or abandonment of service robots; operational complexity, user experience, technical integration and economics were among the relevant factors [S137]. This research supports the adoption questions. It is not evidence of drone-specific customer sentiment or proof that a particular launch failed.

7.11 Synchronize orders with physical production state

The integration problem begins before a robot moves. POS, first-party apps, delivery marketplaces and scheduled orders must resolve into a shared order record with item identity, modifiers, allergen instructions, quantities, channel and promised handoff time. Preserve amendments and cancellations, deduplicate repeated events and record which version was released to production. A payment event is not a reliable observation of a prepared meal.

A useful operating-state model combines those obligations with work in progress, ingredient availability, equipment condition, holding capacity and people available for specific stations. Chef capacity is skill- and task-dependent: preparation, cooking, expediting, replenishment and robot recovery compete for the same minutes. The scheduler should reserve those minutes explicitly and allow authorized staff to correct its state. A roster or scheduled headcount alone is insufficient.

The production scheduler decomposes an order into dependent jobs, allocates station and labor time, and chooses release times that respect preparation duration, available parallel capacity and quality windows. It can batch compatible work while limiting changeovers and keeping allergens and cleaning constraints explicit. A VLA may execute one assigned job; it should not be assumed to infer the restaurant's outstanding promises or allocate a chef's time from camera images.

Toast documents preparation and pickup status transitions separately from order updates, as documented on October 28, 2026. Its fulfillment webhook does not report every amendment, such as an added drink; its orders webhook supplies updates with documented omissions [S151,S152]. An implementation needs both streams and the required permissions. The platform's CLOSED status is not independent confirmation that the right bag reached the right guest.

At execution, require acknowledgment, started, paused, failed and completed states with the order and job identifiers. Quality approval, packing and physical handoff should be separate observations. A model must not mark an order ready merely because it issued a motion command. Return actual durations and exception causes to the shared state so the next schedule reflects the restaurant that is operating now.

Figure 27 and Table 150 specify a GII integration design, not a claim that one vendor already delivers every layer. Existing POS and KDS interfaces provide part of the event infrastructure. Station sensing, equipment adapters, validated robot skills, resource scheduling and recovery procedures complete the physical system.

7.12 Use different scheduling objectives for QSR and table service

A QSR must reconcile a stream of short service promises across counter, drive-through, pickup and delivery. The scarce resource may switch from fryer to assembly, packing or handoff as the order mix changes. Promise times and channel intake should respond to that bottleneck, with transparent customer communication. A large catering order consumes item-level production capacity even if the POS records only one ticket.

A sit-down restaurant must also coordinate courses and the completion of dishes for the same table. The expediter may hold a preparation step so that a short-cook item finishes with a long-cook item. Human judgment about the guest's pace and product condition remains an input. Optimizing the speed of each station independently can produce a worse meal and more rework.

Consider an explicitly hypothetical common-order mix: frying supports 120 completed-order equivalents per hour and packing supports 80. Increasing frying to 160 leaves the system capped at 80 unless packing changes. With one packing station, the effective packing budget is 45 seconds per order equivalent (3,600 / 80). These compatible throughput units are a teaching example, not a restaurant forecast. Table 151 contrasts the objectives and measurements needed for the two service formats.

Starbucks describes Smart Queue sequencing across cafe, drive-through and mobile demand with capacity-dependent pickup windows [S106]. The broader design should similarly adjust permitted promises, reserve critical capacity and monitor delays by channel. Available online-ordering controls vary by integration; Toast's availability events have stated configuration limitations [S153]. A scheduler cannot assume it can throttle every external channel through one universal API.

7.13 Recover from faults without losing the order or the service promise

Integration requires explicit ownership of exceptional states. If an ingredient becomes unavailable after acceptance, the authorized operator decides substitution, cancellation or a revised promise. If a robot stops with food in process, the system records the disposition and remaining work before reallocating the job. Replaying a message must not automatically cook a second meal or repeat an ingredient addition.

Use event identity, timestamps, persistent job state and reconciliation after outages. Where a platform lacks usable version counters, the adapter must handle freshness and conflict resolution without inventing a reliable sequence field. Staff need a visible manual override and a clear record of which jobs have already started. Safety-critical stop and restart authority remains local even when planning or inference uses cloud services.

Permissions and data minimization belong in the interface design. Toast documents omissions from its order webhook [S152]; integrations should request only the additional fields required for fulfillment. Physical quality records and worker-assistance logs also need defined access and retention. A cross-site learning program should export permitted episodes and aggregate operating measures, rather than assuming all customer and worker data can be pooled.

The value of a restaurant network comes from repeating a functioning service contract: compatible events, equipment interfaces, accepted human roles and a support route. Locations amplify learning and service efficiency only when those interfaces can be reused at an acceptable cost. This is the practical link between the task as a learning unit, the site as an economic unit and the network as a scale unit.

8. Cloud, Edge and the Restaurant Computing Architecture

Data center economics matter because they determine the cost, latency, reliability, and strategic dependence of modern AI applications. They do not imply that restaurants should directly construct hyperscale campuses. The operator chooses a workload placement architecture: centralized cloud, regional compute, local edge servers, or controls embedded in equipment. [S08]



Picture 9 | Local compute supports restaurant equipment and operations
AI-generated concept illustration for GII; equipment and settings are illustrative.

8.1 AI infrastructure as an upstream input

The IEA estimates global data-center electricity consumption at 485 TWh in 2025, rising to about 950 TWh in its central 2030 projection. These global macro figures should not be confused with restaurant-specific consumption or restaurant technology spending.

- Classify hyperscaler cloud, private cloud, colocation, enterprise compute and edge systems by revenue and capex ownership.
- Measure GPU inference, CPU orchestration, networking and storage as distinct drivers of per-task computing cost.
- A restaurant AI subscription may purchase cloud capacity indirectly through its vendor; a transaction map is needed to identify the original provider and avoid double-counting.

Sources: [S08]

Table 38 | Data centers and AI infrastructure context

Indicator	Period	Value	Market scope	Source
Global data center electricity	2025	485 TWh	All data centers; not restaurant demand	[S08]
Global data center electricity projection	2030	~950 TWh	IEA central scenario; ~3% global electricity	[S08]
Five leading tech firms combined CapEx	2025	More than $400B	IEA-defined big tech cohort; not pure AI CapEx	[S08]
Projected CapEx growth	2026	~75% additional vs 2025	IEA report forward-looking statement	[S08]
Restaurant hyperscale owned capacity	2026	Not verified	No chain-specific company-owned AI DC capacity established	Research open

High-level context. Do not infer restaurant GPU spending from global hyperscaler power growth.

8.2 Workload placement in the restaurant

Restaurant workloads differ significantly in latency tolerance, bandwidth, computational complexity, and business criticality. Operations that must function during network outages are particularly relevant for local execution.

- Batch forecasting, loyalty recommendations, visual inspection, machine control and payments have different latency and reliability requirements.
- The choice among CPU, NPU, GPU edge and cloud inference depends on the workload, latency tolerance and lifecycle cost.
- Local resilience needs to be evaluated alongside hardware rollout, maintenance and staff or IT support costs.

Table 39 | Cloud, edge and embedded workload placement

Workload	Likely placement	Latency priority	Offline need	Rationale
National demand forecast	Cloud / data platform	Low	Low	Central cross-site batch analytics
Loyalty personalization	Cloud + local cache	Medium	Medium	Global histories and channel personalization
Order workflow sequencing	Edge + cloud	High	High	Must continue during intermittent internet
Equipment monitoring	Embedded + edge	High	High	Local fault detection and control signals
Food vision quality	Edge GPU/NPU when justified	High	Medium–high	Video bandwidth and privacy constraints
Robot motor safety control	Embedded / on-device	Very high	Essential	Safety-critical local closed loop
Multi-store finance analytics	Cloud	Low	Low	Consolidated reporting, not real-time

Illustrative architecture; a validated enterprise IT assessment may choose differently.

8.3 Chick-fil-A engineering as a verified distributed case

First-party technical articles identify chain-wide Kubernetes edge clusters and a later evolution from exactly three nodes to at least three nodes per restaurant, with optional GPU additions. The business benefits and technology unit cost are not publicly disclosed in the sources used here.

- Read the architecture articles as engineering evidence, not as audited proof of productivity improvement.
- Investigate host agent, device telemetry, hardware redundancy and Kubernetes scheduling.
- Use-case-level operational outcomes still require interviews, disclosed measurements or site observation; platform coverage alone does not establish them.

Sources: [S10], [S11]

Table 40 | Chick-fil-A first-party edge platform — verified facts

Attribute	Public evidence	Source / date	Not established
Deployment scale	Platform present in all stores	[S10] Sep 2025	Exact currently active stores with identical node count
Minimum compute nodes	At least three nodes per restaurant	[S10] Sep 2025	Node specifications or capex
GPU expandability	Additional GPU nodes possible by project need	[S10] Sep 2025	How many stores currently have GPUs
Restaurant device data	Kitchen / smart equipment telemetry supported	[S11] Jan 2023	Unit-level margins or sensor adoption effect
Cloud relationship	Cloud platform operates alongside edge	[S11] Jan 2023	Cloud spend or exact site TCO

8.4 Data governance and ownership

Ownership of physical recordings, consumer transaction histories, and restaurant equipment telemetry is contract-dependent. A data supplier cannot assume that a franchisee may freely license employee video or customers' faces to train outside foundation models.

- Data governance requires a record of consent, permitted purpose, storage, vendor access and deletion rights for each data type.
- Workplace observations and customer identity information need controls matched to their sensitivity and permitted use.
- Large chains and individual franchisees may have different bargaining power over prices, access and data rights.

Table 41 | Physical data rights and value distribution

Data type	Typical collector	Potential business use	Consent / governance issue	Value claimant
POS/order details	Restaurant POS platform	Demand/accuracy and operating insight	Customer data rights and vendor contracts	Restaurant, platform, customer
Equipment sensor stream	OEM / operator	Predictive service, energy, quality	Ownership, device compatibility	Operator and equipment OEM
Kitchen video	Operator or AI vendor	Quality or line-flow interpretation	Worker and bystander privacy	Operator, worker, vendor
Head-worn work recordings	Worker / data contractor	Robotics model training	Labor consent, compensation, retention	Worker and model developer
Robot motion logs	Robot provider	Safety, navigation and fleet learning	Customer worksite / worker traces	Operator and robotics provider
Loyalty / location	Brand app	Recommendation, retention	Consumer choice and privacy	Customer and restaurant platform

Qualitative governance research checklist, not legal conclusions.

8.5 Cloud-edge TCO example

Investment approval requires an explicit total cost calculation. An edge appliance may reduce communication requirements but introduce fleet operations, on-site replacements, security patching, and repair logistics.

- Assemble one transparent illustrative five-year ownership cost example.
- A complete TCO comparison includes compute, installation, managed software, networking, outage exposure, replacement and end-of-life removal.
- Normalize financial outcomes to equivalent model quality and service-level requirements.

Table 42 | Illustrative annual restaurant computing TCO

Cost item	Cloud-only /year	Edge-only /year	Hybrid /year	Required real-world quote
Hardware amortization	0	1,400	900	Actual device capex and useful life
Cloud inference/services	3,000	500	1,400	Token volume / platform fee
Support / security	500	1,200	1,000	Fleet support / patching
Connectivity / data transfer	600	200	300	Traffic measurement
Risk-weighted downtime	400	150	150	SLA and avoided outage cost
Illustrative total	$4,500	$3,450	$3,750	Not a reliability-adjusted final NPV

Illustrative annual costs; these are not vendor quotations.

GII pedagogical assumptions. No conclusion about actual lowest-cost placement without matching task performance, installation and reliability.

8.6 Architecture maturity: installation is not equivalent to intelligence

Restaurant edge nodes enable local compute, offline resilience, and potentially high-bandwidth vision processing. Chick-fil-A documents its minimum three-node Kubernetes architecture and the ability to expand with GPUs, which is first-party engineering evidence for infrastructure maturity. It does not publicly establish that every store operates autonomous kitchen robots or has an independently verified AI economic return. [S10]

Similarly, McDonald's September 2026 disclosure of thousands of Google-enabled edge deployments marks considerable progress in hardware and application distribution, yet the full system-wide operating and franchisee profitability claims remain objectives. A rigorous architecture-to-value bridge should measure the share of stores energized/installed, actively processing a named application, triggering operational actions, and producing a measured economic outcome. This resembles data center underwriting stages: capacity announced, capacity energized, capacity leased, capacity utilized, then cash flow.

- New data fields: installed edge nodes, sites running inference, monthly inference activity, direct action/decision frequency, uptime versus counterfactual, and proven labor or quality outcomes.

8.7 Five layers of restaurant intelligence and their measurable unit economics

Physical AI investment should be deconstructed into the data-gathering layer, a business context layer, a computing layer, an action layer, and an outcome layer. The unit-cost economics of each layer differ. A $200 sensing system can make traditional operating procedures more effective without any large language model; a vision-equipped robot may require costly hardware, safety engineering, integration and ongoing employee intervention. The economically optimal configuration depends on the physical task and scale, not brand recognition of the technology supplier.

Table 43 | Data-to-action architecture: economics and governance

Layer	Representative evidence	Required technical capability	Primary KPI	Risk and control
Physical observation	Weights, temperature, equipment fault, KDS event	Sensors / events / existing logs	Data accuracy per dollar	Consent, security, false positives
Operational context	Menu recipe, work order, store layout, staffing	POS-KDS-device identity resolution	Event-to-order linkage completeness	Master-data errors; privacy
Model & compute	Demand models, vision inference, optimization	Cloud CPU/GPU or local edge NPU	Cost / actionable prediction	Model drift, uptime, latency
Human / machine action	Manager alert, batch size, equipment command	Permissions, fail-safes and human handoff	Action completion; exception rate	Safety, employee workload
Economic measurement	Remakes, waste, quality, speed, payroll, repeat visits	Comparable panel data and cost attribution	Incremental profit / job quality / guest outcome	Counterfactual, rebound effects

The architecture is an analytical proposal, not a claim that all restaurants currently deploy a closed-loop AI platform.

8.8 Restaurant Physical AI maturity is a closed-loop capability, not a device count

This analysis formalizes a distinction that improves both technical credibility and industrial-economics interpretation. Many restaurant technology installations provide workflow visibility or employee assistance without autonomous physical control. Those systems may still create meaningful value. The full loop of sensing, inference, action and feedback should be reserved for applications that actually influence real-world execution, whether by equipment or a worker with authority to accept or reject AI advice.

Measurable restaurant output—not stage number alone—determines capital allocation. Edge computing is an enabling architecture. An employee tablet with helpful instructions may produce higher ROI than a robotic device with weak reliability; neither should be scored as superior without matching operating conditions.

Table 44 | Physical-intelligence stages and verification

Stage	Technical capability	Example restaurant application	Evidence needed	Economic test
L0: Digitized	Digital order and operating records	POS/KDS event trace	Reliable timestamps and store coverage	Visibility and administrative cost
L1: Sensing	Timely physical-state data	Fryer temperature/weight/vision sensor	Calibration, false alarms and data rights	Defects prevented per $ of data cost
L2: Recommendation	AI predicts or advises human	Forecast prep demand; recipe assistant	Human use, changes in decisions	Accuracy and staff time saved
L3: Human-in-loop action	Recommendations translated into operating task	Supervisor approves adaptive prep schedule	Actual compliance and exceptions	Contribution margin, work intensity
L4: Automated control	Actuator/equipment changes physical process	Oven adjusts cycle; robotic makeline	Safety, uptime, failure protocols	Task cost, quality and utilization
L5: Adaptive feedback	Measured outcomes update decisions	Learning within authorized control limits	Validation and change-control logs	Sustained causal outcome

Author-defined research classification. It is not a validated maturity scale. Stages should not be assigned to a company without use-case-specific evidence.

8.9 A credible cloud–edge–machine deployment boundary

Computing location should follow operational constraints. Central cloud systems are well suited to model training, fleet analytics, administrative optimization and nonurgent customer intelligence. A local edge server can support time-sensitive coordination, sensor fusion, inference caching and operational continuity when external connectivity is degraded. Embedded controllers remain responsible for fast machine control and safety interlocks. The design test is the required response time, data sensitivity, operating availability and total lifecycle cost of each workload—not whether a restaurant owns a GPU.

Chick-fil-A published a technical description of its Kubernetes edge clusters, originally three nodes per restaurant with provision for additional GPU nodes based on individual location needs [S105]. This is strong evidence of a distributed computing architecture, but it does not prove that general-purpose restaurant world models already operate on those nodes. The economics require including node installation and replacement, vendor support, physical tamper controls, remote management, WAN costs and energy alongside model inference requirements.

An infrastructure partner should ask which information must leave the store. Raw staff video, task traces and customer-identifiable records may involve stricter contractual and privacy constraints than aggregate equipment fault counts. Privacy-preserving feature extraction and restricted edge storage can help, but governance is ultimately defined by contracts, permissions, retention rules and incident procedures. An operator should not accept a claim that processing data locally automatically makes a system compliant.

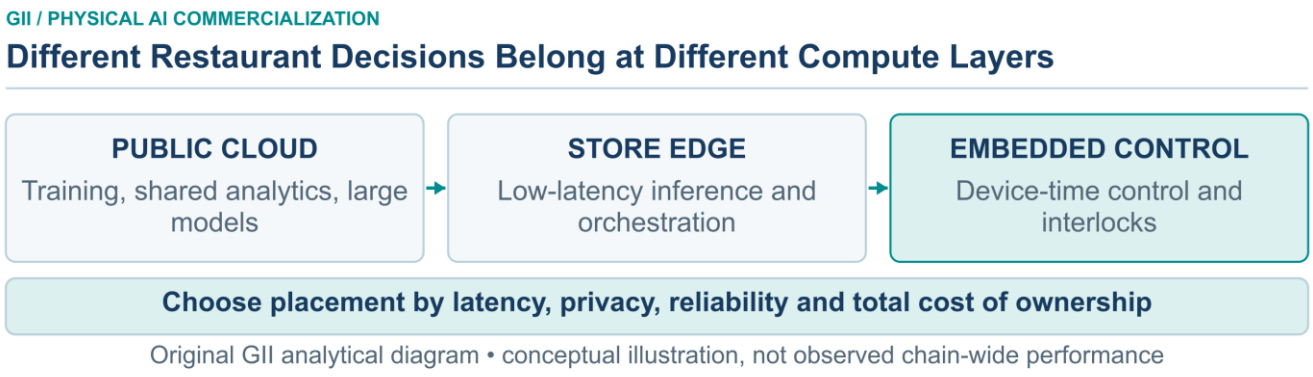


Figure 17 | Cloud, store edge and embedded equipment controls serve distinct workloads

GII proposed placement model. Engineering evidence for store edge is from Chick-fil-A [S105]; no deployment claim for restaurant-specific World Models.

9. Robotics, Automation and Human Adoption in Restaurants

This chapter is about technological fit, work design and economics—not a broad robot popularity contest. Mobile delivery robots, kitchen automation and AI-driven workflow decisions should be considered competing solutions to different physical tasks. Claims that staff universally dislike robots or that restaurant robot companies routinely buy helmet-mounted employee recordings must be examined empirically, not asserted as industry facts.

9.1 Bear, Pudu and Keenon as commercialization cases

The three firms publicly report substantial cumulative cross-industry shipments, but the available claims cannot establish current U.S. restaurant installed base. The correct comparison is contract terms, active deployment, service life, uptime, customer satisfaction and achieved operating benefits.

- Confirm the denominator and reporting date for each reported shipment or site total.
- Investigate capital raised, ownership structure and revenue, using only corroborated disclosures.
- Survey U.S. restaurateurs with active systems as well as those who discontinued trials.

Sources: [S22]–[S25]

Table 45 | Robot company comparison and missing financial data

Company	Public scale claim	Use-case diversity	Ownership / funding evidence	Missing for investment analysis
Bear Robotics	16K+ shipped; 5K+ sites; Sep 2026	Serving, cleaning, industrial	LG 2024 $60M /21%; 2025 control path to 51%	Restaurant-specific active sites, churn, GM
Pudu Robotics	130K+ cumulative shipped	Service, cleaning, industrial, humanoid	Private-company cap table/valuation not verified	Annual U.S. restaurant sales, gross margin
Keenon Robotics	100K+ cumulative shipped	Serving, cleaning, humanoid	Private-company funding value not verified	Operating install base, U.S. penetration

Sources: [S22]–[S25]

9.2 Why a robot deployment may not save wages

A robot can take a plate down a hallway but still require an employee to load it, unload it, greet customers, remove obstructions and intervene during faults. The restaurant may have the same minimum staffing configuration afterwards.

- Before-and-after task mapping and time-and-motion measurement can identify whether work is removed, accelerated or shifted to another person.
- Operator productivity, paid-hour savings, risk reduction and staff welfare are separate outcomes with different evidence requirements.
- Utilization needs to be measured across peak, normal and off-peak demand rather than inferred from a successful demonstration.

Table 46 | Full-task restaurant serving robot economic test

Workflow step	Human-only baseline	Robot-assisted condition	Measure
Assemble plated meal	Kitchen worker	Kitchen worker	Seconds and quality
Load tray	Server	Server / expo	Paid time and ergonomics
Transport to table	Server	Mobile robot	Route and cycle duration
Unload / explain order	Server	Server or guest	Actual staff requirement
Handle exceptions	Server	Server + technical response	Incidents per 100 trips
Clear / sanitize robot	N/A	New maintenance work	Daily burden
Support / training	Conventional training	Robot onboarding and remote support	TCO and turnover

Comparative task audit template; does not assert outcomes for any named vendor.

9.3 Purpose-built kitchens and the Sweetgreen transaction

The $186.4 million Sweetgreen sale of Spyce to Wonder on December 29, 2025 combined $100 million of cash with preferred equity valued at $86.4 million. Sweetgreen retained contractual access to Infinite Kitchen. This is an unusually useful corporate development case because technology access and technology ownership became separable. [S17], [S18]

- The ownership analysis separates acquisition cost, engineering economics, transaction consideration and retained deployment rights.
- Transaction proceeds do not establish realized technology IRR; returns also depend on historical investment, subsequent costs and retained rights.
- Investigate store-level operating outcomes before and after the strategic transaction.

Table 47 | Sweetgreen–Wonder corporate development fact pattern

Field	Disclosed fact	Caution / follow-up
Transaction closing	December 29, 2025	Completed sale, not prospective offer
Total consideration	**$186.4M**	**Value includes non-cash preferred stock**
Cash consideration	$100.0M	Cash proceeds alone ≠ net profit
Wonder Series C preferred stock	$86.4M implied	Valuation tied to preferred financing price
Retained restaurant access	Licensing + supply/services agreement	Commercial fee and terms not fully public
Unit-level Infinite Kitchen ROI	Not provided here	Requires deployment/financial detail

Sources: [S17],[S18]

9.4 Workers, training data and consent

Human-motion recordings may help train robotics models but may impose uncompensated distraction or surveillance burdens on restaurant workers and customers. Revenue to the model company does not automatically imply value created in the restaurant. Research needs to document actual procurement practices before generalizing about helmet-worn cameras.

- Worker task documentation for internal process improvement has a different purpose and rights boundary from vendor model-training data collection.
- Examine consent, retention, access controls, paid participation and consumer visibility.
- Employee choice, training and bargaining asymmetries affect whether a franchise workplace can adopt the proposed data-collection arrangement.

Table 48 | Four-way economic incidence of physical data

Party	Potential benefit	Possible cost	Measurable safeguard
Restaurant owner	Better yields, standardized processes	Integration expense; exposure	ROI and independent productivity evaluation
Employee	Safer tasks, enhanced training	Monitoring, extra work, data-use concerns	Consent, workload survey, compensation
Customer	Speed, order accuracy, quality	Loss of privacy and perceived service	Transparent notice, opt-out, satisfaction
Technology investor/vendor	Better models, revenue, data advantage	Model liability, service and trust risks	Contracts, renewal, validated deployment outcomes

9.5 Adoption testing and counterevidence

Employee resistance should be approached as a hypothesis conditioned on job redesign and leadership. The report must actively seek deployments in which employees welcome robotic assistance and those in which robots increase operational burden.

- Formulate interview protocols that avoid leading questions.
- Adoption evidence is more informative when it includes active users, discontinued deployments and non-adopters.
- Investigate whether robot-enabled tasks meet independent ROI hurdles and improve employee satisfaction.

9.6 Operational adoption evidence: the measurable task, not the robot headline

The IFR 2026 report covers nearly 250,000 professional service robot sales in 2025 reported by 238 surveyed suppliers. Transport and logistics accounted for about 117,500 units; the publication also noted about 7,000 full-size humanoids. IFR expressly warns that service robot sample data are not extrapolated to the entire industry and that cross-edition comparisons are discouraged. The service-robot shipment data are therefore an industry context, not a direct count of restaurant robots or U.S. restaurant robot penetration. [S06]

A more useful restaurant adoption measure is the complete workflow: number of table deliveries, percentage of loads needing human help, avoided carrying distance, actual payroll hours and overtime, table turn-time, maintenance events, guest ratings, and subscription renewal. A delivery robot can reduce staff fatigue without lowering labor cost; conversely,

system-level kitchen orchestration can improve throughput without moving a physical robot. This distinction is essential when evaluating supplier claims and the economics of the customer.

Table 49 | Service and production robotics adoption evidence

Measurement layer	Necessary restaurant observation	Misleading proxy to avoid	Financial/human decision variable
Procurement	Signed units, installation & go-live date	Vendor shipments across all industries	Actual commissioned restaurant fleet
Utilization	Completed in-service tasks / scheduled shift	Marketing demos	Economic activity per robot
Human handoffs	Loading, unloading, errors, recovery	Autonomous navigation alone	Effective labor redeployment
Operating benefit	Actual quality, wait, labor & injury metrics	Distance traveled	Cash benefit plus worker well-being
Costs	Lease/RaaS, maintenance, repairs, integration	Robot purchase price only	Lifecycle TCO / NPV
Retention	Renewal, churn, removals, repeat purchase	Cumulative shipment count	Sustainable market demand

GII evaluation design. No row is a claim that any particular vendor failed; evidence must be gathered at participating locations.

9.7 Human-centered Physical AI: augmentation, surveillance and the real task burden

The 2026 NRA staffing research indicates that 94% of respondents had not permanently eliminated jobs due to technology investments over the previous two to three years. That statistic is compatible with useful automation: restaurant managers may redeploy workers to peak service, customer care, training, or quality inspection. Conversely, it does not prove staff prefer new technology. Physical AI diligence should measure task redistribution, failures requiring intervention, perceived autonomy, scheduling predictability, and the handling of recorded employee data. [S55]

An employee-facing AI assistant and a head-mounted data-collection device are economically and ethically distinct products. For the former, success may require improved access to recipes and fewer confusing exceptions; for the latter, the primary beneficiary may be a robotics training-data buyer. Benefits, compensation, retention period, access rights, worker consent, and customer notice require explicit documentation rather than assuming that collecting more motion data is inherently desirable.

Table 50 | Employee-side Physical AI measurement protocol

Dimension	Baseline instrument	Expected benefit	Hidden burden / risk	Required study design
Physical strain	Steps, lift weight, time at station	Less carrying and repetitive handling	Robot loading/clearing still manual	Matched shifts and employee interviews
Cognitive workload	Order switching, rework, alerts	Reduced information search/decision friction	Alert fatigue, poor UI, false alarms	Validated workload scale + workflow logs
Autonomy and fairness	Schedule autonomy, task choice	More predictable operations	Continuous video monitoring	Anonymous worker survey + governance review
Training and retention	Time-to-proficiency, turnover	Faster onboarding, role confidence	Training overhead, deskilling	Cohort comparisons and role segmentation
Economic value	Paid shifts, overtime, quality and output	Higher output or realized lower total cost	Claimed "labor savings" without payroll change	Payroll reconciliation and contribution model

No facial-expression model or emotion recognition should be presumed necessary to improve employee well-being.

9.8 The hidden task economy: what autonomous systems displace and what they leave behind

The cost baseline for autonomous service is the full activity chain. A serving robot may reduce walking while loading, unloading, guest interaction, exception recovery, cleaning, connectivity and maintenance remain human or vendor work. A preparation system may accelerate one step while food finishing, quality assurance and handoffs remain unchanged. Those residual tasks belong in the operating and financial comparison.

An adoption study should compare identical high-volume shifts for a treatment and control group and collect actual paid hours, item defects, guest satisfaction, staff exertion and downtime. Employee acceptance is a dependent operating outcome, not a marketing assumption. Privacy questions about wearable cameras or headsets require explicit contract and consent review even if the vendor describes collection as model training.

Table 51 | Full-workflow robotics trial measurements

Workflow component	Baseline measurement	Automation follow-up	Include in financial return?
Transport / portioning task	Active worker minutes by peak period	Robot uptime and automated task count	Only realized hours or added capacity
Loading, unloading, cleaning	Minutes and exceptions	Additional human intervention minutes	Include incremental labor burden
Quality and consumer rework	Error frequency / remakes	Defects and complaint changes	Include avoided costs with attribution
Employee workload	Self-reported task burden and turnover	Paired surveys and voluntary response	Report welfare separately from payroll
Maintenance and resets	Historical maintenance	Vendor RaaS + downtime + technician cost	Include full vendor and site TCO
Customer experience	Wait time and service satisfaction	Same-context outcome measures	Report welfare and contribution change

Prospective methodology; no result is asserted for Bear, Pudu, Keenon or any other individual supplier.

9.9 Design human–robot choreography rather than substitute-task slogans

A kitchen automation claim is incomplete unless it describes loading, supervision, unloading, error clearing, cleaning, batch changeover and end-of-shift storage. A mobile service robot may move trays along a corridor but still require employees to arrange plates and perform table-side service. A robotic arm can execute a portioning action but may demand more frequent replenishment, cleaning and inspection. The appropriate productivity measure is the entire station or shift, not the isolated robotic movement.

A site pilot should record who owns each normal and exceptional step. Workers need clear authority to stop a system, safely recover a jam, correct an order and bypass the robot during a peak. Vendors need a response-time commitment, spare-parts availability and escalation path. Food-contact and workplace safety reviews should be treated as stage gates, not assumed consequences of a favorable AI benchmark. This is particularly important where the physical system interacts with hot surfaces, liquids, knives or confined service corridors.

The long-term opportunity remains attractive because restaurants present recurring, high-frequency workflows under real human and physical constraints. But reliability must be measured over shifts, weeks and locations, including failures and unplanned interventions. A claim of labor substitution requires a payroll or productive-capacity bridge; a claim of worker augmentation requires observable improvements in workload, quality or retention. These are distinct hypotheses and should be tested separately.

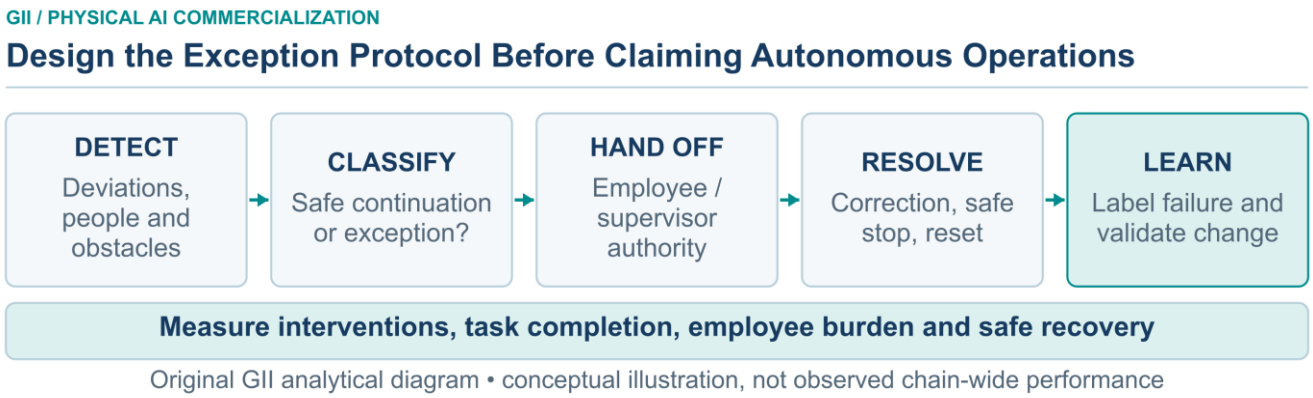


Figure 18 | A human-approved exception pathway around bounded machine execution
GII workflow design. It is not a certified safety architecture for a particular robot or kitchen.

9.10 Skylark and the implementation capacity behind a fleet

Skylark's operator history records completion of approximately 3,000 floor-service robots across 2,100 restaurants in December 2022. Its DX account describes testing three models, training seventeen instructors and using operational data and store visits to improve adoption. These disclosures establish a historical network rollout and an implementation process. They do not establish an October 2026 active-fleet count or independently measured causal financial returns [S114–S115].

The case is particularly useful because the restaurant operator describes the work needed around the equipment. For a robotics supplier, repeatable installation depends on an operating organization that can commission the route, teach staff and revisit low-utilization sites. A developer may reduce local configuration effort through better perception or planning, but the commercial value of that improvement should be measured against the existing rollout process.

GII proposes separating three units of observation. The robot record would identify tasks, movement, interruptions and availability. The site record would include guest demand, staffing and route conditions. The rollout record would capture instructor time, travel, local changes and time to ordinary use. These levels allow a team to ask whether a new model improves task execution, shortens deployment, or simply shifts work between the operator and supplier.

A fleet also creates a data-governance question. Recurring navigation and service events may help identify failure modes, but a large quantity of successful trips is not by itself evidence of transferable manipulation knowledge. A useful study would define the learning objective before deciding which telemetry is collected or reused. The restaurant's authority over operations and workers' data protections remain relevant when a supplier seeks to improve a broader model.

The economic channel should be observed at the location. Reduced walking can allow staff to perform other tasks, support a different shift pattern or improve service quality; those outcomes should be measured separately. A change in paid hours requires a staffing bridge. A change in sales requires evidence that the site could fulfill additional demand and retain contribution after incremental costs. Operator-described productivity effects should keep their original period and population labels.

For a network deployment, the relevant cost includes the people and service capacity required to keep locations using the equipment. GII proposes tracking installation cohorts, active use, support interventions, local modifications and discontinued units. These measurements would help distinguish a shipment from an adopted service and identify whether geographic density improves support economics. They are proposed analytical fields, not additional reported Skylark results.

Skylark complements the U.S. restaurant cases as an international implementation example. Its rollout should not be added to the U.S. Top-20 unit denominator or used to infer the maturity of food-handling arms. Table 140 makes the human role explicit, while Table 142 connects implementation measures to customer and supplier economics.



Picture 10 | Transport automation retains a human service handoff
AI-generated concept illustration for GII; equipment and settings are illustrative.

9.11 Kura Sushi and integration in a constrained service route

RobotLAB's April 5, 2026 Kura Sushi case reports an average of 116 active KettyBots across more than sixty locations in Q1 2026. It describes equipment selection for narrow aisles, floor mapping, training and local support. Kura's own FY2025 filing separately describes robotic service and its dependence on automated equipment. The operator filing does not independently certify the integrator's later fleet metrics [S122–S123].

The vendor reports more than 4,000 monthly fleet work-hours and more than 12,600 across Q1 2026, but also claims roughly 354 monthly hours per location across sixty-plus sites. The quarterly total implies over 4,200 hours per month for the whole fleet; the per-location claim does not reconcile without different definitions or denominators. It is therefore excluded from the economic analysis. The vendor's $590,000 annual labor-value estimate is not treated as realized payroll savings [S122].

The implementation lesson remains useful with those limits visible. GII proposes testing an actual service route under normal aisle occupancy, then recording loading, waiting, handoff and recovery. A robot can complete a route while a guest or worker still performs part of the service. The deployment record should identify the responsible person at each transfer and the safe fallback when the route is obstructed.

For a developer, the acceptance test should include ordinary variation in people and furniture, rather than only a cleared demonstration route. For a commercial team, the comparison should include mapping, staff training, local support and changes needed after opening. Table 140 treats Kura as an integration case, while Table 143 retains the quantitative qualification beside the source. No vendor labor-value estimate is inserted into the financial models.

9.12 Human handoff and remote work in service robotics

LG's Arirang K case describes an increase from two to four CLOi ServeBots at a Korean barbecue restaurant, with staff loading and handing off dishes. The account provides a named small-site example and customer testimony rather than a causal financial evaluation [S129]. It is useful for showing that transport capacity and table-side service can remain separate parts of the same job.

GII proposes measuring the entire service episode: preparation at the loading point, waiting for a route, movement, transfer at the table and return. A reduction in carrying work could improve a worker's experience even when paid hours do not fall. Conversely, repeated loading or recovery tasks could offset part of that gain. The employee and customer measures in Table 50 should therefore accompany any task-time result.

DAWN in Tokyo uses a different allocation of human authority. Its operator describes a permanent experimental cafe in which remote human pilots provide service through OriHime robots; trained pilots also operate the OriHime and NEXTAGE Tele-Barista arrangement [S125]. The service is classified here as telepresence and teleoperation. Conversation with a robot in that setting is not evidence that an autonomous language or action model performs the worker's role.

The relevant opportunity is access to participation and remote work. A proposed evaluation would examine the pilot's control, training, workload and ability to complete a service interaction, alongside the customer's experience and the assistance needed on site. It should account for the remote worker's paid time as well as local support. Neither human contribution should disappear from the cost boundary merely because it is less visible to the guest.

DAWN also prevents an overly narrow definition of commercial value. Robotics can support a service people choose because it connects them with another person. That objective deserves its own measures and should not be forced into a labor-substitution claim. This experimental arrangement clarifies human authority and job design, while its economic transfer to an ordinary high-volume restaurant remains unestablished.



Picture 11 | A remote human remains the operator in telepresence service
AI-generated concept illustration for GII; equipment and settings are illustrative.

9.13 Pudu and Sushi Express illustrate a division of service tasks

Pudu's vendor-hosted Sushi Express case in Taiwan assigns guest guidance and plate-return assistance to KettyBot Pro, and kitchen-to-table transport to BellaBot Pro [S132]. The account makes a multi-robot workflow concrete, but does not provide a dated store census, independently measured paid hours saved or restaurant financial returns. It is an international comparator outside the U.S. Top-20 sample.

The commercial question is how that division affects people at each site. Repeated transport may be suitable for automation while seating judgment, loading, clearing, guest requests and recovery still require employees. Assess customer experience, employee workload and manager willingness to continue separately. A supplier's description of service efficiency is not proof that all three groups prefer the resulting arrangement.

10. What Leading Restaurant Chains Are Actually Deploying

The primary study universe is the top 20 brands in the 2026 QSR 50 (2025 data). That sample captures scale and operating diversity, but it does not represent full-service restaurants or every U.S. outlet. Technology disclosure varies sharply. The audit below records verified evidence without interpreting non-disclosure as non-adoption. [S04]

10.1 Sample frame, parents and capital allocation

Map each ranked brand to the group responsible for its technology and franchise investment decisions. Taco Bell and KFC remain Yum! brands; Pizza Hut ex-China transferred to LongRange Capital in September 2026, while Byte services were expected to continue [S40,S67]. Shared software and current brand ownership therefore require separate records. Table 52 preserves the historical 2025 operating sample alongside the dated ownership change.

- Create ownership and brand matrices to prevent counting shared enterprise technology three times.
- Normalize U.S. footprint against parent annual global restaurant count.
- Franchisor technology fees, franchisee CapEx and local operating benefits accrue to different parties and require separate measures.

Table 52 | Top 20 brand–parent strategy crosswalk

Brand	Parent / group	Units	U.S. system sales $B	Business model qualifier
McDonald's	McDonald's	13,706	55.06	QSR franchise/owned split in Table 8
Starbucks*	Starbucks	16,860	30.25	QSR franchise/owned split in Table 8
Chick-fil-A*	Chick-fil-A	3,287	23.92	QSR franchise/owned split in Table 8
Taco Bell	Yum! Brands	7,784	17.25	QSR franchise/owned split in Table 8
Dunkin'	Inspire Brands	9,999	13.11	QSR franchise/owned split in Table 8
Wendy's	Wendy's	5,969	11.90	QSR franchise/owned split in Table 8
Chipotle	Chipotle	3,938	11.68	QSR franchise/owned split in Table 8
Burger King	Restaurant Brands International	6,649	11.07	QSR franchise/owned split in Table 8
Domino's	Domino's Pizza	7,186	9.95	QSR franchise/owned split in Table 8
Subway*	Subway	18,773	9.20	QSR franchise/owned split in Table 8
Panda Express	Panda Restaurant Group	2,607	6.41	QSR franchise/owned split in Table 8
Panera*	Panera Brands	2,214	5.89	QSR franchise/owned split in Table 8
Popeyes	Restaurant Brands International	3,196	5.70	QSR franchise/owned split in Table 8
Raising Cane's	Raising Cane's	913	5.49	QSR franchise/owned split in Table 8
Wingstop	Wingstop	2,586	5.34	QSR franchise/owned split in Table 8
Sonic Drive-In	Inspire Brands	3,412	5.24	QSR franchise/owned split in Table 8
Pizza Hut	LongRange Capital since 2026-09-01 (previously Yum!)	6,307	5.09	2025 rank/sales historical; current parent changed in Sep 2026 [S40]
Dairy Queen	International Dairy Queen	4,115	5.01	QSR franchise/owned split in Table 8
KFC	Yum! Brands	3,523	4.68	QSR franchise/owned split in Table 8
Whataburger	Whataburger	1,161	4.31	QSR franchise/owned split in Table 8

Parent labels are organizational research tags; validate private-equity ownership/control separately for final publication. Sources: [S04] + source-assisted group mapping

The supplemental implementation cases in Chapters 6 and 9 and Appendix L use a deliberately separate case universe. International operators, independent concepts, suppliers and airport formats are not added to the U.S. Top-20 sample, its unit count or its FY2025 issuer comparison. They provide evidence about particular deployment decisions rather than a representative adoption rate.

10.2 Full Top 20 technology disclosure audit

The following matrix is deliberately conservative. Documented use of cloud software is not by itself evidence of an owned hyperscale data center or closed-loop Physical AI. Technologies for other brands require primary verification.

- Record the strongest first-party public proof for each brand, and the precise disclosure date.

- The disclosure audit uses these codes: verified scaled (VS); verified limited deployment (VL); announced/pilot (AP); historic only (H); not independently verified (NV). These describe the cited evidence rather than certify undisclosed outcomes.
- Create a source-linked deployment matrix with location count, contractual ownership, and quantified outcomes where available.

Table 53 | Top 20 technology disclosure matrix, October 2026

Rank	Brand	Code	Supported public evidence or gap	Sources
01	McDonald's	VS	2026 Investor Day: Google edge in thousands of restaurants across 7 major markets; U.S. completion planned in 2027; ArchIQ efficiency target is not achieved ROI.	[S38],[S39]
02	Starbucks*	VL	2026: Green Dot Assist (worker AI), Smart Queue order orchestration; company plans to extend AI signals to connected equipment. Store coverage not fully disclosed.	[S46]
03	Chick-fil-A*	VS	All-store edge Kubernetes; minimum three nodes and optional GPUs described in 2025	[S10]
04	Taco Bell	VL	Byte by Yum! elements used; NVIDIA AI pilots announced at select U.S. sites	[S12],[S13]
05	Dunkin'	NV	Potential third-party kitchen forecasting uses require first-party verification	Open
06	Wendy's	VL	FreshAi national expansion; order volume reported by company in May 2025	[S15]
07	Chipotle	AP	Autocado and augmented makeline restaurant tests	[S16]
08	Burger King	AP	BK Assistant / Patty headset trial reported at ~500 U.S. locations	[S26]
09	Domino's	VS	DOM OS and PULSE; >85% U.S. retail sales from digital channels 2025	[S19]
10	Subway*	NV	No verified chain-wide restaurant Physical AI architecture in this release	Open
11	Panda Express	NV	No verified chain-wide restaurant Physical AI architecture in this release	Open
12	Panera*	NV	Digital systems likely present; specific Physical AI/edge scale unverified	Open
13	Popeyes	VL	RBI brand context; brand-specific physical AI deployment unverified	RBI group; open
14	Raising Cane's	NV	No verified chain-wide physical AI deployment in this release	Open
15	Wingstop	VS	2026: nationwide Smart Kitchen KDS + AI-assisted demand forecasting; supplier reports 2,500+ restaurants; AI ROI not independently reported.	[S47],[S48]
16	Sonic Drive-In	NV	Parent shared tech stack not independently classified	Open
17	Pizza Hut	VL	Former Yum! brand, sold to LongRange Sep 2026; 2026 Yum disclosure says Byte services continue post-sale. Brand-specific Physical AI ROI unverified.	[S40],[S12]
18	Dairy Queen	NV	No verified physical AI rollout in this release	Open
19	KFC	VL	Byte platform and AI restaurant coach described at group level	[S12]
20	Whataburger	NV	No verified restaurant Physical AI rollout in this release	Open

VS = verified scaled within the described solution; VL = verified limited or cross-brand platform; AP = announced/pilot; H = historical; NV = not independently verified in this release. Does not assert other companies have no AI. Sources: [S09]–[S16], [S19], [S26]

10.3 McDonald’s and Chick-fil-A: two models for local intelligence

McDonald’s describes Edge moving Google-supported computing into U.S. restaurants and accuracy scales supporting order checks. Chick-fil-A provides more detailed engineering disclosure of a fleet of local Kubernetes clusters. These offer technically comparable but strategically different cases of enterprise digital infrastructure.

- The comparison separates franchise and operating models, architecture ownership and local hardware rollout.
- Actual quality and uptime measurements are distinct from aspirational benefits and management targets.
- Investigate whether site-level GPU acceleration is required or only task-specific.

Sources: [S09]–[S11]

10.4 Yum! and the platform economics of a franchise group

Yum! introduced Byte by Yum! in February 2025 and stated that 25,000 global restaurants used at least one product then, with over 300 million annual U.S. digital transactions flowing through group products. A subsequent NVIDIA partnership introduced pilots and a targeted rollout plan. These are not synonymous with a complete adoption of computer vision or an AI agent in 25,000 locations.

- Group-wide software penetration and locations participating in an individual agent pilot have different denominators.
- Group economies of scope may persist across brands through software-service agreements even after brand ownership changes.
- Technology fees and cross-brand data access depend on the commercial arrangements among the operator, franchisor and platform provider.

Sources: [S12], [S13]

10.5 Wendy's, Chipotle, Starbucks, Domino's and RBI

Each provides a distinctive commercial lens: Wendy's voice ordering, Chipotle's selective kitchen mechanization, Starbucks equipment data, Domino's long-standing operating software, and Burger King employee-facing AI. Together they show why Physical AI is not reducible to service robots.

- For Wendy's, distinguish company order-volume statements from net incremental financial benefit.
- For Chipotle, assess Autocado task cost and Hyphen augmented makeline deployment as pilots.
- For Starbucks, distinguish a 2019 technology demonstration from 2026 company-wide deployment.
- For Domino's, measure the link between digital customer interfaces and physical production/dispatch.
- For Burger King, evaluate employee assistance versus workforce surveillance and consent.

Sources: [S14]–[S16], [S19], [S26]

Table 54 | Anchor cases — source-backed operational signals

Operator	Confirmed evidence	Date / scale	Financial outcome disclosed?	Next research question
McDonald's	2026 Investor Day: edge in thousands across seven markets; NEXT/ArchIQ strategy, targeted 250bp gross efficiency	2026-09-23; U.S. rollout planned 2027	Company TARGET: ~$100k annual cash benefit per avg U.S. restaurant; not observed AI-only return	Verify actual franchisee CapEx, margin realization and deployment timeline [S38],[S39]
Chick-fil-A	At-least-three-node restaurant Kubernetes clusters	Sep 2025: all stores	No per-store payback disclosed	TCO vs resilience benefit
Yum! Brands	Byte by Yum! 25k global restaurants use ≥1 product	Feb 2025 group disclosure	Qualitative benefit assertions	Platform fee, franchisee P&L
Wendy's	FreshAi voice ordering processes tens of thousands orders/day	May 2025 company statement	No audited incremental profit in release	Net ordering errors and abandonment
Chipotle	Autocado / Hyphen makeline trials	Sep 2024 site tests	No fleet-wide ROI established	Full task and maintenance cost
Domino's	PULSE / DOM OS, 85%+ U.S. retail sales digital	FY2025 10-K	Digital mix disclosed; no Physical AI ROIC	Operational conversion to throughput
Starbucks	Green Dot Assist, Smart Queue, proposed equipment AI	Updated 2026-06; complete scale not reported	No incremental site AI ROI public	Measure speed-of-service and employee experience [S46]
Burger King	Patty headset AI test	2026: ~500 U.S. locations per AP	No independent employee experience	Consent and work-conditions change

Sources: [S09]–[S16],[S19],[S26]

10.6 Disclosure quality and competitive-position implications

Public technical disclosure may indicate internal engineering capability, but it is an incomplete measure of deployment maturity. Non-disclosure does not establish technological lag.

- Source confidence and deployment maturity are scored separately: a reliable announcement can still describe an early pilot.
- Gather incremental evidence via SEC filings, technical presentations, supplier customer cases and executive interviews.

- Intellectual property, customer-data rights, active deployment scale and measured outcomes are distinct diligence fields.

10.7 September–October 2026 updates: real deployment and real operating constraints

McDonald's September 23, 2026 Investor Day described a unified data foundation, Restaurant > NEXT, ArchIQ and franchisee support. Primary disclosures target about 250 basis points of gross restaurant-level efficiency and roughly $100,000 of gross annual cash-flow benefits per average U.S. restaurant [S38,S58]. The official release also estimates approximately four-year franchisee payback after partnering support [S38]. These forward-looking measures cover a broad modernization program; they are not realized returns on edge compute or AI alone.

Table 55 | McDonald's NEXT targets and evidence gaps

Management metric	Disclosed amount / status	Period or scope	Investment interpretation	Needed verification
Restaurant-level gross efficiency	~250 basis points TARGET	U.S. and International Operated Markets at full rollout	Gross efficiency, not necessarily incremental net profit	Baseline margin and decomposition of savings
Average U.S. store cash benefit	**~$100,000 annually TARGET**	**At implementation of NEXT elements**	**Aspirational, multi-element program economics**	**Actual matched-store cash flow**
Franchisee payback	~4 years TARGET after partnering	Investments within NEXT	Requires cost, subsidies and timing	Franchisee CapEx net of rent/capital support
Partnering support	$8.5B total planned through 2036	Rent relief and capital support	Not AI-only CapEx	Project and geography breakdown
Near-term support	~$5B through 2030 planned	Portion of $8.5B total	Aggregate corporate capital allocation	Cash spending vs relief
Global company network	46,000+ restaurants	Sep 2026	System scale; not count of ArchIQ installations	Country/store installation mapping
Google edge deployment	Thousands, seven major markets	Sep 2026 CIO statement	Hardware/app deployment evidence	US penetration & AI workload usage
U.S. Google edge completion	2027 PLANNED	Future operational target	Not achieved as of Oct 2026	Completion announcement and installed-base audit

Sources: [S38] official release for NEXT targets and approximately four-year franchisee payback after partnering; [S58] FAQ for investment and benefit assumptions; [S39] secondary transcript for the seven-market edge statement. Targets require realization tests. AUV × efficiency is an arithmetic cross-check, not independent AI attribution.

The economic interpretation should distinguish the operator's observed value from vendor monetization. If a franchisee could eventually benefit by $100,000 per site annually, that does not justify paying any single AI supplier the same amount. Benefits arise from menu, equipment, restaurant redesign, operations, training, and digital systems jointly. Franchise investment decisions depend on incremental costs and benefits compared with alternative improvements, execution dates, and management estimates of payback.

Two 2026 disclosures strengthen the counterargument that AI infrastructure alone cannot repair weak demand. Wingstop reported domestic comparable sales declining 7.5% in Q2 2026 while highlighting its Smart Kitchen investments. Wendy's U.S. same-restaurant sales fell 7.0% in Q2 2026 while the company was implementing a wider turnaround. These outcomes do not establish that the technology caused the decline or failed; they show why revenue performance and technology adoption must be evaluated independently. [S48],[S49]

Table 56 | Restaurant technology, demand and scale, 2026

Company	Latest observed operating fact	Physical AI / digital claim	Caveat / gap	Source
McDonald's	46,000+ worldwide units (Sep 2026)	ArchIQ planned; Google edge at thousands in 7 markets	Target benefits not achieved ROI	[S38],[S39]
Starbucks	2026 public plan centers barista workflow and customer experience	Green Dot Assist; Smart Queue; equipment AI planned	Full active location count for these tools not stated	[S46]
Wingstop	2,728 U.S. units at Q2 2026; domestic comps -7.5%	Smart Kitchen nationwide; supplier reports 2,500+ deployment	Technology-specific profitability unreported	[S47],[S48]
Wendy's	5,724 U.S. units Q2 2026; U.S. comps -7.0%	FreshAI / voice ordering in prior disclosures	No verified causal contribution to comps	[S49]
Chipotle	1,500 Chipotlanes milestone Sep 29 2026	Physical format + efficiency; separate from AI robotics pilots	Chipotlane is NOT itself Physical AI	[S50]
Yum! Brands	Pizza Hut divested Sep 2026	Byte services expected to continue for Pizza Hut ex-China	Brand ownership not equal operating-platform ownership	[S40]

Mixed reporting periods shown explicitly; 2025 ranked brands are not restated to 2026 unit counts. Company and supplier claims identified separately.

10.8 Corporate control versus technology-platform continuity

On September 1, 2026 Yum! Brands completed the Pizza Hut ex-China divestiture to LongRange Capital, alongside its prior Pizza Hut China sale to Yum China. The aggregate disclosed transaction consideration was approximately $2.7 billion, subject to adjustments. Earlier Yum SEC disclosure stated that it intended to keep providing Byte by Yum! technology to Pizza Hut ex-China after the transaction. This separation provides a strong corporate-development case: an enterprise operating system can become a contractual service even when the franchised brand changes ownership. [S40]

A related restaurant software platform case is Thoma Bravo's approximately $2.0 billion equity-value acquisition of Olo in September 2025. This measures the purchase price of an established company, NOT the annual restaurant AI market or the economic value of any robotics training data. Comparing Olo, Pizza Hut and Sweetgreen/Spyce illustrates the difference among control of restaurant demand, ownership of software, and ownership of physical automation IP. [S45],[S17]

10.9 From deployments to attributable value: the chain-level evidence gate

New operator adoption data provide a context for the Top 20 brand audit: brand-level investment intentions are common, but meaningful returns are still poorly disclosed. For each restaurant chain, the database should distinguish digital/AI in marketing, management assistance, predictive kitchen planning, device sensing, and autonomous physical actuation. This preserves the difference between operational intelligence and robots; it also avoids assigning high maturity merely because a corporation has hired a major technology provider.

Table 57 | Proposed deployment-to-value diligence standard

Stage	Minimum evidence	Finance status	Representative public evidence	Investor question
Announced	Official strategic partnership	No monetized impact assumed	Named supplier and target problem	What is the contracted commitment?
Prototype	Device/software tested in controlled environment	R&D / experimental	Demo or lab results	Does it work in rush-hour conditions?
Pilot	Defined sites/dates and operational task	Pilot cost measurable	Store count and disclosed use case	What is matched no-AI control?
Scaled rollout	Active sites, uptime, support and support fees	Per-site budget measurable	Store-level production deployment	Who finances, maintains and renews?
Verified operating benefit	Repeatable KPIs and credible controls	Incremental FCF model possible	Time, quality, waste and labor changes	Are improvements causal and recurring?
Valuation-relevant returns	Net income, cost of capital, life-cycle CapEx	NPV, IRR, ROIC defensible	Independent / financial verification	What is the expected return at scale?

No completed company case is automatically assigned all six stages; source-backed facts must govern stage assignment.

10.10 McDonald's > NEXT: technology investment as restaurant underwriting

McDonald's September 23, 2026 NEXT FAQ describes a broad restaurant investment program. For a traditional U.S. drive-through, management estimates about $800,000 of incremental investment, about 250 basis points of gross restaurant-level efficiency and roughly $100,000 of gross annual cash-flow benefits for an average U.S. location [S58]. The official release estimates approximately four-year franchisee payback after partnering [S38]. Support includes rent relief and capital contributions, with the full unit-level allocation undisclosed [S58,S59].

The simple ratio of $800,000 / $100,000 equals eight years, but it is not a contradiction of the company's four-year franchisee payback claim: the numerator is gross incremental project investment while franchisees receive partnering support; the denominator is a gross benefit, some of which may be reinvested; timing and ramp-up also matter. No defensible standalone franchisee IRR or AI-attributable ROI can be calculated without unit-level partnering and net-cash-flow detail.

The useful investment distinction is that McDonald's NEXT combines real estate modernization, operations, software, voice AI, food and beverage execution, and training. The entire $800,000 should not be assigned to AI, nor can the targeted $100,000 be counted as realized technology savings. The package is best treated as a system-level investment thesis with operational milestones requiring later audit.

Table 58 | McDonald's NEXT facts and financial targets

Metric	Value / description	Status	Underwriting caveat
Incremental U.S. store investment	~$800k per traditional drive-thru	Management estimate	Not pure AI capex; phasing / store mix vary
Gross annual efficiency benefit	~$100k per average U.S. store	Management target	Not achieved incremental FCF
Gross restaurant-level P&L efficiency	~250 bps	Post-deployment target	Scope covers Restaurant > NEXT elements
Franchisee simple payback	~4 years after partnering	Management estimate [S38]	After partnering support; net investment and benefit attribution required
Voice AI order accuracy	>90% English and Spanish	Management-reported tool test	Not proof of full-store autonomous deployment
Voice AI labor opportunity	At least 50 hours/week as it scales	Forward-looking expectation	Hours freed != payroll savings
Automated inventory	~5 hours/week and 15% lower waste expected	Forward-looking estimates	Attribution and baseline not disclosed
Predictive equipment operations	~50% improvement in key availability	Management expectation	Specific reliability denominators undisclosed
Corporate partnering support	~$8.5B through 2036; ~5B by 2030	Company plan	Includes rent relief and capital; not additive
Capital partnering subset	~$1.5B–$2B cumulatively through 2030	Company estimate	Subset of partnering; not annual capex

Primary sources: McDonald's official NEXT release, FAQ and SEC release, September 23, 2026 [S38,S58,S59]. Voice, inventory and equipment expectations are management remarks in secondary transcript [S39]. Four-year franchisee payback is a management estimate after partnering support, not standalone AI ROI or an independently audited restaurant outcome.

Table 59 | Illustrative McDonald's cash yield and payback

Calculation	Arithmetic	Interpretation	Not inferable
Unadjusted gross benefit / gross cost	$100k / $800k = 12.5%	Simple pretax gross annual yield after full benefit realization	Net operator FCF or IRR
Unadjusted gross simple payback	$800k / $100k = 8.0 years	Illustrative no-support benchmark	Company's post-support payback
Reported payback after partnering [S38]	~4.0 years	Financing incidence materially changes economics	Exact franchisee net investment
If all $100k retained and fully immediate	4 yr implies $400k net investment	Pure algebraic thought experiment	Any actual per-store subsidy amount; not disclosed here
If only 70% of gross retained and immediate	4 yr implies $280k net investment	Shows sensitivity to reinvestment	Actual cash capture or capital split

Illustrative calculations are intentionally not a reconstruction of McDonald's proprietary underwriting. Company says majority of benefit expected in bottom line over time; does not disclose exact cash capture. [S58,S59]

10.11 Wingstop Smart Kitchen: scaled process intelligence versus systemwide demand

Wingstop and Perfect Company offer an unusually observable example of restaurant process intelligence operating at chain scale. Perfect Company announced in May 2026 that customized KitchenPro systems had rolled out to more than 2,500 Wingstop U.S. restaurants in fewer than twelve months, incorporating visual workflow displays and AI-driven demand forecasting. Wingstop's July 2026 quarterly filing reported 2,728 U.S. units as of June 27, while digital sales represented 71.6% of system sales and domestic same-store sales were negative 7.5%. [S60,S61,S62]

A vendor case study cites enhanced service throughput and customer satisfaction, and restaurant trade reporting describes measurable improvements in on-time production and order accuracy. The sample, measurement denominator and counterfactual are not publicly described in sufficient detail to attribute those gains solely to Smart Kitchen. Moreover, declining comparable sales can coexist with improved throughput when consumer demand weakens. No causal claim should be drawn from this correlation in either direction.

Table 60 | Wingstop deployment and economic evidence

Indicator	Disclosed figure	Source type	What remains unknown
Smart Kitchen rollout	More than 2,500 U.S. sites as of May 2026	Technology supplier [S61]	Per-site fees, cohort ROI, independently audited uptime
Actual U.S. store count	2,728 restaurants June 27, 2026	Issuer Q2 report [S60]	Exact aligned installation denominator by date
Digital mix	71.6% of systemwide sales in Q2 2026	Issuer reported [S60]	Marginal fee/economics of Smart Kitchen
Domestic comparable sales	-7.5% Q2 2026	Issuer reported [S60]	Causal impact of technology; macro and traffic controls
Average U.S. unit volume	**~$1.893M Q2 2026 annualized measure**	**Issuer reported [S60]**	**Store-level dispersion and cohort comparability**
Improved service speed	Vendor/trade reports material improvement	Vendor/trade claims [S62,S73]	Time denominator, matched controls, calendar bias
Customer satisfaction gain	Reported +17 percentage points for delivery	Vendor/trade claims [S62,S73]	Response rates, observed sample, attribution

Information dated May-July 2026. Never present vendor-reported improvements as independently verified causal results. U.S. restaurant count and earlier 2,500+ rollout relate to different dates.

10.12 Starbucks: employee-centered Physical Intelligence before robotics

Starbucks' public AI strategy illustrates the value of small, well-placed information interventions rather than necessarily installing visible robots. In June 2025 the company said Green Dot Assist was being tested in 35 coffeehouses, enabling employees to retrieve recipes and operating guidance through store iPads. In its January 2026 statement, updated June 2026, Starbucks described Smart Queue as sequencing cafe, drive-thru, mobile and delivery orders, with equipment diagnostics and additional forecasting capabilities contemplated or under development. [S64,S65]

This is best characterized as employee-facing decision support and operating workflow intelligence, not evidence of autonomous kitchen actuation. Its economic value should be tested through peak-period throughput, employee-reported friction, order handoff consistency and customer satisfaction, with an explicit control for concurrent labor investments and operational changes.

Table 61 | Starbucks disclosed AI applications

Application	Official disclosed status	Physical operating problem	Evidence needed for ROI
Green Dot Assist	35-coffeehouse pilot in June 2025	Recipe and procedural guidance during busy shifts	Question-response time, mistakes, confidence, active use
Smart Queue	Described as deployed capability in 2026	Sequence simultaneous digital and physical channels	Arrival-to-ready distribution and lost orders
Equipment condition intelligence	Planning extension to equipment	Unexpected machine downtime	Downtime/failure records before-after, sensors/false alerts
Forecasting and scheduling	Forward-looking applications	Out-of-stock and mismatch of staffing and demand	Waste, shortages, labor churn, customer experience

Starbucks primary sources [S64,S65]. Do not infer nationwide rollout or profit impact from the disclosure.

10.13 Chipotle: stage-gated automation and the difference between a press photo and fleet deployment

Chipotle's September 2024 announcement identified two physical restaurant tests: Autocado at a Huntington Beach, California location and the Hyphen Augmented Makeline at Corona del Mar, California. These are company-confirmed trials with explicit addresses, not disclosed chainwide deployment. The company described an approximately 26-second autonomous avocado preparation step and stated that roughly 65% of digital orders were bowls or salads—an operational reason to examine the makeline. [S63]

The official Chipotle release documents the pilot sites, dates, vendors and tasks. Those 2024 observations do not establish a current 2026 fleet count. The photographs remain external source links under Appendix H's rights policy; this paper uses GII-authored analytical figures. Any later republication would require a separately documented permission or applicable license and a check of the depicted deployment's current status.

Table 62 | Chipotle disclosed implementation cases

Device	Site / disclosure	Task and claimed capability	Commercial scale classification	Evidence gap
Autocado	Huntington Beach, CA; Sep 2024	Cut/peel/core avocado; company says ~26 sec per fruit	Identified working restaurant trial	Current installation count, unit cost, waste and labor effects
Augmented Makeline / Hyphen	Corona del Mar, CA; Sep 2024	Dispenses bowls/salads while employees handle other items	Identified working restaurant trial	Reliability, paid hours, quality and capex
Cultivate Next investments	Vebu and Hyphen, company disclosed	Strategic minority technology investments	Corporate investment, not deployment	Investment size per issuer, valuation, acquisition rights

Chipotle original news release and associated official high-resolution photo files [S63,S72]. Do not count venture investment as additional restaurant sites.

10.14 FY2025 comparative public-company financial baseline

These 2025 issuer figures are the first repeatable basis for restaurant-company comparisons. The accounting margins are deliberately shown beside business-model differences. They indicate available corporate cash-generation capacity and operating pressures—not incremental ROIC earned on Physical AI. Starbucks' fiscal year ends in September, while most peers report calendar-year or late-December fiscal years. Several companies record royalties instead of full customer food sales; therefore the consolidated operating margins are not a measure of restaurant-level AI readiness. [S75–S81]

Table 63 | Restaurant issuer financial comparison — FY2025 (USD millions)

Issuer	Business model	Reported revenue	GAAP op. income	GAAP op. margin	Interpretive note	Source
McDonald's (MCD)	Franchise + property	~26,900	~12,400	46.1%	System sales ~$139.4B; operating cash ~$10.6B	[S75]
Starbucks (SBUX)	Mostly company-operated	37,184.4	2,936.6	7.9%	Restructuring charges; Sep FY end	[S76]
Chipotle (CMG)	Company-operated	~11,900	1,935.8	16.2%	Restaurant-level margin 25.4%; 2025 comp -1.7%	[S77]
Yum! Brands (YUM)	Mostly franchised	8,214.0	2,574.0	31.3%	Includes company sales, royalties and advertising services	[S78]
Wendy's (WEN)	Mixed, franchise-heavy	2,176.9	343.5	15.8%	U.S. co-operated restaurant margin 14.2%	[S79]
Wingstop (WING)	Mostly franchised	696.9	179.3	25.7%	System sales $5.3B; 2025 U.S. comp -3.3%	[S80]
Sweetgreen (SG)	Company-operated	679.5	(139.3)	(20.5%)	Restaurant-level profit margin 15.2%; comp -7.9%	[S81]

SOURCE / LIMITATION McDonald's and Chipotle revenue/operating income rounded as disclosed in primary releases; all other amounts $M. GAAP consolidated income does not equal operator net cash return. Sweetgreen restaurant-level margin is non-GAAP. Starbucks FY ended 2025-09-28. Economic comparison must be performed within like business models.

10.15 Restaurant operating performance cannot be inferred from technology rollout

Three counterexamples constrain any aggressive investment conclusion. Chipotle reported 2025 revenue growth supported by new openings even as comparable sales declined 1.7%. Wingstop reported $5.3B systemwide sales for 2025 and 12.1%

growth, but its U.S. comparable-store sales declined 3.3%; the 2026 Q2 comparable-store decline was 7.5% despite a reported scaled Smart Kitchen program. Sweetgreen's 2025 company revenue increased only 0.4% while comparable sales declined 7.9%; its operating losses widened, even as it retained licensed use of automation after selling the Spyce business. None of these observations alone identifies the effect of AI, positive or negative. [S77,S80,S81,S84]

Table 64 | Technology and financial outcomes by cohort

Company	Technology observation	Issuer financial observation	Unsupported causal inference	Required proof
McDonald's	Edge / accuracy platforms; Restaurant > NEXT targets	2025 system sales $139.4B; 46.1% corp op margin	That new investments already delivered 250bp efficiency	Store-level before/after and subsidy-adjusted cash flows
Chipotle	Autocado / Hyphen store-specific pilots	2025 comp -1.7%; restaurant-level margin 25.4%	That pilots caused weaker traffic or achieved ROI	Matched pilot stores, downtime, labor hours, yield
Wingstop	Supplier-reported Smart Kitchen chain rollout	2025 comp -3.3%; Q2 2026 comp -7.5%	That AI rollout caused demand decline or reversed it	Cohort contribution, quote times, sales mix controls
Sweetgreen	Infinite Kitchen; Spyce divestiture	2025 comp -7.9%; restaurant margin 15.2%	That asset sale proves automation failure	Pre/post store economics, sale rationale, licensing economics
Starbucks	Green Dot Assist and queue tools	FY2025 op margin 7.9%; turnaround expenses	That pilot changed corporation-wide margin	Barista task time, store cohorts and quality outcomes
Yum! Brands	Byte platform / vendor partnerships	FY2025 op margin 31.3%	That franchisor margin measures restaurant ROI	Franchisee economics, royalties and data ownership

SOURCE / LIMITATION Sources [S58,S75] MCD; [S63,S77] Chipotle; [S61,S80,S84] Wingstop; [S81] Sweetgreen; [S64,S76] Starbucks; [S78] Yum. Targets, pilots, and issuer financials are separately dated.

10.16 Industry bargaining power and control points

Industrial organization requires analyzing where switching costs and data rights reside, not simply counting stores with cameras. A multi-brand franchisor may aggregate demand and set platform standards. A technology supplier may control the operational interface, service contracts, billing and maintenance. An OEM controls equipment performance. The franchisee often bears retrofit expense and bears the cost of staff disruption. Bargaining power changes as systems become embedded in daily restaurant execution.

Table 65 | Restaurant technology contracts and bargaining power

Value-chain layer	Potential control point	Possible network or scale advantage	Buyer bargaining risk	Diligence metric
Brand / franchisor	Franchise standards, data and procurement	Many stores share one specification	Franchisee misaligned incentives	Fee allocation; vendor substitution rights
POS / KDS platform	Ordering events, payments and integrations	Deep daily workflow integration	Multi-vendor interoperability demands	Net retention; switching time; integration cost
Kitchen AI software	Decision rules, alerts and optimization	Cross-store benchmarks and model learning	Tools become commodity features	Verified savings/site; renewal rate
Computer vision/IoT	Visual, temporal and machine data	Installed sensors and labels	Privacy, hardware cost, data portability	False positives; cost per verified useful event
Robotics OEM / RaaS	Hardware, safety, maintenance, task data	Fleet operations and device reliability	Incomplete workflow and difficult sites	Task-level minutes saved net of exceptions
Cloud / edge infrastructure	Runtime, network and compute placement	Economies of shared capacity	Commodity compute and procurement scale	Five-year risk-adjusted TCO

SOURCE / LIMITATION Analytical structure based on transaction-cost and industrial organization reasoning; evidence of actual market share, pricing power or switching costs requires separate contractual/issuer documentation.

10.17 Two different routes to intelligent restaurant operations

Large-chain disclosures show at least two commercialization pathways. The first is a software-and-infrastructure path: reliable store computing, digital ordering, kitchen displays, employee assistance and production orchestration. The second is a physical-mechanization path: specialized preparation cells, cobots, mobile logistics or automated delivery. A chain may

pursue both, but rollout evidence for software should not be translated into robot unit sales. Task-specific robotics pilots may establish feasibility without demonstrating enough economics for a systemwide rollout.

Starbucks' Green Dot Assist was described as a 35-store generative AI pilot supporting baristas [S98], while the company's Smart Queue addresses multiple order channels and time windows [S106]. Chipotle named actual Autocado and Hyphen restaurant pilot locations in its 2024 disclosure [S99]. Chick-fil-A has described an edge platform present across stores [S105]. These demonstrate different product categories and evidence strengths; none alone validates a restaurant-wide autonomous world model. Operators and investors should keep the evidence stage visible next to each company example.

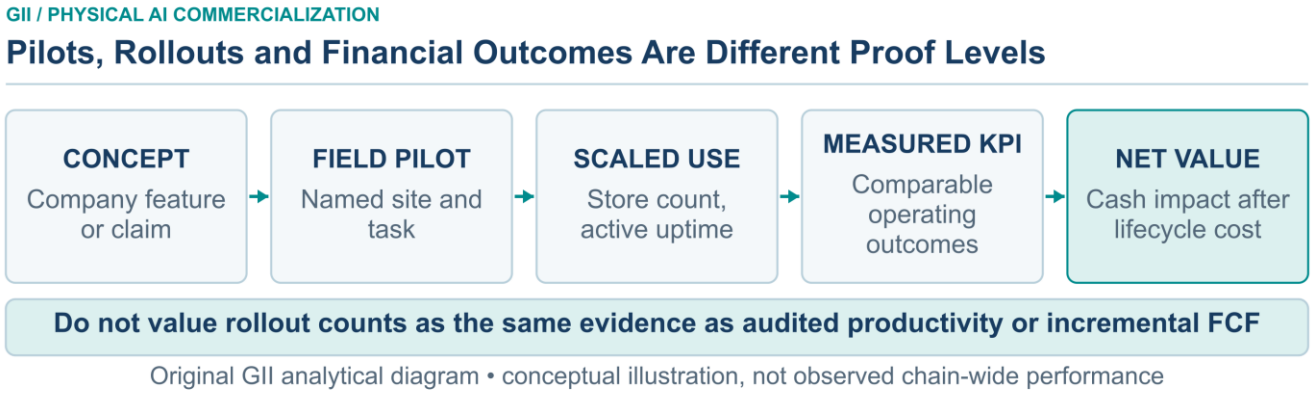


Figure 19 | A taxonomy for interpreting pilots, scaled implementations and demonstrated economics
GII evidence ladder; selected company context from [S98–S99], [S105–S106].

10.18 The strategic lesson from changes in technology ownership

In December 2025 Sweetgreen completed the sale of Spyce, including its Infinite Kitchen technology business, to Wonder, with disclosed cash and equity consideration. Sweetgreen stated that it would continue to use the technology through contractual arrangements [S110]. This is an instructive corporate-development case: the economic value of using automation in restaurants is distinct from the capital and organizational requirements of owning a robotics development business.

A buyer of restaurant technology should therefore compare internal R&D, technology acquisition, commercial partnership and licensed deployment as separate options. A Robotics AI vendor should anticipate governance questions about intellectual property, foodservice service contracts, product liability, technical support, upgrades and access to future restaurant-generated training data. The ultimate winner may not be the entity that first demonstrates a machine, but the organization that reliably operates and services a repeatable solution across the restaurant network.

11. Technology Supplier Markets and Industry Competition

Technology market sizing should distinguish a supplier revenue market from the economic value of problems solved. The service robotics sample from the IFR is informative about shipments, but it is not a census of all global companies and it does not publish a direct restaurant-only market revenue estimate. The 2026 edition expressly warns that the 2025 figures reflect 238 responding suppliers and that annual report samples can differ. [S06], [S34]

11.1 Product-market taxonomy

The market is layered: data collection, restaurant management software, AI inference, in-store compute, sensors, specialized connected equipment, robot systems, integration, and services. The chain of payments creates a material double-counting risk.

- A representative transaction ledger identifies the restaurant purchaser, point of sale, vendor, cloud subcontractor and equipment supplier.
- Differentiate software revenue, payment processing fees, hardware units sold, installed asset value, and cloud provider revenue.
- Produce a non-overlapping Revenue Pool definition for restaurant-focused physical intelligence.

Table 66 | Physical AI and restaurant technology taxonomy

Layer	Observable product	Market metric	Possible double count
POS/commerce	Ordering, payments, online channels	Software fees + payment net take	GPV and gross processing revenue
Operating systems	KDS, labor, inventory, forecasts	SaaS ARR / annual service revenue	POS bundled modules
Sensors and connected machines	Temp, equipment, vision, weight	Hardware sales + upkeep	OEM equipment bundles
Cloud/inference	Hosted databases and model tokens	Spend / supplier revenue	Bundled SaaS cloud charges
Edge hardware	Restaurant processors and NPU	New units / replacement value	OEM-integrated chips
Automation	Food preparation and dedicated tools	Capex + recurring support	Integrator resale vs OEM sale
Service robotics	Mobile serving / cleaning	Unit sales, RaaS, maintenance	Robot hardware and lease charges
Integration	Design, connectivity, security	Professional services spend	Resold vendor licenses

Sources: GII taxonomy; evidence anchors [S12],[S20],[S06]

11.2 Most recent robotics statistics

The latest IFR materials describe 2025 global professional service robot shipments near 250,000, approximately 117,500 transport/logistics robots, and **around 7,000 full-size humanoids**. Industrial robot installations surpassed 600,000 in 2025, with worldwide operating stock near five million. These are different categories and methodologies; the service-robot annual reports should not be mechanically chained into a long-run CAGR. [S06], [S07]

Table 67 | Reported robotics-market volumes

Metric	Period	Published value	Interpretation / caution	Source
Professional service robots	2025	Nearly 250,000	IFR sample-based international shipments; ~238 suppliers	[S06],[S34]
Transport/logistics robots	2025	117,500	47% of reported professional service robot shipments	[S06]
Hospitality category	2025	2nd-largest application; exact public count not verified	Guidance/information/telepresence included; not restaurant-specific	[S06]
Full-size humanoids	2025	About 7,000	Commercial/professional, >140 cm; early market	[S06]
Industrial robots installed	2025	More than 600,000	Global annual industrial installations	[S07]
Industrial robot operating stock	2025	About 5 million	Global factories, not restaurant robotics	[S07]
Hospitality robots	2024	More than 42,000	Earlier sample; do not compare 2024→2025 as homogeneous	[S37]; IFR 2025 edition

The IFR says service-robot statistics are supplier sample data and explicitly warns against comparing World Robotics editions due to changes in sample composition.

11.3 Publicly observable restaurant-technology economics

Toast offers one of the more detailed public financial references for a restaurant-centered software and payments platform. Reported GPV represents processed transaction volume, while total ARR includes recurring revenue run-rate from payments and subscription services and is not GAAP revenue.

- Public supplier comparisons require segmented GAAP revenue, recurring metrics, gross profit, active locations and payment volume, each with its own definition.
- Trace trends separately by subscription, fintech services, equipment, and professional services.
- Investigate PAR Technology and other specialized suppliers after comparing accounting definitions.

Sources: [S20],[S21]

Table 68 | Toast reported operating scale

Metric	FY2025 / 2025-12-31	2026 update	Revenue type / caution	Source
GAAP consolidated revenue	$6,153M	Not assembled	Includes significant gross fintech processing revenue	[S20]
Subscription-services GAAP revenue	$936M	Not assembled	Narrower than ARR	[S20]
Financial technology GAAP revenue	$5,037M	Not assembled	Gross reporting, payments-related	[S20]
Hardware / services GAAP revenue	$180M	Not assembled	Sales and services	[S20]
Recurring run-rate / ARR	$2,047M	$2.4B as of Q2 2026 (rounded)	Includes payment-related run-rate; not a revenue forecast	[S20],[S21]
Locations	~164,000	Q2 2026 net adds +~9,500	Toast-defined live locations, not every U.S. restaurant	[S20],[S21]
Gross payment volume	$195.1B	—	Transaction throughput, not supplier revenue	[S20]

2026 Q2 ARR figure is a rounded issuer update. This is NOT a standalone restaurant Physical AI market size.

11.4 Supplier commercialization and ownership

Robot market share is particularly difficult to infer because published shipment counts often span restaurants, hotels, industrial buildings and other sectors. Ownership events may be more informative about investor strategy than bare deployment claims.

- Cumulative robot shipments across industries, active customer sites and individual restaurant pilots measure different stages of commercialization.

- The LG/Bear ownership path is relevant to supplier consolidation; transaction economics require separately sourced consideration and control terms.
- Investigate customer retention, RaaS contracts, device service lives, and fleet utilization.

Table 69 | Service robotics suppliers and strategic position

Supplier	Cumulative unit claim	Sector scope	Capital/ownership event	Confidence
Bear Robotics	16,000+ robots shipped as of Sep 2026	Cross-sector; 5,000+ sites / 13 verticals	LG 51% control path in 2025; earlier $60M for 21%	Vendor shipment claim; LG event primary
Pudu Robotics	130,000+ shipped	Global cross-sector / 85+ countries	Funding, investors and valuation to verify	Company-stated shipment count
Keenon Robotics	100,000+ shipped	Cross-sector / 70+ countries	Funding, investors and valuation to verify	Company-stated shipment count

Cumulative shipments are NOT active installed base, U.S. restaurant penetration, or comparable market shares. Claims have distinct dates and include different products. Sources: [S22]–[S25]

11.5 Constructing a restaurant-addressable market

The critical gap is a public, comprehensive source for U.S. restaurant-specific spending on physical intelligence. This report therefore defines the bottom-up pricing, eligible-unit and adoption fields needed to calculate defensible ranges.

- Create product revenue pools based on the end buyer, to avoid hardware resale and cloud spend duplications.
- A restaurant-addressable revenue model requires subscription or RaaS prices, hardware ASPs, recurring support fees and deployment eligibility by format.
- Triangulate estimates with SEC supplier segment revenue, credible industry studies and unit installs.

Research limitation: restaurant-only revenue by segment, comparable U.S. supplier receipts, price samples and active paid installations remain incomplete.

Table 70 | Market revenue pool — data collection status

Market	Revenue measure	Current supported benchmark	Current status / next evidence
POS + payment platforms	Separate software fees and processing gross/net revenue	Toast 2025 company revenue breakdown only	Partial; broader market unverified
AI kitchen software	U.S. restaurant SaaS spend	No common audited market total	Research open
Restaurant edge compute	U.S. hardware + managed service spend	Chain architecture but no spending totals	Research open
Connected equipment / IoT	New restaurant systems and service fees	Historical Starbucks use case only	Research open
Food-prep automation	Hardware new installs and recurring maintenance	Chipotle pilots, Sweetgreen transaction	Research open
Serving/service robots	New units and RaaS in U.S. restaurants	IFR all-sector shipments, vendor cumulative counts	Research open; no US restaurant TAM

TBD means not disclosed or not yet validated, not market size = $0.

11.6 Verified 2026 restaurant technology supplier revenue and installed-base benchmarks

Financial results from public restaurant-technology vendors provide a defensible starting point for market value-chain research, even though none reports an audited U.S. restaurant Physical AI revenue category. Toast monetizes restaurant and adjacent commerce through subscriptions, payment processing and services; its gross reported revenue differs greatly from subscription revenue and ARR. PAR reports subscription-plus-payments ARR on active sites. NCR Voyix serves both retail and restaurant customers, with the restaurant segment disclosed separately. SoundHound reports cross-industry voice and agentic AI revenue. The four business models cannot be added to create a restaurant AI TAM. [S41]-[S44]

Table 71 | Restaurant technology supplier financial observations

Company	Q2 2026 GAAP revenue	Recurring metric / period	Installed-base metric	Sector coverage / caveat	Source
Toast	$1,908M	$2,409M ARR at Jun 30; subscriptions $1,210M, payments $1,199M	~180,000 live locations	Restaurants plus adjacent retail/hospitality; fintech revenue gross	[S41]
PAR Technology	$133.4M	$338.0M ARR at Jun 30	174,300 active sites	Subscription and transaction processing; site overlap possible	[S42]
NCR Voyix	$523M total; $158M restaurant segment	$435M consolidated quarterly recurring revenue; $211M recurring software	Not established in Q2 release	Restaurant & retail; hardware model shifted	[S43]
SoundHound AI	$61.9M	No comparable restaurant-only ARR disclosed	Not disclosed as verified active U.S. restaurant total	Enterprise voice and agents across multiple industries	[S44]
Olo (private)	$2.0B 2025 transaction EQUITY VALUE, not 2026 revenue	Private after Sep 2025 acquisition	Not used as current site count	Restaurant digital platform; company valuation != annual demand	[S45]

Fiscal Q2 2026 for operating figures unless noted; figures in USD. Toast and PAR ARR include payments-related economic streams. NCR recurring metrics are quarterly GAAP categories, not ARR. Olo equity transaction value is deliberately shown as a different measurement, not a comparable revenue figure.

Two mechanical ratios illustrate comparability pitfalls. Toast's $2.409 billion ARR divided by approximately 180,000 locations equals roughly $13,383 of annual run-rate per reported location; PAR's $338 million ARR divided by 174,300 active sites equals roughly $1,939 per active site. Neither ratio is an observed per-unit price, restaurant software ARPU, or a relative profitability measure. Product mix, payment economics, active-site definitions, international coverage and multi-vendor use differ. They should not be used as Physical AI unit pricing inputs without direct vendor validation.

These results still establish that mature enterprise operating software and digital-commerce platforms form an observable upstream economic infrastructure for restaurant intelligence. The research must then isolate the incremental sensor, local compute, model-inference, maintenance, and integration expenditures associated with closed-loop Physical AI applications. Existing enterprise software revenue is not itself evidence of Physical AI monetization.

11.7 AI adoption estimates: three different denominators, not one penetration rate

A recurring market-sizing trap is to label every positive response as technology penetration. A 2026 NRA survey found 26% of broad restaurant operators used tools that incorporate AI. A Toast vendor survey of 676 U.S. operators with 16 or fewer sites found 86% were already experimenting either directly, via vendors, or through both methods. Qu surveyed 168 QSR and fast-casual brands with at least 20 sites; 51% said they were actively investing in AI and 22% expected to begin that year. These findings cannot be pooled into a single national adoption rate because their survey universes, definitions, and question wording differ materially. [S53],[S54],[S55]

The Qu report itself has a source-quality ambiguity: its introduction says participating brands operate 91,000 locations, while the methodology section displays both 94K and 91,000 locations. This report retains the 168-brand denominator and reports the location base as approximately 91,000–94,000 pending publisher clarification. Neither figure should be used to calculate national restaurant AI penetration. [S53]

Table 72 | 2026 restaurant AI adoption surveys: comparability audit

Source	Sample / universe	Survey measure	Reported statistic	What may be inferred	What cannot
NRA [S55]	U.S. operators; all restaurant formats	Actually using tools that incorporate AI	26%; FSR 28%; LSR 24%	Broad self-recognized operator use	Nationwide Physical AI or in-kitchen robot penetration
Toast [S54]	676 U.S. respondents; operators with <=16 locations	Currently experimenting with AI	86% = vendor only 42% + own only 19% + both 25%	Interest/experimentation in small-operator survey	Paid production deployment or realized ROI
Toast [S54]	Same 676	Comfortable using AI	87%	Attitudinal acceptance among respondents	Actual purchase or physical automation
Qu [S53]	168 QSR/fast-casual brands; >20 locations	Already investing in AI	51%	Brand-level technology capital prioritization	Number or share of stores actively operating AI
Qu [S53]	Same 168	Will start investing in 2026	22%	Planned future adoption in same cohort	Committed dollars or deployed devices
Qu [S53]	Same 168	Meaningful / transformational impact	9%	Self-reported mature outcomes are uncommon	Hard causal evidence of actual margin improvement

Samples and questions differ; the range 26%–86% is not a time-series growth rate or an estimate of the national adoption interval.

11.8 Enterprise AI execution gap and application priorities

Among Qu respondents, the top reported AI investment areas were marketing/CRM/personalization (53%), predictive analytics (40%), voice ordering (39%), inventory and demand (35%), kitchen automation (23%), and kitchen computer vision (16%). These are multi-select responses, not mutually exclusive expenditure shares. A technology supplier cannot multiply its segment TAM by these percentages as if they represented revenue allocation. [S53]

The same survey ranked operational execution (55%) and system/data fragmentation (37%) ahead of technology novelty as barriers to consistent experience. These responses suggest that interoperability and human workflow design may precede advanced robotics in the commercial adoption sequence. [S53]

Table 73 | Qu 2026 AI priorities and reported value

Operator-reported indicator	Respondents (%)	Survey classification	Economic interpretation	Caveat
Actively investing in AI	51	Adoption stage	Existing investment allocation decision	Brand-level, not unit-level
Plan to start in 2026	22	Future intention	Possible incremental software/hardware demand	Not executed purchase orders
Meaningful transformational impact	9	Self-reported benefit	Caution on economic evidence maturity	No accounting-level proof
Limited impact	43	Self-reported benefit	Integration/quality limitations may remain	Do not infer negative IRR
Emerging impact	33	Self-reported benefit	Deployment still early	Do not infer positive IRR
Marketing / CRM / personalization	53	Use case multi-select	Demand and customer-data emphasis	Not share of AI spend
Predictive analytics	40	Use case multi-select	Forecasting and analytical demand	Not physical-loop autonomy
Kitchen automation	23	Use case multi-select	Lower reported focus than CRM	Includes varying technology levels
Kitchen computer vision	16	Use case multi-select	Early physical sensing adoption signal	Not confirmed camera-per-store count
Order-flow improvement priority	62	Operational objective	Kitchen throughput central economic issue	Not measured financial return

Qu report: 168 multi-unit quick-service and fast-casual brands, 2026. Percentages may not total 100 due to question wording and multi-select design. [S53]

11.9 Updated 2026 robotics market boundary and commercialization discipline

The IFR World Robotics 2026 release reports nearly 250,000 professional service robots sold in 2025 by its respondent supplier sample, with 117,500 transport/logistics units and approximately 7,000 full-size humanoids. Hospitality was the

second-largest professional service application category, but much of that category consists of guidance, information-point and telepresence machines; it is not a restaurant-only delivery category. IFR specifies that the service-robot report reflects 238 respondents, is not projected to the entire industry, and should not be naively compared with previous survey waves because participants change. [S56]

Table 74 | IFR 2026 service-robot sample limitations

Indicator	Published 2025 value	Scope	Permitted inference	Invalid inference
Professional service robot shipments	Almost 250,000 units	Global IFR service robot survey; 238 suppliers	Non-industrial applications have meaningful unit volume	Global census of all robot manufacturers
Transport/logistics category	117,500; 47% reported share	Broad logistics and material transport	Largest commercial category in supplier survey	Restaurants bought 117,500 delivery robots
Hospitality robots	Second-largest application category	Mobile guidance, information points, telepresence prominent	Hospitality market is nontrivial	Restaurant serving robot annual shipments
Full-size humanoids	Approx. 7,000 units	>140 cm, commercial/professional global units	Humanoid market is small by units relative to entire service robotics sample	Robots running productive restaurant shifts
RaaS mode	Growing; leading in U.S. per IFR release	Survey observation	Pricing/ownership should be studied	Universal US restaurant RaaS share
Historical comparability	Not recommended	Varying respondent group by wave	Use same-source same-period categories	2024–2025 growth using two different report cohorts

2026 IFR release cautions against projecting surveyed sales to industry totals; all categories are global, not U.S. restaurant revenue. [S56]

11.10 Revenue-pool validation and market concentration: what counts as restaurant intelligence spending?

Public restaurant-technology vendors report several incompatible economic measures, including GAAP revenue, subscription ARR, gross payment volume, processing revenue and serviced locations. The value of a restaurant AI infrastructure market should be inferred only from incremental purchases and identifiable supplier receipts, not from gross restaurant sales or a technology vendor's payment throughput. A useful investor question is whether the AI feature has a standalone price, is embedded in a paid tier, or is a loss-leading feature supporting payment retention.

A non-overlapping category ledger assigns each invoice line to the final buyer and original economic provider. Vendor ARR, gross payment volume and cloud fees embedded in the same SaaS contract cannot be added as separate restaurant spending. Until such a ledger is populated, supplier totals provide market context rather than a validated Physical AI TAM.

Table 75 | Revenue recognition and market double-count controls

Reported metric	What it can establish	Prohibited inference	Follow-up evidence
Systemwide sales	Restaurant consumer demand and operating scale	Restaurant-technology vendor revenue	POS invoices, fee splits, payment economics
Payment gross volume	Transaction activity processed	Incremental Physical AI revenue	Net take rate and processing economics
Software ARR	Period-end recurring supplier revenue run-rate	All AI software revenue or realized customer ROI	AI-specific subscriptions, cancellations, contract terms
Installed robot units	Physical deployment/shipments if confirmed	Active restaurant fleet and service revenue	Active retention, net deployment, RaaS fee
Data center capacity / MW	Power and processing infrastructure potential	Restaurant-specific AI inference demand	Workload mix and compute utilization
Technology vendor EV	Capital market price of a business	Market annual revenue/TAM	Comparable profit, growth, margins, capital structure

Source context: restaurant technology vendor filings and prior tables [S27-S50]. Definitions are GII analytic controls; no unsupported aggregate estimate is calculated.

11.11 Public supplier economics: gross profit, revenue mix and the limits of AI TAM

The investable market for restaurant intelligence sits inside several existing revenue streams. Toast provides a useful counterexample to simplistic market sizing: its 2025 payments/financial-technology revenue was much larger than its

subscription revenue, but gross margins differed materially. PAR Technology sells software subscriptions, hardware and professional services with a different mix and a considerably smaller revenue base. Directly summing both companies' total revenues would neither isolate AI, establish non-overlapping market demand nor distinguish processor pass-through from software economics. [S82,S83]

Table 76 | Restaurant technology revenue economics, 2025

Issuer	Period	Revenue	Subscriptions	Other revenue mix	GAAP gross margin	Source
Toast (TOST)	FY2025	$6,153.0M	$936.0M (15.2%)	$5,037.0M fintech (81.9%)	25.9% total GAAP	[S82]
PAR Technology (PAR)	FY2025	$455.5M	$291.2M (63.9%)	$106.4M hardware (23.4%); $58.0M services	43.5% total GAAP	[S83]

SOURCE / LIMITATION Amounts are issuer GAAP reported revenue, not pure AI sales; not comparable to platform GPV or annual recurring run-rate (ARR). GM computed from reported gross profit / total revenue. Rounding may affect totals.

Table 77 | Toast and PAR gross-profit economics

Issuer / revenue stream	FY2025 sales ($M)	FY2025 gross profit ($M)	Stream gross margin	Interpretation
Toast subscriptions	936.0	672.0	71.8%	Software economics; not necessarily AI
Toast financial technology	5,037.0	1,146.0	22.8%	Transaction-processing economics and payment costs
Toast hardware / services	180.0	(220.0)	(122.2%)	Hardware/services may be subsidized by other streams
Toast consolidated	**6,153.0**	**1,593.0**	**25.9%**	**Mix-sensitive GAAP result**
PAR subscription services	291.2	159.1	54.7%	Software and platform base
PAR hardware	106.4	24.4	22.9%	Hardware margin; not pure robotics
PAR professional services	58.0	14.5	25.0%	Installation/integration economics
PAR consolidated	**455.5**	**198.0**	**43.5%**	**Includes acquired operations and non-AI business**

SOURCE / LIMITATION Toast stream gross margins calculated from 2025 cost-of-revenue lines [S82]; PAR streams from 2025 audited operating statement [S83]. The negative Toast hardware/services gross profit includes its reported cost allocation; it is not a restaurant-device-level loss or all attributable to AI hardware.

Table 78 | Issuer profitability and industrial-economics metrics

Issuer	FY2025 GAAP operating profit	Operating margin	Other reported measure	Valuation implication
Toast	$292.0M	4.7%	$608M FY2025 FCF; $2.047B year-end ARR; $195.1B GPV	GPV is payment volume, not Toast revenue. Assess subscription and payment-processing revenue and margins separately; EV/total sales alone distorts software value.
PAR Technology	($68.8M)	(15.1%)	$291.2M subscription revenue; customer concentration disclosed	Recurring revenue quality and operating cash conversion require further analysis

SOURCE / LIMITATION Toast reported ARR includes a payments component and cannot be treated as subscription revenue, and GPV represents processed merchant volume. PAR 2025 GAAP operating loss includes acquisition and other accounting costs; do not equate EBITDA with GAAP profit. [S82,S83,S85]

Competition and pricing: platform control versus measurable customer value

Two-sided competitive pressure is emerging: large restaurant groups can build internal operating platforms and spread fixed software costs across a large installed base, while outside vendors can spread product development across many brands and provide specialized capabilities. The economic winner is not predetermined. A supplier can win despite a chain's proprietary software if it owns mission-critical integrations and service capacity; conversely, headquarters can bargain down vendor fees and centralize customer data. Market share cannot be inferred from one vendor's disclosed locations or processed GPV without matching geographic and customer denominators. [S82,S83]

11.12 Where robotics suppliers can capture—and lose—economic value

Physical AI value may be split among robot hardware vendors, compute providers, control software companies, systems integrators, original equipment manufacturers, enterprise restaurant platforms and distributors. Their gross-margin profiles

differ because a hardware invoice contains material and installation costs while recurring software depends on active usage, support and retention. A supplier cannot infer an attractive SaaS multiple from total restaurant technology gross billings or payment flows.

Data rights and integration position may become bargaining levers. A vendor with a technically superior manipulator but no connection to the operator's order, equipment or service systems may face expensive custom engineering for each chain. An incumbent restaurant operating platform may have customer distribution and transaction visibility but lack direct control of real-world sensors and actuators. Strategic partnerships can accelerate commercialization if the parties allocate data access, model updates, pricing, uptime obligations and termination rights clearly.

The central competition may therefore be over the restaurant operating interface rather than the standalone robot. Technology developers should describe what they uniquely own—task data, machine interfaces, deployment playbooks, safety certification, fleet operations or restaurant customer relationships—and identify which dependencies are supplied by others. The financial result must still be proven after recognizing OEM discounts, installation labor, warranty, field service, sales expense and customer concentration.

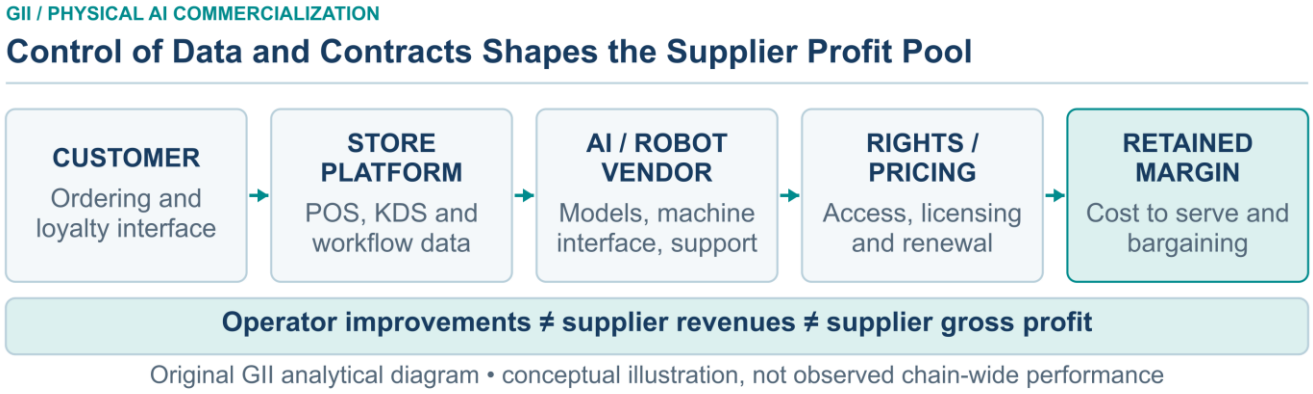


Figure 20 | Contract and data-rights interfaces determine platform bargaining power

GII industrial-organization hypothesis; DoorDash's 2025 SevenRooms closing [S109] provides an example of customer-relationship platform investment.

12. Restaurant AI TAM, SAM, SOM and Investment Underwriting

The central market-sizing principle is to define what is being sold. Addressable waste avoided across restaurants is not vendor revenue. Estimated robot hardware shipments, cloud consumption, SaaS ARR, and enterprise valuations cannot be totaled without a non-overlapping transaction ledger.

12.1 Top-down and bottom-up market reconciliation

Top-down restaurant macro measures set context; bottom-up eligible restaurant units and spending per unit set the technology opportunity. Both must be carefully scoped.

- Reconcile revised NRA sales, QSR Top 20 unit counts, Census establishments and vendor/customer segmentation.
- Create annual revenue pools for software, hardware, installation, and services separately.
- Publicly reported customer sites, price sheets and investor filings can validate parts of the bottom-up estimate; gaps remain explicit.

Sources: [S01]–[S04], [S20]

Table 79 | Market-sizing definitions and available evidence

Measure	Unit / formula	Verified example	Critical caveat
Consumer restaurant spend	$ annual gross foodservice sales	NRA 2026E $1.36T revised	Includes broader foodservice, not software demand
Chain brand systemwide sales	Customer sales across stores	QSR Top20 $246.558B in 2025	Not franchisor corporate GAAP revenue
Technology supplier revenue	GAAP recognized services/sales	Toast GAAP 2025 $6.153B	Includes gross fintech processing
Technology recurring run-rate	ARR annualized at period end	Toast $2.047B Dec 2025	Not GAAP full-year revenue
Robot shipments	Physical units newly shipped	Pudu lifetime >130K	Cross-sector cumulative, not annual restaurant units
Restaurant-installed service base	Active paid locations × service	Not fully public	Must be verified or modeled
Enterprise value	Market cap + debt - cash + etc	No comparable current EV dataset yet	Not TAM or sales revenue

Sources: [S01],[S04],[S20],[S24]

12.2 Bottom-up site-based scenario

The QSR Top 20 year-end sample provides 124,185 U.S. stores for a bounded scenario of applications suitable for national limited-service chains. Its scope excludes much of the broader restaurant technology market.

- Scenario eligible shares 25%, 50%, 75%; annual price $2,400/$3,600/$4,800. Compute the FULL eligible-site recurring potential.
- Add adoption 10%, 25%, 40% separately to estimate revenue actually captured under each scenario.
- Specify hardware spending separately with unit capex and replacement cycles.

Table 80 | Hypothetical Top 20 SaaS scenarios

Scenario	Eligible %	Annual fee/site	Eligible units	Eligible pool $M	Adopted sites	Revenue $M
Cautious	25%	$2,400	31,046	74.5	3,105	7.5
Illustrative base	50%	$3,600	62,092	223.5	15,523	55.9
High adoption	75%	$4,800	93,139	447.1	37,256	178.8

Scenario arithmetic is explicitly hypothetical and is not observed restaurant AI spending.

Non-overlapping one-product illustration, based only on QSR Top20 units. Annual fees and adoption rates are analyst-chosen assumptions, not observed market prices or forecasts. Eligible pool is theoretical annual spend if all eligible stores bought the product. Sources: [S04] store denominator; GII hypothetical inputs

12.3 TAM/SAM/SOM methodology

TAM, SAM and SOM estimates require a defined product, an authorized buyer and economic eligibility assumptions. Assigning a share of restaurant sales to Physical AI does not establish what customers can or will pay.

- Application-level TAM is annual spending among eligible sites, adjusted for bundled products and overlapping supplier receipts.
- Define SAM as reachable by the actual product, installation resources, customer format and region.

- Define SOM from installation capacity, competition, customer onboarding, retention and commercial budget.

Table 81 | Application TAM/SAM/SOM build specification

Application	Unit denominator	Eligibility inputs	Annual fee / one-time price	Status
Computer vision kitchen QA	Kitchen sites / cameras	Camera rights, lighting, ROI	Vendor quote required	Incomplete
Edge software orchestrator	Restaurant units	Connectivity, critical workflows	Cloud/edge SaaS quote required	Incomplete
Connected equipment control	Compatible ovens/fryers	OEM access, code/safety, volume	Device + service quote required	Incomplete
Dedicated portion automation	High-volume repetitive makelines	Floor area, volume, staffing	Hardware price + TCO required	Incomplete
Mobile serving robot	Accessible table service sites	Circulation, task completion	RaaS / unit selling price required	Incomplete
Worker-assisted AI headset	Participating employee shifts	Consent, role fit, privacy	Licensing/seat cost required	Incomplete

These should NOT be aggregated until product overlap and pricing are reconciled.

12.4 Unit-level DCF and deployment cash flows

Alternative A below separates revenue contribution, savings, recurring costs and replacement cash flows in a five-year unit model.

- Its illustrative inputs are $80,000 initial CapEx, $3 million restaurant revenue, 1% sales uplift at 35% incremental contribution, $18k realized labor benefit, $12k waste benefit and $10k annual technology spend.
- Benefits reach 60% in Year 1 and full run-rate thereafter. Cash tax is 25% of positive incremental operating income after $16k annual depreciation; a Year-1 EBIT loss creates no current cash-tax credit.
- The scenario deducts $12k of Year-4 reinvestment, discounts at 12% and credits no terminal value. All inputs and outputs are hypothetical.

Table 82 | Illustrative DCF A with depreciation tax shield

Year	Benefit ramp	Gross benefit	EBITDA Δ	Cash tax	Reinvest	Free cash flow
1	60%	$24,300	$14,300	$0	$0	$14,300
2	100%	$40,500	$30,500	$3,625	$0	$26,875
3	100%	$40,500	$30,500	$3,625	$0	$26,875
4	100%	$40,500	$30,500	$3,625	$12,000	$14,875
5	100%	$40,500	$30,500	$3,625	$0	$26,875

Alternative A retains its original five-year after-tax Physical AI DCF assumptions, including depreciation tax treatment; it is distinct from Module II.

Initial cash flow = -$80,000. Calculated NPV @12% = $-1,976; implied IRR = 11.0%. Scenario-specific hypothetical assumptions only, not an observed restaurant result. Sources: GII illustration; explicit assumption register

FINANCIAL MODEL RECONCILIATION Table 82 is the earlier illustrative Alternative A: 60% first-year benefit ramp, full run-rate from Year 2, $16,000/year straight-line tax depreciation, and $12,000 Year-4 reinvestment, producing approximately -$1,976 NPV under those assumptions. New Tables 90-94 are a separately defined, more conservative Alternative B: 50%/80%/100% base benefit ramp, full recurring costs from Year 1, no assumed depreciation tax shield and $10,000 Year-4 renewal. Alternative B produces approximately -$22,299 NPV. The difference is explained by input/tax assumptions; neither scenario represents observed restaurant returns or a direct comparable. Both are retained intentionally to demonstrate model sensitivity and the need to specify tax accounting and adoption timing. For the Executive Summary and investment decision figures, the principal comparison is the fully specified Module II case (Tables 117–121); Alternatives A and B should not be compared as if they use identical assumptions or are verified restaurant returns.

Table 83 | Financial assumptions and validation needs

Input	Illustrative value	Source status	Priority for external verification
Annual restaurant sales	$3,000,000	Analyst assumption	Store P&L by format
Sales uplift	1.0% × 35% contribution	Analyst assumption	Matched controls; ticket and labor data
Realized labor benefits	$18,000/year	Analyst assumption	Actual payroll records; not task time
Waste benefits	$12,000/year	Analyst assumption	Waste log and ingredient cost
Incremental recurring spend	$10,000/year	Analyst assumption	Vendor quote
Initial capital spending	$80,000	Analyst assumption	Vendor/contractor quote
Tax and depreciation	25%; 5 years SL	Analyst simplification	Entity tax treatment/asset lives
Discount rate	12%	Analyst assumption	Franchisee cost of capital
Additional Year-4 CapEx	$12,000	Analyst assumption	Equipment maintenance schedule
Residual value	$0	Analyst assumption	Useful life and secondary market

12.5 Chain-level rollout economics

Scaling from one successful site to a franchise system creates program-management costs, technology churn, help desks, integration variations, supplier concentration, and retraining. Annualized unit economics must therefore be reconciled against group-level rollout capex and observed deployment speed.

- The 100-, 1,000- and 5,000-site scenarios separate local installation from central program costs and rollout coordination.
- Differentiate franchisee payment from parent-level technology subsidy.
- Consider cross-store data benefits, remote support, and hardware replacement cycles.

Table 84 | Illustrative rollout capital requirements

Site count	Unit CapEx	$ total site CapEx	Central program budget	Total starting deployment
100	$80,000	$8.0M	$0.5M	$8.5M
1,000	$80,000	$80.0M	$2.5M	$82.5M
5,000	$80,000	$400.0M	$9.0M	$409.0M

Illustrative capital commitments, not announced company rollout plans.

Static arithmetic with assumed initial site investment and assumed central integration budgets. Excludes working capital, financing and annual fees.

12.6 No-AI counterfactual

The investment committee should compare every AI proposal with a feasible low-tech or existing-software alternative. If the traditional intervention produces more predictable quality and a higher NPV, the proper decision may be to defer AI hardware.

- Construct minimum viable alternatives: reconfigure kitchen, train staff, replace aging equipment, standardize recipes, optimize schedules.
- Comparable interventions need a common operating objective and consistent treatment of CapEx, staff disruption, quality, risk, uptime and discounted cash flow.
- Monitor actual gains via phased or matched-control rollout.

Table 85 | Alternative solutions before incremental AI capex

Problem	Low-tech alternative	AI alternative	Required test
Inconsistent portions	Portion SOP, training and scales	Vision / automated portioning	Blind quality + food cost vs capex
Kitchen backlog	Station redesign and labor move	Queue optimization software	P95 ticket and paid hours
Failed equipment	Preventive service schedule	Predictive sensor alerts	True downtime cost and false positives
Long customer queue	Menu simplification and pickup design	AI ordering/forecasting	Abandonment and net margin
Worker burnout	Job redesign and staffing stability	AI assistant / dedicated automation	Opt-in employee outcomes and churn

12.7 Market-value bridge: from technology spend to investable supplier economics

To link market sizing and investment underwriting, the study must separate technology payments made by restaurants, supplier-recognized revenue, supplier gross profit, cash flow, and investment enterprise value. These are not synonymous

and generally differ because of hardware pass-through, payments processing, reseller arrangements, bundled subscriptions, integration fees, and financing. Restaurant transaction volume is an underlying commerce flow, not the revenue pool available to physical AI vendors.

Table 86 | Value-chain measures and permitted interpretations

Observed measure	Published example	Annual recurring?	What can be inferred	What cannot be inferred
Customer foodservice spending	$1.36T U.S. 2026E NRA	No	Macro restaurant output	Physical AI SaaS TAM
Restaurant brand system sales	$246.558B Top 20 2025	No	Large-chain spending base	Corporate GAAP revenue
Platform ARR	$2,409M Toast Jun 2026	Yes, run-rate	Digital-commerce monetization scale	AI-only market share
Platform ARR	$338M PAR Jun 2026	Yes, run-rate	Subscription/payments platform scale	Price per AI-equipped store
Operating vendor revenue	$158M NCR Voyix restaurant Q2 2026	No	Restaurants segment sales across products	Annual edge/robotics revenue
Cross-sector voice AI revenue	$61.9M SoundHound Q2 2026	No	Enterprise agentic/voice business size	Restaurant-only AI spend
Completed equity M&A	$2.0B Olo 2025 acquisition	No	Sponsor valuation of restaurant digital platform	Annual restaurant IT market size
Future unit economic target	$100k per avg McD U.S. site	Not realized	Potential operating productivity hurdle	Supplier price ceiling or AI-alone benefit

Sources [S01],[S04],[S38],[S41]-[S45]. These values overlap and differ in geography/definition; they are deliberately not summed.

12.8 Unit-level sensitivity: separating attributable benefits from technology headlines

The initial illustrative restaurant DCF assumes $80,000 initial technology investment and $40,500 annual gross operating improvements (at full ramp) before $10,000 recurring spend. The model remains illustrative. The next research gate is to replace every labor, waste, uplift and infrastructure input with actual restaurant observations or vendor quotations, and compare each AI case with a process-redesign-only case.

For the McDonald's NEXT analysis, the company's ~$100,000 annual average-site target applies to the specific bundle of stated future operational improvements; it is not interchangeable with a software vendor fee or the single-site $40,500 illustrative benefit in the separate GII DCF. The study will maintain independent model layers and report net financial gain after ramp-up, installation interruption, wages actually paid, taxes, and reinvestment.

12.9 Operational value pools are not supplier revenue TAM

An estimated $157 billion of surplus food value attracts attention, but an investment analyst must subdivide that number into: (a) a measured or modeled economic-loss pool; (b) the portion that is technically preventable; (c) the portion operators can actually monetize after modifying menu, staff and inventory behavior; and (d) technology suppliers' possible revenue capture. The fact that most surplus is plate waste underscores that consumer preferences, dish quality, portion decisions and behavior may dominate the effectiveness of a kitchen production robot. [S51],[S52]

The scenario below is explicitly a mathematical sensitivity illustration—not a prediction, not an audited addressable market, and not a claim that foodservice companies can collect the estimated ReFED economic value as cash. It simply shows the extraordinary differences between problem value and realistic technology spend assumptions. An investment case requires observed store results, 2024-specific relevant loss categories, vendor pricing, and a nonoverlapping set of customer payments.

Table 87 | Illustrative surplus-value sensitivity, 2024

Illustrative avoidable portion	Gross modeled value	Provider capture assumption	Hypothetical annual supplier receipts	Classification
0.5%	$0.785B	5%	$39.3M	Illustrative formula; NOT forecast or TAM
0.5%	$0.785B	10%	$78.5M	Illustrative formula; NOT forecast or TAM
0.5%	$0.785B	20%	$157.0M	Illustrative formula; NOT forecast or TAM
1.0%	$1.570B	5%	$78.5M	Illustrative formula; NOT forecast or TAM
1.0%	$1.570B	10%	$157.0M	Illustrative formula; NOT forecast or TAM
1.0%	$1.570B	20%	$314.0M	Illustrative formula; NOT forecast or TAM
2.0%	$3.140B	5%	$157.0M	Illustrative formula; NOT forecast or TAM
2.0%	$3.140B	10%	$314.0M	Illustrative formula; NOT forecast or TAM
2.0%	$3.140B	20%	$628.0M	Illustrative formula; NOT forecast or TAM

Sensitivity only using the 2024 baseline; not a forecast or supplier-revenue TAM.

Formula: $157B x assumed preventable portion x assumed vendor share. This ignores execution costs, double counting, customer concentration, and the fact that modeled value is not necessarily monetizable. [S51]

12.10 Recommended controlled-store operating trial

To turn a promising application into evidence strong enough for investor underwriting, test incremental performance at the restaurant unit rather than depending entirely on case-study claims. A feasible pilot is a staggered rollout across matched restaurants with comparable baseline sales, menu, staffing rules, operational formats and seasonal patterns. Record pre-intervention outcomes, monitor execution quality, and identify the contemporaneous no-AI condition. A credible primary study may support a difference-in-differences estimate; small convenience samples should not be presented as causal proof for the industry.

Table 88 | Physical AI field-test design

Domain	Control-variable considerations	Primary outcome	Financial mapping	Disqualifying confounder
Kitchen waste	Store format, foot traffic, menu, promotion	Grams and $ waste / 1k orders	Avoided ingredient purchases net of fees	New menu or supply-price shift
Peak throughput	Daypart, staffing, demand mix, promotions	Orders per station hour; p95 lead time	Incremental contribution from retained demand	Unobserved marketing-induced demand
Employee burden	Role mix, manager, training, shift	Repeated workload scale, steps, interventions	Overtime, attrition, retention; noncash welfare	Staffing changes unrelated to AI
Food and customer quality	Recipe, supplier, guest demographics	Quality defects, remakes, repeat visits	Remake spend, retained contribution	Menu/pricing change during trial
Compute efficiency	Internet quality, local hardware, vendor SLA	Inference latency, uptime, $/site	Five-year TCO and avoided outage loss	Different hardware procurement terms

Illustrative prospective study plan only. No sites recruited and no experiment completed as of October 26, 2026.

12.11 The non-robot alternative as a required investment comparator

Across the highest-value restaurant operating opportunities, automation must compete with actions that often require less new capital: simpler menus, smaller prep batches, standardized recipes, more effective training, better shift allocation, accurate timestamps, sensor-only monitoring, or conventional inventory software. ReFED specifically recommends appropriate portion sizes, waste tracking, low-waste menu planning, and demand forecasting. A final restaurant Physical AI market model should exclude operators who can achieve the same outcome with cheaper, simpler interventions. [S52],[S57]

12.12 A decision-useful market model: minimum information before assigning a dollar TAM

Physical AI market reports often create high estimates by applying a technology penetration percentage to an enormous installed restaurant base. This can be valid only when the underlying restaurant need, buyer, annual budget, real implementation cost and the vendor's portion of value capture are well defined. The report therefore follows two independent estimations: (a) what restaurants could rationally spend given measurable savings and outcomes and (b) what suppliers could actually invoice given installation capacity and market competition.

The 124,185 QSR Top20 locations in the 2025 sample are useful for bounded scenarios (Table 13), but not an estimate of restaurant Physical AI TAM. Disaggregation by high-volume QSR, beverage formats, drive-through, full-service, and

independents is necessary. Similarly, the total restaurant industry sales baseline is evidence of consumer-market scale, not a pool of spending already allocated to robots or computer vision.

Table 89 | Evidence for restaurant Physical AI market estimates

Sizing element	Preferred data source	Core uncertainty	Publication rule
Total addressable stores	**NRA/Census and defined unit categories**	**Foodservice vs restaurant, outlet vs firm**	**No mixing definitions**
Economically eligible stores	Site/format/task screening and threshold payback	Gross sites overstate viable demand	Show clear exclusion funnel
Incremental annual technology fee	Contracts, vendor recognized net revenue, quotes	Bundles and hardware double count	Split one-time and recurring
Actual paid penetration	Vendor customers, issuer deployments, installations	Pilot vs commercial, active vs shipped	Date-stamp verified site counts
Market concentration	Active vendor revenue by application	Unreported private data and resellers	Range estimates with unknown bucket
Scenario growth	Replacement, churn, deployment and price sensitivity	Jevons effects and competition	Downside/base/upside only
Asset / enterprise value	Issuer EV & financial metrics	Negative EBITDA, valuations detached from sales	Not a proxy for market size

Industrial organization modeling standard; indicates critical data collection steps before publishing any restaurant-specific Physical AI revenue forecast.

12.13 Re-underwriting the restaurant AI unit case: GAAP versus cash and the no-AI comparator

Alternative B provides an explicit after-tax five-year free-cash-flow scenario for one hypothetical $3.0 million AUV restaurant. It shows why an apparently favorable pre-tax simple payback can coexist with a negative NPV after the timing of benefits, recurring platform costs, taxes, refresh CapEx and the cost of capital. None of these amounts are presented as achieved vendor savings. This is a reproducible illustrative decision model, not a valuation of any public restaurant chain.

Table 90 | Illustrative AI restaurant project assumptions

Input	Model assumption	Interpretation / limitation
Restaurant annual sales	$3,000,000	Same in each case; illustrative AUV
Upfront integrated AI/site investment	$80,000	Year 0; physical installation and software integration
Discount rate	12%	Illustrative project hurdle rate, not issuer WACC
Tax rate on positive incremental cash operating gains	25%	Simplified cash taxes; no amortization/depreciation tax shield
Model life	5 years	Zero residual value; no subsequent major expansions
Device/major renewal	Year 4: $10,000	Treat as incremental cash CapEx
Annual AI operations expense	Case-specific	Full expense begins in first year, not ramped
Base benefits ramp	50%, 80%, 100%, 100%, 100%	Applies to gross operating benefits, not recurring cost

Table 91 | Illustrative scenario economics and returns

Scenario	Sales uplift	Annual gross benefit @ steady state	Annual recurring cost	5Y NPV @12%	IRR	Decision
Downside	0.0%	$10.0K	$12.0K	$-100.0K	N.M.	Does not pass 12% hurdle
Base	1.0%	$40.5K	$10.0K	$-22.3K	1.2%	Does not pass 12% hurdle
Upside	2.5%	$80.0K	$8.0K	$+87.4K	47.7%	Passes illustrated 12% hurdle

Table 92 | Hypothetical five-year after-tax cash flows

Case	Initial (Y0)	Y1	Y2	Y3	Y4	Y5
Downside	$-80.0K	$-7.0K	$-4.5K	$-2.0K	$-12.0K	$-2.0K
Base	$-80.0K	$+7.7K	$+16.8K	$+22.9K	$+12.9K	$+22.9K
Upside	$-80.0K	$+36.0K	$+48.0K	$+54.0K	$+44.0K	$+54.0K

Table 93 | Investment discipline: break-even and sensitivity

Metric	Reproducible calculation	Illustrative result	What would alter the conclusion
Base static pretax full-run-rate gain	$40.5K gross benefit − $10K annual recurring	$30.5K/year	Realized labor and gross contribution, not nominal minutes
Undiscounted simple payback	$80K initial / $30.5K steady pretax	2.62 years	Ignores ramp, tax, refresh, discounting
Discounted base project NPV	NPV(12%, [-80K, Y1..Y5 after-tax FCF])	$-22.3K	Five-year life and year-4 renewal assumptions
Break-even steady annual gross benefit	**Solve NPV=0, holding base cost/ramp/taxes**	**$50.4K**	**Discount, tax, risk or lower CapEx**
Extra gross annual benefit needed	$50.4K − $40.5K	$9.9K	Could come from quality, sales, labor, yield
Equivalent incremental sales at 35% CM	Extra gross benefit / 35%	$28.3K revenue	Requires sufficient demand and capacity

SOURCE / LIMITATION The financial model is illustrative: simple cash taxes on positive gains; depreciation tax shields, working capital, financing, and the option value of later upgrades omitted; losses create no usable current cash-tax benefit. Existing cash flows use scenario-specific benefit ramps: downside 50%/75%, base 50%/80%, upside 70%/90% in Years 1–2, then 100%. Returns use unrounded cash flows. These are analyst assumptions, not company-reported ROI.

Conclusion: in the alternative legacy scenario described immediately above, the base case has negative five-year NPV despite an attractive simple pretax payback. This differs from the primary Module II scenario, which uses a $45,000 mature gross benefit and produces approximately +$1,507 NPV. These are two explicitly different illustrative assumption sets, not conflicting estimates of one observed restaurant installation. The comparison demonstrates why vendors should show cash timing, refresh and a non-AI counterfactual rather than quote unqualified “hours saved.”

Table 94 | Illustrative store-level rollout economics

Sites	Initial site CapEx @ $80K	Full steady pretax benefit @ $30.5K/site	Aggregate base NPV @12% (before central cost)	Additional risks
100	$8.0M	$3.05M/year	$-2.23M	Regional install, training, vendor SLA
1,000	$80.0M	$30.5M/year	$-22.30M	Fleet quality control, franchisee opt-in
5,000	$400.0M	$152.5M/year	$-111.5M	Central rollout, cybersecurity, change-management costs

This illustration scales both site value and model errors; it is not a company forecast.

SOURCE / LIMITATION Economies of scale, multi-site volume discounts, shared platform costs, and centralized CapEx are omitted. The table is a multiplication exercise to demonstrate why scale is not value creation; it is not a forecast of a named restaurant chain.

12.14 The market-sizing unit is an eligible site with an economically feasible task

An investment-grade robot TAM cannot begin with nationwide restaurant sales multiplied by an assumed AI adoption percentage. The commercially addressable unit is a site–task–buyer combination: a particular workflow with enough value at a location that can physically accommodate the system, plus a purchaser with the authority and budget to adopt. An addressable-establishment forecast should reconcile chain-owned versus franchised units, overlapping tasks, hardware replacement cycles, payment structures and operating lives.

For each technology category, estimate eligible sites first; then scenario penetration, contracted average recurring revenue, hardware sale or lease revenue, installation services, warranty burden and retention. Separately estimate gross profit after hardware and service delivery, recognizing that a subscription vendor and a capital-equipment vendor capture different shares of site-level value. Do not add a restaurant's avoided labor cost to a technology vendor's sales, or add a marketplace's gross bookings to the restaurant's sales as though they were independent revenues.

The market will be bounded by physics and institutions as much as demand. Narrow cooking spaces, menu variability, food-contact regulations, unreliable utilities, franchisee decision rights and scarce field-service capacity can create an adoption ceiling below the available store count. Those constraints should be modeled explicitly and sensitivity-tested rather than dismissed as temporary engineering inconveniences.

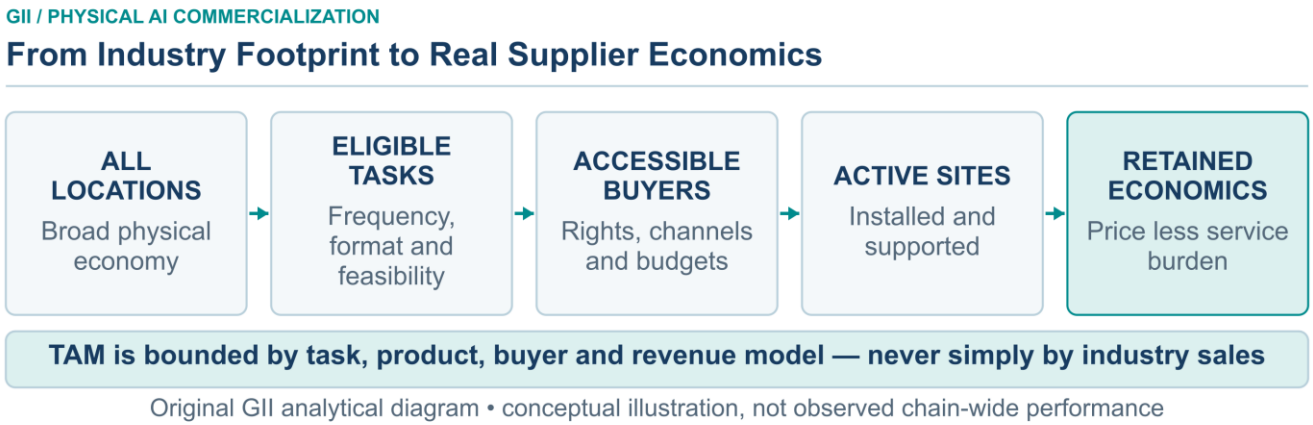


Figure 21 | The narrowing funnel from industry scale to active, retained supplier economics

GII original TAM/SAM/SOM methodology. All funnel stages are conceptual; no unverified restaurant AI market dollar estimate is claimed.

12.15 From robot task performance to incremental free cash flow

Robot vendors should build a bridge from task-level measurements to the restaurant P&L. First measure time saved or additional output per completed task. Then test whether the change improves a sold-out production bottleneck, reduces paid staffing hours, avoids a genuine rework cost, protects product quality or changes consumer behavior. Only the monetizable effects belong in the incremental benefit case. Next subtract ongoing equipment service, downtime, consumables, software, worker training and site changes, and account for initial capital, taxes and future replacement.

The illustrative GII five-year model provides a reproducible discipline rather than a universal restaurant ROI estimate: an $80,000 initial investment, $10,000 annual operating cost, phased operating benefits, refresh expense in Year 4, 25% assumed tax rate and 12% discount rate yield approximately $44,325 as the mature annual gross-benefit threshold for zero NPV. Actual projects require independently supported assumptions and a matched no-AI alternative. Every claimed chain rollout NPV should additionally test site heterogeneity and who pays the capital.

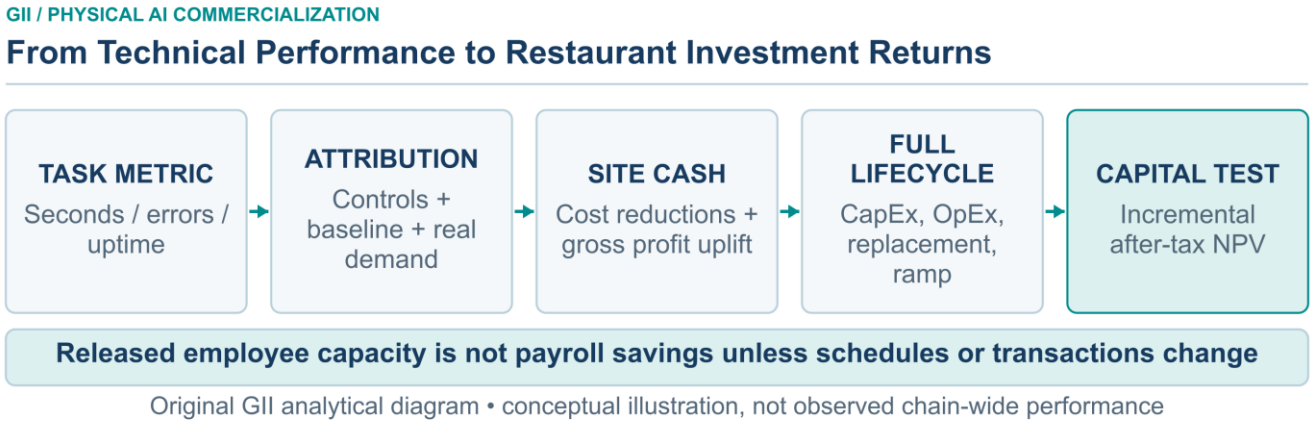


Figure 22 | The economic bridge from measured tasks to site investment returns

GII Module II model; illustrative assumptions, not realized restaurant ROI.

13. Value Capture, Stakeholder Economics and Corporate Development

13.1 Customer and workforce value

A four-party framework separates restaurant, worker, customer and supplier outcomes. Restaurant technology should improve operating economics and the human experience of service. A larger robotics dataset or more management monitoring alone does not establish value for the people who operate or use the restaurant.

- Treat customer welfare, employee well-being, operator margins, and capital-provider returns as separate measured outcomes.
- A balanced pilot report includes at least one quality outcome and one financial outcome.
- Gains to one constituency may create costs for another; the stakeholder ledger records both sides.

Table 95 | Four-stakeholder measurement framework

Group	Primary gains	Primary cost/risk	Validated outcome KPI
Customers	Food quality, convenience, reliable service	Pricing, errors, privacy	Repeat visits; food quality; P95 wait
Employees	Less strain, clearer work, safer routines	Surveillance, displaced hours, exception work	Turnover, paid hours, opt-in workload
Operators	Revenue, waste and staffing productivity	CapEx, complexity, vendor dependence	Restaurant contribution Δ; uptime
Investors	Scalable FCF and capital efficiency	Overpaying for unproven assets	NPV, IRR, ROIC, adoption retention

13.2 Value-capture and industrial-organization mechanisms

Industry market structure determines who captures the value created by customer and restaurant data. The strongest technology providers may control integration points, switching costs, software distribution, or proprietary labeled physical-action data, rather than the most visible robot design.

- Bargaining power depends on the chain's, franchisee's, OEM's, cloud platform's and startup's contractual alternatives and dependencies.
- The source of a productivity gain and the party charging for it may differ.
- Defensible supplier advantages require evidence beyond commodity sensor or compute resale.

Table 96 | Value capture / contracting problem

Economic asset	Who may control	Who may pay	Potential conflict
Restaurant customer relationship	Restaurant brand or platform	Guest / franchisee / chain	Access and personalization rights
Equipment telemetry	OEM / operator / cloud vendor	Restaurant	Supplier data monopoly
Physical action footage	Worker / restaurant / robotics company	Model developer or employer	Compensation and consent
Restaurant AI model	Brand / SaaS vendor / cloud provider	Restaurant or franchise system	Switching cost and IP rights
Edge equipment	Operator / lease provider / chain	Franchisee / central IT	Asset refresh and local support
Automation device	Restaurant / lessor / OEM	Operator / RaaS customer	Maintenance and residual value

13.3 Corporate development: build, buy, partner, invest, acquire

Corporations can purchase technology services without owning product IP. Sweetgreen's Spyce transaction illustrates this distinction, while LG's Bear ownership is an example of strategic vertical integration driven by broader product strategy.

- Build, buy and partner decisions trade control, capital needs, obsolescence, implementation speed and integration risk.
- Licensing and vertical integration suit different needs; the relevant comparison includes retained rights, support obligations and the cost of maintaining the capability.
- An acquisition rationale needs the target's standalone cash flows and a specific strategic fit.

Table 97 | Build, buy, partner and invest decisions

Option	Capital burden	Control	Key investment hazard	When attractive
Build internally	High / repeated R&D	High	Slow, costly, talent concentration	Unique data and system-wide advantage
Buy SaaS / device	Low–medium	Limited	Vendor lock-in and price escalation	Commodity functions with proven benefits
Strategic partnership	Negotiated	Shared	Commercial and IP misalignment	Joint product commercialization
Minority venture investment	Investment ticket	Limited + information rights	Illiquidity and minority governance	Option value with strategic learning
Acquire technology vendor	High + integration	High	Overpayment and integration failure	Differentiated strategic asset and clear synergies
RaaS / lease	Lower initial cash	Limited	Higher lifetime cost, contract exit	Uncertain demand / rapid technical refresh

13.4 Diligence and interview methodology

A buyer or investor needs to know which operating, contractual and financial facts would change the capital-allocation decision. The interview guide below focuses evidence collection on those decisions.

- Standardized interviews with operators, workers, technology leaders and investors can document disagreements and practical deployment constraints.
- Market-entry templates, adoption scorecards and contract questions make those interviews comparable without treating them as representative causal evidence.
- Construct a source-linked discovery log and avoid treating vendor testimonials as control-group evidence.

Table 98 | Priority primary interview guide

Participant	Core question	Evidence requested
Restaurant GM	Which physical process fails most often and why?	Task log, quality returns, P95 service time
Frontline worker	Which technology made tasks easier or harder?	Opt-in daily-work examples; turnover
Chain CIO / CTO	What must run locally versus cloud?	Architecture, offline requirement, vendor costs
Franchise operator	Who pays for platform rollout and who benefits?	Fee and capital budgets; approval rights
Robotics product lead	Where does task completion end and staff work begin?	Device uptime, interventions, active installs
Investor / corp dev	Which use case passes your capex hurdle?	NPV/IRR assumptions, integration plan
Facilities manager	What site design prevents installation?	Plans, panel schedules, retrofit costs

13.5 Transaction economics and post-ownership technology rights

Physical AI investment research becomes more valuable when it considers contract rights and corporate boundaries. A franchisor can sell a brand while continuing to supply technology. A restaurant operator can sell robotics intellectual property while retaining access under license or supply contracts. A private-equity buyer can purchase an operating software provider primarily for recurring digital-commerce cash flow, not for robots. The location of data access, customer relationships, and post-closing obligations may matter as much as the visible brand or equipment.

Table 99 | Restaurant technology transaction control points

Transaction	Date / published value	Asset/control change	Continuing technology or revenue question	Source
Pizza Hut ex-China / LongRange	2026-09; ~$1.5B subject to adjustments	Brand and franchise business transferred out of Yum!	Byte by Yum! continuing service; economics of transitional contracts	[S40]
Pizza Hut China / Yum China	Closed 2026-08; part of aggregate $2.7B	China brand rights/control	Interplay with KFC/Taco Bell operations	[S40]
Olo / Thoma Bravo	2025-09; ~$2.0B EQUITY value	Restaurant digital SaaS provider taken private	Valuation of recurring enterprise platform vs AI IP	[S45]
Spyce automation / Wonder	2025-12; source documents transaction consideration	Restaurant robot/automation IP separated from Sweetgreen	Continued Infinite Kitchen use under commercial terms	[S17],[S18]
Bear Robotics / LG Electronics	2025-01 announcement of majority-control path	Robotics company control and vertical distribution	Corporate-backed commercialization vs independent growth	[S23]

Transaction values are consideration/equity values, not comparable multiples without net debt, EBITDA, closing adjustments and perimeter normalization. Source dates vary.

13.6 A welfare and finance bridge for the restaurant-first investment thesis

Four distinct stakeholder outcomes must remain visible in the final recommendation. A faster kitchen may increase owner contribution margin without improving dining quality; a worker-friendly job design might improve retention without reducing current payroll; a food waste reduction program might lower ingredient consumption while leaving customer satisfaction unchanged; and recording employee movements might create valuable training data for a robotics vendor without transferring profits back to the restaurant. The analytical task is to quantify direct effects, plausible spillovers and hidden costs separately, then make the distribution of value transparent.

Table 100 | Stakeholder welfare and financial value

Proposed initiative	Customer value metric	Employee value metric	Restaurant cash-flow metric	Investor/financing test	Governance caveat
Freshness/ temperature alerts	Quality consistency, complaints	Fewer crisis reworks	Avoided discard and remake costs	Positive NPV after hardware/software	Ingredient and food-safety accountability
Order-flow optimization	Wait times and accuracy	Peak workload balance	Incremental served orders and labor efficiency	Attributable contribution margin	Do not assume higher capacity equals demand
Smart equipment maintenance	Reliable product availability	Fewer breakdown interventions	Repair spend, downtime losses	Risk-adjusted payback	Human override and equipment safety
Wearable training cameras	Awareness/consent if customers filmed	Voluntary data rights and compensation	Potential vendor fee or productivity gain	Signed revenue-sharing and cost allocation	Retention, purpose limits, deletion, transparency
Dedicated preparation robot	Consistent product/price	Safer tasks, exception handling	Valid avoided hours or higher capacity	Hardware life-cycle IRR	Employee acceptance, uptime and fallbacks

This table proposes KPI measurement; no customer, worker or investor result is being claimed for a particular vendor or restaurant.

13.7 Financial value is not the same as who captures it

Multi-party systems distribute benefits and costs unevenly. A franchisor may fund a common platform, franchisees may bear local CapEx, suppliers earn fees, and employees and customers experience changes in work or service. The analysis therefore separates total surplus from its allocation. A compelling return for one party alone is insufficient to establish a durable deployment.

The distinction is particularly important when restaurant operators allow cameras, worker-worn devices or employee interactions to become training data for robotics companies. Contracts should clarify who may retain the data, whether individual employee consent is meaningful, who may sell derivative models, and what operational benefit the restaurant obtains.

Table 101 | Stakeholder value appropriation

Value event	Possible financial gain	Who may capture value	Risk or missing transfer
Fewer remakes	Food cost and labor savings	Restaurant operator	Customer sees no quality gain if savings not reinvested
Faster order completion	Additional peak sales margin	Operator; franchisor via royalties	Employee intensity could rise
AI software fees	Subscription margin	Technology vendor	Vendor capture exceeds real operator benefit
Reduced repetitive walking	Worker exertion benefit	Employees / operator	No payroll reduction, but job quality matters
Personalization / ordering data	Higher frequency, customer convenience	Brand, vendor and consumer	Privacy, price discrimination, control
Employee image/motion data	Potential training-model asset	Vendor/model owner	Consent and compensation unclear
Rent/CapEx support	Lower operator upfront financing	Franchisee; franchisor indirectly	Value shifted between franchisor and operator

Analytical scenarios; each row requires contract and field evidence before imputing a dollar value.

13.8 Who owns restaurant intelligence and who monetizes it?

Value created by restaurant physical intelligence can be separated from value captured. An automated portion-control system can save operator food expense but require the franchisee to finance equipment, the franchisor to impose standards, the technology provider to collect subscription fees, and the employee to manage exceptions. The contract determines cash distributions; the customer judges food quality, not whether the model ran on a GPU. This makes ownership and service-level terms necessary inputs to investment underwriting.

Table 102 | Stakeholders, contracts and Physical AI value

Value event	Customer benefit	Restaurant/operator benefit	Vendor/franchisor benefit	Financial diligence
Order accuracy improvement	Correct meal; less frustration	Lower remake/waste; repeat sales	Possible SaaS renewal; platform stickiness	Quantify errors, margin, fees and refunds separately
Predictive fry station	Freshness and consistency	Lower rejects, fewer station bottlenecks	Machine/vendor service fee, sensor sales	Measure temperature, overtime and waste
AI employee copilot	Reliable service	Less search/training time	Subscription and support revenue	Realized staff outcomes; training cost; no assumed job cuts
Edge resilience	Fewer interrupted orders	Continuity at local network outage	Infrastructure/maintenance contract	Avoided downtime × gross contribution net of support
Delivery robot	Potentially timely service	Reduced walking or transport burden	Hardware or RaaS receipts	Net labor minutes incl load/unload; customer acceptance
Motion data licensing	Must preserve privacy	Only if revenue share/useful process insights	Potential robot model training value	Contractual data rights, consent, compensation

SOURCE / LIMITATION Illustrative value-allocation framework. The presence of a potential benefit is not empirical proof that every stakeholder receives it. Employee and consumer outcomes require direct evidence.

13.9 Location data and the limits of vertical integration

Access to a restaurant's physical environment does not confer ownership of its customers, staff work product, robot-training data or the underlying building. Those rights can be distributed among restaurant brand, franchisee, landlord, software platform, equipment manufacturer and robotics vendor. The contracting map should define operational use, training use, cross-client model improvements, intellectual-property ownership, portability at termination and responsibilities for incidents. Buyer control of physical assets is not equivalent to the vendor's control of an interface.

Vertical integration may reduce coordination costs when the same party designs robot hardware, policy, equipment connection and field support. It may also create costly organizational scope, capital intensity and integration risk. The right question is which boundary lowers total deployment and service cost while retaining enough supplier bargaining power to invest in future model and product improvements. The Sweetgreen–Wonder ownership transfer [S110] is a concrete reminder to distinguish continued technology use from continued ownership of an automation developer.

13.10 Translate technical productivity into attributable, monetizable site value

Technical efficiency is an input to the underwriting, not the answer. Suppose an intervention eliminates 18 seconds of routine handling from each of 160 tasks per day. Across 350 operating days, the arithmetic equals 280 employee-hours of potential annual capacity. At an illustrative fully burdened labor cost of $22 per hour, that is $6,160 of theoretical capacity value. Cash savings are zero unless schedules, overtime, staffing additions or other paid costs actually change. If the same freed capacity instead serves new demand, a separate incremental-contribution calculation is required; the same minutes cannot be counted both as payroll reduction and additional sales.

Table 103 | Illustrative saved-motion cash-flow bridge

Illustrative mechanism	Transparent inputs	Arithmetic result	Recognition requirement
Work time made available	18 sec × 160 tasks/day × 350 days	280 hours/year	Capacity, not automatically cash
Labor equivalent upper bound	280 hours × $22 per hour	$6,160/year	Paid hours or overtime genuinely avoided
Potential throughput contribution	2 extra orders/hour × 5 hours × 300 days × $4	$12,000/year	Document unmet demand and incremental margin
Field service burden	0.5 hours/week × 52 × $100/hour	$2,600/year cost	Include provider / operator cost allocation

All input figures are deliberately hypothetical. Examples are alternatives, not benefits to sum. Field service cost is a separate potential deduction. Excludes initial capital, tax, financing, capital replacement and error costs.

This is the narrow financial link between World Model performance and restaurant investment. A better policy may reduce rework, interventions or uncertain order preparation, but a restaurant buyer should request signed-off baselines and identify the specific line of unit economics that changes. Benefits must be adjusted for downtime, incremental cleaning, quality rejects, utilities, maintenance, insurance, software, network reliability and site modifications. Retained employee capacity can be valuable even without payroll reduction, but should be reported as an operating or workforce outcome until observed customer contribution or avoided cash expense supports monetization.

On the vendor side, a high deployment count can coexist with weak economics if each new restaurant requires custom software, weekly field visits or bespoke equipment. The buyer-side five-year NPV test in Module II and the vendor-side gross-profit per active site in Module III must both be positive under defensible assumptions. Successful model performance is necessary in selected tasks; it is not sufficient for investment-grade market sizing.

14. Commercialization Risks and the 2030 Outlook

The 2030 outlook separates analyst scenarios from forecasts. Physical AI may grow through low-cost sensing, kitchen software and dedicated equipment while general-purpose restaurant humanoids remain economically unsuitable for many formats. Outcomes should be evaluated empirically rather than by the most ambitious technology demonstration.

14.1 Market scenarios

Scenarios should be built from eligible sites, supplier pricing, technology performance, integration friction, and contract economics. Assign scenario probabilities only when supported by calibrated evidence.

- Downside: adoption slows due to integration, workforce concerns and low positive-NPV share.
- Base: modular restaurant sensing, operating software and selective automation gain traction.
- Upside: affordable compute, standardized devices and verified pilot outcomes increase rollout speed.

Table 104 | Scenario drivers and observable signals

Driver	Downside	Base	Upside	Monitoring evidence
Restaurant unit demand	Stagnant traffic	Stable/modest growth	Higher store-level activity	Real sales and transaction counts
AI hardware costs	Slow cost declines	Gradual efficiency	Rapid performance/watt gains	Vendor quotes, TCO
Adoption obstacles	Integration slow / workforce distrust	Selective wins	Modular interfaces and trust	Active installs vs pilots
Vendor economics	Heavy services burden	Repeatable deployments	Growing software share	Margins, churn, support load
Operator returns	Uncertain payback	Positive NPV in eligible sites	Multiple repeatable use cases	Controlled rollout cash flows

14.2 Principal investment and commercialization risks

Real-world restaurant technology risks include operating quality deterioration, restaurant IT downtime, supplier concentration, privacy incidents, underutilized robot assets, equipment obsolescence, franchisee resistance and overbuilt expectations of TAM.

- Create separate technical, operating, economic, regulatory and perception risk categories.
- Link each risk to a measurable monitoring metric and mitigant.
- Vendor funding success does not establish product–market fit or recurring customer value.

Table 105 | Physical AI commercialization risk register

Risk	Leading indicator	Operational mitigation	Financial sensitivity
Weak paid demand	Low expansion beyond pilots	Refine tasks/price and eligible formats	Active locations / attach rate
Equipment downtime	High manual intervention per 100 cycles	Modular service/support	Maintenance costs, lost orders
Franchise conflicts	Delayed upgrade approvals	Subsidy and gain-sharing	Net cash flow by payer
Workforce opposition	Workarounds and turnover	Co-design and employee choice	Wages, retraining, replacement
Privacy/security	Complaints and exposure incidents	Minimal collection, controls	Claims, reputation and penalties
Cloud/edge over-spend	Idle compute and duplicate systems	Workload-based placement	Infrastructure TCO
Vendor concentration	Proprietary data and interfaces	Exit rights, portability	Pricing escalation and churn

14.3 Decisions for investors and operators

The purpose of this white paper is to support better decisions: identify valuable restaurant tasks first; collect minimum necessary data; deploy the lowest-cost reliable solution; evaluate employees and consumers; and confirm returns through incremental free cash flow.

- The board or investment committee checklist should connect each proposed adoption to an operating baseline, buyer, cash-flow case and stopping rule.
- The same evidence can support corporate-development, consulting and capital-allocation decisions when the intended decision and its limits are explicit.

- Name specific gaps that must be solved by fieldwork before recommending a technology.

The restaurant is a physical economic platform. Its future intelligence should be measured by food quality, worker experience, customer value and return on invested capital—not by number of robots, quantity of data or megawatts consumed.

14.4 2026 evidence-based industry conclusions and questions for the next release

The verified 2026 evidence supports a narrower and more credible narrative than a robotics-led vision. One U.S. restaurant chain discloses thousands of physical edge deployments and explicit franchisee payback targets for a much broader restaurant modernization bundle; another documents local edge clusters across its estate; another reports a large-scale intelligent kitchen-screen rollout, and Starbucks emphasizes barista assistance and sequencing. These models coexist with weak same-store demand at some adopters. None of these facts proves that all restaurants require dedicated GPUs or expensive mobile robots.

The next research priority is continued use at the site. How much of the benefit survives staff intervention and software expense? Which formats qualify? Who funds the installation and controls the resulting data? Answers grounded in cash returns and food quality would add more to the commercial case than another aggregate robot count.

Table 106 | Highest-value evidence priorities for publication

Priority	Research dataset	Minimum verified field	Economic output	Best source
P0	McDonald's NEXT cohort rollout	Installed sites; franchisee cost net of support; achieved savings	NPV/payback against four-year management target	Investor disclosures + franchisee interviews
P0	Top 20 chains 2026 technology audit	App-level site penetration & evidence code	Format-level adoption denominator	Operator engineering + SEC + vendors
P0	Enterprise restaurant software revenue pools	U.S. restaurant-only subscription vs payment revenue	Non-overlapping annual market revenue	Toast/PAR/NCR/Olo + contracts
P0	Wingstop Smart Kitchen & Starbucks Smart Queue	Kitchen throughput, labor hours, actual rollout	Kitchen operations economics	Operator and vendor site data
P1	Cloud versus store-edge commercial quotes	Hardware, managed services, egress, uptime costs	Five-year site TCO	Provider quotes + site engineering
P1	Restaurant robot active fleets	Sites, device utilization, renewal, churn	Robot revenue penetration and customer ROI	Fleet records; vendor/restaurant interviews
P1	Consumer and employee outcomes	Wait time, quality, workflow burden, survey responses	Stakeholder surplus and retention	Matched store/worker evidence

The priorities are research gates, not a statement that these datasets have been acquired. Public data alone do not verify restaurant-level AI ROI.

14.5 Investment conclusions strengthened by the 2026 evidence

Three implications now appear sufficiently supported to warrant deeper investigation: (1) the core economic unit of analysis is the restaurant operating site and its customer experience, not the robot shipment; (2) emerging AI adoption rates should be viewed alongside the much lower share of operators reporting substantial benefits; and (3) independent operational metrics are necessary to translate theoretical physical losses into an investable recurring technology market. ReFED food waste, Qu technology deployment and NRA workforce surveys make these three implications tangible without pretending to establish a final Physical AI winner. [S51]–[S56]

Table 107 | Investor diligence by value at risk

Rank	Critical diligence question	Necessary evidence	Decision to support
1	Is the problem material and preventable?	Unit loss dollars and root-cause decomposition	Exclude immaterial applications
2	Is AI economically better than conventional process change?	Matched no-tech alternative and total costs	Choose lowest-cost effective intervention
3	Is the proposed market measurable without double counting?	Buyer spend, non-overlapping categories, actual paid deployments	Reliable TAM/SAM/SOM
4	Does the system improve food and customer outcomes?	Taste/quality proxy, remakes, returns, complaint trend	Approve only consumer-compatible rollouts
5	Does staff burden fall or merely shift?	Interventions, training, safety, employee responses	Rework job design and training
6	Who owns data and physical assets?	Contracts, privacy, IP, lease, vendor SLAs	Set partner/acquisition structure
7	What is realized incremental cash flow?	Cohort-level margin, payroll, waste, renewals	Underwrite return on invested capital

Pending independent restaurant-level results, these are decision screens, not recommendations to buy securities.

14.6 Investment discipline: operational performance without market growth

The contrasting 2026 cases support an important but non-causal observation: AI-enabled productivity initiatives can be rational even when same-store sales are weak, but management should not claim that a stronger kitchen alone creates consumer demand. Wingstop's Q2 domestic comparable sales were negative even as Smart Kitchen remained an operational priority. By contrast, McDonald's NEXT explicitly links operating productivity to hospitality, menu quality and repeat visits, combining demand generation with productivity rather than treating computing as a stand-alone strategy. [S58,S60]

An investor therefore needs two income statements: a demand-side analysis (customer counts, menu preference, realized price, reorder behavior) and a supply-side analysis (labor, waste, throughput, equipment availability and technology spend). The project meets a high underwriting standard only when the combined benefits outweigh capex and operating risk, and stakeholder distribution is acceptable.

Table 108 | Physical AI investment evidence gates

Gate	Disqualifying weakness	Minimum decision-quality evidence	Return metric
Market size	Units counted without willingness to pay	Eligible sites, buyer budget, vendor revenue	Price x retained installed base
Food/customer value	Technology has no measurable service benefit	Product quality, error, wait, retention outcomes	Incremental contribution and welfare
Employee benefit	Extra monitoring/work with weak staff support	Staff survey, paid hours, task burden	Full labor/welfare accounting
Technology feasibility	Reliability unproven at peak load	Field uptime, fallback, integration test	Total cost and service risk
Financial return	CapEx justified by vendor quote only	No-tech comparator, deployment cohorts, NPV	IRR/NPV/cash payback
Ownership/contract	Data value transferred without agreed rights	Vendor/brand/franchisee/employee permissions	Risk-adjusted net capture

GII investment committee protocol for evaluating applications, not a recommendation about any specific security.

14.7 Evidence-based investment conclusions and falsification tests

The following evidence-based propositions guide diligence and strategy; they do not establish company-specific investment recommendations. Current filings provide enough information to distinguish attractive economics from inadequate disclosure, but do not yet justify a public-equity price target, a definitive physical-AI TAM figure, or claims of realized enterprise-wide project ROI. The next evidence gate is company- or store-level attributable before/after cash flow. [S75–S85]

Table 109 | Investment conclusions by evidence quality

Conclusion	Current verified basis	Unproven critical step	Investor action	Would falsify the case
Large-chain contracts are valuable and concentrated	PAR: ~36% revenue from 3 named restaurant accounts	Segment/client contribution and remaining contract term	Underwrite top-account exposure and platform rights	Customer concentration falls with diversification or accounts churn
Vendor recurring revenue is not Physical AI TAM	Toast: $5.037B fintech; $936M subscriptions in 2025	Share of either stream from physical-world AI features	Use incremental paid product cohorts and gross profit	No paid retention, bundling, or incremental fees
Franchise economics govern effective deployment	MCD and YUM have high franchised share and high corp margins	Who funds store upgrades and keeps savings	Model three cash-flow recipients separately	Franchisees reject costs or benefits fail to accrue
Deployment does not establish causal earnings impact	Wingstop comps down while Smart Kitchen vendor reports scale	Matched store cohort operating P&L effects	Do controlled difference-in-differences test	No material lift over non-AI operating redesign
Task economics favor right-sized automation	Chipotle pilots and Sweetgreen asset ownership changes	Robot/service uptime and true total install economics	Compare sensors/SOP/automation/robot independently	Full-workflow cost exceeds incremental customer value
Restaurant AI need not imply dedicated data centers	Chick-fil-A edge and MCD cloud-edge evidence	Site workload/latency/power/TCO and privacy requirements	Choose compute placement via lifecycle TCO	Centralized/cloud solution meets SLA at lower cost

Ranking is by evidence quality, not expected stock return.

SOURCE / LIMITATION Qualitative investment conclusions with direct evidence and explicit unproven causal links. Investor implications are about research diligence and corporate strategy, not securities recommendations. Sources [S58,S63,S75]–[S85].

Valuation readiness: what cannot responsibly be stated yet

Public-market multiples and discounted equity values require a common trading date, diluted share count, debt, cash, lease treatment and normalized financial measures. Those inputs have not been established in the current snapshot, so this report assigns no target prices, stock-return rankings or Physical AI valuation premiums. A later valuation study would need a synchronized basis and separate peer groups for franchisors, restaurant operators and technology suppliers.

14.8 Geography and density: the location-based autonomy hypothesis

Restaurant Physical AI should be investigated as a location-based economy. The useful geography is not an aggregate map of all restaurants; it is a map of eligible equipment footprints, transaction density, delivery catchments, permitted operating areas, utility availability and nearby field-service resources. This matters for fixed kitchen robotics and especially for mobile or aerial fulfillment. Robot deployment may become more attractive in clusters because trained technicians, spare parts, model-updating protocols and customer support can be shared across nearby sites.

DoorDash's September 2026 DoorDash Air announcement explicitly emphasizes first- and last-ten-feet infrastructure, packaging, loading systems and restaurant handoffs [S102]. Its public support description also notes weather, zone, payload and operating limits [S108]. These disclosures point to a geography-specific service design, not a general claim that drones are always cheaper than couriers. A model should distinguish the physical feasibility of a particular customer route from the unit economics of sustained service at a given order density.

The location-network opportunity combines production, QSR pickup and local fulfillment with the data and service infrastructure needed by Physical AI developers. The new drone cases remain proofs of bounded feasibility or limited early services. Customer, employee and store-manager acceptance, property fit and sustainable channel contribution must be tested before extrapolating to large store counts. Appendix M records the distinctions.

The 2030 outlook should therefore use conditional scenarios rather than one adoption curve. The constructive case is a geographically concentrated network of standardized tasks, well-specified action data, interoperable kitchen equipment and field service that enables reliable multi-site economics. The downside case is technical fragmentation: customized pilots, variable layouts, hidden support cost, uncertain data rights and a lack of incremental margin. The decisive evidence is active, renewed deployments with independent before-and-after operating measures—not press announcements or model benchmark rankings alone.

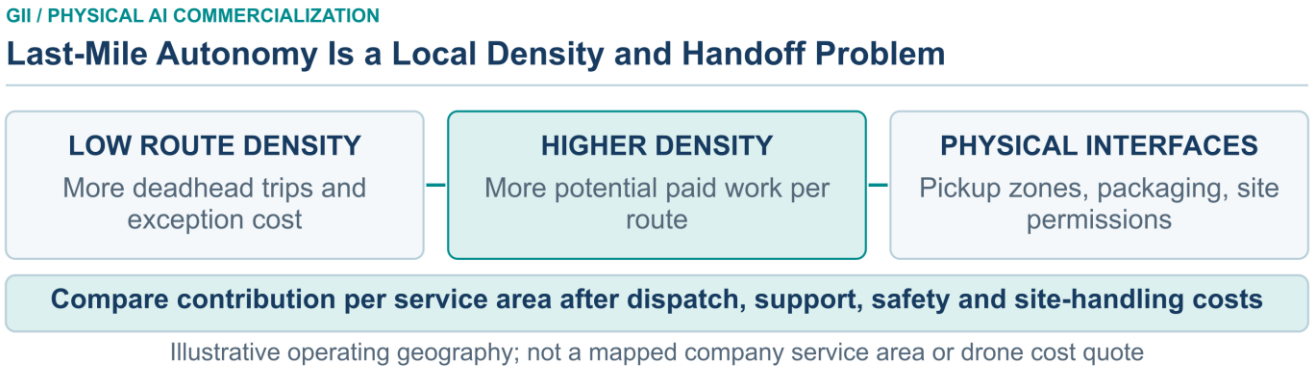


Figure 23 | Local service density and operating zones determine fulfillment feasibility

GII conceptual site-density framework. DoorDash disclosures [S102], [S108] describe selected operational constraints; no cost claims are extrapolated.

14.9 Current deployment judgment and what would change it

The current evidence supports investment in bounded production and service tasks with a defined human recovery path. Learned manipulation is advancing, but a foundation-model benchmark, a lab upgrade and a commercial pilot carry different operating implications. Table 147 makes those stages visible. Restaurant-wide autonomy requires coordination of orders, equipment, staff, quality and handoff beyond the competence of one successful robot skill.

A task has to fit the site’s layout, peak load and permitted operating conditions, then produce a cash benefit the operator can realize. Network value comes from repeating that result with less local adaptation and dependable service. Persistent custom integration, heavy intervention or discontinued use would weaken the case for scale.

For builders, the next useful evidence is accepted output per peak hour and cost to commission the next site. For operators, it is dependable service and incremental cash flow against the best feasible alternative. For World Model researchers, it is transfer measured before local retuning. These conclusions favor practical execution and reusable interfaces while leaving room for a generalist system to outperform the bounded baseline.

15. A Market-Entry Playbook for World Model and Robotics AI Companies

PART VI | PRACTICAL COMMERCIALIZATION MARKET ENTRY

A credible restaurant entry strategy begins with work that an operator will pay to improve. Define the task, observe the current process, secure appropriate data access and identify the budget owner. The five-year economic test should be credible before the supplier presents a chain-wide sales opportunity.

15.1 Five commercialization gates

Table 110 | Proposed commercialization stage gates

Gate	Questions to answer	Minimum evidence	Decision
1. Task and market	What high-frequency job causes material losses or friction?	Eligible restaurant formats and task frequency	Reject low-impact work
2. Data and model	Which observations, actions, states and rights are required?	Permitted data protocol and held-out task test	Proceed only if lawful and learnable
3. Physical deployment	Does the whole system work in actual kitchen constraints?	Safety, uptime, staff intervention, cleaning, IT	Pilot in a controlled restaurant
4. Buyer and adoption	Who authorizes capital and recurring contracts?	Headquarters/franchisee/vendor responsibility matrix	Secure sponsor and workflow owner
5. Unit economics	Does it beat workflow, sensing and ordinary software?	After-tax NPV, payback, worst-case cash needs	Scale, reprice or discontinue

GII proposed task-to-cash gates are an investment-underwriting heuristic, not validated product certification.

15.2 A two-sided commercialization agreement

Supplier and restaurant agreements should specify operational data rights, on-site access, model-improvement permissions, safety and liability, measurable KPIs, uptime or service remedies, trial costs, employee training, ongoing maintenance, removal rights and the division of economic benefits. For franchises, the restaurant brand and location operator may be different economic purchasers; the agreement should reflect the legal and financial structure rather than a logo.

15.3 Research applications across the existing case universe

The appropriate field test depends on task and format. Chick-fil-A can illuminate local edge architecture; McDonald's scale can test franchise systems and quality verification; Starbucks illustrates demand-to-barista coordination; Chipotle illustrates specialist manipulation; Yum! shows integrated restaurant operating software; Sweetgreen shows changes in ownership of automation development; Bear, Pudu and Keenon illustrate service-mobility economics; DoorDash and Uber illuminate handoff and marketplace power. No single case proves the entire opportunity.

15.4 Investment committee test

A go decision requires evidence that the system completes the task repeatedly at a cost both parties can sustain. Review staff workload and food quality alongside paying sites, deployment expense and supplier gross profit after support. Customer concentration, data licensing and franchise approvals can limit the business even when the robot performs well.

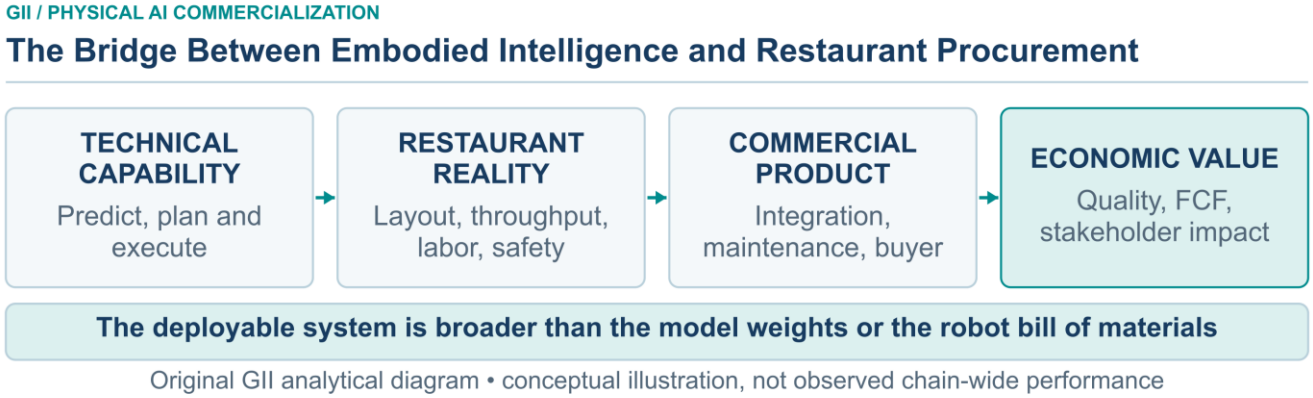


Figure 24 | GII Restaurant Physical AI Commercialization Bridge

Original investment synthesis; conceptual framework, not observed operating performance.

15.5 A disciplined first hundred days in an operating restaurant

The first phase is task discovery, not a product demonstration. Observe the restaurant at different dayparts, including the busiest shift, and document actual workflow, labor allocation, current software, equipment make, kitchen geometry, sanitation protocol, throughput constraints and who owns the budget. Ask management what operational result would justify spending money, and ask employees where the system is likely to interrupt or complicate service. Produce a task thesis with a measurable operational denominator and a non-robot alternative.

The second phase is a consent-based learning specification. Identify which states and actions must be represented; which cameras, sensors, equipment signals or demonstrations are permitted; the owners of recordings and model derivatives; and the rights granted to the vendor. Define evaluation tasks, success thresholds, unacceptable failures and a safe human takeover procedure. Make data deletion, privacy and incident reporting explicit before collecting the first identifiable recording.

The third phase is a bounded field pilot with independent measurement. Install at a site representative of the intended market, not simply the most convenient flagship. Measure uptime, successful output per operating hour, human interventions, cleaning and setup time, product conformity, injury or safety events, and customer delays. Record the counterfactual comparison, staff training, operator feedback, site downtime and all third-party charges. Stop if the system fails agreed safety or performance gates, even if its model benchmark appears promising.

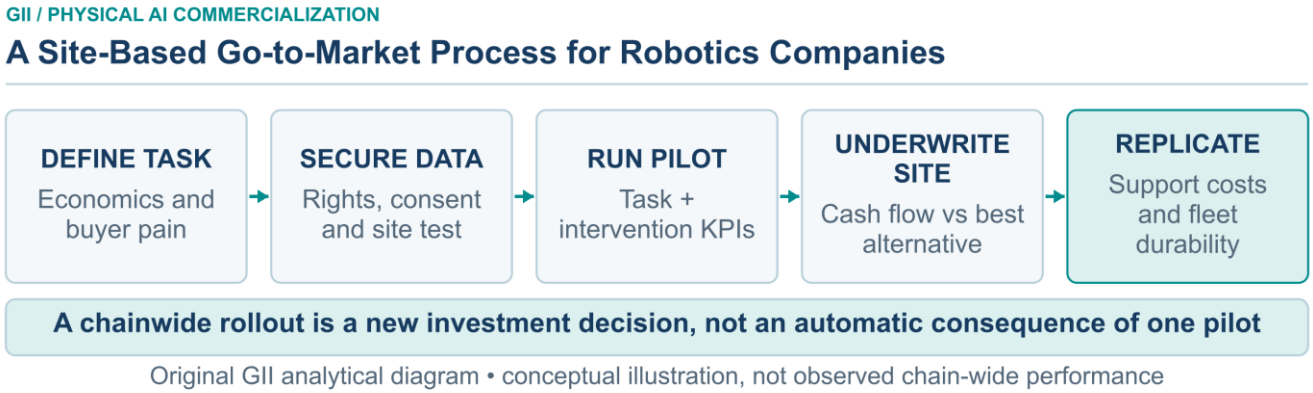


Figure 25 | A staged go-to-market process from problem definition to active site and fleet economics

GII original commercialization process. Milestones are proposed, not an established universal restaurant procurement standard.

15.6 Move from a successful pilot to a repeatable hardware and software business

After the field pilot, reconcile the restaurant's incremental free cash flow with the supplier's recurring economics. The restaurant side includes initial construction or retrofit costs, saved paid work, incremental served demand, quality, service disruptions, site cost and exit flexibility. The supplier side includes equipment gross profit, deployment labor, field service, software support, customer success, warranty, spares and ongoing model training. A site can show a positive technical result while creating negative economics for either party.

The first network rollout should be a controlled cluster of comparable restaurants with documented site classifications. Use repeated installation instructions, a permitted data schema, tested software rollback, spare-parts inventory, standardized operator training, and the same measurement protocol across sites. A product is commercially scalable only when the cost and duration of the next installation become predictable and the vendor can honor uptime without custom engineering in every location.

Robotics hardware builders can use the report to select end effectors, sensing envelopes, power requirements and cleanable mechanisms around proven site needs. Robot software builders can identify the order, KDS, equipment and event interfaces required for orchestration. World Model researchers can define task-scoped data and sim-to-real benchmarks. Corporate development teams can evaluate whether to acquire data rights, integration capacity, service networks or specialist equipment suppliers rather than merely pursuing a larger robot demonstration.



Picture 12 | A service network supports repeatable restaurant deployments
AI-generated concept illustration for GII; equipment and settings are illustrative.

15.7 Investment committee questions for a location-based Physical AI venture

Model capability, restaurant value and supplier value require different evidence. A useful skill may earn too little for the restaurant, or consume too much support time for the vendor. Ask for a cohort of active sites showing renewals, service hours, profitability by format and customer concentration. Deferred service obligations belong in that review as well.

The near-term portfolio should reflect technical maturity. Order orchestration and connected appliances can build on existing interfaces. Specialized manipulation needs enough volume and suitable geometry; service robots need workable routes and human handoffs. World Model pilots need permissioned data and credible transfer tests. Each route can justify investment on its own evidence and schedule.

Table 111 | Pilot-to-fleet commercialization gates

Gate	Go evidence	No-go signal
Site qualification	Measurable task value, fit, budget owner	Unworkable layout or no buyer
Action data	Permissioned state/action/outcome dataset	Consent or rights unresolved
Safe field pilot	Representative uptime, quality and interventions	Uncontrolled failures / unacceptable downtime
Store ROI	Positive risk-adjusted NPV versus counterfactual	Benefit not monetizable or transfer of burden
Fleet economics	Repeatable install cost and retained vendor profit	Excess customization or unserviceable geography
Customer renewal	Adoption beyond pilot champion	No usage, renewal or support capacity

Source: GII proposed investment-screen framework, not a published standard or documented acquisition policy.

15.8 Pilot-to-fleet evidence contract: what builders must prove at each site

A robotics firm seeking restaurant-scale commercialization should first negotiate a measurable service contract rather than a promotional prototype announcement. The contract should define an approved task, accepted output, safety and sanitation process, required availability windows, human intervention, installation responsibilities, cloud and data rights, support response and what happens when the machine stops. Each participating location is simultaneously an operating customer, a data-governance subject and a physical facility with its own constraints.

The sequence of technical and economic evidence should be cumulative. First, the machine can accomplish a bounded task with documented failure recovery. Second, trained staff can use it throughout a realistic shift. Third, performance generalizes to deliberately different locations without a custom engineering engagement for every store. Fourth, the restaurant can substantiate incremental operating value against a non-AI alternative. Finally, the supplier can earn recurring gross profit from an active fleet after deployment labor, parts, field service and warranty obligations.

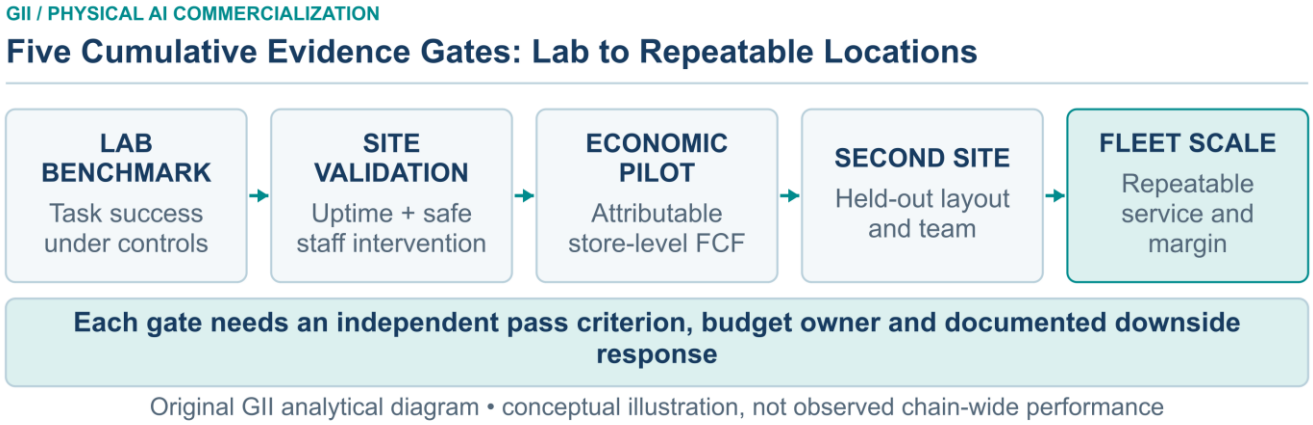


Figure 26 | Five evidence gates connecting robot capability to repeatable location economics

GII original commercialization framework; suggested site gates, not a universal robotics certification or a demonstrated chain adoption outcome.

Table 112 | Commercial evidence for a robotics venture

Stage	Primary unit	Evidence / owner	Stop or redesign if
Task pilot	Action episode	Robot team: output quality, hazards, exception records	Unsafe failures or no measurable task value
Restaurant trial	Operating shift	Restaurant ops: uptime, staff impact, quality	Human rescue and setup overwhelm benefit
Held-out location	Store / site type	Joint: transfer with fixed deploy protocol	Custom retrofit or data demand explodes
Buyer economics	Net cash per site	Restaurant finance: incremental NPV versus alternative	Negative NPV at buyer's true cost basis
Supplier cohort	Active paying site	Vendor finance: GM/site, retention, maintenance	Service cost scales faster than receipts

GII proposed technical-to-finance evidence ladder. Pass thresholds should be agreed with each site and validated by appropriate experts.

Location economics then governs the first commercial cluster. A dense cluster of stores with comparable kitchens can reduce travel time for field engineers, spare-parts inventory and commissioning effort. But this benefit is conditional on actually sharing hardware, integration interfaces, menu workflows and site permits. Restaurants in the same brand may differ sufficiently that a uniform installation price is economically misleading. Track time-to-commission, on-site support hours, critical parts usage and active recurring gross profit by site cohort, rather than using signed brand logos as the scaling metric.

The proposed research collaboration is deliberately two-sided. Restaurants should contribute clearly authorized access to tasks, data and operating feedback in exchange for a credible path to safer or more productive work. Robotics developers supply the model, hardware, field integration and evidence transparency; platform providers provide supported interfaces where authorized; investors supply capital only after task, location and fleet economics reconcile. The exciting opportunity is precisely that reliable, transferable physical intelligence could reduce the friction of operating many locations. The hard constraint is that technology must earn that scalability one accepted site at a time.

15.9 Build a network of accepted and economically viable locations

For a World Model or robotics company, the restaurant opportunity begins with an operating network rather than a list of potential robot buyers. Select a bounded task and a site format whose demand, equipment, buyer and support requirements are understood. Include the inside-the-four-walls production process and the outside-the-four-walls handoff in the product boundary. A preparation robot, pickup interface and delivery platform can create interdependent costs and benefits even when different firms supply them.

The first cluster should prove more than technical replication. Customers must receive food and service they choose to use; employees need workable routines and credible recovery support; managers need controllable operating demands and acceptable incremental returns. Record rejection, discontinued use and changes requested by each group. A successful demonstration with a willing champion is an entry point, not evidence of broad acceptance.

A large franchise and property network can then amplify deployment capability through repeatable equipment interfaces, installation protocols, staff training, maintenance routes and permitted task data. For research teams, varied sites provide a way to test transfer with held-out layouts and operating conditions. For operators and landlords, those same sites connect production capacity to local demand and potential sales beyond the dining room. The shared opportunity exists only where these learning and service advantages survive the costs and preferences of the people using the location.

This is the contribution of a restaurant, franchise, real estate and technology-adoption perspective to Physical AI: identify the eligible location, the people who must accept the change, the permitted physical task and the cash-flow recipient before describing a network-scale market. Broad rollout should follow evidence of continued use and repeatable site economics. Store count, flight count, model performance and merchant eligibility answer different questions.

15.10 Five views of the restaurant location network

The following concept renderings connect the report's production, learning, handoff, property and network themes. They illustrate possible arrangements, not the named drone cases or evidence of installed scale, approved layouts or human acceptance.



Picture 13 | A restaurant kitchen as a supervised food-production node
AI-generated concept illustration for GII; generic equipment and settings, not a named deployment or approved operating layout.



Picture 14 | Simulation and field feedback test transfer between sites

AI-generated concept illustration for GII; generic equipment and settings, not a named deployment or approved operating layout.



Picture 15 | Separate handoff areas support several fulfillment channels

AI-generated concept illustration for GII; generic equipment and settings, not a named deployment or approved operating layout.



Picture 16 | Shared property infrastructure can support local delivery

AI-generated concept illustration for GII; generic equipment and settings, not a named deployment or approved operating layout.



Picture 17 | Restaurant networks connect production with local demand

AI-generated concept illustration for GII; generic equipment and settings, not a named deployment or approved operating layout.

Investment Decision Tools — Feasibility, Unit Economics and Supplier Value Capture

Module I — Physical AI Commercial Feasibility

I.1 Industrial-economics framework: commercialization before technical novelty

A restaurant is a constrained production and service system in which the economic value of automation is limited by the binding bottleneck, the quality of the product and customers' willingness to return. An AI system that automates a non-bottleneck step may raise local productivity while leaving restaurant throughput and payroll unchanged. Conversely, a modest sensor and workflow intervention may generate substantial benefit without visible robotics. The correct unit of analysis is the complete activity chain, not the promoted device or a single task.

The commercialization sequence should begin with a precise job-to-be-done and measurable baseline, then verify physical integration, adoption, unit economics and scale. Restaurants should reject claims that infer annual payroll savings from "minutes saved" before confirming whether scheduled paid hours, overtime, labor availability, revenue capacity or service quality change. Worker fatigue reductions can be valuable even without payroll reductions, but should be disclosed as workforce benefits, not attributed cash savings.

I.2 Five commercial feasibility gates

Table 113 | Five-stage commercial feasibility gates

Gate	Decision test	Minimum public or analyst evidence	Failure / stop condition
G0. Problem	Material recurring customer or operating pain	Ticket-time distribution; quality defects; waste; safety; staff pain; estimated dollar exposure	The task does not affect customer outcomes or the binding bottleneck.
G1. Function	Technology performs safely and repeatedly	Published test conditions; availability; exception rates; human assists; operator sign-off	Demo depends on unrealistic space, labor, connectivity or inputs.
G2. Restaurant fit	Physically and organizationally deployable	Site dimensions; power; cleaning; network; shift/traffic flows; training; data governance	Retrofit or staff burden exceeds realistic operating constraints.
G3. Economics	Positive incremental value against cheaper alternatives	CapEx/OpEx estimates, ramp, maintenance, refresh, tax, NPV/IRR sensitivity	Insufficient attainable cash benefit or poor customer/worker outcome.
G4. Scale	Replicable beyond a controlled pilot	Operational variance by format; service network; franchise permissions; contractual ownership	Vendor cannot maintain quality or support dispersed restaurant locations.

Method / source: Gates are proposed analyst diligence tests, not actual certification standards.

I.3 Comparative evidence: what is actually disclosed?

The strongest public cases support different, complementary claims: McDonald's provides explicit restaurant-level targets but bundles AI with kitchen and design changes; Chick-fil-A provides architectural evidence of distributed computing but not a verified per-site ROI; Chipotle describes geographically named restaurant pilots and stage-gated automation; Starbucks describes an employee-support AI pilot; restaurant technology vendors describe broad deployments while control-group economic results remain largely undisclosed. The appropriate conclusion is about deployment readiness, not investment return.

Table 114 | Selected Physical AI cases and investment evidence

Company / case	Publicly established state	Value proposition	Missing evidence for ROI
McDonald's > NEXT	2026 management plan: approx. $800k incremental traditional U.S. drive-through investment; 250bp gross efficiency target; estimated four-year payback after partnering via official release [S38]; other targets [S86].	Kitchen / order accuracy / inventory / employee workflow, plus broader modernization.	Investment contribution specifically attributable to AI; measured net incremental FCF; actual cohort results.
Chick-fil-A edge clusters	2025 engineering article: Kubernetes platform in all stores; originally three nodes per store [S88].	Local reliability, orchestration and support of operational applications.	Cost/store, outage reduction and contribution margin compared with cloud-only architecture.
Chipotle Autocado / Hyphen	2024 restaurant pilots in Huntington Beach and Corona del Mar; staff/guest feedback part of stage gate [S87].	Reduce prep friction and improve order preparation.	Commercial rollout economics, real labor change, upkeep, quality and customer acceptance.
Starbucks Green Dot Assist	June 2025 company-reported initial 35-coffeehouse AI assistant pilot [S89].	Employee access to operating knowledge; task assistance.	Reduced turnover or customer value relative to conventional training/process controls.
Wingstop Smart Kitchen	Supplier-reported 2026 rollout to >2,500 U.S. sites [prior S47]; company comp-sales data separate [prior S84].	Kitchen sequence optimization and wait-time consistency.	Independent site-level savings and attributable same-store economics.

Method / source: Dates and disclosures are not contemporaneous; FY2025/2026 progress is not extrapolated beyond publicly disclosed scale.

I.4 Application triage: where does sensing or autonomy genuinely matter?

Table 115 | Restaurant problems and physical actions

Physical operating constraint	Minimum data / workflow	Potential action	Commercial disqualifier
Cold food / inconsistent cooking	Product temperature, equipment state, time since preparation	Crew alert or connected equipment adjustment	No proof that intervention reduces reject or quality failure rate.
Long peak-time queues	POS timestamps, station workload, prep completion and pickup time	Shift prep priority; redirect work; improve staging	Kitchen labor or downstream station remains the true bottleneck.
Unplanned equipment downtime	Fault codes, energy/temperature logs, service history	Predictive maintenance dispatch	Failure frequency too low to justify data and service cost.
Food overproduction	Demand forecast, prep yield, inventory, plate-waste attribution	Batch-size advice or automatic production control	Most identified loss is plate waste rather than preventable prep overproduction.
Employee physical overload	Human-centered time and motion study; anonymous staff feedback	Change task allocation, workflow or targeted equipment	Monitoring reduces autonomy or induces additional exception-handling.
Service robot transport	Load/unload events, actual staff paid hours, aisle constraints	Automated noncritical transport	No reduction in total work, inadequate walkway or high field support.

Method / source: Analyst examples; any real-world claim requires a before/after or other evidence design. ReFED national totals are not AI-capturable TAM [S92].

I.5 Public-data substitutes for unavailable control groups

The evidence hierarchy must be explicit. Published field results with independent checks provide the highest confidence. Formal issuer plans and company engineering disclosures establish intent and system design, not causal gains. Vendor case studies are hypotheses awaiting validation. Analyst cost calculations should be presented as feasibility screens, not "measured restaurant improvements." A practical alternative to matched cohorts is a pre-specified baseline and multi-case triangulation, with wide uncertainty intervals and counterfactual alternatives.

Table 116 | Operating KPI definitions and audit protocol

KPI	Measurement definition	Source / frequency	Attribution safeguard
Food consistency	Defective or remade items / completed orders	KDS + manual audit; daily	Hold menu mix, timing and audit method constant.
Peak capacity	Fulfilled orders per 15-minute peak interval	POS + KDS timestamp logs	Check pickup bottleneck and abandoned orders.
Staff work	Paid hours, overtime and worker-assist minutes per shift	Payroll + shift observation	Distinguish redeployed minutes from eliminated paid hours.
Customer value	P50/P95 wait, order accuracy, repeat visits	POS, app feedback, survey	Adjust for promos, price, traffic and seasonality.
Equipment reliability	Downtime minutes / critical machine / week	IoT fault and maintenance logs	Avoid treating scheduled maintenance as outages.
Technology cost	Lifecycle expense including help desk, edge/cloud, update cycle	GL allocation and vendor invoice (if available)	Count installation downtime and security overhead.

Method / source: A proposed field protocol for future diligence; no operator-led proprietary data has been obtained.

I.6 Market opportunity should be bounded by the feasible job, not robot counts

An investor screening model should first exclude restaurants where the operating pain is too small, the kitchen format is unsuitable, worker and customer acceptance is uncertain, or the installation competes with a cheaper process intervention. That filter matters more than multiplying every U.S. restaurant by a hypothetical robotics selling price. Furthermore, a supplier’s sales opportunity is only the portion of the economic gain that restaurants are willing and able to pay for after risk-sharing, integration and maintenance.

Module II — Restaurant Unit Economics and Investment Underwriting

II.1 Illustrative five-year discounted cash-flow design

This working model prices a hypothetical restaurant-level intelligent operating system, not any named company's actual project. Capital is committed at t=0; gross benefits increase with a 60%/75%/90%/100%/100% realization schedule; costs are expensed annually; initial investment is depreciated over five years for a simple tax assumption; a separate cash equipment refresh occurs in Year 4; no terminal value is credited. Tax credits are not assumed when project-level taxable income is negative. The model excludes financing, salvage value and working-capital movements.

The purpose is to identify a break-even commercial threshold. A restaurant operator can substitute quoted implementation costs, wage rates, actual error rates and forecast volumes when those data become available. For this publication, every dollar in this subsection remains a transparent analyst assumption rather than a disclosed franchise result.

Table 117 | Five-year AI operating system assumptions — illustrative only

Input	Base assumption	Economic treatment	Evidence class
Initial installation + integration	$80,000 at t=0	Capitalized initial cash investment; straight-line depreciation over five years	Analyst assumption
Full-run-rate annual gross operating benefit	$45,000/year	Benefits include only attributable realized financial gains; no separate time-saved cash credit	Analyst assumption
Benefit realization by year	60% / 75% / 90% / 100% / 100%	Reflects rollout, training and process learning	Analyst assumption
Annual cloud, software, support and maintenance	$10,000/year	Recurring cash OpEx from Year 1 to Year 5	Analyst assumption
Extra technology refresh / capital maintenance	$10,000 in Year 4	Deducted as cash in Year 4; no added terminal tax shield modeled	Analyst assumption
Tax rate / discount rate	25% / 12%	Cash taxes only when project EBIT positive; unlevered discount rate	Analyst assumption
Terminal value and disposal	$0	Conservative five-year project boundary	Analyst assumption

Method / source: Module II is the principal GII case and differs intentionally from Alternatives A (Table 82) and B (Tables 90–94). It assumes no incremental tax shield for the Year-4 refresh and no NOL carryforward. These are simplifying scenario assumptions, not a determination of entity-specific tax treatment.

Table 118 | Illustrative after-tax restaurant cash flow

Year	Benefit realization	Gross benefit	Recurring OpEx	Cash taxes	Refresh	Net annual FCF
t=0	—	—	—	—	—	-$80,000
1	60%	$27,000	$10,000	$250	$0	$16,750
2	75%	$33,750	$10,000	$1,938	$0	$21,812
3	90%	$40,500	$10,000	$3,625	$0	$26,875
4	100%	$45,000	$10,000	$4,750	$10,000	$20,250
5	100%	$45,000	$10,000	$4,750	$0	$30,250

Illustrative model cash flows are reproducible, not audited operating results.

Method / source: Annual tax=max(0, (gross benefit − OpEx − $16,000 depreciation))×25%. NPV at 12% = $1,507; IRR = 12.7%. Returns use unrounded cash flows; displayed dollars are rounded. Assumptions, not audited chain results.

II.2 Scenario investment returns and threshold economics

Table 119 | AI operating system investment scenarios

Scenario	Gross benefit / mature yr	Five-year NPV	IRR	Year 1 FCF	Year 5 FCF
Downside	$28,000	-$39,644	-9.0%	$6,800	$17,500
Base	$45,000	$1,507	12.7%	$16,750	$30,250
Upside	$65,000	$46,164	31.4%	$25,750	$45,250

Method / source: All cases share $80,000 t=0, $10,000 annual OpEx, $10,000 Year-4 refresh, five-year depreciation, 25% tax, 12% hurdle.

Base-case break-even. With these exact timing, tax and replacement assumptions, the system requires approximately $44,325 of mature-year gross attributable operating benefits to achieve zero NPV. At an illustrative $3.0 million AUV, that

represents about 1.48% of sales; it is not a prediction about a particular chain. A $45,000 annual benefit clears the threshold narrowly, so uncertainty in benefits, installation cost and ramp can overturn the decision.

Table 120 | Five-year NPV sensitivity, USD thousands

Initial CapEx	Benefit $25k	Benefit $35k	Benefit $45k	Benefit $55k	Benefit $65k
$40,000	-11.0	+12.0	+34.3	+56.6	+79.0
$60,000	-29.0	-4.6	+17.9	+40.2	+62.6
$80,000	-48.0	-21.9	+1.5	+23.8	+46.2
$100,000	-68.0	-40.0	-15.6	+7.4	+29.8
$120,000	-88.0	-58.5	-32.9	-9.2	+13.4

USD thousands; varies initial CapEx and mature gross benefit while retaining the other stated assumptions.

Method / source: NPV figures are in USD thousands, + denotes value creation; cost, refresh, ramp, taxes and hurdle held fixed as in Table 117.

II.3 A required benchmark: no-AI process redesign and sensing-only investment

The relevant alternative is not always a higher-performance robot. A manager may improve throughput through different food batching, better workstation allocation or scheduling without investing in inference hardware. Similarly, a sensing-first project may address the dominant failure mode with much less CapEx. A capital allocator should compare mutually exclusive or partially substitutable options using the same cash-flow methodology. The outcomes below illustrate the principle; they do not rank real products.

Table 121 | Hypothetical alternative interventions

Intervention	Initial CapEx	Mature gross benefit / yr	Recurring OpEx / yr	NPV, five years	IRR
No-AI workflow redesign	$15,000	$17,000	$4,000	$14,848	42.2%
Sensing-first system	$30,000	$26,000	$7,000	$11,995	25.5%
Edge/AI operating system	$80,000	$45,000	$10,000	$1,507	12.7%
Robotics-intensive option	$180,000	$75,000	$18,000	-$43,476	2.4%

Method / source: all four interventions use the same benefit ramp, tax-rate convention and discount rate, with initial CapEx depreciated over five years. The Year-4 cash refresh inputs of $0 / $4,000 / $10,000 / $20,000, respectively, were reconstructed to reproduce the earlier comparison; they remain unconfirmed. Benefits are uncalibrated analyst assumptions. The ranking is conditional on the inputs and comparable task scope; it is not evidence that a particular technology is superior.

II.4 Benefit recognition: translating physical tasks into cash flow

Time saved should be tracked in two parallel ledgers. Ledger A records physical operating improvements (fewer steps, lower fatigue, fewer queue exceptions). Ledger B records realizable cash-flow effects (fewer paid hours, reduced overtime, incremental order contribution, fewer remakes, lower supplies expense). Adding both as cash benefits double-counts the same intervention. Benefits should also be net of crew checking, cleanup, maintenance and customer-recovery time. This accounting discipline connects workforce welfare to project finance without treating employees as replaceable cost units.

Table 122 | Physical improvement and cash-benefit attribution

Claimed operating outcome	Economic conversion formula	What not to assume	Possible proof source
45 min/shift walking eliminated	Only realized paid-hour or overtime change × loaded wage; or attributable incremental sales margin	45 minutes saved = 45 minutes of reduced payroll	Shift rosters, timekeeping, queue observations
Additional 10 peak orders/day	10 × operating days × contribution per incremental order × incremental share	All fulfilled orders are incremental; no cannibalization	POS, aborted orders, contribution mix
Food prep waste falls 10%	Eligible preventable waste base × 10%; avoid plate waste	10% of total national foodservice waste is supplier TAM	Production logs, plate waste split, ReFED definitions
Equipment uptime improves	Avoided downtime × contribution lost per hour less catch-up sales	Every downtime minute loses full restaurant sales	Maintenance logs, capacity study
Worker strain declines	Independent welfare or turnover valuation, with stated method	Health/workload gain counted again as wage savings	Anonymous employee survey, injury and turnover data

Method / source: Use ReFED for physical loss context, not an AI savings claim [S92].

II.5 Franchisor/franchisee financing and multi-site scale

System-level value and franchisee cash returns are not equivalent. Consider an illustrative 100-store program requiring $80,000 per unit: total installation capital is $8.0 million. If a franchisor funds 40% ($3.2 million) and franchisees fund 60%

($4.8 million), the system has still spent $8.0 million; the transfer only reallocates cash requirements and private returns. A four-year franchisee payback cannot be equated to system payback without the amount and timing of corporate support, benefits, maintenance and rollout cadence. This is particularly important when discussing McDonald's NEXT [S86].

Table 123 | Illustrative 100-site franchise financing

Item	Total system	Franchisor	Franchisees
Installation: 100 × $80,000	$8.0M	$3.2M funded	$4.8M funded
Sponsor support rate	40% assumed	$32,000 per store	Reduces initial franchisee cash exposure
Full-run-rate gross benefit	100 × $45,000 = $4.5M/year	No incremental royalty modeled	Assumed captured at restaurant
Recurring OpEx	100 × $10,000 = $1.0M/year	Contract allocation unknown	Assumed paid by operating units
Year-4 replacement	100 × $10,000 = $1.0M	Funding subject to contract	Not in initial subsidy assumption

Method / source: Hypothetical financing transfer; do not interpret as McDonald's actual franchise contract or booked financial results.

Module III — Technology Supplier Value Capture and Industry Competition

III.1 Separating software economics from payment throughput and hardware resale

Two audited technology providers offer a useful lesson in the structure of restaurant-adjacent technology revenue. Toast bundles subscriptions with substantial financial-technology processing and device/service activities. PAR derives a higher reported revenue share from subscription service, but that category includes SaaS, related support, managed services and transaction-based payment processing. Neither company's total revenue is a measure of sales of Physical AI. Comparisons should focus on revenue definitions, gross margin, customer concentration, customer economics and the actual incremental product being evaluated.

Table 124 | Toast and PAR SEC revenue, FY2025, USD millions

Firm / audited FY2025 category	Revenue ($M)	% consolidated revenue	Measurement caution
Toast — subscription services	936.0	15.2%	Subscription does not equal Physical AI sales.
Toast — financial technology solutions	5,037.0	81.9%	Payment-processing solutions; many gross processing costs.
Toast — hardware and professional services	180.0	2.9%	Hardware and related services, not purely AI.
Toast — total	**6,153.0**	**100.0%**	**Company revenue, not restaurant AI TAM.**
PAR — subscription service	291.170	63.9%	Includes SaaS, support, managed and some payment-processing services.
PAR — hardware	106.410	23.4%	POS/drive-thru/other equipment sales.
PAR — professional service	57.967	12.7%	Installation/support/implementation.
PAR — total	**455.547**	**100.0%**	**Acquisitions and mix affect growth comparability.**

Method / source: Source: 2025 SEC Form 10-K for Toast [S90] and PAR [S91]. Revenue groups from different firms are not additive estimates of a unique market.

III.2 Gross margin matters more than headline technology revenue

Table 125 | Reported GAAP segment gross margins, FY2025

Operating category	Revenue ($M)	Cost of revenue ($M)	Gross profit ($M)	Gross margin
Toast: subscriptions	936	264	672	71.8%
Toast: fintech	5,037	3,891	1,146	22.8%
Toast: hardware/services	180	400	-220	-122.2%
Toast: consolidated	**6,153**	**4,560**	**1,593**	**25.9%**
PAR: subscription	291.17	132.027	159.143	54.7%
PAR: hardware	106.41	82.044	24.366	22.9%
PAR: professional services	57.967	43.45	14.517	25.0%
PAR: consolidated	**455.547**	**257.521**	**198.026**	**43.5%**

Reported GAAP segment gross margins are accounting measures, not margins attributable solely to AI.

Method / source: Toast gross profit by segments includes additional $5M acquired intangible amortization in consolidated COGS; hence component GP does not sum perfectly. PAR subscription includes payment processing; gross margins not directly interchangeable.

The economics diverge sharply: Toast's 2025 subscription service gross margin was about 71.8%, its fintech gross margin about 22.8%, and its hardware/professional services category carried a substantial gross loss. PAR's subscription gross margin was approximately 54.7%, while consolidated GAAP gross margin was about 43.5%. These are disclosed accounting outcomes. They do not prove the profitability, margin, or market attractiveness of any specific Physical AI application. The next diligence question is whether a new AI function raises customer willingness to pay faster than its inference, support and integration costs.

III.3 Customer concentration, bargaining power and platform control

The data also reveal industrial concentration. PAR reported that McDonald's, Yum! Brands and Dairy Queen accounted for about 21%, 8% and 7% of total 2025 revenue respectively; together roughly 36%, with percentages rounded [S91]. This establishes real bargaining-power exposure, but it is not evidence of imminent customer losses or a franchisee technology contract being renegotiated. A large restaurant chain might improve its negotiating position through scale, proprietary data

and standardized deployment. Conversely, changing operationally embedded software can create switching costs and migration risk.

Table 126 | PAR customer concentration, FY2025 derived dollars

Customer	Share of 2025 revenue	Approx. revenue attributable ($M)	Interpretation
McDonald's	21%	95.7	Material contract and renewal exposure.
Yum! Brands	8%	36.4	Large restaurant portfolio and cross-brand bargaining power.
Dairy Queen	7%	31.9	PAR voluntarily reports this customer in its three-year customer table; 7% is below 10% in FY2025.
Top three sum	36%	164.0	Based on rounded shares; not separate named AI contracts.
All other customers	64%	291.6	Other revenue is diversified but may include other large accounts.

Method / source: Derived as 2025 PAR revenue of $455.547M × rounded reported customer percentages. Does not equal ARR at risk from any specific technology.

III.4 Supplier-side recurring-revenue model

A restaurant AI supplier can announce thousands of contracted locations and still experience weak economics if active deployment is delayed, support costs are high or a single chain controls the negotiation. The following model makes the revenue/usage distinction explicit. One thousand contracted sites yield only 700 active paid locations at 70% activation. At $250 per active store per month, that is $2.1 million of annual subscription revenue—not $3.0 million. After illustrative $90 per location monthly delivery cost and $1.2 million central product operations, the remaining profit contribution is narrow and excludes sales, general corporate overhead, tax and financing.

Table 127 | Hypothetical supplier recurring-revenue model

Input / result	Value	Formula / decision implication
Contracted restaurant locations	1,000	Signed access does not establish billing-active sites.
Activation rate	70%	700 active locations; not necessarily technical uptime.
Subscription price / month	$250/site	Hypothetical incremental software-only charge.
Variable cost / month	$90/site	Inference, support, hosting and routine service assumption.
Central fixed product operations / year	$1,200,000	Illustrative technical delivery overhead; excludes SG&A/R&D overhead beyond assumption.
Annual recurring subscription revenue	**$2,100,000**	**700 × $250 × 12.**
Variable annual service cost	$756,000	700 × $90 × 12.
After fixed delivery contribution	$144,000	Revenue less variable costs and specified central fixed costs.
Active-store break-even requirement	625 locations	Fixed / ($250 − $90) / 12; about 625 active locations.

Hypothetical supplier underwriting, not a real company's reported results.

Method / source: All assumptions illustrative; contribution after selected costs is not GAAP EBITDA, net income or audited Physical AI ARR.

III.5 Supplier downside tests: customer loss, rollout delay and price compression

Table 128 | Supplier unit economics: annual contribution sensitivity

Stress case	Paid active sites	Revenue	Variable cost	After $1.2M fixed delivery costs
Base activation	700	$2,100,000	$756,000	$144,000
30% active-site loss	490	$1,470,000	$529,200	-$259,200
Price -20%	700	$1,680,000	$756,000	-$276,000
Support cost +50%	700	$2,100,000	$1,134,000	-$234,000
Full activation	1000	$3,000,000	$1,080,000	$720,000

Method / source: Central fixed costs held flat; no one-time hardware, installation revenue, corporate SG&A, sales commissions, taxes or balance-sheet costs modeled.

III.6 Bounded chain-scale monetization: adoption is not TAM

The existing research master assembled 124,185 U.S. locations across its Top 20 limited-service brand sample (see the Top 20 sample). That total is a bounded research universe, not all U.S. restaurants and not a verified customer addressable market for a particular Physical AI product. Applying assumed annual software fees and penetration rates yields an analytical annual revenue sensitivity, only if these payments are incremental, commercially feasible and non-overlapping with existing subscriptions. It is not a formal industry estimate.

Table 129 | Illustrative Top 20 software-fee receipts

Eligible paid penetration	Illustrative paid sites	$1,200/site/year	$2,400/site/year	$4,800/site/year
10%	12,418	$14.9M	$29.8M	$59.6M
25%	31,046	$37.3M	$74.5M	$149.0M
50%	62,092	$74.5M	$149.0M	$298.0M

Bounded Top 20 software-fee illustration; not an industry TAM.

Method / source: Site count 124,185 is a compiled Top 20 sample. Scenarios are pure multiplication and exclude overlapping POS contracts, existing vendors, nontechnical eligibility and affordability constraints.

III.7 Who can retain the economics?

Table 130 | Physical AI industry value-capture and control points

Layer	Potential pricing mechanism	Competitive control	Economic failure mode
Restaurant operator / franchisor	Net savings, new orders or franchise royalties	Demand access, workflow IP, site permissions, brand/data rights	Benefits cannot be realized across staff, quality and capex.
Restaurant SaaS platform	Subscription, premium feature attach, integration fees	POS/KDS workflow embedding, installed base, proprietary datasets	AI premium included in existing SaaS price or copied by platform.
Cloud / edge operator	Compute/storage fees, capacity or support contract	Availability, developer tools, inference unit cost, security	Price declines / substitution / low utilization.
Equipment / robot supplier	Hardware ASP, leasing, RaaS, maintenance	Reliable task execution, safety, service network, patent/IP	Low fleet uptime, costly onsite support, incomplete task substitution.
Data / model developer	Training services, licensing, model inference	Exclusive licensed data and relevant prediction accuracy	Restaurant unwilling to share data; collection cost exceeds gains.
Franchisee / property owner	Operating margin or asset cash flow	Local execution, lease and facility rights	Franchisor mandates investment without sufficient unit-level benefit.

Method / source: Industrial-organization analysis; the table describes mechanisms, not verified company-level margin forecasts.

III.8 The investment diligence decision

Supplier value capture will be durable only when a restaurant customer can demonstrate an economically important operating job, when integrations and contractual data rights create real deployment advantages, and when the provider's gross profit exceeds all delivery, implementation and renewal costs. High ARR growth is insufficient if most reported revenue is payments, acquired activity, bundled legacy services or concentrated enterprise contracts. Conversely, restaurant chains may benefit from partnering rather than owning a software vendor or data center. The correct conclusion is contingent on installed economics and bargaining, not on unverified market narratives.

Table 131 | Cross-module investment decision standard

Diligence question	Minimum acceptable answer	What the evidence cannot claim
Does it solve a physical restaurant pain point?	Named task, baseline KPI, measurable intervention and safety acceptance.	A demo or press release is not proof of operational value.
Is restaurant unit economics positive?	After-tax discounted scenario; break-even gross benefit; no-AI comparator.	Analyst simulation is not actual store ROI.
Can the supplier monetize it?	Specific incremental fee, gross-profit pathway and active-store activation.	Combined processor revenue or hardware shipments are not Physical AI ARR.
Can chains deploy at scale?	Store format and physical infrastructure fit, support network, contractual risk-sharing.	Pilot count is not national penetration.
Who captures the improvement?	Franchisee, parent, customer, employee and vendor incidence independently identified.	Franchise subsidy does not add to total system value.
What would falsify the investment case?	Repeated negative FCF, customer churn, customer/worker rejection, lower-cost superior alternative.	Neither sector enthusiasm nor high CapEx proves an investable return.

Method / source: Use as a professional decision template for high-level investment and advisory discussions, not investment advice for a particular security.

III.9 Hardware service economics must be tested alongside software revenue

A hardware/RaaS supplier carries installed capital and field-service obligations that the earlier software illustration does not model. Chef's service description and Ricoh's Hyphen partnership make these obligations concrete [S146,S144]. The quotation must identify who funds equipment, installation, spare parts, cleaning support, replacements and removal. A software margin cannot be applied to the entire hardware invoice.

Table 153 introduces a separate GII hardware-service illustration. It leaves every earlier financial model unchanged. Its $1,500 monthly fee covers supplier-funded hardware recovery and field service. Table 127's $250 software-service subscription has a different cost base, including its stated routine assistance cost; these illustrative fees are not comparable vendor price quotations. The example distinguishes accounting contribution from cash recovery of installed capital and tests both the minimum paid unit count and the technician's capacity. It is not a vendor quotation or a forecast. A route with enough customers to cover cost can still fail the response-time promise if travel or breakdowns exceed the assumed workload.

Appendix A — Research Data Quality and Evidence Protocol

Table 132 | Evidence quality coding

Code	Meaning	Disclosure rule
P1	Audited/regulatory data or official revised series	Link source and financial/period definitions
P2	Company engineering and filings	Technical fact, not inferred profit
C1	First-party company-reported outcome	Label as management/vendor report
I1	Independent industry or academic research	Disclose coverage, selection and year
M1	Analyst model / calculated arithmetic	Show all input assumptions and equations
N/A	Not public / not yet verified	Never substitute zero
HIST	Historical claim	Do not present as current operating status
PILOT	Pilot or announced target	Never equate with national rollout

Every numerical table should retain the following columns in its editable source ledger even if some are omitted from the publication layout: metric name, value, unit, reporting entity, product/market scope, observation period, source title, retrieval URL, observed-versus-estimated flag, relevant denominator, underlying arithmetic, and confidence classification.

Some documents carry multiple source IDs because appendices were compiled in successive research passes. Exact URL aliases are mapped to their earliest registered ID in the source crosswalk supplied with the verification records; distinct releases, filings and model documents retain separate identities. S143 and S147 are alias slots for S130 and S97, respectively.

Appendix B — Primary Source Register [S01–S37]

The source library is intentionally concentrated on official economic releases, direct engineering disclosures, SEC filings, and trade-statistical publications. Older material is retained as background with a HIST label. All URLs below are clickable.

[S01] National Restaurant Association — Restaurant sales volume revised lower based on benchmarked government data
Date: 2026-10-19 | Source role: Primary; 2026 benchmark revision

[S02] National Restaurant Association — Restaurant and Foodservice Sales: October 2026 revised chart
Date: 2026-10 | Source role: Primary; amounts are chart-rounded

[S03] National Restaurant Association — 2026 State of the Restaurant Industry (original February outlook)
Date: 2026-02-11 | Source role: Historical release; superseded sales levels

[S04] QSR Magazine — QSR 50 2026: Top 50 Fast-Food Chains, Ranked by Sales
Date: 2026-08-03 | Source role: Industry ranking; some figures QSR-estimated

[S05] QSR Magazine — QSR 50 2025: Top 50 Fast-Food Chains, Ranked by Sales
Date: 2025-08-01 | Source role: Historical ranking; 2024 baseline

[S06] International Federation of Robotics — Global Sales of Professional Service Robots Surge 24%
Date: 2026-09-30 | Source role: IFR 2026 service robotics; sample-based, cross-year caution

[S07] International Federation of Robotics — Five Million Robots Now Operate in Factories Globally
Date: 2026-09-24 | Source role: IFR 2026 industrial robotics

[S08] International Energy Agency — Key Questions on Energy and AI, executive summary
Date: 2026-04-16 | Source role: Primary energy outlook

[S09] McDonald's Corporation — Where Innovation Meets Scale: Digital Transformation
Date: 2025-08-14 | Source role: Primary operator disclosure

[S10] Chick-fil-A engineering — How Our Edge Kubernetes Platform Has Evolved
Date: 2025-09-09 | Source role: First-party engineering documentation

[S11] Chick-fil-A engineering — Enterprise Restaurant Compute
Date: 2023-01-10 | Source role: First-party engineering documentation

[S12] Yum! Brands — Introducing Byte by Yum!
Date: 2025-02-06 | Source role: Primary; company deployment claims

[S13] Yum! Brands — Collaboration with NVIDIA
Date: 2025-03-18 | Source role: Primary; announced plans vs pilots

[S14] Microsoft / Starbucks — Starbucks turns to technology to brew more personal connections
Date: 2019 | Source role: Historical IoT/Azure case; not current rollout verified

[S15] Wendy's — Transforming the Ordering Experience: FreshAi Update
Date: 2025-05-02 | Source role: First-party; company-reported scale

[S16] Chipotle — Chipotle Debuts Autocado and Augmented Makeline
Date: 2024-09-16 | Source role: Primary; limited pilots

[S17] Sweetgreen / SEC — Sweetgreen completes sale of Spyce to Wonder
Date: 2025-12-29 | Source role: Primary SEC exhibit; transaction

[S18] Sweetgreen / SEC — 2025 10-K Note: Sale of Spyce
Date: 2026-02 | Source role: Primary SEC accounting disclosure

[S19] Domino's / SEC — 2025 Form 10-K
Date: 2026-02-23 | Source role: Primary Form 10-K; digital ordering and PULSE

[S20] Toast / SEC — 2025 Form 10-K
Date: 2026-02 | Source role: Primary filing; revenue categories/ARR/locations

[S21] Toast Investor Relations — Second Quarter 2026 Results
Date: 2026-08 | Source role: Primary; later ARR business metrics

[S22] Bear Robotics — Physical AI Robots Proven in the Real World
Date: 2026-09 metrics | Source role: Vendor-reported cumulative cross-sector deployments

[S23] LG Electronics — LG Acquires Majority Stake in Bear Robotics
Date: 2025-01-24 | Source role: Primary transaction disclosure

[S24] Pudu Robotics — About PUDU Robotics
Date: accessed 2026-10-26 | Source role: Vendor-reported cumulative cross-sector shipments

[S25] KEENON Robotics — About KEENON Robotics
Date: accessed 2026-10-26 | Source role: Vendor-reported cumulative cross-sector shipments

[S26] Associated Press — Burger King tests AI headsets in approximately 500 U.S. restaurants

Date: 2026-02 | Source role: Independent journalism; program not generalized outcomes

[S27] U.S. Census Bureau — County Business Patterns
Date: ongoing | Source role: Official industry location and establishment counts; reconciliation needed

[S28] U.S. Bureau of Labor Statistics — Food Services and Drinking Places Industry
Date: ongoing | Source role: Official employment/labor series

[S29] NBER — Working Papers and Publications
Date: ongoing | Source role: Economics methods and productivity literature

[S30] U.S. Securities and Exchange Commission — EDGAR Company Filings
Date: ongoing | Source role: Company financials and material transactions

[S31] Wendy's — Global Next Gen format operating and facilities claims
Date: 2025 | Source role: Vendor/operator stated design targets

[S32] Circana — Circana Announces Top 50 U.S. Restaurants for 2026
Date: 2026 | Source role: Alternative consumer spending ranking; different universe

[S33] National Restaurant Association — 2026 State of Industry Executive Summary
Date: 2026-02 | Source role: Historical survey; 42% operators reported not profitable in 2025

[S37] International Federation of Robotics — Service robots see global growth boom: IFR 2025 figures
Date: 2025-10 | Source role: Previous edition; older sample, not directly comparable to IFR 2026

[S34] IFR syndication / Ritzau — 2026 service robot press release with sampling caveat
Date: 2026-09-30 | Source role: Primary IFR press release syndication; sample base=238

[S35] Wendy's — Smart irrigation and restaurant water savings
Date: accessed 2026-10 | Source role: Primary example of physical sensing and operating value

[S36] NRA source chart — October 2026 revised sales series graphic
Date: 2026-10-19 | Source role: Data transcription from official labeled chart

Appendix C — Source Register [S38–S50]

The following entries extend the existing [S01]–[S37] register. Each date refers to the original publication or corporate reporting date, not the date of this research document. Primary source URLs are included so that future updates can verify numbers rather than repeat untested summaries.

[S38] McDonald's — 2026 Investor Day NEXT and Restaurant Efficiency Targets Primary link

Publication: 2026-09-23 | Interpretation: Primary corporate forward-looking financial and franchisee targets; not realized AI-only outcomes

[S39] McDonald's — Investor Day 2026 Transcript (Brian Rice) Secondary transcript

Publication: 2026-09-23 | Interpretation: Third-party transcript reproducing management remarks on thousands of edge sites, seven markets, 2027 U.S. plan

[S40] Yum! Brands — Completed Sale of Pizza Hut to LongRange Capital Primary link

Publication: 2026-09-01 | Interpretation: Primary completed deal and changed ownership. Continuing Byte services described in 2026 Q2 10-Q

[S41] Toast Inc. — Q2 2026 Financial Results (SEC Exhibit 99.1) Primary link

Publication: 2026-08-04 | Interpretation: Primary GAAP Q2 revenue, subscriptions, fintech, ARR and live-location definitions

[S42] PAR Technology — Q2 2026 Financial Results (SEC Exhibit 99.1) Primary link

Publication: 2026-08-06 | Interpretation: Primary revenue, ARR, active-site data and 2026 KPI-definition change

[S43] NCR Voyix — Q2 2026 Earnings Release Primary link

Publication: 2026-08-05 | Interpretation: Primary retail versus restaurant segment revenue; company restructuring comparability caution

[S44] SoundHound AI — Q2 2026 Financial Results (SEC) Primary link

Publication: 2026-08 | Interpretation: Primary cross-sector enterprise AI revenue, not restaurant-only

[S45] Thoma Bravo — Acquisition of Olo Completed Primary link

Publication: 2025-09-12 | Interpretation: Primary transaction announcement; $2.0B equity value, not recurring revenue

[S46] Starbucks — Supporting the Moments That Matter with Artificial Intelligence Primary link

Publication: 2026-01-29; updated 2026-06 | Interpretation: Primary operating description; deployed employee AI, queue management and prospective equipment AI

[S47] Perfect Company — Wingstop Smart Kitchen Nationwide Rollout Primary link

Publication: 2026-05-06 | Interpretation: Vendor-reported 2,500+ deployment; not independent financial ROI validation

[S48] Wingstop — Fiscal Q2 2026 Earnings Release Primary link

Publication: 2026-07-29 | Interpretation: Primary number of U.S. stores, comparable sales and management Smart Kitchen prioritization

[S49] Wendy's — Q2 2026 Financial Results Primary link

Publication: 2026-08-07 | Interpretation: Primary unit counts and negative U.S. comps; not causal evidence concerning AI

[S50] Chipotle — Reaches 1,500 Chipotlanes Primary link

Publication: 2026-09-29 | Interpretation: Primary physical restaurant format and site milestone; Chipotlane not itself Physical AI

Appendix D — Source Register [S51–S57]

This addendum extends the original [S01]–[S50] source register. Source dates are preserved, and statistical scope is recorded next to each entry to prevent subsequent citations from treating survey percentages as population-wide adoption rates.

[S51] ReFED — Food Waste by Sector: Foodservice, 2024 data Primary link

Date: 2026 fact sheet, covering 2024 | Measurement / limitation: 12.5M tons and $157B of sector surplus value; not restaurant technology revenue

[S52] ReFED — Restaurants and Foodservice: Waste Causes and Recommendations Primary link

Date: Accessed October 2026; data year 2024 | Measurement / limitation: Plate waste ~69.6%; overproduction 11.9%; broad foodservice scope

[S53] Qu — 2026 State of Digital & Beyond Restaurant Technology Benchmark Primary link

Date: 2026 report | Measurement / limitation: 168 brands, ~91K–94K sites as inconsistently shown on page; vendor-run voluntary survey; adoption/impact self-reported

[S54] Toast — How Restaurants Are Using AI in 2026: 676 Operators Primary link

Date: Published 2026-09-14; surveyed 2026-04-03 through 04-20 | Measurement / limitation: U.S. restaurants with 16 or fewer stores; experimentation and attitudes, not Physical AI production

[S55] National Restaurant Association — Research Insight: Hiring & Staffing (2026) Primary link

Date: April 2026 | Measurement / limitation: 26% AI use, 94% no permanent job elimination, staffing pressure and time-to-productivity; primary operator survey

[S56] International Federation of Robotics — Global Sales of Professional Service Robots Surge 24% Primary link

Date: Published 2026-09-30, measures 2025 | Measurement / limitation: 238 supplier respondents, global units; 2025 sample should not be projected to whole market or compared naïvely to prior sample

[S57] ReFED — Solutions Database, Foodservice Filter Primary link

Date: Accessed October 2026 | Measurement / limitation: Modeled intervention benefits and investment requirements; not achieved vendor returns or audited Physical AI TAM

Appendix E — Additional Sources and Evidence Notes [S58–S74]

Research cutoff: October 26, 2026. New source IDs [S58]–[S74] continue the existing register. Each URL refers to a publicly accessible primary or identified secondary source. The full text of the press release and original figure is the controlling source, not a secondary headline.

Table 133 | Additional public-source registry

ID	Source and dated evidence	URL / publication domain
[S58]	McDonald's, McDonald's > NEXT FAQ (September 23, 2026)	Open source · corporate.mcdonalds.com
[S59]	McDonald's, September 23, 2026 Investor Update / SEC Exhibit 99.1	Open source · sec.gov
[S60]	Wingstop, Q2 2026 financial results (July 29, 2026)	Open source · ir.wingstop.com
[S61]	Perfect Company, Wingstop Smart Kitchen nationwide rollout (May 6, 2026)	Open source · prnewswire.com
[S62]	Perfect Company, Smart Kitchen deployment and case-study claims	Open source · perfect.tech
[S63]	Chipotle, Autocado and Augmented Makeline in operating restaurants (September 16, 2024)	Open source · newsroom.chipotle.com
[S64]	Starbucks, Supporting the Moments that Matter with AI (updated June 2026)	Open source · about.starbucks.com
[S65]	Starbucks, Green Dot Assist pilot (June 10, 2025)	Open source · about.starbucks.com
[S66]	Chick-fil-A Tech, How Our Edge Kubernetes Platform Has Evolved (September 9, 2025)	Open source · medium.com
[S67]	Yum! Brands, completed Pizza Hut sale (September 1, 2026), SEC Form 8-K	Open source · d18rn0p25nwr6d.cloudfront.net
[S68]	Associated Press, Burger King Patty headset trial (2026)	Open source · apnews.com
[S69]	McDonald's, Digitizing the Arches (August 14, 2025)	Open source · corporate.mcdonalds.com
[S70]	Wingstop Investor Relations, official Smart Kitchen media assets	Open source · ir.wingstop.com
[S71]	Bear Robotics, official Servi Q product and media page	Open source · bearrobotics.ai
[S72]	Chipotle, Hyphen digital makeline 2023 press images and operating concept	Open source · newsroom.chipotle.com
[S73]	QSR Magazine, Wingstop operations and field-reported results (2026)	Open source · qsrmagazine.com
[S74]	Restaurant Business, Wingstop Smart Kitchen commercial case	Open source · restaurantbusinessonline.com

Source links are supplied for audit and editorial use. Media links are not evidence of republication permission. Company and vendor claims remain attributed to their source.

Appendix F — Fiscal 2025 Public Issuer Filing Sources [S75–S85]

[S75] McDonald's Corporation FY2025 Form 10-K | 2026-02 (FY ended 2025-12-31). Consolidated revenue ~$26.9B; operating income ~$12.4B; operating margin 46.1%; FCF ~$7.2B Official source

[S76] Starbucks FY2025 Q4 and full-year results | 2025-10-29 (FY ended 2025-09-28). Revenue $37,184.4M; GAAP operating income $2,936.6M; margin 7.9%; restructuring affects comparison Official source

[S77] Chipotle FY2025 full-year results | 2026-02-03 (FY ended 2025-12-31). Revenue ~$11.9B; GAAP operating income $1,935.8M; restaurant-level margin 25.4%; comp -1.7% Official source

[S78] Yum! Brands FY2025 annual report | 2026 annual report (FY ended 2025-12-31). Revenue $8,214M; operating profit $2,574M; franchise and advertising revenue embedded Official source

[S79] Wendy's FY2025 fourth-quarter/full-year results | 2026-02 (FY ended 2025-12-28). Total revenue $2,176.9M; operating profit $343.5M; U.S. company-operated restaurant margin 14.2% Official source

[S80] Wingstop FY2025 fourth-quarter/full-year results | 2026-02 (FY ended 2025-12-27). Revenue $696.9M; operating income $179.291M; systemwide sales $5.3B; U.S. comp -3.3% Official source

[S81] Sweetgreen FY2025 earnings release filed on EDGAR | 2026-02 (FY ended 2025-12-28). Revenue $679.5M; operating loss $139.3M; restaurant-level profit margin 15.2%; comp -7.9% Official source

[S82] Toast FY2025 Form 10-K | 2026-02 (FY ended 2025-12-31). Revenue mix: subscription $936M; fintech $5,037M; hardware/services $180M; total $6,153M; operating income $292M; GPV $195.1B Official source

[S83] PAR Technology FY2025 Form 10-K | 2026-03 (FY ended 2025-12-31). Revenue $455.547M; subscription revenue $291.170M; operating loss $68.771M; named-customer concentrations 21%, 8%, 7% Official source

[S84] Wingstop Q2 2026 quarterly results | 2026-07/08 (quarter ended 2026-06-27). Domestic comparable sales -7.5%; digital sales 71.6% of systemwide sales; demand vs deployment distinction Official source

[S85] Toast investor quarterly and FY2025 results | 2026-02 (FY ended 2025-12-31). FY2025 free cash flow $608M; adjusted EBITDA $633M non-GAAP; 30K net locations added Official source

Source-to-Metric Verification Ledger

Table 134 | Source-linked calculation audit

Derived metric	Reported components	Calculated value	Source / limitation
McDonald's FY2025 op margin	$12.4B / $26.9B (rounded disclosures)	46.1% issuer stated	[S75]; cannot replicate exactly from rounded inputs
Yum! FY2025 op margin	$2,574M / $8,214M	31.3%	[S78]; franchise/advertising model
Wingstop FY2025 op margin	$179.291M / $696.9M	25.7%	[S80]; franchisor margin, not unit margin
Toast FY2025 GAAP total gross margin	$1,593M / $6,153M	25.9%	[S82]; payment-heavy mix
Toast FY2025 subscription GM	($936M − $264M)/$936M	71.8%	[S82]; not AI-exclusive
PAR FY2025 subscription GM	$159.143M / $291.170M	54.7%	[S83]; acquisition mix considerations
PAR top three customer concentration	21% + 8% + 7%	~36% of FY2025 revenue	[S83]; rounded reported percentages
PAR 10% MCD account attrition	0.10 × 0.21 × $455.547M	~$9.6M sales at risk	[S83]; mechanical sensitivity, not a forecast
Alternative B project NPV	5-year after-tax FCF, 12% hurdle, $80K initial	$-22.3K	Alternative B, Tables 90–94; simplified cash taxes; distinct from Module II

Appendix G — Decision-Tool Sources [S86–S95] and Model Verification

The evidence below supports the three investment decision tools. URLs point to official issuer documentation or publicly released company technical materials. Each source states a date and a reporting scope. Data for calculations in Tables 117–131 are generated from the explicitly stated numerical assumptions or reported SEC values; no proprietary franchise records were obtained.

Table 135 | Sources and permitted investment conclusions

ID	Primary evidence / reporting period	Permissible usage
[S86]	McDonald's, "FAQ: McDonald's > NEXT," 2026-09-23.	Official management targets: $800k incremental U.S. drive-thru investment, ArchIQ, 250bp expected efficiency, franchisor support.
[S87]	Chipotle, Autocado and Augmented Makeline restaurant pilots, 2024-09-16.	Physical named-site restaurant pilots, not fleet ROI.
[S88]	Chick-fil-A Tech, "How Our Edge Kubernetes Platform Has Evolved," 2025-09-09.	Restaurant-local distributed computing deployed across chain; architecture source only.
[S89]	Starbucks, "Meet Green Dot Assist," 2025-06-10.	Official 35-store initial generative AI assistant pilot.
[S90]	Toast, Inc., Form 10-K FY ended 2025-12-31, filed 2026.	GAAP segment revenue, COGS, revenue mix, calculated gross margins.
[S91]	PAR Technology Corporation, Form 10-K FY ended 2025-12-31, filed 2026.	GAAP category revenue, costs and named-customer concentration.
[S92]	ReFED, "Food Waste by Sector: Foodservice," model based on 2024.	Aggregate food surplus/waste and plate-waste segmentation; NOT AI-addressable supplier revenue.
[S93]	National Restaurant Association, "The Hiring and Staffing Dividend," 2026.	Staffing findings; outlet footprint from the release's "About" section.
[S94]	McDonald's, official 2026-09-23 investor update (SEC exhibit).	SEC: NEXT and capital plan; ~$100k gross benefit via FAQ [S86]; four-year post-partnering payback via official release [S38].
[S95]	Chipotle, Hyphen digital makeline official initial test, 2023-10-03.	Task definition and limitations: human completion remains part of workflow.

Method / source: Use original company/SEC reports for attribution; hypothetical unit-level financials are derived analyst calculations, not vendor results.

[S86] McDonald's, "FAQ: McDonald's > NEXT," 2026-09-23. Primary source

Use in report: Official management targets: $800k incremental U.S. drive-thru investment, ArchIQ, 250bp expected efficiency, franchisor support.

[S87] Chipotle, Autocado and Augmented Makeline restaurant pilots, 2024-09-16. Primary source

Use in report: Physical named-site restaurant pilots, not fleet ROI.

[S88] Chick-fil-A Tech, "How Our Edge Kubernetes Platform Has Evolved," 2025-09-09. Primary source

Use in report: Restaurant-local distributed computing deployed across chain; architecture source only.

[S89] Starbucks, "Meet Green Dot Assist," 2025-06-10. Primary source

Use in report: Official 35-store initial generative AI assistant pilot.

[S90] Toast, Inc., Form 10-K FY ended 2025-12-31, filed 2026. Primary source

Use in report: GAAP segment revenue, COGS, revenue mix, calculated gross margins.

[S91] PAR Technology Corporation, Form 10-K FY ended 2025-12-31, filed 2026. Primary source

Use in report: GAAP category revenue, costs and named-customer concentration.

[S92] ReFED, "Food Waste by Sector: Foodservice," model based on 2024. Primary source

Use in report: Aggregate food surplus/waste and plate-waste segmentation; NOT AI-addressable supplier revenue.

[S93] National Restaurant Association, "The Hiring and Staffing Dividend," 2026. Primary source

Use in report: Staffing survey findings. The broad outlet footprint is the Association's background estimate in the release's "About the National Restaurant Association" section.

[S94] McDonald's, official 2026-09-23 investor update (SEC exhibit). Primary source

Use in report: NEXT/capital plan [S94]; ~$100k gross benefit: FAQ [S86]; four-year post-partnering payback: official release [S38].

[S95] Chipotle, Hyphen digital makeline official initial test, 2023-10-03. Primary source

Use in report: Task definition and limitations: human completion remains part of workflow.

G.1 Computational Model Checks

Reproducibility checks for the base AI model: five annual project cash flows plus the t=0 investment reconcile to NPV and IRR; the breakeven full-run-rate gross benefit is solved by setting project NPV to zero with the same tax and ramp rules. Supplier-segment gross margins reconcile to reported SEC revenues and cost of revenues. PAR customer exposure is multiplied by rounded disclosed revenue proportions; only approximations should be published.

Table 136 | Model checks and reporting traceability

Check	Verified calculated output	Reconciliation path
AI t=0 capital expenditure	-$80,000	Table 117 initial investment
AI Year 1 through Year 5 FCF	$16,750, $21,812, $26,875, $20,250, $30,250	Direct formula; cash taxes when EBIT positive
Base AI five-year NPV at 12%	$1,507	Table 119 Base; direct discounted sum
Base AI IRR	12.70%	IRR of six period flows
Zero-NPV mature gross annual benefit	$44,325	Root solving on full taxable cash-flow model
Toast subscriptions GAAP GP	$672M	$936M − $264M
PAR subscriptions GAAP GP	$159.143M	$291.170M − $132.027M
PAR top-three approx customer dollars	$163.997M	$455.547M × (21%+8%+7%)

Method / source: Checks reproduce Tables 117–129 and the additional illustrations in Tables 152–153 from report inputs and do not constitute an external audit of underlying company reporting.

Additional checks: Table 152 reproduces four NPVs (−$77,746, −$34,469, +$1,507 and +$68,493) and a $44,325 zero-NPV annual gross-benefit threshold. Table 153 reproduces $716.67 monthly unit contribution, 12 paid units to cover technician cost, capacity of 56 units per technician, and 36.67 months of simple cash payback at 20 units. Full-precision calculations are in the verification records.

Appendix H — Publication Photography and Reproduction Rights Review

Rights baseline: October 26, 2026; original concept-illustration inventory updated through the October 28 supplement. The publication supports research, professional positioning and possible consulting outreach. Public access to a photograph does not establish republication rights. The current figure program comprises 27 analytical figures, including financial charts, and 17 AI-generated concept illustrations; externally owned restaurant photographs remain excluded unless a specific permission, applicable license or separately reviewed legal basis is established. The rights register below is retained. This review is not a legal opinion.

Table 137 | Corporate photography reproduction-rights review

Source	Material	Terms / rights evidence	Circulation status	Editorial action
McDonald's	Corporate media library, restaurant exteriors	Editorial-only media terms; personal/commercial use restricted.	NOT CLEARED	Keep source reference; request specific use authorization.
Starbucks	Green Dot Assist and media library	Media assets stated to be editorial only, with commercial use prohibited.	NOT CLEARED	Link to official project story; obtain rights before embedding.
Chipotle	Autocado and Hyphen official installation photos	General terms reserve reproducing company graphics absent express written consent.	NOT CLEARED	Retain site-specific official press links; seek written permission.
Yum! Brands	Byte by Yum! platform press graphic	Terms restrict copying graphics beyond personal use without written permission.	NOT CLEARED	Use factual paraphrase and original analytical chart instead.
Bear Robotics	Servi Q promotional imagery/CGI	No reusable license or express republication grant verified.	NOT CLEARED	Remove CGI from manuscript; refer readers to original product page.
Wingstop	Smart Kitchen media photographs	Company provides media images; broad commercial reprint license not verified.	NOT CLEARED	Link to media assets; ask investor relations for permission.

Official source/terms URLs (for rights confirmation):

McDonald's: https://corporate.mcdonalds.com/corpmcd/en-us/CorpNewsroom/Media/media-assets/terms-and-conditions.html

Starbucks: https://about.starbucks.com/press-center/

Chipotle: https://www.chipotle.com/terms-of-use

Yum! Brands: https://www.yum.com/wps/portal/yumbrands/Yumbrands/footer-pages/terms-and-conditions

Bear Robotics: https://www.bearrobotics.ai/servi-q

Wingstop: https://ir.wingstop.com/news-and-events/media-assets/

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Appendix I — World Model and Restaurant Commercialization Sources [S96–S102]

[S96] NVIDIA Developer Blog (2026-05-31). Develop Physical AI Reasoning, World, and Action Models with NVIDIA Cosmos 3. Official source

Technical model architecture; not restaurant deployment evidence.

[S97] Google DeepMind (2026-07-30). Gemini Robotics On-Device 2 model card. Official source

Description and limited-access state; no restaurant production proof.

[S98] Starbucks (2026). Supporting the moments that matter with artificial intelligence. Official source

Company description of Smart Queue, employee assistance and future equipment applications.

[S99] Chipotle (2024-09-16). Chipotle Debuts Autocado and the Augmented Makeline by Hyphen in Restaurants. Official source

Official restaurant pilot; not chainwide deployment.

[S100] Uber Investor Relations (2026-10-06). Uber to Acquire ezCater as It Brings Catering to Uber Eats. Official source

Announced $2.3B acquisition, pending closing; gross bookings not acquired revenue.

[S101] DoorDash (2026-09-30). DoorDash Introduces DashOS: One Platform to Help Restaurants Grow. Official source

Operating platform launch; vendor-reported performance evidence.

[S102] DoorDash (2026-09-30). DoorDash Air Unveils Drone Delivery System for Local Businesses. Official source

Pilot and delivery-system announcement; no universal unit economics demonstrated.

Appendix J — Additional Physical AI Architecture, Deployment and Location Sources [S103–S111]

This appendix records sources used in the expanded location-economics, world-model engineering, physical execution and field commercialization analysis. Source owner claims do not independently establish site-level financial benefits. Retrieval and research cutoff: October 26, 2026.

[S103] NVIDIA Newsroom (2026-05-31). NVIDIA Launches Cosmos 3, the Open Frontier Foundation Model for Physical AI.

Original source: Open official publication

Evidence status: Official technical platform claims regarding reasoning, generation and action prediction; not restaurant deployment proof.

[S104] Google DeepMind (2026-07-30). Introducing Gemini Robotics ER 2.

Original source: Open official publication

Evidence status: Official embodied-reasoning capability description, not a verified commercial restaurant installation.

[S105] Chick-fil-A Tech (2025-09-09). How Our Edge Kubernetes Platform Has Evolved.

Original source: Open official publication

Evidence status: Company engineering documentation of per-store edge clusters and optional GPU expansion.

[S106] Starbucks (2026). Making It Easier to Plan Your Starbucks Visit.

Original source: Open official publication

Evidence status: Official Smart Queue and scheduled pickup description; operating claims, not independent labor ROI.

[S107] Luckin Coffee Inc. (2026 filing). Annual Report on Form 20-F for fiscal 2025.

Original source: Open official publication

Evidence status: Company-reported cashier-less app environment and 99.1% pickup share of self-operated stores at December 31, 2025.

[S108] DoorDash Customer Help Center (accessed 2026-10-26). Drone Delivery.

Original source: Open official publication

Evidence status: Current public eligibility, payload, weather and delivery-zone constraints for selected drone operations.

[S109] DoorDash (2025-06-13). DoorDash Completes Acquisition of SevenRooms.

Original source: Open official publication

Evidence status: Official completed acquisition and customer-relationship platform context.

[S110] Sweetgreen / SEC (2025-12-29). Sweetgreen Completes Sale of Spyce to Wonder, Exhibit 99.1.

Original source: Open official publication

Evidence status: Issuer-reported transaction and continued technology use under commercial agreements.

[S111] Perfect Company (2026-05-06). KitchenPro Suite for Wingstop Smart Kitchen deployment.

Original source: Open official publication

Evidence status: Vendor-reported deployment of kitchen displays in more than 2,500 locations; not independently verified buyer-level ROI.

Appendix K — Critical Technical and Commercial Claim Verification

This addition targets source statements most likely to be challenged by world-model engineers, commercial restaurant operators and investment committees. Each record separates the cited party's statement from conclusions that cannot be inferred. Information cutoff: October 26, 2026; future changes require a new verification date.

Table 138 | Technical disclosures and evidence boundaries

Source / date	Affirmatively documented	Do not infer	Manuscript treatment
[S103] NVIDIA (May 31, 2026)	Cosmos 3 physical AI reasoning, simulation and action-generation model announcement	Chainwide restaurant installation or food-contact safety	Technical possibility only
[S104, S112] DeepMind (July 30, 2026)	ER 2 embodied reasoning; model card lists text outputs; lower-level VLA executes motor actions	ER 2 itself certifies or directly controls restaurant actuation	Explicit reasoning/policy/control split
[S105] Chick-fil-A (Sept 9, 2025)	Enterprise restaurant Kubernetes edge platform across stores; 3+ nodes and optional GPU nodes	Specific AI workload return on investment	Architecture evidence
[S106] Starbucks (Apr 22, 2026)	Scheduled mobile pickup windows and company-described Smart Queue sequencing	Independently established labor savings or causal NPV	Real order-to-kitchen use case
[S87] Chipotle (Sept 16, 2024)	Named in-restaurant Autocado and Hyphen pilots, locations disclosed	Fleetwide commercial scale or verified payback	Task-scoped pilot evidence
[S100] Uber (Oct 6, 2026)	Agreement to acquire ezCater for $2.3B, subject to conditions	Closed deal, owned subsidiary, buyer economics	Pending transaction
[S102] DoorDash (Sept 30, 2026)	First-party aircraft and upcoming Northern California pilot details	Universal active service or restaurant unit ROI	Pilot/announcement evidence
[S110] Sweetgreen (Dec 29, 2025)	Spyce sale to Wonder; long-term use and service arrangements	Technology is discontinued or software supplier is unchanged	Ownership / financing transition

Independent interpretation of eight cited commercial and technical disclosures; source statements do not independently establish restaurant ROI.

The table verifies the interpretation of source statements, not independently realized customer ROI or undisclosed contract economics. Some previous sources contain user-facing press statements and should not be treated as audited engineering or investment outcomes.

[S112] Google DeepMind. (July 30, 2026). Gemini Robotics ER 2 Model Card. https://deepmind.google/models/model-cards/gemini-robotics-er-2/. The card states ER 2 is a vision-language model producing text outputs. Google's related model page describes its handoff to a lower-level vision-language-action model for motor execution. https://deepmind.google/models/gemini-robotics/embodied-reasoning/. This source establishes the official developer taxonomy, not restaurant-specific validated performance.

Financial-model reconciliation: the positive Module II base NPV (approximately +$1,507) and the earlier negative-NPV legacy sensitivity use different analyst assumptions. Neither is a real customer result; do not select the attractive example as evidence of sector profitability. Cross-reference Chapter 12 and the explicit Module II cash-flow table before quoting a number.

Appendix L — Restaurant Robotics Implementation Cases and Sources [S113–S130]

This supplemental case universe connects public implementation evidence with the site, task and network framework. It is separate from the U.S. Top-20 sample. Tables 139–143 distinguish reported activity, operating boundaries and proposed measurements. Chipotle and Sweetgreen retain their existing case treatment and source keys.

Dated deployment claims retain the October 26, 2026 evidence cutoff. Undated live locators and process pages were accessed October 27 to identify configurations and sources; they are not used to reconstruct an October 26 active fleet or to claim a new deployment event. Operator, issuer, integrator and product statements retain their source classes.

L.1 Dated deployment evidence

Table 139 | Supplemental robotics deployment evidence

Case	Public evidence	Evidence stage	Builder decision
Miso [S113]	10 third-generation units across three customers; March 31, 2026	Issuer-reported installed fleet	Bounded fry task; lifecycle service and equipment generations
Skylark [S114–S115]	Completed network rollout; December 2022	Operator-documented historical scale	Training, telemetry and local implementation capacity
Donatos [S116–S118]	Named airport system and partners; launch and host records	Named retail configuration	Complete order, production and pickup transaction
Artly [S119–S121]	Company cafe activity and demonstration-learning description	Commercial presence plus vendor technical claims	Versioned action data and held-out transfer tests

Sources: [S113–S121]. Source statements retain their evidence classes; see L.6 for exact locators and permitted use.

L.2 Service tasks and human roles

Table 140 | Service robotics tasks and human roles

Case	Machine contribution	Human role or proposed measurement	Evidence boundary
Skylark	Floor transport in a network rollout	Operator describes instructors and store-level improvement	Historical deployment; no causal cash-return inference
Kura / RobotLAB	Vendor-described navigation and delivery	Measure loading, handoff, route obstruction and support	Fleet averages are integrator reports; per-site hours unresolved
Arirang K / LG	Transport of multiple dishes	Staff load and transfer dishes to diners	Small-site vendor case and testimony
DAWN / OryLab	Telepresence and remotely operated service	Remote human pilots; local assistance remains in scope	Experimental cafe; no autonomous conversational-AI inference

Sources: [S114–S115], [S122–S123], [S125], [S129]. Source statements retain their evidence classes; see L.6 for exact locators and permitted use.

L.3 Preparation and beverage boundaries

Table 141 | Preparation systems and operating boundaries

Case	Physical setting	Learning or control evidence	Unresolved commercial test
Artly	Robotic beverage preparation	Vendor describes demonstrations and VLA; latest hand not assumed fleetwide	Quality-equivalent transfer and per-site service burden
Richtech / ADAM	Supplier-operated beverage flagship	Company describes dual-arm and vision-based execution	External customer adoption and contribution after full operating cost
iWok	Wok workflow and separate boba equipment	Firsthand reporting; supplier/control stack unconfirmed	Prepared inputs, residual work and comparable active sites
Cafe X	Airport robotic coffee retail	Host lists a named service configuration	Occupancy, replenishment, uptime and consumer handoff
Chipotle / Hyphen	Existing named restaurant trial	Purpose-built ingredient assembly; retain Table 62 classification	Measured deployment economics and full-task labor boundary
Skild / Mitsui	Named commercial-kitchen pilot	Company describes S1-powered general-purpose robots	Task results, setup effort and returns at identified kitchens

Sources: [S87], [S99], [S119–S121], [S124], [S126–S128], [S130]. Source statements retain their evidence classes; see L.6 for exact locators and permitted use.

L.4 Proposed operating and financial measurements

Table 142 | Proposed robotics case measurements

Measurement	Required unit and comparison	Customer decision	Supplier decision
Task completion and quality	Accepted tasks, failures and quality criteria over complete shifts	Can the site rely on the service?	Which failures require product or policy changes?
Human work	Paid and unpaid preparation, supervision, handoff, recovery and cleaning time	Is work removed, shifted or improved?	How much training and support is required?
Deployment and uptime	Commissioning hours, ordinary-use date, interruptions and active paid units	What is the cost of installing and maintaining use?	Does each installation become easier to deliver?
Cash realization	Attributable contribution or paid-hour change against a feasible alternative	Does the project improve site cash flow?	Can pricing retain value after service obligations?
Lifecycle cost	Installation, recurring fees, parts, refresh and removal by responsible party	What is the full commitment over the site-use period?	What margin survives warranty and field service?
Transfer and data rights	Versioned held-out conditions, retuning effort and permitted uses	Can the system adapt without undue disruption?	Can learning be reused under the actual agreements?

GII proposed measurement plan; no named-company results or new model assumptions are asserted.

L.5 Source classes and claim qualifications

Table 143 | Supplemental case sources and qualifications

Case and sources	Source class	Supported use	Required qualification
Miso [S113]	Issuer filing	Dated fleet and product/service structure	Combined customers; no measured operator ROI
Skylark [S114–S115]	Operator reports	Historical rollout and implementation process	Current active fleet and causal cash effects not established
Donatos [S116–S118]	Brand and airport host	Named configuration and partnership	Plans, locator and observed performance are distinct
Artly [S119–S121]	Company and technology supplier	Commercial activity and described learning method	No independent model validation or uniform hardware version
Kura [S122–S123]	Integrator and operator filing	Integration account; operator equipment reliance	Later metrics remain vendor claims; hours need reconciliation
ADAM [S124]	Issuer release	Own-concept activity	No inference of outside-chain procurement or profit
DAWN [S125]	Operator description	Remote human service arrangement	Experimental teleoperation; retain human labor
iWok [S126–S127]	Firsthand press and operator locator	Named workflow and addresses	Control stack and active-location status remain qualified
Cafe X [S128]	Airport host locator	Named robotic-coffee retail setting	No dated fleet, uptime or unit-return inference
Arirang K [S129]	Vendor case/testimony	Small-site transport and handoff	No causal financial evaluation
Skild [S130]	Founder announcement	Named kitchen pilot	No named restaurant fleet or disclosed kitchen returns

Sources: [S113–S130]. Source statements retain their evidence classes; see L.6 for exact locators and permitted use.

L.6 Sources S113 to S130

Access date for the following source checks is October 27, 2026. Dates below identify the publication or observation period where available. An issuer filing or vendor description is primary evidence of that party's statement, rather than independent validation of every operating outcome.

[S113] Miso Robotics. FY2025 Form 1-K, business and product disclosures. FY ended 2025-12-31; fleet dated 2026-03-31. Issuer filing. Source link

Permitted use: Dated combined third-generation fleet, sales/lease structure and service obligations; prospective benefits remain management claims.

[S114] Skylark Holdings. Integrated report DX section. Report extract in 2024 collection; publication day unstated. Restaurant operator report. Source link

Permitted use: Implementation process, instructors and operational checking; no independent causal financial evaluation.

[S115] Skylark Holdings. Integrated report company history. Report extract in 2024 collection; event December 2022. Restaurant operator report. Source link

Permitted use: Completed historical rollout; not an October 2026 active-fleet census.

[S116] Donatos Pizza. Donatos Pizza To Open Fully Autonomous Restaurant in Partnership with Appetronix. 2025-05-27. Restaurant brand announcement. Source link

Permitted use: Named partners, proposed June opening and airport configuration; announcement is distinct from operating performance.

[S117] Donatos Pizza. Franchise frequently asked questions. Undated live page; accessed 2026-10-27. Restaurant brand locator and process description. Source link

Permitted use: Identifies airport Pepp-Tron and central dough preparation. Conventional restaurant franchise costs are not robotic-kiosk costs.

[S118] Columbus Regional Airport Authority. Board pre-read materials. 2025-04-29. Airport host planning record. Source link

Permitted use: Proposed ordering and pickup configuration; planned cycle time is not measured throughput.

[S119] Artly AI. Artly debuts mini Barista Bot at National Restaurant Association 2025 show. 2025-05-17. Company-issued release distributed by PR Newswire. Source link

Permitted use: Company-reported cafe/cup counts and demonstration-trained VLA description; no independent model benchmark or site return.

[S120] Artly. Artly Presents Natural Robotic Hand Upgrade for Barista Bot at Nvidia GTC 2025. 2025-03-17. Company product announcement. Source link

Permitted use: Announced upcoming hand upgrade; does not establish installation of that hand at every cafe.

[S121] MANUS. Artly AI approach to scalable embodied AI for real-world tasks. Publication day unstated; accessed 2026-10-27. Technology supplier account. Source link

Permitted use: Motion-capture data workflow described by a commercial partner; not independent validation.

[S122] RobotLAB. Kura Sushi Integrates KettyBot to Elevate Service and Scale Smarter. 2026-04-05; Q1 2026 fleet averages. Integrator case study. Source link

Permitted use: Named integration account and reported fleet. Per-site hours require reconciliation; estimated labor value is not payroll savings.

[S123] Kura Sushi USA. FY2025 Form 10-K. Filed 2025-11-06; FY ended 2025-08-31. Restaurant operator filing. Source link

Permitted use: Corroborates robotic service and equipment dependence; does not independently confirm the integrator Q1 2026 metrics.

[S124] Richtech Robotics. ADAM Surpasses 16000 Drinks Served at Flagship Las Vegas Location. 2025-06-17. Issuer release. Source link

Permitted use: Company-reported activity at its own flagship; not third-party fleet adoption or store profitability.

[S125] OryLab. BUNSHIN Robot Cafe DAWN operating and Tele-Barista descriptions. Undated live page; accessed 2026-10-27. Cafe operator description. Source link

Permitted use: Human remote operation and experimental service arrangement; no autonomous conversational-AI claim.

[S126] Houston Chronicle. New fast-casual Asian fusion spot to open with robots instead of a chef. 2026-01-16. Firsthand reporting and operator interview. Source link

Permitted use: Observed cooking/boba setup and planned opening; staffing intentions are not measured operating results.

[S127] iWok. Official locations. Undated live page; accessed 2026-10-27. Restaurant operator locator. Source link

Permitted use: Named addresses only; inconsistent live Katy status prevents an active-fleet inference.

[S128] San Francisco International Airport. Cafe X retail listing. Undated live page; accessed 2026-10-27. Airport host locator. Source link

Permitted use: Named airport robotic-coffee configuration; not a dated network count or uptime/profit test.

[S129] LG Business Solutions. Arirang K case study. Publication day unstated; describes June 2022 deployment. Vendor case with customer testimony. Source link

Permitted use: Small-site plate-transport and human-handoff account; no disclosed causal financial result.

[S130] Skild AI. The Hidden Pillar of Robotics. 2026-09-10. Founder/company announcement. Source link

Permitted use: Named Mitsui commercial-kitchen pilot; no restaurant fleet or unit-return disclosure. Partner meal volume is not robot throughput.

Appendix M — Location Networks, Drone Cases and Adoption Evidence

Supplement checked October 28, 2026. The October 27 ODSC report informed GII's research framing; technical and deployment statements below cite their controlling primary sources. Earlier source entries retain their original dates and evidence cutoff. This supplement does not add a nationwide active-store census or certify all inherited sources.

M.1 U.S. drone delivery evidence

Table 144 | U.S. restaurant drone cases and evidence limits

Case / source date	Documented scope	Evidence stage	Unresolved adoption and returns
Wendy's / Wing / DoorDash 2024-03-21 [S133]	One named Christiansburg restaurant	Single-store pilot; bounded proof of concept	Current store count, customer preference, staff burden, manager continuation and site returns not established here
Chipotle / Zipline 2025-08-21 [S134]	Named Rowlett early-access launch	Limited early service; proof of handoff feasibility	Broader rollout and acceptance by customers, employees and store managers not established here
Flytrex / DoorDash 2025-06-26 [S135]	Limited parts of Little Elm / Frisco; named restaurant partners	Local service following a pilot	Reachable households and merchant participation are not installed-store counts or independent satisfaction results
Brookfield / Wing / DoorDash 2024-12-18 [S136]	Two named Texas shopping-center hosts	Property-based launch / trial announcement	50+ eligible merchants do not establish 50 restaurant launch sites, tenant acceptance or property-level returns

Company announcements support the dated scope above. They are not a current nationwide store census. Proof of concept and limited service do not establish broad restaurant rollout or social acceptance.

M.2 Restaurant location functions

Table 145 | Location functions and value-realization tests

Location function	Potential operating value	Site and human constraint	Evidence to collect
Food production / kitchen automation	Consistency or useful production capacity	Ventilation, equipment fit, cleaning, replenishment and employee recovery work	Successful complete tasks, food quality, paid work and net incremental contribution
Dining and customer service	On-site sales and repeat demand	Guest preferences, aisle use and service quality	Complaints, repeat visits, order accuracy and effects on existing guests
QSR pickup point	Fast first-party or marketplace collection	Frontage, holding space, security and crowding	Pickup delays, remakes, channel switching and contribution after fees
Courier / ground-robot handoff	Reach beyond the dining room	Loading, staff walking, access and failed collection	Full handoff work, quality at arrival and incremental paid orders
Drone dispatch interface	Bounded local aerial fulfillment	Controlled space, site permissions, loading work and operating limitations	Eligible routes, completed deliveries, refusals, downtime and full channel cost
Network learning and service	Potential lower replication and support cost	Data rights, layout variation, human acceptance and service coverage	Held-out-site tests, next-site install cost, renewal and retained supplier margin

GII proposed framework. These are possible functions and measurements, not verified benefits, universal permitted uses or new numerical model assumptions.

M.3 Acceptance and continued use

Table 146 | Proposed customer and store-team acceptance register

Decision-maker / user	Possible burden or objection	Observation and comparison	Continuation evidence
Customers	Collection restrictions, noise, privacy concerns, poor food condition or loss of human service	Voluntary choice, repeat use, complaints and failed-order recovery versus available delivery alternatives	Users continue to choose the service without concealed quality or access trade-offs
Employees	Extra packing, loading, walking, monitoring, rescue work or concern about job roles	Complete-shift work logs, training needs and confidential feedback; include those who decline	Workload and safety are acceptable; unresolved extra tasks are assigned and resourced
Store managers	Peak-hour disruption, unclear responsibility, new costs and service escalations	Peak versus off-peak outcomes, paid work, refunds, support response and net margin	Managers choose continued use after normal operating experience, not only a subsidized demonstration
Franchisee / property owner	Space opportunity cost, access conflicts and misaligned capital obligations	Identify payer, property rights, tenant effects and recurring support obligations	The responsible owner accepts the site economics and operating arrangements
Robotics / delivery supplier	Bespoke installation, support visits and inactive contracted locations	Installation effort, successful service, retention and full support cost by site cohort	Repeatable deployment and retained margin accompany accepted customer use

GII proposed adoption tests, informed by hospitality continuance research [S137]. Potential objections are hypotheses; this table does not claim that participants in the named drone cases were dissatisfied.

M.4 Sources S131 to S138

[S131] NVIDIA. Alpamayo-R1: Bridging Reasoning and Action Prediction for Generalizable Autonomous Driving in the Long Tail. 2026-01-07. Source link

Technical primary source; automobile reasoning/action research, not restaurant deployment or safety certification.

[S132] Pudu Robotics. Sushi Express multi-robot service case. Undated; accessed 2026-10-28. Source link

Vendor-reported Taiwan workflow; no dated store census, independent financial result or general acceptance measure.

[S133] DoorDash. DoorDash and Wing announce U.S. drone delivery pilot. 2024-03-21. Source link

Company announcement; one named Wendy's pilot, not national restaurant rollout evidence.

[S134] Chipotle. Introducing Zipotle: aerial delivery with Zipline. 2025-08-21. Source link

Operator announcement; named Rowlett early-access service; no established broad acceptance or store returns.

[S135] DoorDash. Flytrex drone delivery launch in Dallas–Fort Worth. 2025-06-26. Source link

Company-reported limited-area service; access, deliveries and merchant participation are different denominators.

[S136] Wing. Brookfield shopping-center drone delivery with DoorDash. 2024-12-18. Source link

Company launch/trial announcement at two named properties; merchant eligibility is not installed restaurant-site count.

[S137] Tuomi, Ashton, Schmidt and Ling. After the Hype: Managerial Impacts of Planned (Dis)continuance of Service Robot Usage. 2025-06-18. Source link

Original hospitality interview research; 8 suppliers and 10 decision-makers; not evidence about acceptance of these four drone cases.

[S138] Global Innovation Institute. From Autonomous Cars to Intelligent Restaurant Robots: ODSC Marco Pavone research report, v0.2. Event 2026-10-27; reviewed 2026-10-28.

Author-supplied unpublished working report. Restaurant applications are GII interpretation; partial event records are not a complete speaker transcript. Primary claims are cited directly above.

Appendix N — Restaurant Robotics Engineering and Production Integration

Technical supplement checked October 28, 2026. Original cutoffs and source IDs remain. The exhibits distinguish original disclosures, developer evaluations, GII engineering judgments and new analyst scenarios; they do not add a restaurant fleet census. Existing financial assumptions are preserved.

N.1 Technology and implementation evidence

Table 147 | Restaurant robotics technology and implementation stage

System / source	Technical approach	Current evidence stage	Next operating test
Miso vision upgrade [S139]	3D recognition; local GPU; vision-guided fry station	October 2026 lab demonstration; December shipments expected	Installed generation; peak accuracy; cleaning and fault recovery
Artly [S140,S141]	Demonstrations, skill library and vendor-described VLA	Commercial beverage setting; internal performance not independently tested here	Same-quality drinks under changed tools; assistance and retuning
Skild S1 / Mitsui [S142,S130]	Video-context foundation policy	Developer tests plus commercial-kitchen pilot disclosure	Whole-job success at service speed; pilot site denominator
Hyphen / Ricoh [S144]	Digital-order assembly and installation/service support	October 2026 partnership; rollout outcomes not established	Commissioning time; uptime; human replenishment and service cost
Chef Robotics [S145,S146]	Learned food manipulation and RaaS	Prepared-food production comparator; not restaurant-fleet evidence	Ingredient variability; changeover; cleanability; usable output
Google / PI [S97,S148–S150]	Local VLA or hosted model adaptation/inference	Model access, developer evaluations and beta offerings	Task validation; latency/outage response; embodiment transfer

Technical approach and deployment maturity are separate fields. Sources checked October 28, 2026; claims retain their original dates. No row certifies causal restaurant returns.

N.2 Robot embodiment and site constraints

Table 148 | Robot base choice and restaurant operating constraints

Requirement	Fixed manipulation cell	Tray-delivery robot	Wheeled manipulator	Bipedal humanoid
Execution	Bounded station task	Transport; people may load and unload	Arms plus mobile workstation access	Whole-body movement and manipulation
Access	Fixture, utilities and dedicated footprint	Suitable floors, routes and aisle clearance	Also docking and arm reach at each workstation	Potential steps; foot placement and recovery validation
Precision	Calibrated tools and repeatable geometry	Stable transport and correct destination	Base calibration and stable arm operation	Coordinate arm forces with balance and changing support
Hot loads	Containment and validated transfer	Acceleration, braking and tray stability	Also grip control and liquid slosh	Also footfall motion and balance-failure consequences
Maintenance	Tools, guards and fixtures	Chassis, charging and load surfaces	Adds arms, tools and docking	Adds legs, feet and whole-body recovery
Shared gates	Food quality, safe recovery and cleanability	Accepted workflow and residual human work	All relevant quality, hygiene and task gates	Human-like form grants no food-contact approval

GII engineering assessment. Research examples: wheeled bimanual Mobile ALOHA [S154] and Google whole-body work [S148]. Compare the installed configuration, not the humanoid label.

N.3 Engineering release scorecard

Table 149 | Engineering scorecard for restaurant deployment

Measure	Required denominator	Record together	Release interpretation
First-pass task success	Accepted attempts / all attempted tasks	Failed attempts, retries and product rejection	A corrected completion is not first-pass success
Unassisted job completion	Complete accepted jobs without help / all jobs	Time limit, step count and intervention policy	Benchmark steps do not equal complete restaurant orders
Peak throughput	Accepted output / scheduled peak hour	Order mix, packing, replenishment and changeover	Validate the system bottleneck, not fastest arm speed
Intervention burden	Human minutes / accepted output	Load/unload, resets, cleaning and remote assistance	Measure displaced work and new work separately
Availability and recovery	Usable time / required service window	Peak outages, safe stop and restart duration	Annual average may hide unacceptable rush-hour failures
Food quality and hazards	Defects / exposed attempts, by severity	Temperature, portion, contamination and spill checks	Critical hazards require independent acceptance criteria
Site transfer	Held-out sites before local tuning	Hardware/policy version; data and retuning hours	Count the cost to reproduce the result at the next site

GII acceptance protocol; no universal numeric release limit is asserted. Limits follow the permitted task, hazard assessment, service promise and economics.

N.4 Order and physical-state interface

Table 150 | Order and physical-state interface contract

State / event	Minimum operational content	Source / owner	Failure to prevent
Order and amendment	Order/item IDs; quantities; modifiers; channel; promise; freshness	POS/channel adapter; authorized order record	Lost changes, duplicate meals, unauthorized substitution
Resource availability	Station skill; staff assignment; equipment mode; ingredients	KDS, equipment adapter and authorized staff	Double-booked chef; unavailable machine or ingredient
Job release and acknowledgment	Job ID; order version; station; release time; accepted state	Scheduler and executor	Command sent treated as physical work completed
Started / paused / failed	Actual start; cause; in-process food disposition; remaining work	Human station or bounded machine controller	Blind replay after a partial action or outage
Completed and quality approved	Physical completion; accepted quality; rework disposition	Station and quality/expediter authority	Failed product sent to packing
Packed and handed off	Item reconciliation; package identity; verified collection event	Expediter / pickup interface	Wrong bag, missing item or premature ready message
Stop / restart / reconcile	Local stop authority; durable job state; manual recovery	Validated controller and authorized operator	Cloud or network fault defeating the safe state

GII proposed system contract. Toast [S151–S153] supplies examples of available events and limitations, not the full architecture. Quality and handoff require their own observations.

N.5 Service format and scheduling

Table 151 | QSR and table-service scheduling requirements

Decision	QSR and pickup	Sit-down restaurant	Common physical constraint
Service objective	Short promises by channel; peak output	Courses and dishes synchronized to table pace	Accepted food within quality and holding windows
Production release	Batch compatible demand without starving a channel	Fire short- and long-cook items to a coordinated finish	Station time, recipe dependencies and changeovers
Labor capacity	Prep, cook, assemble, pack and recover robots	Chef specialties, expediting, runners and guest pacing	One worker's minutes cannot be assigned twice
Demand overload	Revise available promises or permitted channel intake	Adjust seating/ordering pace and communicate waits	No scheduler can create missing machine or chef capacity
Completion	Correct package ready at promised handoff	Complete course ready for coordinated service	Command acknowledgment is not quality approval
Success measurement	Ready-at-promise, dwell, remakes and net contribution	Table completion spread, holds, rework and guest experience	Staff burden, outages and rejected output remain visible

GII operating design. Starbucks [S106] provides a disclosed sequencing example. The teaching bottleneck in Section 7.12 assumes compatible order equivalents and fixed mix.

N.6 AUV sensitivity with explicit scaling

Table 152 | AUV sensitivity under a stated benefit-ratio assumption

Annual sales	Mature benefit at 1.5%	Five-year NPV at 12%	Required mature benefit / sales
$1,000,000	$15,000	-$77,746	4.43%
$2,000,000	$30,000	-$34,469	2.22%
$3,000,000	$45,000	$1,507	1.48%
$5,000,000	$75,000	$68,493	0.89%

New GII sensitivity only. All Module II cost, tax, depreciation, ramp and refresh assumptions are retained; mature benefit alone is assumed to scale with sales. Actual labor, waste and revenue benefits need separate drivers. Required gross benefit is $44,325; the example is not a store-eligibility census.

N.7 Hardware RaaS and service density

Table 153 | Separate hardware RaaS illustration and service capacity

Input / result	Assumed or calculated value	Treatment	Operating implication
Capital per installed robot	$30,000 hardware + $3,000 installation	Supplier-funded; $33,000 initial cash	Not the earlier restaurant $80,000 model
Fee and other monthly cost	$1,500 fee; $200 other direct cost	Parts/monitoring reserve included in $200	No vendor quotation or achieved margin claim
Life and installation allocation	60-month hardware life; 36-month installation allocation	Zero residual; $500 + $83.33 per unit/month	Accounting allocation; actual replacement cash separate
Technician and capacity	$8,000/month; 140 service hours available	1.5 on-site + 1.0 travel hours per unit/month	One technician supports at most 56 units on these averages
Cost break-even	12 active paid units	Unit contribution before technician $716.67/month	Requires route feasibility and contract response times
Illustrative route at 20 units	$6,333.33/month delivery contribution	20 × $716.6667 less $8,000 technician	Excludes corporate overhead, sales, financing and tax
Capital recovery at 20 units	36.67 months simple cash payback	Cash per unit $900/month after technician allocation	No discounting, churn, late activation or renewal guarantee

All values are new analyst assumptions or derived outputs, not company data. Monthly accounting contribution before technician = fee − other cost − hardware depreciation − installation allocation. Required units = ceiling(8,000 / 716.6667); capacity = floor(140 / 2.5). At 20 units, simple cash payback = 33,000 / (1,500 − 200 − 8,000/20).

N.8 Restaurant production coordination architecture

The scheduler allocates restaurant resources. Human and machine executors return observed status; quality and handoff return confirmed outcomes. Local controllers retain safety and recovery authority.

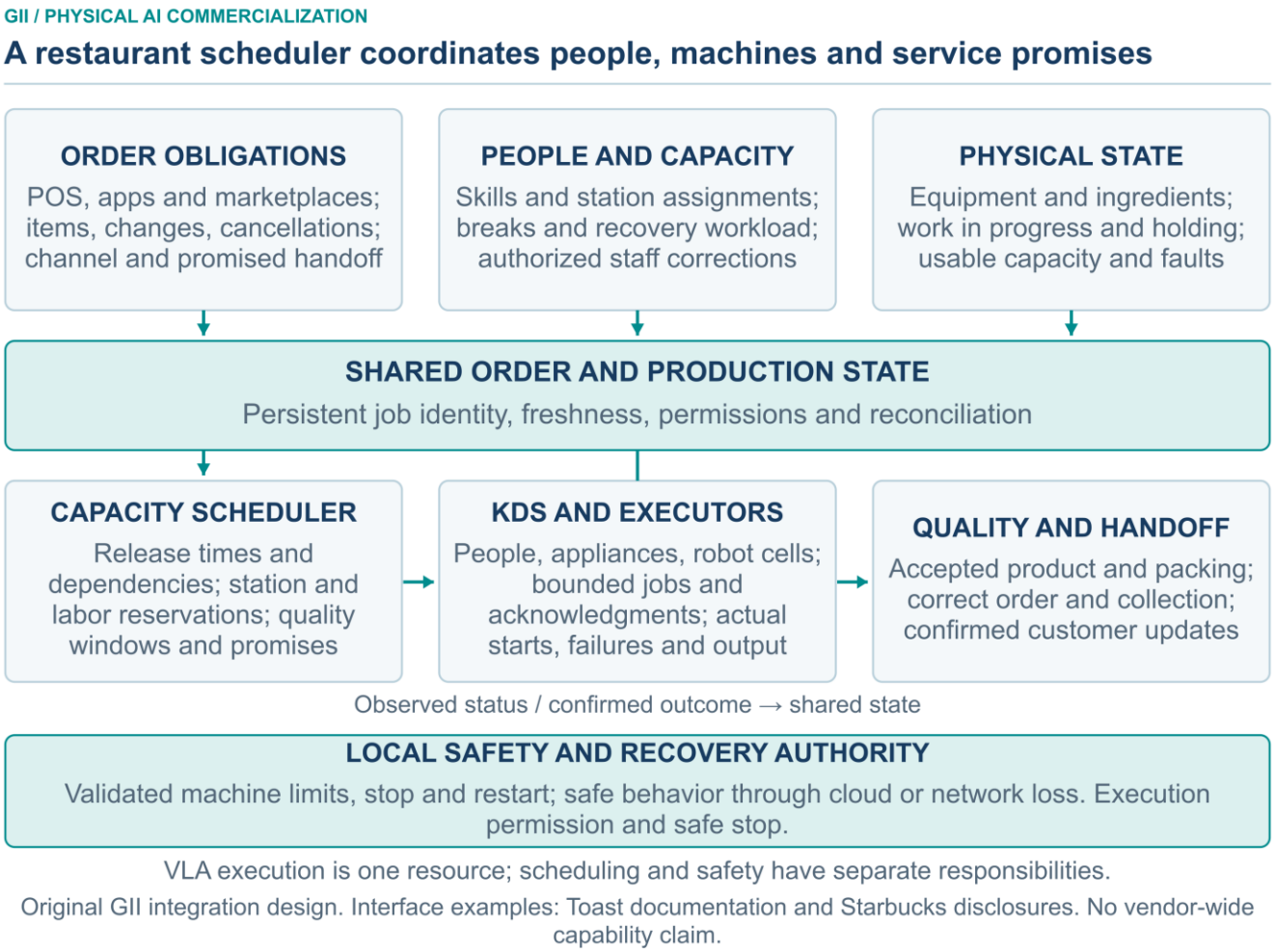


Figure 27 | Coordinating orders, people, equipment and confirmed fulfillment

GII original integration design; interface evidence [S151–S153] and Starbucks sequencing [S106]. The full architecture is proposed, not a disclosed capability of one vendor.

N.9 Sources S139 to S154

[S139] Miso Robotics. Flippy Can Now See in 3D. 2026-10-20. Source link

Company-hosted announcement of an October 15 laboratory demonstration; December shipments were expected. The new vision upgrade is separate from the existing installed fleet.

[S140] Artly AI. Barista Bot. Undated; checked 2026-10-28. Source link

Vendor description of demonstration learning, skill composition and VLA use; no independent cross-site success or intervention denominator.

[S141] Artly AI. Robot School. Undated; checked 2026-10-28. Source link

Vendor architecture covering multimodal teleoperation, embodiment adapters and staged learning. Announced architecture is not an installed-version census.

[S142] Skild AI. Introducing S1 In-Context Learning for Robotics. August 2026. Source link

Developer metric: “average of cumulative per-step success rate across all tasks.” At 100,000 pretraining hours: approximately 96% seen and 66% unseen; 4–8-minute tasks. Recovery interventions primarily concern the VLA baseline. Not restaurant first-pass order completion.

[S143] Alias of [S130]. Skild AI. The Hidden Pillar of Robotics. 2026-09-10. Source link

Company disclosure of S1 commercial-kitchen pilots with Mitsui; partner meal volume is not robot production or an installed-kitchen count.

[S144] Ricoh USA. Ricoh and Hyphen partner to drive robotic automation in food service industry. 2026-10-15. Source link

Issuer announcement of makeline installation and lifecycle support. Partnership capability is not achieved chainwide rollout.

[S145] Chef Robotics. Artificial intelligence Food Foundation Model. Undated; checked 2026-10-28. Source link

Vendor food-manipulation description. Production counters and training-volume labels on this live page are inconsistent; no volume is imported.

[S146] Chef Robotics. Physical AI for the food industry and Robotics as a Service. Undated; checked 2026-10-28. Source link

Food-manufacturing comparator and vendor service scope; supplier-reported NSF/ANSI 169 certification names module C-001748 and is configuration-specific, not blanket restaurant approval.

Sources S147 to S154 continued

[S147] Alias of [S97]. Google DeepMind. Gemini Robotics On-Device 2 Model Card. 2026-07-30. Source link

VLA numerical-action outputs, local execution and trusted-tester access; no restaurant production certification.

[S148] Google DeepMind. Gemini Robotics 2 brings whole body intelligence to robots. 2026-07-30. Source link

Developer demonstrations and evaluations of whole-body control and dexterity. ER reasoning and motor-action models have different roles.

[S149] Physical Intelligence. Frontier Models for Every Robot Announcing our API beta. 2026-10-13. Source link

Managed fine-tuning and hosted inference beta; announcement acknowledges connectivity issues. API availability is not deployment readiness.

[S150] Physical Intelligence. A Steerable Model with Emergent Capabilities. 2026-04-16. Source link

Original developer work on multimodal prompting, skill composition and kitchen-appliance tasks; not commercial restaurant validation.

[S151] Toast. Guest order fulfillment status webhook. Live developer documentation; checked 2026-10-28. Source link

Preparation/pickup/closure/void transitions; does not report every item amendment. CLOSED is a platform status, not sensor-confirmed physical handoff.

[S152] Toast. Orders webhook. Live developer documentation; checked 2026-10-28. Source link

Order creation/update messages contain full order information with documented omissions; event identity, timestamps and privacy limits matter.

[S153] Toast. Restaurant availability webhook. Live developer documentation; checked 2026-10-28. Source link

Online/channel availability events with configuration limitations. Does not supply a complete station or labor-capacity model.

[S154] Zipeng Fu, Tony Z. Zhao and Chelsea Finn. Mobile ALOHA Learning Bimanual Mobile Manipulation with Low-Cost Whole-Body Teleoperation. CoRL 2024. Source link

Original wheeled bimanual mobile-manipulation research, including cooking tasks. Laboratory task results are not ordinary restaurant readiness.